

Pharmacokinetics in Aging: Optimizing Drug Therapies for Geriatric Patients

Pharmacokinetics in Aging: Optimizing
Drug Therapies for Geriatric Patients



A. Elma Edwina

A. Elma Edwina

KU Leuven
Biomedical Sciences Group
Faculty of Medicine
Department of Department of Public Health and Primary Care
Gerontology and Geriatrics Research Unit



Pharmacokinetics in Aging: Optimizing Drug Therapies for Geriatric Patients

Angela Elma EDWINA

Promotor	: Prof. Dr. Jos Tournoy
Co-promotors	: Prof. Dr. Isabel Spriet Prof. Dr. Lorenz Van der Linden Prof. Dr. Erwin Dreesen
Chair examining committee	: Prof. Dr. Patrick Callaerts
Chair public defence	: Prof. Dr. Fabianne Dobbels
Jury members	: Prof. Dr. Karel Allegaert Prof. Dr. Pieter Annaert Prof. Dr. Mirko Petrovic Prof. Dr. Birgit Koch

Dissertation presented in
partial fulfilment of the
requirements for the
degree of Doctor in
Medical Sciences

Leuven, 19 February 2025

Wednesday, 19 February 2025, 17h00

Aula van de Tweede Hoofdwet
Thermotechnisch Instituut
Kasteelpark Arenberg 41
3001 Heverlee
Belgium

Gerontology and Geriatrics Unit
Department of Public Health and Primary Care
UZ Herestraat 49 – box 7003
3000 Leuven
Belgium

Contents

Words of Thanks.....	i
List of abbreviations	iv
General Introduction	1
1. Geriatric patients and the lack of their representation in clinical trials	2
2. Age-related (patho)-physiological changes altering pharmacokinetics and pharmacodynamics in geriatric patients.....	3
2.1. Intraluminal pH.....	4
2.2. Gastrointestinal transit time	5
3. Frequent morbidities in geriatric patients and their treatment	6
3.1. Atrial fibrillation and edoxaban.....	6
3.2. Gram-Positive bacterial infections and vancomycin	8
4. Pharmacometrics modeling.....	11
4.1. Model-informed precision dosing: combining therapeutic drug monitoring and population modeling.....	18
4.2. The potential importance of pharmacometrics modeling in geriatric patients 18	
References	20
Objectives	27
Part 1. Gastrointestinal physiology	31
Chapter 1. Gastrointestinal transit and intraluminal pH dynamics in aging populations using wireless motility capsule.....	33
Abstract	34
1. Introduction	35
2. Methods	36
2.1. Study participants.....	36
2.2. Study procedures.....	36
2.3. Data collection	37
2.4. Historical healthy young controls data	37
2.5. Data processing and statistical analysis	38
3. Results	41
3.1. Study participants.....	41
3.2. Transit time and intragastric pressure	41
3.3. Intraluminal pH.....	44
4. Discussion.....	45
5. Conclusion	51

References	52
Appendix	55
Part 2. Pharmacokinetics of Direct Oral Anticoagulants	61
Chapter 2. Insights into the Pharmacokinetics and Pharmacodynamics of Direct Oral Anticoagulants in Older Adults with Atrial Fibrillation: A Structured Narrative Review	63
Abstract	64
1. Introduction	65
2. Methods	66
2.1. Eligibility Criteria and Study Selections	66
2.2. Data extraction	68
3. Results	68
3.1. Search and included studies.....	68
3.2. Exposure metrics in studies including older adults	69
3.3. PK parameters of DOACs	71
3.4. Characteristics related to pharmacokinetics of DOACs in older adults.....	71
3.5. The relationship between DOACs concentrations and clinical outcomes	83
3.6. Coagulation biomarkers as PD endpoints	84
4. Discussion: unmet needs and future directions	88
4.1. Does advanced age influence DOACs PK/PD?	88
4.2. Is DOACs dosing optimal in this special population with AF?	90
4.3. The potential role of therapeutic drug monitoring as tool to optimize DOACs therapy in older adults	92
4.4. Future direction	92
4.5. Limitations	93
5. Conclusion	94
References	96
Appendix	102
Chapter 3. Fixed-Dose Edoxaban in Frail Geriatric Patients: A Population Pharmacokinetics Study	123
Abstract	124
1. Introduction	125
2. Methods	126
2.1. Study Design and Population	126
2.2. Edoxaban and Factor Xa Bioassay.....	127
2.3. Population PK Modelling and Simulation	127

2.4. Establishing The Relationship Between Edoxaban Concentrations and Anti-Factor Xa Activity.....	128
2.5. Software.....	128
3. Results	131
3.1. Study Population.....	131
3.2. Population Pharmacokinetics	134
3.3. The correlation between edoxaban concentrations and anti-Xa activity values	134
3.4. Dosing simulation	134
4. Discussion.....	136
References	141
Appendix.....	144
Part 3. Pharmacokinetics of Vancomycin	153
Chapter 4. Decreased kidney function explains higher vancomycin exposure in older adults	155
Abstract	156
1. Introduction	157
2. Methods	158
2.1. Exposures and study endpoints	158
2.2. Local vancomycin dosing guideline	158
2.3. Patient characteristics	159
2.4. Statistical analysis	159
2.5. Software.....	160
3. Results	161
4. Discussion.....	166
5. Conclusion	169
References	171
Appendix	174
Chapter 5: Optimizing vancomycin dosing in older patients: predictive performance of pharmacometrics models.....	187
Abstract	188
1. Introduction	189
2. Methods	190
2.1. Clinical data	190
2.2. Candidate models	190
2.3. Model evaluations.....	190

2.4.	Multi-model approaches	191
2.5.	Predictive performance evaluations	191
2.6.	Robustness analysis, sensitivity analysis and Bland–Altman analysis.....	192
2.7.	Software.....	192
3.	Results	194
3.1.	Clinical data	194
3.2.	Population pharmacokinetics models of vancomycin	194
3.3.	Model evaluations.....	194
3.4.	Influence of the models in the multi model approaches	194
3.5.	Predictive performances evaluations	197
3.6.	Robustness analysis, sensitivity analysis and Bland–Altman analysis.....	197
3.7.	Evaluation of the predicted AUC_{0-24h}	200
4.	Discussion.....	200
	References.....	203
	Appendix.....	205
	General discussion.....	235
1.	Main findings and implications for clinical practice	236
2.	Limitations.....	242
3.	Future perspectives.....	243
	References.....	246
	Summary	249
	Curriculum Vitae	253
	Acknowledgements, Personal Contribution and Conflict of Interest Statements	259
	Acknowledgements	260
	Personal contribution.....	261
	Financial supports	261
	Conflict of interest	261

Words of Thanks

"It always seems impossible until it's done." – Nelson Mandela

Pursuing a PhD was one of the best decisions I've ever made. It has not only been an academic journey but also one of tremendous personal growth. Of course, there were many moments of doubt when I questioned my abilities and wondered if I was truly capable of completing this work. Yet, step by step, I kept moving forward until the finish line came into view.

Completing this PhD in 3.5 years would have been impossible without the unwavering support of my supervisory team, who were always there when I needed their advice and guidance.

I would like to express my gratitude to my promotor, Prof. Jos Tournoy. You believed in me even when I doubted myself. You are always interested in my work particularly in modeling for geriatric patients' data which kept me motivated throughout this journey and gave me confidence that this research is impactful. From you, I have not only learned about research and clinical studies but also about the importance of diplomacy and self-accountability. Thank you for your guidance, for always support my passion, for giving me the freedom to grow, and for always promptly answer my questions and help me, even with your busy schedule.

I also thank my co-promotor Prof. Isabel Spriet, who is my inspiration as a woman in science. I am very grateful for our many discussions and the insights you shared. Your enthusiasm for my research updates always encouraged me to stay motivated and keep moving forward.

I would like to thank my co-promotor, Prof. Lorenz Van der Linden, for sharing your knowledge, especially in cardiovascular medicine. Your good humor, uplifting emails, and our lively discussions—many of which happened spontaneously in the hallways—have made this journey both enjoyable and memorable.

Thank you to my co-promotor, Prof. Erwin Dreesen. Your hands-on help with modeling has been so valuable, and you have been my go-to person for anything related to modeling and statistics. You have a very nice way of making difficult topics easier to understand and apply, which has inspired me to follow your path as a pharmacometrician.

To my internal jury members, Prof. Pieter Annaert and Prof. Karel Allegaert, thank you for being there throughout my PhD journey, giving me invaluable inputs for my work and challenge my critical thinking during yearly evaluations.

To my external jury members, Prof. Birgit Koch and Prof. Mirko Petrovic, thank you for your time to assess my thesis, providing constructive feedback, and being part of this important milestone in my academic journey.

I would like to also thank Prof. Patrick Callaerts to act as a chair of the examining committee and Prof. Fabienne Dobbels who chairs the PhD defense.

Dr. Mirko Koziolk and Dr. Álvaro López Mármol, thank you for your patience supervising me during my internship at AbbVie. Because of you, I got a chance to learn many new things and experiment with various in vitro tests, it was very nice experience also to learn about pharmaceutical industry.

"Many hands make light work." – John Heywood.

The works presented in this thesis are the result of collective efforts from many individuals.

Thanks to the patients/study participants and their caregivers. Without you, this research would not have been possible. My gratitude is immense. It is truly remarkable to see that, despite everything, you have not lost sight of the others and future patients.

I thank my students, Louise and Hanne, for your help with data collection and practicality of clinical studies. All those early mornings starting at 6 AM have truly paid off. Your efforts have not only contributed significantly to this work but have also helped me grow as both a researcher and a teacher.

I also thank the clinicians, geriatrician supervisors, head nurses, nurses, and dieticians at the geriatric wards for always being available to answer our questions and provide assistance whenever needed during the inclusion of our patients.

Thank you also to Dr. Jürgen Burhenne and Dr. Kathrin Foerster for your valuable work in analyzing our samples.

Thanks to all other co-authors in our articles: Prof. Peter Verhamme, Prof. Thomas Vanassche, Dr. Matthias Gijzen, Nada Dia, Dr. Nele Steenackers, Prof. Tim Vanuytsel, Prof. Christophe Matthys, Cleo Demeester, Donnia Robins, Prof. Patrick Augustijns, Dr. Ibrahim Ince, Dr. Andreas Lehmann, Prof. Maria Vertzoni, Dr. Jan Schlender.

I am just thankful I am surrounded by great people.

I would like to thank the research team at Gerontology and Geriatrics. Laura Vercauteren, thank you for always helping me with the practicalities of the clinical study. To PhD fellows, Jolan, Laurence, Nadja, Sigrid, Lisa, Bertha, Tim, Louana, and Marie, thank you for our discussions during the G&G research meetings and the enjoyable moments we shared during after-work get-togethers. Thanks to Laura Hellemans and Julie who always be available for my questions about clinical study and RedCap.

To PhD fellows at Leuven Pharmacometrics Group Wannee, Zhigang, Omar, Nada, Yannick, Zoë, and Luong, Juma, Wei, Emmanuel, I always enjoyed our discussion at journal clubs and research meetings, as well as our fun activities outside works. Learning pharmacometrics is always fun with you all. Thank you for welcoming me in the group as a friend and a colleague, and for all of our happy moments.

To my fellow AGePOP PhDs—Henriette, Dora, Gabriël, Ekaterina, Adam, Dorota, Fritz, Jonas, Donnia, and Cleo—thank you for the friendship and the scientific discussions during our AGePOP weeks and trainings. I also extend my gratitude to the AGePOP consortium for trainings, scientific meetings, and networking opportunities that have supported my journey.

To Indonesians in Leuven (yoga and running club, Gowes Leuven, PhD student association), thank you for making us feel like home when far from home.

Terima kasih kepada Mama dan Papa yang telah membesarkan, mendidik, serta memberikan akses pendidikan berkualitas, dan selalu mendukung cita-cita anak-anaknya untuk meraih pendidikan tinggi. Terima kasih juga kepada Bapak dan Ibu atas cinta serta doa tulus yang selalu diberikan kepada kami.

Lastly, greatest gratitude goes to my super husband, Filipus, who quit his promising career to come with me to Belgium to support my PhD journey, taking care of me, from cooking to driving to Germany to deliver study samples. You are a strict supervisor. When I complained, and you would sit in front of me and made sure that I finish my work. You are the driving force of the success of this PhD. I am so excited for our future adventures and more cycling miles.

I would like to extend my heartfelt gratitude to those whose names I cannot mentioned individually. Thank you for being this part of milestones in my life. I am truly gratitude with your help, kindness, beautiful moments we shared.

Elma Edwina
February 2025

List of abbreviations

AF	Atrial Fibrillation
AIC	Akaike Information Criterion
AI	Artificial Intelligence
AKI	Acute Kidney Injury
ALM	Appendicular Lean Mass
ALT	Alanine Aminotransferase
ANOVA	Analysis of Variance
ARISTOTLE	Apixaban for the Prevention of Stroke in Atrial Fibrillation Trial
AST	Aspartate Aminotransferase
AUC _{24h}	24-hour Area Under the Curve
AUC _{24h} /MIC	24-hour Area Under the Curve to Minimum Inhibitory Concentration ratio
BCS	Biopharmaceutical Classification System
BID	Twice Daily
BMI	Body Mass Index
BIS	Berlin Initiative Study equation
BUN	Blood Urea Nitrogen
CAG	Chronic Atrophic Gastritis
CCI	Charlson Comorbidity Index
CFS	Clinical Frailty Scale
CHA2DS2-VASc	Congestive heart failure, Hypertension, Age ≥ 75 years, Diabetes mellitus, Stroke, Vascular disease, Age 65–74 years, Sex category (female)
CI	Confidence Interval
CKD-EPI	Chronic Kidney Disease Epidemiology Collaboration
CL	Clearance
CrCL	Creatinine Clearance
CTT	Colonic Transit Time
C _{max}	Maximum Concentration
C _{trough}	Trough Concentration
CVVH	Continuous Veno-Venous Hemofiltration
DOACs	Direct Oral Anticoagulants
EC	Ethical Committee
EC ₅₀	Half Maximal Effective Concentration
EHRA	European Heart Rhythm Association
EMA	European Medicines Agency
ENGAGE AF	Effective Anticoagulation with Factor Xa Next Generation in Atrial Fibrillation trial
ElderCARE-AF	Edoxaban in Elderly Patients with Atrial Fibrillation trial
E _{max}	Maximum Effect
eCrCL	Estimated Creatinine Clearance using Cockcroft–Gault Equation
eGFR	Estimated Glomerular Filtration Rate
eGFR _{BIS}	Estimated Glomerular Filtration Rate using Berlin Initiative Study equation

eGFR _{CKD-EPI}	Estimated Glomerular Filtration Rate using Chronic Kidney Disease Epidemiology Collaboration equation
eGFR _{FAS}	Estimated Glomerular Filtration Rate using Full Age Spectrum equation
FDA	Food and Drug Administration
GFR	Glomerular Filtration Rate
GGT	Gamma-Glutamyl Transferase
GI	Gastrointestinal
GOF	Goodness-of-Fit
HASBLED	Hypertension, Abnormal liver/renal function, Stroke, Bleeding history, Labile INR, Elderly (age ≥65), Drugs/alcohol concomitantly
IDSA	Infectious Diseases Society of America
IIV	Interindividual Variabilities
IQR	Interquartile Range
ka	Absorption Rate Constant
LBM	Lean Body Mass
MAA	Model Averaging Algorithm
MDRO	Multidrug-Resistant Organism
MIC	Minimum Inhibitory Concentration
MIPD	Model-Informed Precision Dosing
MLE	Maximum Likelihood Estimate
ML	Machine Learning
MRSA	Methicillin-Resistant Staphylococcus Aureus
MSA	Model Selection Algorithm
mL/min	Milliliters per Minute
μmol/L	Micromoles per Liter
NIHSS	National Institutes of Health Stroke Scale
NPDE	Normalized Prediction Distribution Errors
OFV	Objective Function Value
OR	Odds Ratio
P-gp	P-glycoprotein
pcVPC	Prediction-Corrected Visual Predictive Check
PD	Pharmacodynamics
PK	Pharmacokinetics
popPK	Population Pharmacokinetics
PPI	Proton Pump Inhibitor
PT	Prothrombin Time
QD	Once Daily
RE-LY	Randomized Evaluation of Long-term Anticoagulation Therapy Trial
rBias	Relative Bias
rRMSE	Relative Root Mean Square Error
ROC	Receiver Operating Characteristic
ROCKET AF	Rivaroxaban Once Daily Oral Direct Factor Xa Inhibition Compared with Vitamin K Antagonism for Prevention of Stroke and Embolism Trial in Atrial Fibrillation
SARC-F	Strength, Assistance with walking, Rise from a chair, Climb stairs, and Falls questionnaire
SD	Standard Deviation

SITT	Small Intestinal Transit Time
TDM	Therapeutic Drug Monitoring
TIA	Transient Ischemic Attack
Tlag	Lag Time
Tmax	Time to Maximum Concentration
UPLC-MS/MS	Ultra Performance Liquid Chromatography–Tandem Mass Spectrometry
Vd	Volume of Distribution
VIF	Variance Inflation Factor
VKA	Vitamin K Antagonist
Vp	Volume of Peripheral Compartment
VPC	Visual Predictive Check
WGTT	Whole Gut Transit Time
WMC	Wireless Motility Capsule

General Introduction

A part of this chapter (subsections 2.1 and 2.2) was based on the following publication:

Demeester C†, Robins D†, Edwina AE†, Tournoy J, Augustijns P, Ince I, Lehmann A, Vertzoni M, Schlender JF. *Physiologically based pharmacokinetic (PBPK) modelling of oral drug absorption in older adults - an AGePOP review*. Eur J Pharm Sci. 2023 Sep 1;188:106496. †shared first authors

1. Geriatric patients and the lack of their representation in clinical trials

Gap:

Geriatric patients—older adults at risk for frailty, reduced organ function, cognitive impairment, multimorbidity, and polypharmacy—face a high health and medication burden. Yet, they remain underrepresented in pivotal clinical trials, leaving only limited data to support their care.

The global population is undergoing major demographic changes, primarily due to aging, with the proportion of people aged 65 and older projected to rise from 9.3% in 2020 to 16.0% by 2050 (1). Life expectancy is expected to continue rising in many regions (1). In the European Union, the percentage of people aged 65 and older—commonly classified as ‘older adults’ (2)—is projected to increase from 20.8% in 2021 to 31.3% by 2100 (3). In Belgium, life expectancy is projected to rise from 79.6 years for men and 84.3 years for women to 88.3 and 89.9 years, respectively, by 2070 (4). As of 2022, there were 1,065,054 Belgian residents aged 75 years or older, accounting for 11% of the total population (4).

Out of this age group, geriatric patients are distinguished as a subgroup. Even though most geriatric patients are 75 years or older, their classification is not strictly determined by age. Instead, it is based on a broader set of factors known as the geriatric syndrome, which includes frailty, reduced organ function, cognitive impairment, multiple chronic conditions (multimorbidity), and the use of multiple medications (polypharmacy) with on average 10 medications per day (5-7). Despite the significant health needs of geriatric patients, their representation in clinical drug trials is lower than that of younger age groups (8). Although older adults making up 72% of estimated Alzheimer’s disease patients and being significantly affected by conditions like atrial fibrillation, venous thromboembolism, and osteoporosis, they account for only 22% of study participants (9-11). They were also excluded from more than 50% of COVID-19 clinical trials (12). In trials unrelated to age-specific conditions, such as those for epilepsy, depression, and bipolar disorder, the participation rate for those over 75 dropped to just 1% (9, 10). As a result, the population that uses medications most frequently has limited supporting data from clinical research.

2. Age-related (patho)-physiological changes altering pharmacokinetics and

Gaps:

- An unexplained high pharmacokinetic variability of commonly used agents such as paracetamol has been observed in geriatric patients, especially in the absorption phase.
- Geriatric patients experience increased sensitivity to medication action and their side effects due to PD alterations.
- Changes in gastrointestinal transit and pH remain poorly understood in geriatric patients.
- The impact of these physiological changes and pathophysiological associated with geriatric profiles on pharmacokinetics and pharmacodynamics remains unclear.

The difference in demographic between study populations and the target geriatric population limits the accurate prediction of efficacy and safety of medications for geriatric patients when based on clinical trials conducted in younger individuals (13, 14). Results from trials involving participants aged 18–64 years cannot always be reliably extrapolated to geriatric patients. A key reason for this limitation is the age-related changes in (patho-)physiology, which can profoundly affect drug pharmacokinetics (PK) and pharmacodynamics (PD). This PK variability partially explained the increased susceptibility of older adults to adverse drug effects (15).

The interplay between geriatric syndromes and altered PK is complex, indicating that their impact goes beyond normal age-related changes. For instance, while age-related declines in organ function are common, specific conditions associated with geriatric syndromes can further impair renal and hepatic drug clearance, necessitating careful dose adjustments (16, 17). Cancer drug trials have shown an association between sarcopenia or frailty and dose-related toxicities, resulting in treatment delays and dose reductions (18). However, clear evidence of a direct causal relationship between sarcopenia and PK changes is still lacking. It is believed that lower lean body mass, which reduces the volume of distribution, increases drug exposure and contributes to these toxicities, but this hypothesis remains inconclusive (19). These PK challenges are further complicated by the presence of multimorbidity and polypharmacy, which substantially increase the risk of drug-drug and drug-disease interactions, further compounding the risk of adverse outcomes (20, 21). Geriatric patients experience increased sensitivity to medication action and their side effects as a result of PD alterations (22). For example, they are more susceptible than younger individuals to adverse reactions from antidepressants and benzodiazepines,

including falls, bleeding, cardiovascular events, cognitive deficits, and increased overall mortality (23, 24).

Research conducted by our group has revealed a high PK variability of paracetamol, a commonly used medication, among geriatric patients, particularly during the drug absorption phase (25, 26). This variability remained unexplained by any of the covariates we tested. A potential source of this variability, which was not explored in the previous study, might be the alteration in gastrointestinal (GI) physiology. These changes in GI physiology, including variations in GI transit time and intraluminal pH, may play a crucial role in the observed differences in drug absorption. The subsections 2.1 and 2.2 focus on exploring current knowledge of these specific GI physiological changes in older adults.

2.1. Intraluminal pH

Gastrointestinal secretions and regional pH variations are crucial for the solubility of solid drugs (27). For a long time, it was assumed that gastric secretions, similar to many other physiological functions, decline with age (28-31). However, previous publications have reported that gastric acid secretion is dependent on chronic atrophic gastritis (CAG) rather than age, as similar rates of gastric acid secretion were found between young and older adults when those with CAG were excluded (32-34). The presence of *Helicobacter pylori* (*H. pylori*) was associated with CAG and have shown significantly reduced peak acid output. As a result, gastric acid secretion may appear to decline with age when older adults with CAG and *Helicobacter pylori* (*H. pylori*) infection are included.

A previous study showed that after a solid meal, older people needed 89 and 154 minutes to return to a pH of 3 and 2, respectively. Compared to 42 and 100 minutes in younger adults, this is considerably prolonged (35). However, conflicting results regarding pH in older adults were found. When comparing the fasted gastric median pH of 1.7 (IQR=1.4–2.1) in healthy young men, as studied by Dressman et al. (36), to the fasted gastric median pH of 1.3 (IQR=1.1–1.6) in older men, as reported by Russel et al. (35), it appears that gastric pH decreases slightly with aging. On the other hand, Pedersen et al. (37) found no tendency toward pH differences between the three different age groups they included. Beside gastric pH, duodenal pH showed also contradictory results in the literature (35, 36, 38).

pH is an important factor for the bioavailability, particularly for Class II drugs in the Biopharmaceutics Classification System, which are poorly soluble basic compounds with high permeability. For example, plasma ketoconazole concentrations were significantly lower in achlorhydric older adults with a fasted gastric pH greater than 5 compared to non-achlorhydric older adults with a pH less than or equal to 4.5 after ketoconazole administration. When fasted achlorhydric older adults took ketoconazole crushed in acidic juice, the PK profile was again similar to that of the fasted control group (27). Similarly, when older adults with achlorhydria took dipyridamole, they had significantly lower T_{max} , C_{max} , and absorption rate constant than the control low pH older adults (39).

2.2. Gastrointestinal transit time

Gastric emptying is another highly variable and disputed factor, as it not only reflects muscle function but depends on hormonal and neuronal activity as well (40, 41). Data published on gastric emptying rates in older adults compared to younger adults are conflicting and are studied with various methods, which makes it difficult to compare. Some studies detected slower rates in older adults while others revealed no change in gastric emptying (42-47). Even less literature is available for changes in small intestinal transit time in the older population. Madsen *et al.* (48) and Kagaya *et al.* (49) observed similar transit times in the small intestine for both younger and older adults. Other studies in presumably healthy, older adults report no significant differences in colon transit time compared to the young (50).

Despite the conflicting results regarding any age-associated differences between healthy older adults compared to younger ones, several comorbidities as well as specific drugs do have an impact on gut transit. Medications such as anticholinergics, levodopa, and diseases including chronic liver disease, and Parkinson's disease are all examples that are associated with delayed gastric emptying (51-53). Persons with chronic diseases such as diabetes mellitus tend to experience reduced motility and have therefore a slower small intestinal transit time (54). Additionally, there is a higher prevalence of constipation in the older population which translates to longer colonic transit times (48, 55). It is difficult to quantify the independent factor of aging on colon transit time because constipation can also be caused by certain drugs, a lack of dietary fibre, or physical activity. Altered transit times may alter the absorption and disposition of drugs as they can lengthen or shorten the available time for processes like dissolution and passive transport.

3. Frequent morbidities in geriatric patients and their treatment

Gaps:

- Atrial fibrillation, which requires anticoagulation such as edoxaban for stroke prevention, occurs in 50% of the geriatric population.
- Infections in hospitalized older patients are often treated with vancomycin.
- Despite the frequent use of these high-risk and narrow therapeutic window medications, we lack reliable data on PK/PD in an at-risk patient population.

Among the various morbidities affecting geriatric patients, cardiovascular diseases and infections are very prevalent among hospitalized patients and pose significant challenges, largely due to the complexity of their pharmacological management. Conditions such as atrial fibrillation (AF) and bacterial infections not only demand careful therapeutic strategies but also require adjustments for age-related (patho)-physiological changes that can affect drug efficacy and safety.

3.1. Atrial fibrillation and edoxaban

3.1.1. Atrial fibrillation

One of the most common conditions in geriatric patients is AF (56). The incidence of AF increases with age, with nearly one in eight individuals over 80 years old being affected, and it peaks in the >85 years age group (57). In hospital populations, the prevalence of AF in geriatric patients is almost 50% (58).

AF is characterized by irregular electrical activity in the atria, disrupting normal atrial contractions. This leads to the reduction of blood flow to the ventricles and a decreasing heart pumping efficiency. Intracardiac blood clots can form, primarily in the left atrial appendage, potentially leading to serious complications such as ischemic events from clot embolization (59, 60). If these clots migrate to the brain, they can result in a stroke (60). Generally, AF increase the risk of stroke fivefold. This risk increases with age, with individuals 65 and older having a threefold higher risk (RR: 3.3; 95% CI: 1.92–5.81) (61). A meta-analysis of randomized controlled trials by van Walraven *et al.* showed that age independently raises the risk of ischemic stroke by 45% per decade (95% CI: 1.26–1.66), even after adjusting for other factors (62).

3.1.2. Edoxaban

Anticoagulants reduce the stroke risk in AF by approximately two-thirds (63-65). Stroke prevention in AF patients poses significant challenges, primarily due to the prolonged action of VKAs, substantial variability in their effects caused by interactions with food, other medications, and genetic factors, and a narrow therapeutic window that requires frequent laboratory monitoring and dose adjustments (66, 67). To address the challenges associated with oral VKAs, direct (non-vitamin K) oral anticoagulants (DOACs) have been developed and are now preferred for stroke prevention in AF due to more reliable PK allowing for fixed dosing without the need for regular laboratory monitoring (68). These include direct factor Xa inhibitors—apixaban, edoxaban, and rivaroxaban—and a direct thrombin inhibitor, dabigatran.

Edoxaban is the newest addition to the DOAC class in Europe and is recommended at a daily dose of 60 mg for stroke and systemic embolism prevention in AF patients. For those with moderate or severe renal impairment (creatinine clearance [CrCL] 15–50 mL/min), low body weight (≤ 60 kg), or those taking certain P-glycoprotein (P-gp) inhibitors, a reduced dose of 30 mg once daily is advised (69). The once-daily dosing of edoxaban contributes to ease of use and may promote adherence to therapy, unlike dabigatran and apixaban that require twice-daily dosing. Although the ENGAGE AF-TIMI 48 study suggests that edoxaban may be a non-superior treatment option due to its safety and effectiveness (70), these findings cannot be directly extrapolated to the geriatric population. The trials predominantly involved non-frail participants, while age-related physiological changes in older adults may increase edoxaban exposure, potentially affecting both its efficacy and safety in this group. Therefore, its outcomes in the geriatric population remain uncertain.

3.1.3. Pharmacokinetics and pharmacodynamics of edoxaban

Edoxaban is classified as a weak base with a pKa value of 6.7. In healthy volunteers, the PK of edoxaban is characterized by rapid absorption (1–3 h) (71, 72). Peak plasma concentrations of edoxaban are achieved at approximately 1.5 h after oral administration (71). The absolute oral bioavailability of edoxaban in healthy subjects is 62% (72). Food intake has a clinically insignificant effect on the absorption of edoxaban and does not affect the overall exposure of edoxaban (73). The steady-state volume of distribution of edoxaban is 107 liters. Its terminal half-life of 10–14 hours allows steady-state concentrations to be

reached within 3 days (72). Edoxaban plasma protein binding is about 55% (69). Its PK primarily involve enzyme-mediated hydrolysis predominantly mediated by carboxylesterase-1 and less by CYP3A4/5, with unchanged edoxaban remaining predominant in the plasma. This process results in the formation of three active metabolites, with M-4 being the main one, although it accounts for less than 10% of total edoxaban exposure (74). Edoxaban's total clearance is estimated at 22 L/hour, with 50% of this clearance occurring through the kidneys as unchanged drug (11 L/hour) (72). Based on the ENGAGE-AF TIMI 48 study, edoxaban's observed plasma trough concentrations range from 12–43 ng/mL (75). Edoxaban concentrations showed only a modest association with stroke prevention. Meanwhile, as concentrations increased, the risk of bleeding rose much more sharply, particularly at higher concentrations. (76). However, there is still limited knowledge about the PK and PD of edoxaban and other DOACs in several special patient populations including geriatric patients. It remains unclear which specific characteristics or covariates in geriatric patients influence the PK and PD of edoxaban and other DOACs. Additionally, although anti-factor Xa activity is routinely used to monitor edoxaban exposure and is directly correlated with plasma concentrations in the general population, this approach has not been validated in geriatric patients. Furthermore, while liquid chromatography tandem mass spectrometry (LC-MS/MS) methods provide a more sensitive and specific measurement of drug concentrations, they are not always available in clinical settings, limiting the ability to ensure accurate monitoring in geriatric patients.

3.2. Gram-Positive bacterial infections and vancomycin

3.2.1. *Gram-positive bacterial infections*

Although chronic conditions have surpassed infectious diseases as the leading cause of mortality among geriatric patients, infections remain a significant threat to this population (77). Infections are the primary cause of hospitalization among nursing home residents, and nearly 14% of hospital admissions of older patients are infection--related (78, 79). As age advances, the incidence of infections also increases. Among those hospitalized for infections, 66.5% were in the oldest-old group (aged 85+), which had the highest hospitalization rate (78).

The increasing risk of infection with age is influenced by several factors beyond chronological age itself, such as the decline in immune function, known as immunosenescence. This process affects all older people to varying degrees and is closely

linked to frailty (80). Many older adults experience mild immunosuppression due to immunosenescence, compounded by age-related organ changes, multimorbidity, geriatric syndromes, frailty, malnutrition, functional impairments, and polypharmacy. All of these negatively impact the prognosis of geriatric patients with infectious diseases and reduce the ability to recover after an acute event such as an infection (81-83). Frail older patients are frequently transferred between facilities, such as nursing homes, assisted living centers, specialty clinics, or hospital wards. These transitions of care can also further add to the susceptibility of geriatric patients to infections as they facilitate the spread of pathogens between different healthcare environments and individuals (84).

Frailty is a significant risk factor for multidrug-resistant organism (MDRO) infections. Studies have shown that reduced functional status, as a surrogate marker of frailty, is independently associated with a higher risk of MDRO infections, including methicillin-resistant *Staphylococcus aureus* (MRSA) (85, 86). Functionally disabled patients are also more likely to have additional risk factors for MDRO colonization, such as prior antibiotic use, presence of chronic wounds, long-term catheter use, and frequent interactions with healthcare systems or environments where MDROs are prevalent, such as hospitals and nursing homes (86, 87).

3.2.2. Vancomycin

Vancomycin is commonly the drug of choice for the treatment of serious Gram-positive infections involving MRSA infections (88). In addition to MRSA, it is effective against other Gram-positive organisms, including methicillin-Resistant *Staphylococcus epidermidis*, *streptococci* (such as penicillin-resistant *Streptococcus pneumoniae*), and *enterococci*. Intravenous vancomycin is indicated for the treatment of several serious infections, including complicated skin and soft tissue infections, bone and joint infections, various types of pneumonia (community- and hospital-acquired, including ventilator-associated), infective endocarditis, catheter-related infections and bacteremia associated with any of these conditions. Additionally, vancomycin is used for perioperative antibacterial prophylaxis in patients at high risk of developing bacterial endocarditis during major surgical procedures (89). Based on the summary of product characteristics of vancomycin, the recommended dose is 15 to 20 mg/kg of body weight every 8 to 12 h, not exceeding 2 grams per dose. For seriously ill patients, a loading dose of 25–30 mg/kg may be administered to achieve target concentrations more rapidly (89).

Vancomycin is effective only against Gram-positive bacteria due to its mechanism of inhibiting cell wall synthesis. The key component of most bacterial cell walls is peptidoglycan, which provides a protective barrier and acts as a selective filter for external molecules (90). This makes peptidoglycan a critical target in antibiotic development. Peptidoglycan is a three-dimensional structure composed of alternating polysaccharides, N-acetylglucosamine and N-acetylmuramic acid, which are crosslinked by pentapeptide side chains. Its synthesis occurs in three stages: starting in the cytoplasm, moving through the cytoplasmic membrane, and ending with the polymerization of the D-alanyl-D-alanine precursor into the peptidoglycan layer within the periplasmic space. Vancomycin binds to this precursor, blocking its transport and preventing the polymerization of peptidoglycan by steric hindrance. Since peptidoglycan is essential for bacterial cell walls, vancomycin is considered bactericidal. Gram-positive bacteria have thick layers of peptidoglycan, while Gram-negative bacteria have only a thin layer, along with an outer membrane containing lipopolysaccharides. As a result, vancomycin is highly effective against Gram-positive but not Gram-negative bacteria, which defines its antimicrobial spectrum.

3.2.3. Pharmacokinetics and pharmacodynamics of vancomycin

Vancomycin is a hydrophilic compound with very limited ability to penetrate fat tissues with fat-to-serum ratio of 0.153 (91). Upon intravenous administration, it undergoes rapid distribution primarily into extracellular fluid compartments. The volume of distribution for vancomycin varies depending on the patient population, typically ranging from 0.4 to 1.0 L/kg in adults (92-96). Vancomycin's protein binding is ranging from 10% to 50%, primarily binding to albumin (97, 98). Because of its hydrophilic characteristics, vancomycin penetration in many tissues is low, with only 0–18% of concentrations in uninflamed meninges, 36–48% in inflamed meninges, a maximum of 41–51% in the lung, and 10–30% in diabetic and normal skin and soft tissues (99). The ability of vancomycin to reach therapeutic concentrations in tissues can be influenced by factors such as inflammation and tissue perfusion, which are relevant mediators in patients with severe infections (100). Vancomycin is primarily eliminated through the kidneys. Approximately 75% of an administered intravenous dose is excreted unchanged in the urine (101). As a result, renal function plays a significant role in determining the drug's clearance. For patients with normal renal function, vancomycin has a total clearance 0.05–0.1 L/kg/h and a half-life of 5–11 hours. However, in patients with impaired renal function, this half-life can be

significantly prolonged, potentially extending up to 7 days in patients with end-stage renal disease (89, 102).

Vancomycin exhibits a clear dose–response relationship. In vitro and in vivo evaluations of PK/PD models for human MRSA infection have demonstrated that bactericidal activity (a 1- to 2-log reduction in bacterial inoculum in animal models) is achieved when the vancomycin AUC_{24h}/MIC ratio reaches or exceeds 400 (99, 103). Additionally, in vitro data suggest that an AUC below 400 increases the risk of developing MRSA resistance and vancomycin-intermediate *S. aureus* strains (104, 105). In a study by Suzuki et al., the relationship between vancomycin AUC and the development of acute kidney injury (AKI) was examined. Patients who developed AKI had mean AUC values ranging from 600 to 800 mg·h/L, compared to 400 to 600 mg·h/L in those who did not develop AKI (P = 0.014) (106). A meta-analysis also showed that increasing vancomycin trough concentrations correlate with a significant rise of nephrotoxicity risk with a cut-off of 15 mg/L (107).

Dosing vancomycin in geriatric patients is challenging due to the significant interindividual pharmacokinetic variability that can occur in this population (25). Given vancomycin’s narrow therapeutic window, even small deviations in drug levels can lead to therapeutic failure or toxicity. These risks emphasize the importance of individualized dosing and careful therapeutic drug monitoring or even (pharmacometrics) model-informed precision dosing in geriatric patients.

4. Pharmacometrics modeling

Gap:
Pharmacometrics modeling can differentiate variabilities and capture the complex interplay between patient characteristics and drug behavior, which is relevant for geriatric patients. However, it has not yet widely studied and implemented in geriatric population.

A pharmacometrics model is a mathematical and statistical construct that quantifies drug, disease, and clinical trial data to aid in efficient drug development and clinical decision-making (108). Among the various methods, population modeling is the most commonly applied approach for population analysis (109). Population modeling is a method that simultaneously analyzes data from all individuals in a population applying nonlinear mixed-effects modeling on unbalanced data(sets), e.g., different trials, combination of dense and sparse data or different sampling times (**Figure 1**) (109). Population models may describe

the relationship between drug concentrations and dosing (population PK) or between drug concentrations and effect (population PD). When dosing, concentrations, and effects are modeled together, it is referred to as population PK-PD modeling. The ability to simulate from pharmacometrics models is a powerful tool for informed decision-making. For example, in the United States, dabigatran etexilate mesylate is labeled to be administered at a dose of 75 mg twice daily in patients with severe renal impairment (creatinine clearance of 15–30 mL/min). This dose was not studied in this patient population but instead resulted from conclusions based on population PK model simulations (110).

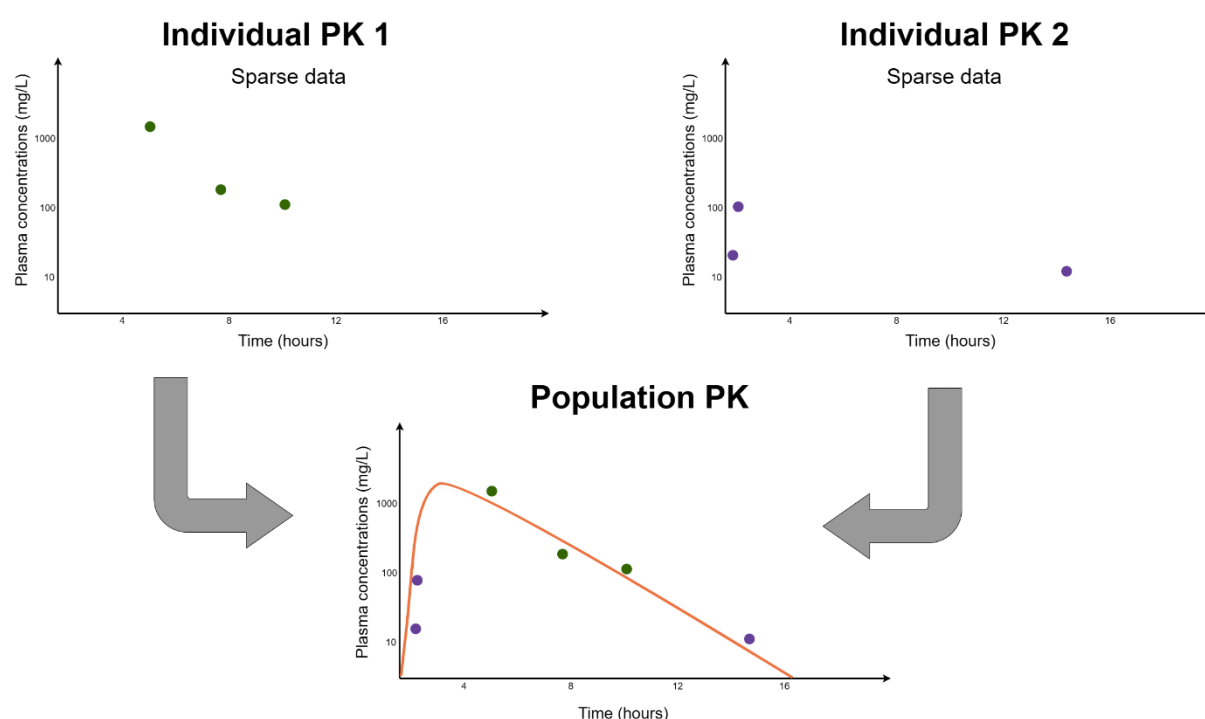


Figure 1. Population data pooling. The sparse individual data might not be sufficient for PK analyses, but the pooled population data are.

The primary function of a model is to represent a system based on observed data. A model typically requires several key components:

- a. Mathematical-statistical structure: This is often derived from a compartmental PK model. For simplicity, the body is initially considered as a single well-stirred compartment. However, since drugs do not distribute uniformly across tissues outside the bloodstream, additional compartments can be introduced to more accurately capture the data. **Figure 2** illustrates the difference between drug

concentrations over time profile between one- and two-compartment model. A PK model is commonly expressed as a system of ordinary differential equations, which describe the mass transfer between compartments. **Equations (1), (2) and (3)** describe the mass transfer for a two-compartment model after oral administration.

Figure 3 express a two-compartment model after oral administration.

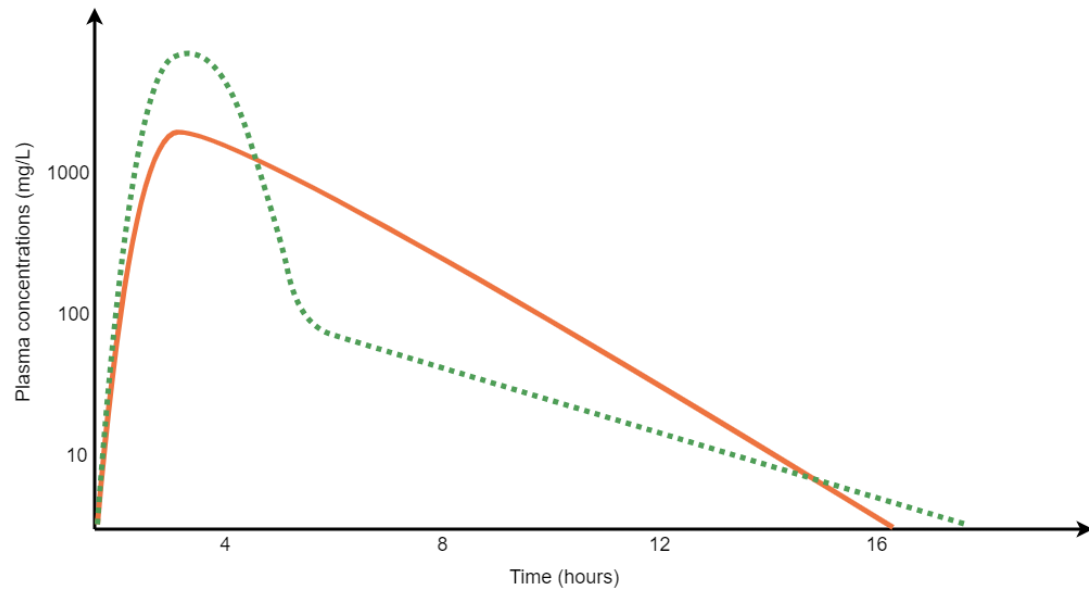


Figure 2. Concentration versus time curve. Solid line: linear one-compartment; dotted line: linear two-compartment

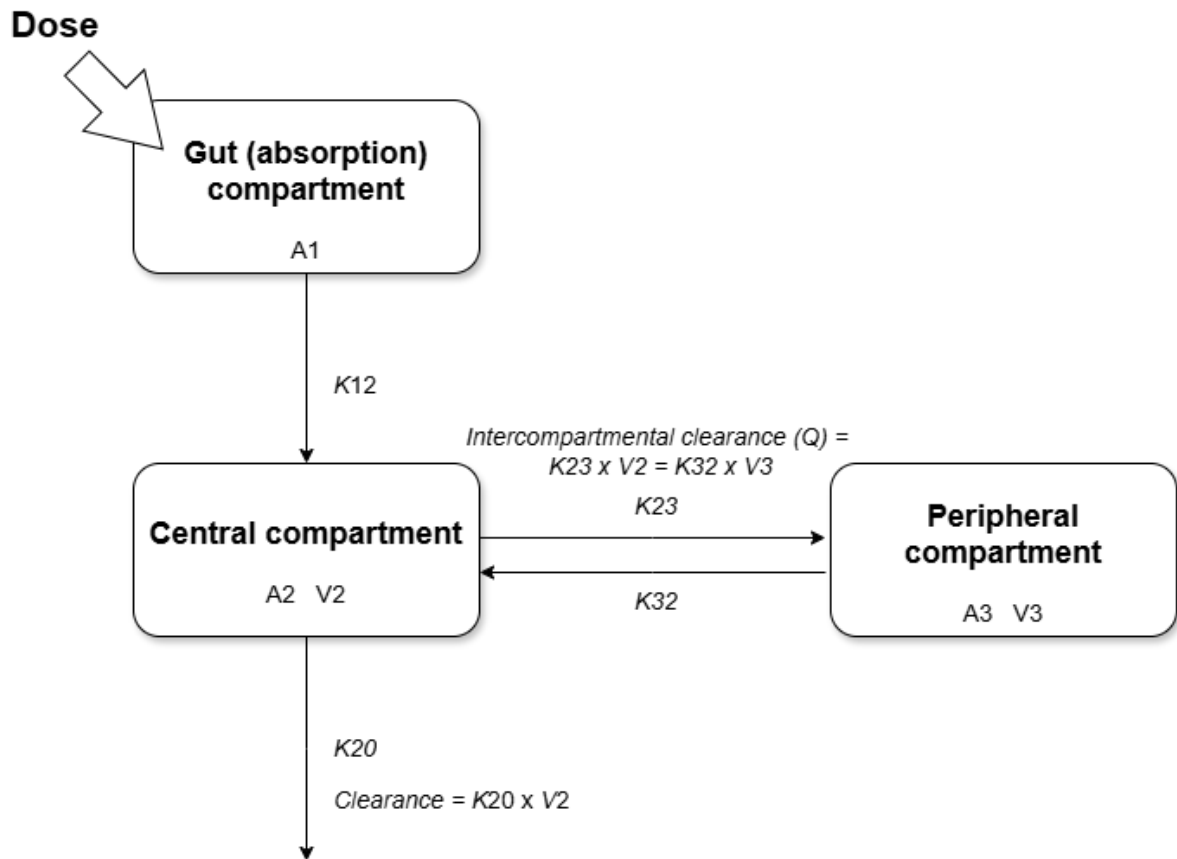


Figure 3. A two-compartment structural model with linear absorption and linear elimination from the central compartment after oral administration. Numberings 1, 2 and 3 refer to the absorption, central and peripheral compartment, respectively. A: amount, V: volume of distribution, k: rate constant

$$\frac{dA_1}{dt} = -k_{12} \times A_1 \quad (1)$$

$$\frac{dA_2}{dt} = k_{12} \times A_1 - k_{20} \times A_2 + k_{32} \times A_3 - k_{23} \times A_2 \quad (2)$$

$$\frac{dA_3}{dt} = -k_{32} \times A_3 + k_{23} \times A_2 \quad (3)$$

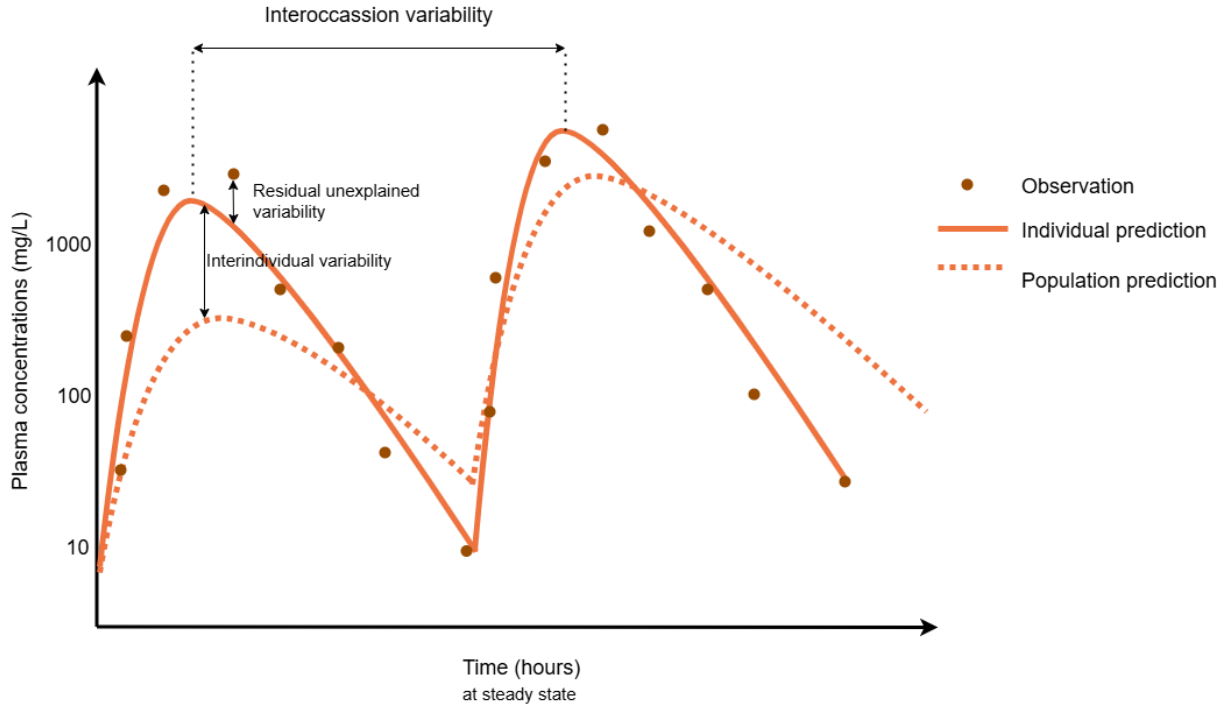


Figure 4. Type of variabilities in population pharmacokinetic model

- b. Parameters of the model: The '*mixed-effect*' refers to the estimation of both fixed and random effects parameters. The fixed effects represent the typical population parameters such as population Ka , CL , V .

$$Ka = \theta_1 \quad (4)$$

$$CL = \theta_2 \quad (5)$$

$$V = \theta_3 \quad (6)$$

where θ_1 , θ_2 and θ_3 are the population values of Ka , CL and V , respectively.

The random effect refers to the variability in these parameters. The different types of variability commonly estimated during a population PK analysis are depicted in **Figure 4**. Variability can be estimated at different levels, including interindividual variability, which is modeled as an exponential equation, given that variability of the parameter is usually log-normally distributed.

$$Ka_i = \theta_1 \times e^{\eta_{i,1}} \quad (7)$$

$$CL_i = \theta_2 \times e^{\eta_{i,2}} \quad (8)$$

$$V_i = \theta_3 \times e^{\eta_{i,3}} \quad (9)$$

The terms $\eta_{i,1}$, $\eta_{i,2}$, $\eta_{i,3}$ represent the deviations from the typical population values for Ka , CL , and V for a specific individual i . The variances and covariances of η are organized into the omega (Ω) $p \times p$ covariance matrix as follows:

$$\Omega = \begin{bmatrix} \omega_1^2 & & & \\ \omega_{1,2}^2 & \omega_2^2 & & \\ \vdots & \vdots & \ddots & \\ \omega_{1,p}^2 & & & \omega_p^2 \end{bmatrix} \quad (10)$$

with ω_p^2 representing the variance of the model parameter. It is important to include variance terms for parameters expected to be influenced by covariates. If variance terms are excluded, the PK parameter defaults to the typical population value or known as ‘complete shrinkage’.

Residual variability represents the unexplained variability between observed data and individual model predictions. This can arise from differences in data collection, timing discrepancies, or inaccuracies in bioanalysis.

$$Cp_{i,j} = f(\vec{\theta}, \vec{\eta}_i, t_{i,j}) + \varepsilon_{i,j,1} + f(\vec{\theta}, \vec{\eta}_i, t_{i,j}) \times \varepsilon_{i,j,2} \quad (11)$$

where Cp is the drug concentration in an individual i at time point j as the nonlinear function of the independent variables including the vector of fixed-effects parameters of population ($\vec{\theta}$), the vector of random effect parameters of individual i ($\vec{\eta}_i$) at time point j . $\varepsilon_{i,j,1}$ represent the additive residual variability and $\varepsilon_{i,j,2}$ represents the proportional residual variability for individual i at time point j , accounting for the unexplained differences between the observed and individual model-predicted values.

- c. Independent variables that improve the ability of the model to describe the data such as covariates: When interindividual variability in parameters is low, concentration–time profiles across patients will be similar, meaning that the required dose to achieve a target concentration will also be similar. However, significant variability in parameters can lead to considerable differences in concentration–time profiles across patients given the same dosage (**Figure 5**). If this variability is systematic and explained by measurable covariates, the dose regimen can be individualized to increase the likelihood of achieving the desired concentration. Conversely, if large interindividual variability is unexplained, predicting individual concentration profiles becomes challenging. Covariate models help explain the relationship between PK

parameters and patient-specific factors, such as kidney function (**Figure 6**). The parameterization of these models depends on the type of covariate data (i.e. continuous or categorical data). The relationship between a parameter and a continuous covariate can take various forms, including linear, hockey-stick (a piecewise linear relationship with two different slopes), exponential, or power models. For categorical covariates, a linear model can be used, with an additional parameter introduced for each category.

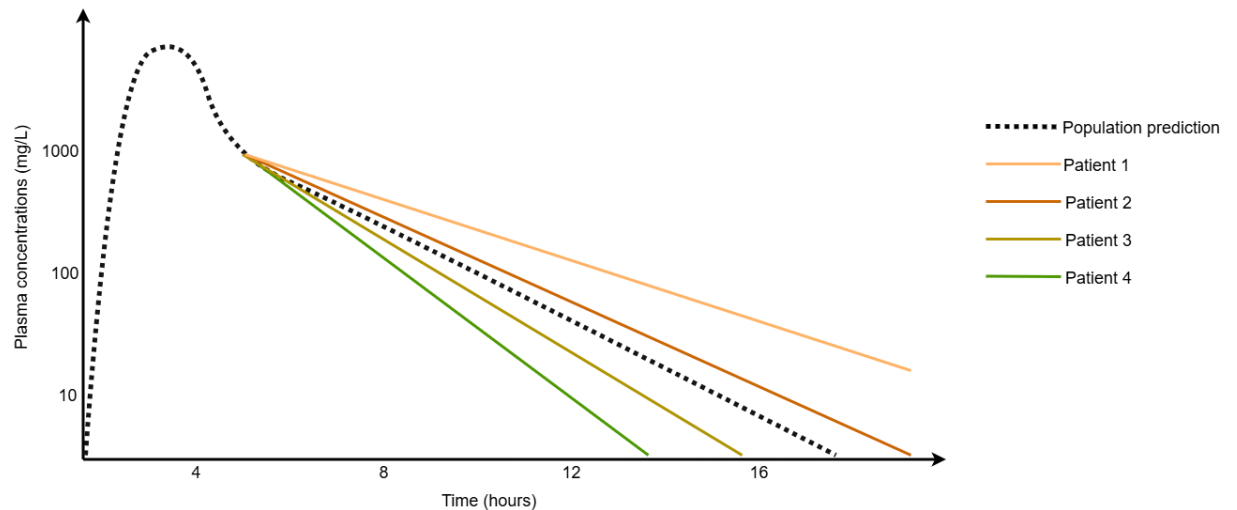


Figure 5. The deviations of individual PK profile in the elimination phase from the typical population value.

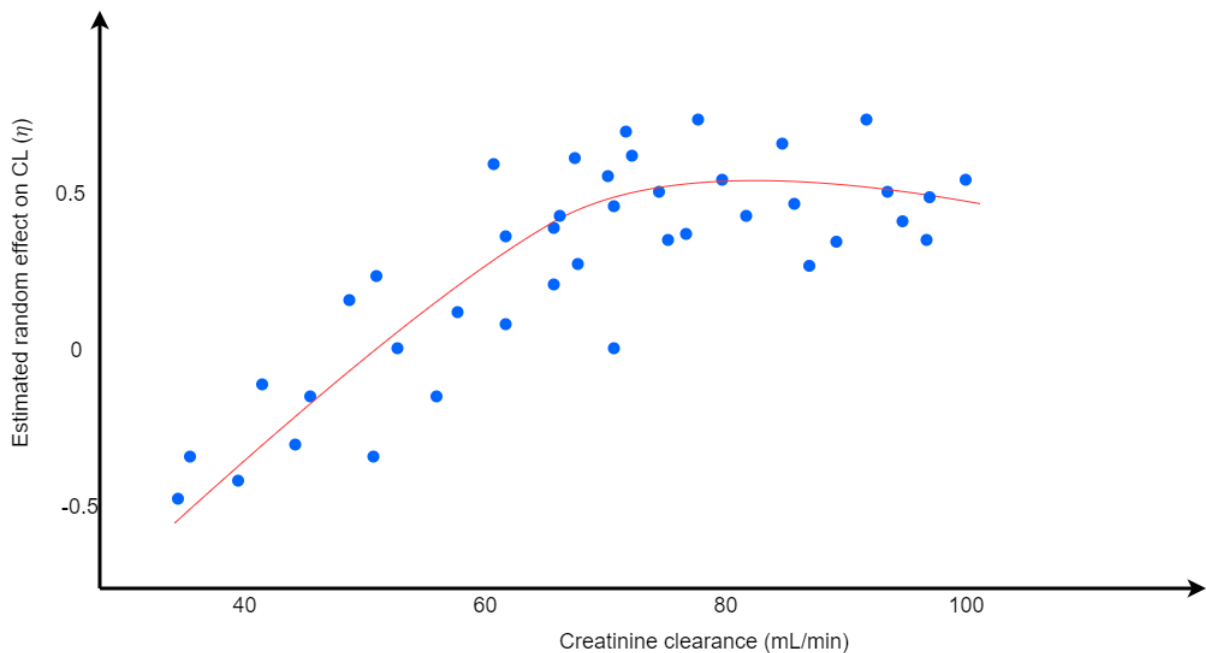


Figure 6. The relationship between creatinine clearance and estimated random effect on CL (η). Red line is the smooth line.

4.1. **Model-informed precision dosing: combining therapeutic drug monitoring and population modeling**

In model-informed precision dosing (MIPD), the popPK model acts as a prior, essential for predicting an individual's PK parameters. This process involves calculating the most likely parameter values for a specific patient based on their characteristics (i.e., covariates) and the assumed prior population model. This approach, known as probabilistic dosing or *a priori* prediction, uses general population data to forecast individual outcomes.

While therapeutic drug monitoring (TDM) is limited to steady-state conditions and relies on trough concentrations and dosing nomograms for adjusting maintenance doses, MIPD enables precise prediction of target attainment for PK/PD or toxicity goals by using popPK models that can be updated to a posterior model by incorporating not only patient-specific covariates that explain PK variability but also previous measurements at any time of that patient's drug concentrations (111). This method, called Bayesian forecasting or a posteriori prediction, refines the predictions by combining prior knowledge with individual data. The patient's individual PK parameters, known as empirical Bayes estimates, are derived from both their covariates and drug concentration data. These estimates are then used to adjust the dosing regimen to ensure the patient achieves the desired drug exposure target.

4.2. **The potential importance of pharmacometrics modeling in geriatric patients**

The aging population is highly heterogeneous, encompassing both older and geriatric individuals. Pharmacometrics modeling plays a crucial role in geriatric pharmacology by quantifying the impact of age-related (patho-)physiological changes on PK and PD through the identification and incorporation of relevant covariates, such as renal function, body composition, and hepatic metabolism. Beside pharmacometrics modeling, physiology-based pharmacokinetic (PBPK) modeling offers a complementary approach by incorporating detailed mechanistic representations of anatomical, physiological, and biochemical processes. PBPK modeling enables the simulation of drug absorption, distribution, metabolism, and excretion in older people by integrating age-specific parameters such as reduced renal function, changes in body composition due to frailty, and drug-drug interaction (112). Together, these modeling approaches provide valuable tools to optimize drug therapy for the highly variable geriatric population. Additionally, it is useful in addressing the effects of drug-drug interactions, which are common in older adults due to

polypharmacy. This personalized approach is particularly important as interindividual variability in drug responses is greater than in younger adults, and the consequences of inappropriate dosing can be more severe. Moreover, population modeling allows for the estimation of PK parameters from sparse sampling, which is especially valuable in geriatric patients. They are physically inactive, frail, or even bedridden. It is known that these frail patients often present with "difficult intravenous access," making it challenging to collect rich blood samples. This approach ensures that despite these limitations, optimal dosing regimens through modeling and simulation can still be determined for this vulnerable population (113).

Although pharmacometrics modeling is a valuable tool in geriatric pharmacology, providing a robust approach to understanding the complexities of drug therapy and PK in older adults, it remains underexplored. A PubMed search using the keywords (geriatric[Title/Abstract]) AND ("pharmacokinetic*" OR "pharmacometrics*") AND (model) identified 109 original articles, of which only 15 involved original pharmacometric modeling. However, 8 of these studies were unclear about whether the population was truly geriatric or simply older, non-frail individuals, as no geriatric assessment was performed, and patients were not specifically recruited from geriatric wards (114-121). This left only 7 studies focused solely on the geriatric population (26, 122-127), emphasizing a significant gap in the research and the need for further investigation in the field of geriatric pharmacology.

References

1. United Nations DoEaSA, Population Division. World Population Ageing 2019: Highlights. 2019.
2. Eurostat. Ageing Europe: Looking at the lives of older people in the EU. Luxembourg: Publications Office of the European Union; 2019.
3. Eurostat. Population structure and ageing. 2022
4. Statbel. Bevolkingsvooruitzichten: Synthese van de demografische vooruitzichten 2023-2070 voor België en de gewesten 2023. 2023.
5. van Dam CS, Labuschagne HA, van Keulen K, Kramers C, Kleipool EE, Hoogendijk EO, et al. Polypharmacy, comorbidity and frailty: a complex interplay in older patients at the emergency department. *Eur Geriatr Med*. 2022;13(4):849-57.
6. Inouye SK, Studenski S, Tinetti ME, Kuchel GA. Geriatric syndromes: clinical, research, and policy implications of a core geriatric concept. *J Am Geriatr Soc*. 2007;55(5):780-91.
7. Van Der Linden L, Decoutere L, Walgraeve K, Milisen K, Flamaing J, Spriet I, et al. Combined Use of the Rationalization of Home Medication by an Adjusted STOPP in Older Patients (RASP) List and a Pharmacist-Led Medication Review in Very Old Inpatients: Impact on Quality of Prescribing and Clinical Outcome. *Drugs & Aging*. 2017;34(2):123-33.
8. Glickman SW, Anstrom KJ, Lin L, Chandra A, Laskowitz DT, Woods CW, et al. Challenges in enrollment of minority, pediatric, and geriatric patients in emergency and acute care clinical research. *Ann Emerg Med*. 2008;51(6):775-80 e3.
9. Beers E, Moerkerken DC, Leufkens HG, Egberts TC, Jansen PA. Participation of older people in preauthorization trials of recently approved medicines. *J Am Geriatr Soc*. 2014;62(10):1883-90.
10. van Marum RJ. Underrepresentation of the elderly in clinical trials, time for action. *Br J Clin Pharmacol*. 2020;86(10):2014-6.
11. Banzi R, Camaioni P, Tettamanti M, Bertele V, Lucca U. Older patients are still under-represented in clinical trials of Alzheimer's disease. *Alzheimers Res Ther*. 2016;8:32.
12. Helfand BK, Webb M, Gartaganis SL, Fuller L, Kwon CS, Inouye SK. The Exclusion of Older Persons From Vaccine and Treatment Trials for Coronavirus Disease 2019-Missing the Target. *JAMA Intern Med*. 2020;180(11):1546-9.
13. Abbasi J. Older Patients (Still) Left Out of Cancer Clinical Trials. *JAMA*. 2019;322(18):1751-3.
14. Watts G. Why the exclusion of older people from clinical research must stop. *BMJ*. 2012;344:e3445.
15. Dubrall D, Just KS, Schmid M, Stingl JC, Sachs B. Adverse drug reactions in older adults: a retrospective comparative analysis of spontaneous reports to the German Federal Institute for Drugs and Medical Devices. *BMC PHARMACOLOGY & TOXICOLOGY*. 2020;21(1).
16. Hilmer SN, Tran K, Rubie P, Wright J, Gnjdjic D, Mitchell SJ, et al. Gentamicin pharmacokinetics in old age and frailty. *Br J Clin Pharmacol*. 2011;71(2):224-31.
17. Johnston C, Hilmer SN, McLachlan AJ, Matthews ST, Carroll PR, Kirkpatrick CM. The impact of frailty on pharmacokinetics in older people: using gentamicin population pharmacokinetic modeling to investigate changes in renal drug clearance by glomerular filtration. *Eur J Clin Pharmacol*. 2014;70(5):549-55.
18. Vergara-Fernandez O, Trejo-Avila M, Salgado-Nesme N. Sarcopenia in patients with colorectal cancer: A comprehensive review. *World J Clin Cases*. 2020;8(7):1188-202.
19. Hilmer SN, Wu H, Zhang M. Biology of frailty: Implications for clinical pharmacology and drug therapy in frail older people. *Mech Ageing Dev*. 2019;181:22-8.
20. Novaes PH, da Cruz DT, Lucchetti ALG, Leite ICG, Lucchetti G. The "iatrogenic triad": polypharmacy, drug-drug interactions, and potentially inappropriate medications in older adults. *Int J Clin Pharm*. 2017;39(4):818-25.
21. Turner RM, de Koning EM, Fontana V, Thompson A, Pirmohamed M. Multimorbidity, polypharmacy, and drug-drug-gene interactions following a non-ST elevation acute coronary syndrome: analysis of a multicentre observational study. *BMC Med*. 2020;18(1):367.

22. Poh Choo How GX. Pharmacological Overview in Geriatrics: Pharmacodynamics, Pharmacokinetics, Laboratory Monitoring. Inpatient Geriatric Psychiatry 2019. p. 47–61.
23. Donna M Fick TPS, Judith Beizer, Nicole Brandt, Robert Dombrowski, Catherine E DuBeau, Woody Eisenberg, Jerome J Epplin, Nina Flanagan, Erin Giovannetti, Joseph Hanlon, Peter Hollmann, Rosemary Laird, Sunny Linnebur, Satinderpal Sandhu, Michael Steinman. American Geriatrics Society 2015 Updated Beers Criteria for Potentially Inappropriate Medication Use in Older Adults. *J Am Geriatr Soc.* 2015;63(11):2227-46.
24. Darowski A, Chambers SA, Chambers DJ. Antidepressants and falls in the elderly. *Drugs Aging.* 2009;26(5):381-94.
25. Hias J, Van der Linden L, Walgraeve K, Gijssen M, Mian P, Koch BCP, et al. Pharmacokinetics of 2 oral paracetamol formulations in hospitalized octogenarians. *British Journal of Clinical Pharmacology.* 2021.
26. Van Der Heijden LT, Mian P, Hias J, De Winter BCM, Tournoy J, Van Der Linden L, et al. Highly Variable Paracetamol Pharmacokinetics After Multiple Oral Dosing in Frail Older People: A Population Pharmacokinetic Analysis. *Drugs & Aging.* 2022;39(1):83-95.
27. Hurwitz A, Ruhl CE, Kimler BF, Topp EM, Mayo MS. Gastric function in the elderly: effects on absorption of ketoconazole. *J Clin Pharmacol.* 2003;43(9):996-1002.
28. Pollard WS, Bloomfield AL. normal standards of gastric function. *J Clin Invest.* 1931;9(4):651-8.
29. Bloomfield AL, Keefer CS. Gastric acidity: relation to various factors such as age and physical fitness. *J Clin Invest.* 1928;5(2):285-94.
30. Vanzant Fr, Alvarez Wc, Eusterman Gb, Dunn Hl, Berkson J. The normal range of gastric acidity from youth to old age: an analysis of 3,746 records. *Archives of Internal Medicine.* 1932;49(3):345-59.
31. Baron JH. Studies of basal and peak acid output with an augmented histamine test. *Gut.* 1963;4(2):136-44.
32. Collen MJ, Abdulian JD, Chen YK. Age does not affect basal gastric acid secretion in normal subjects or in patients with acid-peptic disease. *Am J Gastroenterol.* 1994;89(5):712-6.
33. Goldschmiedt M, Barnett CC, Schwarz BE, Karnes WE, Redfern JS, Feldman M. Effect of age on gastric acid secretion and serum gastrin concentrations in healthy men and women. *Gastroenterology.* 1991;101(4):977-90.
34. Kekki M, Samloff IM, Ihamaki T, Varis K, Siurala M. Age- and sex-related behaviour of gastric acid secretion at the population level. *Scand J Gastroenterol.* 1982;17(6):737-43.
35. Russell TL, Berardi RR, Barnett JL, Dermentzoglou LC, Jarvenpaa KM, Schmaltz SP, et al. Upper gastrointestinal pH in seventy-nine healthy, elderly, North American men and women. *Pharm Res.* 1993;10(2):187-96.
36. Dressman JB, Berardi RR, Dermentzoglou LC, Russell TL, Schmaltz SP, Barnett JL, et al. Upper gastrointestinal (GI) pH in young, healthy men and women. *Pharm Res.* 1990;7(7):756-61.
37. Pedersen PB, Vilmann P, Bar-Shalom D, Müllertz A, Baldursdottir S. Characterization of fasted human gastric fluid for relevant rheological parameters and gastric lipase activities. *Eur J Pharm Biopharm.* 2013;85(3 Pt B):958-65.
38. Annaert P, Brouwers J, Bijnen A, Lammert F, Tack J, Augustijns P. Ex vivo permeability experiments in excised rat intestinal tissue and in vitro solubility measurements in aspirated human intestinal fluids support age-dependent oral drug absorption. *Eur J Pharm Sci.* 2010;39(1-3):15-22.
39. Russell TL, Berardi RR, Barnett JL, O'Sullivan TL, Wagner JG, Dressman JB. pH-related changes in the absorption of dipyridamole in the elderly. *Pharm Res.* 1994;11(1):136-43.
40. Bhutto A, Morley JE. The clinical significance of gastrointestinal changes with aging. *Current Opinion in Clinical Nutrition & Metabolic Care.* 2008;11(5).
41. Rozé C. Neurohumoral control of gastrointestinal motility. *Reprod Nutr Dev (1980).* 1980;20(4b):1125-41.

42. Watson LE, Xie C, Wang X, Li Z, Phillips LK, Sun Z, et al. Gastric Emptying in Patients With Well-Controlled Type 2 Diabetes Compared With Young and Older Control Subjects Without Diabetes. *J Clin Endocrinol Metab.* 2019;104(8):3311-9.
43. Serra-Prat M, Palomera E, Clave P, Puig-Domingo M. Effect of age and frailty on ghrelin and cholecystokinin responses to a meal test. *Am J Clin Nutr.* 2009;89(5):1410-7.
44. Hellstrom PM, Samuelsson B, Al-Ani AN, Hedstrom M. Normal gastric emptying time of a carbohydrate-rich drink in elderly patients with acute hip fracture: a pilot study. *BMC Anesthesiol.* 2017;17(1):23.
45. Gainsborough N, Maskrey VL, Nelson ML, Keating J, Sherwood RA, Jackson SH, et al. The association of age with gastric emptying. *Age Ageing.* 1993;22(1):37-40.
46. Clarkston WK, Pantano MM, Morley JE, Horowitz M, Littlefield JM, Burton FR. Evidence for the anorexia of aging: gastrointestinal transit and hunger in healthy elderly vs. young adults. *Am J Physiol.* 1997;272(1 Pt 2):R243-8.
47. Brogna A, Loreno M, Catalano F, Bucceri AM, Malaguarnera M, Muratore LA, et al. Radioisotopic assessment of gastric emptying of solids in elderly subjects. *Aging Clin Exp Res.* 2006;18(6):493-6.
48. Madsen JL, Graff J. Effects of ageing on gastrointestinal motor function. *Age Ageing.* 2004;33(2):154-9.
49. Kagaya M, Iwata N, Toda Y, Nakae Y, Kondo T. Small bowel transit time and colonic fermentation in young and elderly women. *J Gastroenterol.* 1997;32(4):453-6.
50. Brogna A, Ferrara R, Bucceri AM, Lanteri E, Catalano F. Influence of aging on gastrointestinal transit time. An ultrasonographic and radiologic study. *Invest Radiol.* 1999;34(5):357-9.
51. Hurwitz A, Robinson RG, Herrin WF, Christie J. Oral anticholinergics and gastric emptying. *Clin Pharmacol Ther.* 1982;31(2):168-74.
52. Pfeiffer RF. Gastrointestinal dysfunction in Parkinson's disease. *Lancet Neurol.* 2003;2(2):107-16.
53. Pfeiffer RF, Isaacson SH, Pahwa R. Clinical implications of gastric complications on levodopa treatment in Parkinson's disease. *Parkinsonism & Related Disorders.* 2020;76:63-71.
54. O'Mahony D, O'Leary P, Quigley EMM. Aging and Intestinal Motility. *Drugs & Aging.* 2002;19(7):515-27.
55. Kim SE. Colonic Slow Transit Can Cause Changes in the Gut Environment Observed in the Elderly. *J Neurogastroenterol Motil.* 2017;23(1):3-4.
56. Bertschi D, Waskowski J, Schilling M, Donatsch C, Schefold JC, Pfortmueller CA. Methods of Assessing Frailty in the Critically Ill: A Systematic Review of the Current Literature. *Gerontology.* 2022;68(12):1321-49.
57. Wasmer K, Eckardt L, Breithardt G. Predisposing factors for atrial fibrillation in the elderly. *J Geriatr Cardiol.* 2017;14(3):179-84.
58. Van Der Linden L, Hias J, Walgraeve K, Loyens S, Flamaing J, Spriet I, et al. Factors associated with the number of clinical pharmacy recommendations: findings from an observational study in geriatric inpatients. *Acta Clin Belg.* 2021;76(2):119-26.
59. Prystowsky EN, Padanilam BJ, Fogel RI. Treatment of Atrial Fibrillation. *JAMA.* 2015;314(3):278-88.
60. Nesheiwat Z, Goyal A, Jagtap M. Atrial Fibrillation. *StatPearls.* Treasure Island (FL)2024.
61. Marinigh R, Lip GY, Fiotti N, Giansante C, Lane DA. Age as a risk factor for stroke in atrial fibrillation patients: implications for thromboprophylaxis. *J Am Coll Cardiol.* 2010;56(11):827-37.
62. van Walraven C, Hart RG, Connolly S, Austin PC, Mant J, Hobbs FD, et al. Effect of age on stroke prevention therapy in patients with atrial fibrillation: the atrial fibrillation investigators. *Stroke.* 2009;40(4):1410-6.
63. Risk factors for stroke and efficacy of antithrombotic therapy in atrial fibrillation. Analysis of pooled data from five randomized controlled trials. *Arch Intern Med.* 1994;154(13):1449-57.

64. Hart RG, Pearce LA, Aguilar MI. Meta-analysis: antithrombotic therapy to prevent stroke in patients who have nonvalvular atrial fibrillation. *Ann Intern Med.* 2007;146(12):857-67.
65. Singer DE, Chang Y, Fang MC, Borowsky LH, Pomernacki NK, Udaltsova N, et al. The net clinical benefit of warfarin anticoagulation in atrial fibrillation. *Ann Intern Med.* 2009;151(5):297-305.
66. Holbrook AM, Pereira JA, Labiris R, McDonald H, Douketis JD, Crowther M, et al. Systematic overview of warfarin and its drug and food interactions. *Arch Intern Med.* 2005;165(10):1095-106.
67. Sconce EA, Khan TI, Wynne HA, Avery P, Monkhouse L, King BP, et al. The impact of CYP2C9 and VKORC1 genetic polymorphism and patient characteristics upon warfarin dose requirements: proposal for a new dosing regimen. *Blood.* 2005;106(7):2329-33.
68. Wadhera RK, Russell CE, Piazza G. Cardiology patient page. Warfarin versus novel oral anticoagulants: how to choose? *Circulation.* 2014;130(22):e191-3.
69. Summary of product characteristics: Lixiana (Edoxaban). European Medicines Agency; 2024.
70. Giugliano RP, Ruff CT, Braunwald E, Murphy SA, Wiviott SD, Halperin JL, et al. Edoxaban versus Warfarin in Patients with Atrial Fibrillation. *New England Journal of Medicine.* 2013;369(22):2093-104.
71. Ogata K, Mendell-Harary J, Tachibana M, Masumoto H, Oguma T, Kojima M, et al. Clinical Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of the Novel Factor Xa Inhibitor Edoxaban in Healthy Volunteers. *The Journal of Clinical Pharmacology.* 2010;50(7):743-53.
72. Matsushima N, Lee F, Sato T, Weiss D, Mendell J. Bioavailability and Safety of the Factor Xa Inhibitor Edoxaban and the Effects of Quinidine in Healthy Subjects. *Clin Pharmacol Drug Dev.* 2013;2(4):358-66.
73. Mendell J, Tachibana M, Shi M, Kunitada S. Effects of food on the pharmacokinetics of edoxaban, an oral direct factor Xa inhibitor, in healthy volunteers. *J Clin Pharmacol.* 2011;51(5):687-94.
74. Bathala MS, Masumoto H, Oguma T, He L, Lowrie C, Mendell J. Pharmacokinetics, biotransformation, and mass balance of edoxaban, a selective, direct factor Xa inhibitor, in humans. *Drug Metab Dispos.* 2012;40(12):2250-5.
75. Steffel J, Verhamme P, Potpara TS, Albaladejo P, Antz M, Desteghe L, et al. The 2018 European Heart Rhythm Association Practical Guide on the use of non-vitamin K antagonist oral anticoagulants in patients with atrial fibrillation. *Eur Heart J.* 2018;39(16):1330-93.
76. Ruff CT, Giugliano RP, Braunwald E, Morrow DA, Murphy SA, Kuder JF, et al. Association between edoxaban dose, concentration, anti-Factor Xa activity, and outcomes: an analysis of data from the randomised, double-blind ENGAGE AF-TIMI 48 trial. *Lancet.* 2015;385(9984):2288-95.
77. The state of aging and health in America 2013. Atlanta: Centers for Disease Control and Prevention, U.S. Department of Health and Human Services; 2013.
78. Goto T, Yoshida K, Tsugawa Y, Camargo CA, Hasegawa K. Infectious Disease-Related Emergency Department Visits of Elderly Adults in the United States, 2011-2012. *Journal of the American Geriatrics Society.* 2016;64(1):31-6.
79. Videcnik Zorman J, Lusa L, Strle F, Maraspin V. Bacterial infection in elderly nursing home and community-based patients: a prospective cohort study. *Infection.* 2013;41(5):909-16.
80. Sadighi Akha AA. Aging and the immune system: An overview. *J Immunol Methods.* 2018;463:21-6.
81. Esme M, Topeli A, Yavuz BB, Akova M. Infections in the Elderly Critically-Ill Patients. *Front Med (Lausanne).* 2019;6:118.
82. Elias R, Hartshorn K, Rahma O, Lin N, Snyder-Cappione JE. Aging, immune senescence, and immunotherapy: A comprehensive review. *Semin Oncol.* 2018;45(4):187-200.
83. Tannou T, Koeberle S, Manckoundia P, Aubry R. Multifactorial immunodeficiency in frail elderly patients: Contributing factors and management. *Med Mal Infect.* 2019;49(3):167-72.
84. Montoya A, Mody L. Common infections in nursing homes: a review of current issues and challenges. *Aging health.* 2011;7(6):889-99.

85. Pop-Vicas A, Mitchell SL, Kandel R, Schreiber R, D'Agata EM. Multidrug-resistant gram-negative bacteria in a long-term care facility: prevalence and risk factors. *J Am Geriatr Soc.* 2008;56(7):1276-80.
86. Trick WE, Weinstein RA, DeMarais PL, Kuehnert MJ, Tomaska W, Nathan C, et al. Colonization of skilled-care facility residents with antimicrobial-resistant pathogens. *J Am Geriatr Soc.* 2001;49(3):270-6.
87. Solis-Hernandez PS, Vidales-Reyes M, Garza-Gonzalez E, Guajardo-Alvarez G, Chavez-Moreno S, Camacho-Ortiz A. Hospital-Acquired Infections in Elderly Versus Younger Patients in an Acute Care Hospital. *International Journal of Infection.* 2015;3(1).
88. Kirby WM. Vancomycin therapy of severe staphylococcal infections. *J Antimicrob Chemother.* 1984;14 Suppl D:73-8.
89. Summary of product characteristics: Vancomycin 2022.
90. Hammes WP, Neuhaus FC. On the mechanism of action of vancomycin: inhibition of peptidoglycan synthesis in *Gaffkya homari*. *Antimicrob Agents Chemother.* 1974;6(6):722-8.
91. Payne CJ, Thomson AH, Stearns AT, Watson DG, Zhang T, Kingsmore DB, et al. Pharmacokinetics and tissue penetration of vancomycin continuous infusion as prophylaxis for vascular surgery. *J Antimicrob Chemother.* 2011;66(11):2624-7.
92. Matzke GR, Zhanel GG, Guay DR. Clinical pharmacokinetics of vancomycin. *Clin Pharmacokinet.* 1986;11(4):257-82.
93. Rodvold KA, Blum RA, Fischer JH, Zokufa HZ, Rotschafer JC, Crossley KB, et al. Vancomycin pharmacokinetics in patients with various degrees of renal function. *Antimicrob Agents Chemother.* 1988;32(6):848-52.
94. Rotschafer JC, Crossley K, Zaske DE, Mead K, Sawchuk RJ, Solem LD. Pharmacokinetics of vancomycin: observations in 28 patients and dosage recommendations. *Antimicrob Agents Chemother.* 1982;22(3):391-4.
95. Blouin RA, Bauer LA, Miller DD, Record KE, Griffen WO, Jr. Vancomycin pharmacokinetics in normal and morbidly obese subjects. *Antimicrob Agents Chemother.* 1982;21(4):575-80.
96. Golper TA, Noonan HM, Elzinga L, Gilbert D, Brummett R, Anderson JL, et al. Vancomycin pharmacokinetics, renal handling, and nonrenal clearances in normal human subjects. *Clin Pharmacol Ther.* 1988;43(5):565-70.
97. Ackerman BH, Taylor EH, Olsen KM, Abdel-Malak W, Pappas AA. Vancomycin serum protein binding determination by ultrafiltration. *Drug Intell Clin Pharm.* 1988;22(4):300-3.
98. Bailey EM, Rybak MJ, Kaatz GW. Comparative effect of protein binding on the killing activities of teicoplanin and vancomycin. *Antimicrob Agents Chemother.* 1991;35(6):1089-92.
99. Rybak MJ, Le J, Lodise TP, Levine DP, Bradley JS, Liu C, et al. Therapeutic monitoring of vancomycin for serious methicillin-resistant *Staphylococcus aureus* infections: A revised consensus guideline and review by the American Society of Health-System Pharmacists, the Infectious Diseases Society of America, the Pediatric Infectious Diseases Society, and the Society of Infectious Diseases Pharmacists. *Am J Health Syst Pharm.* 2020;77(11):835-64.
100. Albanese J, Leone M, Bruguerolle B, Ayem ML, Lacarelle B, Martin C. Cerebrospinal fluid penetration and pharmacokinetics of vancomycin administered by continuous infusion to mechanically ventilated patients in an intensive care unit. *Antimicrob Agents Chemother.* 2000;44(5):1356-8.
101. Patel S, Preuss CV, Bernice F. Vancomycin. StatPearls. Treasure Island (FL)2024.
102. Aljutayli A, Marsot A, Nekka F. An Update on Population Pharmacokinetic Analyses of Vancomycin, Part I: In Adults. *Clin Pharmacokinet.* 2020;59(6):671-98.
103. Moise-Broder PA, Forrest A, Birmingham MC, Schentag JJ. Pharmacodynamics of vancomycin and other antimicrobials in patients with *Staphylococcus aureus* lower respiratory tract infections. *Clin Pharmacokinet.* 2004;43(13):925-42.
104. Singh NB, Yim J, Jahanbakhsh S, Sakoulas G, Rybak MJ. Impact of cefazolin co-administration with vancomycin to reduce development of vancomycin-intermediate *Staphylococcus aureus*. *Diagn Microbiol Infect Dis.* 2018;91(4):363-70.

105. Vidaillac C, Gardete S, Tewhey R, Sakoulas G, Kaatz GW, Rose WE, et al. Alternative mutational pathways to intermediate resistance to vancomycin in methicillin-resistant *Staphylococcus aureus*. *J Infect Dis*. 2013;208(1):67-74.
106. Suzuki Y, Kawasaki K, Sato Y, Tokimatsu I, Itoh H, Hiramatsu K, et al. Is peak concentration needed in therapeutic drug monitoring of vancomycin? A pharmacokinetic-pharmacodynamic analysis in patients with methicillin-resistant *staphylococcus aureus* pneumonia. *Chemotherapy*. 2012;58(4):308-12.
107. Bellos I, Daskalakis G, Pergialiotis V. Relationship of vancomycin trough levels with acute kidney injury risk: an exposure-toxicity meta-analysis. *J Antimicrob Chemother*. 2020;75(10):2725-34.
108. Ette EI, Williams PJ. *Pharmacometrics: The Science of Quantitative Pharmacology*. New Jersey 2007.
109. Mould DR, Upton RN. Basic concepts in population modeling, simulation, and model-based drug development-part 2: introduction to pharmacokinetic modeling methods. *CPT Pharmacometrics Syst Pharmacol*. 2013;2(4):e38.
110. Hariharan S, Madabushi R. Clinical pharmacology basis of deriving dosing recommendations for dabigatran in patients with severe renal impairment. *Journal of clinical pharmacology*. 2012;52(1):119S-25S.
111. Minichmayr IK, Dreesen E, Centanni M, Wang Z, Hoffert Y, Friberg LE, et al. Model-informed precision dosing: State of the art and future perspectives. *Adv Drug Deliv Rev*. 2024;115421.
112. Demeester C, Robins D, Edwina AE, Tournoy J, Augustijns P, Ince I, et al. Physiologically based pharmacokinetic (PBPK) modelling of oral drug absorption in older adults - an AGePOP review. *Eur J Pharm Sci*. 2023;188:106496.
113. Mariani D, Ragusa FS, Alongi M, Gugliuzza E, Petta G, Luca A, et al. The Association Between Multidimensional Frailty and Poor Venous Accesses in a Geriatric Population: A Retrospective Study. *Rejuvenation Res*. 2024;27(2):75-80.
114. Soedarsono S, Jayanti RP, Mertaniasih NM, Kusmiati T, Permatasari A, Indrawanto DW, et al. Development of population pharmacokinetics model and Bayesian estimation of rifampicin exposure in Indonesian patients with tuberculosis. *Tuberculosis (Edinb)*. 2023;139:102325.
115. Sanchez JL, Dominguez AR, Lane JR, Anderson PO, Capparelli EV, Cornejo-Bravo JM. Population pharmacokinetics of vancomycin in adult and geriatric patients: comparison of eleven approaches. *Int J Clin Pharmacol Ther*. 2010;48(8):525-33.
116. Maire P, Barbaut X, Girard P, Mallet A, Jelliffe RW, Berod T. Preliminary results of three methods for population pharmacokinetic analysis (NONMEM, NPML, NPEM) of amikacin in geriatric and general medicine patients. *Int J Biomed Comput*. 1994;36(1-2):139-41.
117. Kim R, Jayanti RP, Lee H, Kim HK, Kang J, Park IN, et al. Development of a population pharmacokinetic model of pyrazinamide to guide personalized therapy: impacts of geriatric and diabetes mellitus on clearance. *Front Pharmacol*. 2023;14:1116226.
118. Barends C, den Daas I, Driesens M, Visser A, Absalom A, Colin P. Development of a pharmacokinetic and pharmacodynamic model for intranasal administration of midazolam in older adults: a single-site two-period crossover study. *Br J Anaesth*. 2023;131(2):284-93.
119. Skinner-Robertson S, Fradette C, Bouchard S, Mouksassi MS, Varin F. Pharmacokinetics of Tramadol and O-Desmethyltramadol Enantiomers Following Administration of Extended-Release Tablets to Elderly and Young Subjects. *Drugs Aging*. 2015;32(12):1029-43.
120. Kwa A, Sprung J, Guilder MV, Jelliffe RW. A population pharmacokinetic model of epidural lidocaine in geriatric patients: effects of low-dose dopamine. *Ther Drug Monit*. 2008;30(3):379-89.
121. Ornstein E, Lien CA, Matteo RS, Ostapovich ND, Diaz J, Wolf KB. Pharmacodynamics and pharmacokinetics of cisatracurium in geriatric surgical patients. *Anesthesiology*. 1996;84(3):520-5.

122. Zhou Y, Gao F, Chen C, Ma L, Yang T, Liu X, et al. Development of a Population Pharmacokinetic Model of Vancomycin and its Application in Chinese Geriatric Patients with Pulmonary Infections. *Eur J Drug Metab Pharmacokinet*. 2019;44(3):361-70.
123. Mian P, van Esdonk MJ, Olkkola KT, de Winter BCM, Liukas A, Spriet I, et al. Population pharmacokinetic modelling of intravenous paracetamol in fit older people displays extensive unexplained variability. *Br J Clin Pharmacol*. 2019;85(1):126-35.
124. Eugene AR. Gender based Dosing of Metoprolol in the Elderly using Population Pharmacokinetic Modeling and Simulations. *Int J Clin Pharmacol Toxicol*. 2016;5(3):209-15.
125. Charhon N, Neely MN, Bourguignon L, Maire P, Jelliffe RW, Goutelle S. Comparison of four renal function estimation equations for pharmacokinetic modeling of gentamicin in geriatric patients. *Antimicrob Agents Chemother*. 2012;56(4):1862-9.
126. Cazaubon Y, Bourguignon L, Goutelle S, Martin O, Maire P, Ducher M. Are ciprofloxacin dosage regimens adequate for antimicrobial efficacy and prevention of resistance? *Pseudomonas aeruginosa* bloodstream infection in elderly patients as a simulation case study. *Fundam Clin Pharmacol*. 2015;29(6):615-24.
127. Bourguignon L, Ducher M, Matanza D, Bleyzac N, Uhart M, Odouard E, et al. The value of population pharmacokinetics and simulation for postmarketing safety evaluation of dosing guidelines for drugs with a narrow therapeutic index: buflomedil as a case study. *Fundam Clin Pharmacol*. 2012;26(2):279-85.

Objectives

The overall aim of this PhD thesis is to investigate gastrointestinal physiology and pharmacokinetics of frequently used drugs (i.e. edoxaban and vancomycin) with high risk and narrow therapeutic-toxic margins in geriatric patients and older adults aged **75 years and older** to improve our understanding of drug disposition in this population. By using pharmacometrics modeling and simulation, the ultimate goal is to enhance the safety and efficacy of commonly used, high-risk medications with narrow therapeutic windows in this vulnerable population. This PhD project specifically addresses the following objectives:

Part 1: Gastrointestinal physiology

Chapter 1. *Tracking the journey: gastrointestinal transit and intraluminal pH dynamics in aging populations using wireless motility capsule*

To characterize segmental gastrointestinal pH profiles and transit times in the aging population, including community-dwelling older adults and geriatric inpatients, compared to young, healthy volunteers, using the wireless motility capsule. It was hypothesized that the aging population would have a longer transit in all segments and increased luminal pH of the upper GI tract.

Part 2: Pharmacokinetics of DOACs

Chapter 2. *Insights into the Pharmacokinetics and Pharmacodynamics of Direct Oral Anticoagulants in Older Adults with Atrial Fibrillation: A Structured Narrative Review*

To provide a comprehensive summary of the PK and PD of apixaban, edoxaban, rivaroxaban, and dabigatran in older adults with atrial fibrillation using systematic literature searches, and thus could be used to inform the design of the prospective PK study in Chapter 3. It was hypothesized that, based on the literature, older adults had higher interindividual PK and PD variability than younger adults, which were sourced from their unique characteristics.

Chapter 3. *Assessing the Suitability of Fixed-Dose Edoxaban in Frail Geriatric Patients: A Population Pharmacokinetics Study*

To exploit population PK modeling to describe the dose–exposure relationship of edoxaban in frail geriatric patients and identify patient characteristics that contribute to PK variability. We performed simulations to predict whether the current fixed-dose guideline for edoxaban offers safe and effective stroke prevention for geriatric patients by using target trough concentrations. Additionally, the link between anti-factor Xa activity levels and edoxaban

plasma concentrations was evaluated. We hypothesized that geriatric patients had lower clearance and volume of distribution as well as slower absorption than the population in pivotal trials, which led to higher edoxaban exposures.

Part 3: Pharmacokinetics of Vancomycin

Chapter 4. *Decreased Kidney Function Explains Higher Vancomycin Exposure in Older Adults*

To assess the attainment of therapeutic targets at the first therapeutic drug monitoring measurement of vancomycin in older adults compared to younger patients. Additionally, we explored how various patient characteristics influence vancomycin concentrations. It was hypothesized that older patients had higher vancomycin exposures compared to younger patients when similar doses were applied.

Chapter 5. *Optimizing vancomycin dosing in older patients: predictive performance of pharmacometrics models*

To evaluate the predictive performance of both single-model and multi-model approaches using published population PK models to guide vancomycin dosing in older patients. We hypothesized that multi-model algorithms and popPK models including older patients aged 75 years and above had superior predictive performance.

Part 1. Gastrointestinal physiology

Chapter 1. Gastrointestinal transit and intraluminal pH dynamics in aging populations using wireless motility capsule

This chapter is published as:

Edwina AE, Koziolk M, Steenackers N, Vanuytsel T, López Mármol Á, Matthys C, Dreesen E, Van der Linden L, Spriet I, Tournoy J. *Tracking the journey: gastrointestinal transit and intraluminal pH dynamics in aging populations using wireless motility capsule*. Int J Pharm. 2024 Dec 5;669:125016

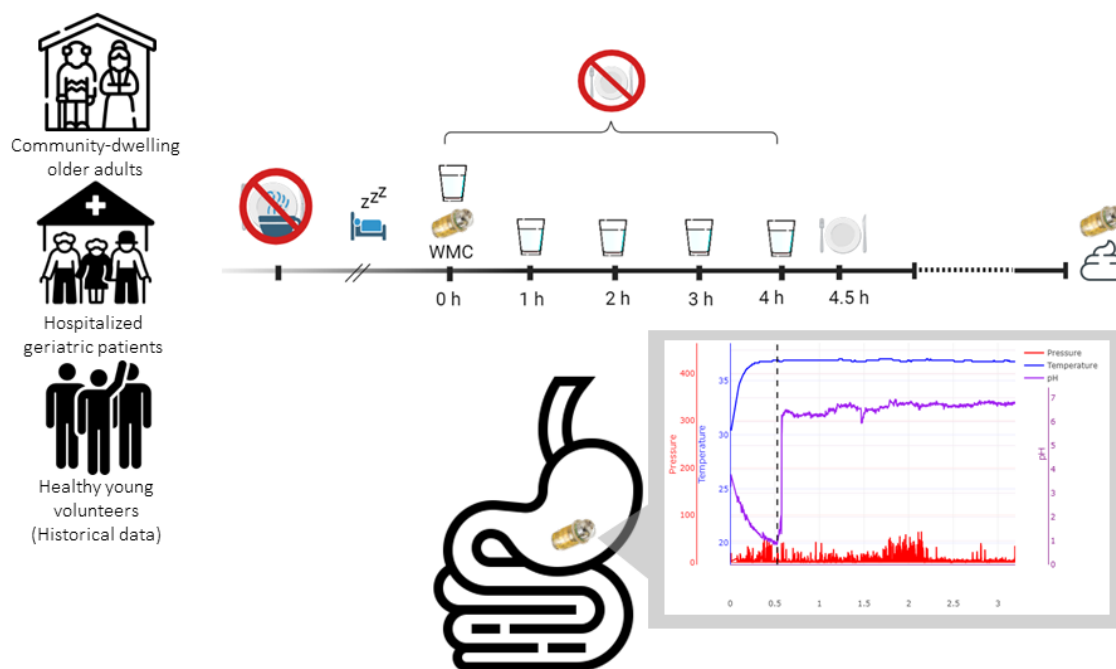
Abstract

Introduction: Gastrointestinal (GI) changes may alter drug absorption, potentially impacting both efficacy and safety of oral pharmacotherapy. However, the GI physiology is rarely studied in the aging population. This study aimed to explore GI transit time and pH in geriatric inpatients and older adults, and compare these findings with those from young volunteers.

Methods: A prospective study was performed in geriatric inpatients and community-dwelling adults aged ≥ 75 years. GI transit and intraluminal pH were assessed by using a wireless motility capsule system. Participants' data were pooled with those from a previous study in healthy young volunteers ($n=11$). Multivariable linear regression models were developed to identify explanatory variables for GI transit time and pH value.

Results: We recruited 30 subjects (median age 80 [interquartile range (IQR)=77–83] years). When compared to young volunteers (median age 21 [20–33] years), geriatric inpatients ($N=8$) exhibited prolonged gastric emptying times (median=0.63 h [IQR=0.32–0.86 h] vs median=2.98 h [IQR=1.50–20.35 h], $p<0.01$). Community-dwelling older adults ($N=22$) did not demonstrate delayed gastric emptying (median=0.42 h [IQR=0.36–0.53 h]; $p=0.85$). Proton pump inhibitors (PPI) use was associated with increased gastric pH (median pH=3.75 [IQR=1.69–4.96] vs. non-PPI median pH=1.28 [IQR=1.01–1.70]; $p<0.01$) in the pooled analysis. The total older cohort had significantly lower small intestine pH compared to young group (median pH=6.8 vs. 7.2; $p<0.01$).

Conclusion: Older age did not necessarily alter GI physiology with respect to gastric emptying time and luminal pH values. Only geriatric inpatients exhibited delayed gastric emptying of the motility capsule as compared to young volunteers, while community-dwelling older adults did not. Prevalent PPI use in the aging population led to higher gastric pH levels, with substantial intersubject variability.



Graphical abstract

1. Introduction

The global population is aging rapidly, particularly in high-income countries, with projections indicating a doubling of the number of older adults over the next three decades (1, 2). This demographic shift presents substantial challenges to healthcare systems worldwide, primarily due to the increased demand for already limited healthcare resources (3). Prescribed medication use increases considerably with age, peaking among the oldest age groups due to high multimorbidity. For instance, 86% of European adults aged 75 and above use prescription medicines (4). Among this demographic, geriatric patients in particular face heightened health risks due to functional disabilities, multimorbidity, and polypharmacy.

Unfortunately, clinical studies to investigate drug therapies predominantly enroll younger adults, thereby underrepresenting older adults. Older adults aged 75 years and older represent only 22% of participants in phase III trials for conditions common in their age group, and only 1% in trials for diseases that are less common among them (5, 6). Recognizing this scarcity of relevant data, the European Medicines Agency (EMA) and the US Food and Drug Administration (FDA) have stressed the critical importance of tailored pharmaceutical evaluations for older adults (7, 8). Such a paradigm entails exploring physiological changes that accompany aging, including those affecting gastrointestinal (GI) physiology, which might significantly alter a drug's pharmacokinetics (PK). This understanding is essential for optimizing drug efficacy and safety in older adults.

Older adults may experience changes in GI physiology that can affect GI transit time and intraluminal pH values. These changes frequently manifest as relevant clinical conditions such as constipation or gastroparesis (9), often worsened by diabetes, thyroid diseases and other age-related comorbidities (10). Additionally, medications such as opioids, glucagon-like peptide 1 receptor antagonists, and anticholinergics might further prolong GI transit time. The widespread use of proton pump inhibitors (PPIs) also modifies intraluminal pH values. Together, these factors can markedly impact the release and absorption of drugs and nutrients (11). Our recent research has revealed a high unexplained interindividual variability in the absorption phase of paracetamol, a commonly used analgesic, among geriatric patients (12, 13). This variability, unaccounted for by known covariates in our analysis, likely stems from these physiological changes, particularly since

paracetamol is highly soluble and highly permeable drug for which absorption is highly dependent on gastric emptying time.

Given the unique characteristics of the geriatric population, there is an urgent need for an improved understanding of GI transit times and pH values. In particular, the detailed characterization of GI physiology in older adults is required, as this information is essential for the development of tailored formulations and the design of predictive biopharmaceutics *in vitro* tools and *in silico* models. Advanced technologies, such as the SmartPill® wireless motility capsule (WMC), provide a minimally invasive assessment of GI transit time, intraluminal pH and motility patterns throughout the GI tract (14). This study aimed to characterize GI transit and intraluminal pH in an aging population, defined as both community-dwelling older adults and geriatric inpatients, by using the WMC technology, and identify factors that influence these GI parameters. Data in older cohort were compared with data from healthy young volunteers.

2. Methods

This prospective exploratory study was performed at the University Hospitals Leuven, Leuven, Belgium. The study procedure was in accordance with the Declaration of Helsinki and approved by the Ethics Committee Research UZ/KU Leuven (EC reference number S66501). All subjects entered the study after providing written informed consent.

2.1. Study participants

Subjects aged 75 years and above were recruited at the hospital's geriatric wards and community-dwelling older population. Exclusion criteria were defined as follows: swallowing problems, Zenker's diverticulum, history of diverticulitis, extensive diverticula, gastric hernia, gastric bezoar, Crohn's disease, strictures, fistulas, known GI obstruction, history of GI surgery except appendectomy, or in case of an implanted or portable electro-mechanical medical device, scheduled for magnetic resonance imaging, and those who were mentally or legally incapacitated based on the physicians' discretion.

2.2. Study procedures

A SmartPill® (Medtronic, Dublin, Ireland) is a single-use wireless motility capsule (WMC) with a size of 26 mm × 13 mm that consists of a pressure, pH and temperature sensor. Study patients were asked to refrain from consuming any foods and drinks overnight for a

minimum of 10 h before WMC administration. The following morning, the WMC was administered with 150 mL of still water at room temperature. This is in line with drug administration in bioequivalence and food effect studies according to EMA (15). Before administration, WMC required activation and calibration using the MotiliGI software and the provided buffer (sodium citrate, pH 6.0). WMC ingestion was supervised with mouth checking and confirmed with the pH change and higher pressure caused by stomach entry, which was recognized through the data receiver. Participants were observed during for 5 h after ingestion. During this timeframe, participants were asked to drink 100 mL of water every hour. A standard lunch consisting of four slices of white bread, 10 g butter, two slices of cheese, two slices of cooked ham and ad libitum water (maximum caloric value: 500 kcal) was served 4.5 h after WMC intake. Subsequently, they were allowed to eat and drink without restrictions (immediately) after lunch. The data receiver needed to be worn closely to the body in order to record the data packages sent by the capsule or must be kept as close as possible, maximum 1.5 m, to minimize data loss. Additionally, subjects were asked to document their sleeping habits, food intake, defecation and periods in which they did not carry their recorder in an event diary until the excretion of the capsule. To ensure participant safety, an abdominal X-ray was performed if the WMC had not passed after seven days. If the X-ray indicated that the capsule remained in the GI tract, or if the patient experienced obstruction symptoms, laxatives were administered to facilitate its passage.

2.3. Data collection

The following patient characteristics were collected during the study visit: age, sex at birth, body weight, height, body mass index (BMI), clinical frailty scale (CFS) (14), Charlson comorbidity index (CCI) (15) and concomitant medications including PPIs and anticholinergic drugs based on the list published by Kiesel et al. (16). The relative potency of PPIs was calculated as omeprazole equivalents, taking into account both the specific PPI and its dose. Relative PPI potency ranged from 4.5 mg omeprazole equivalents (20 mg pantoprazole) to 72 mg omeprazole equivalents (40 mg rabeprazole) (16).

2.4. Historical healthy young controls data

Previously published data from 11 healthy young volunteers with normal weight ($18.5 \text{ kg/m}^2 \leq \text{BMI} \leq 25 \text{ kg/m}^2$) were also pooled in our analysis (17). In terms of in- and exclusion criteria, aside from age, the young cohort differed only in that they did not use any medications one

week before the WMC administration until the end of the study and had no morbidities. This cross-sectional study involved two visits per participant: one under a fasted state protocol and the other under a fed state protocol. For our analysis, we only pooled data from the fasted state, as it followed the same study procedures as in our current study.

2.5. Data processing and statistical analysis

Following the excretion of the capsule, the raw data from the data receiver were extracted and imported into R (version 4.3.1, R Core Team, Vienna, Austria). Temperature-compensated data were segmented into stomach, small intestine, and colon based on the following specific physiological markers (**Figure 7**): the ingestion time point, which correlates with a transition from ambient to body temperature; the gastric emptying time point, characterized by a sudden and consistent pH increase of at least 3 pH units, indicating a move from acidic to neutral environment; and the ileocecal passage time point, marked by a decrease in pH of at least 1 unit. The capsule excretion was defined by a notable drop in temperature or data loss (18). These landmarks were established through consensus among three authors (AE, MK, JT).

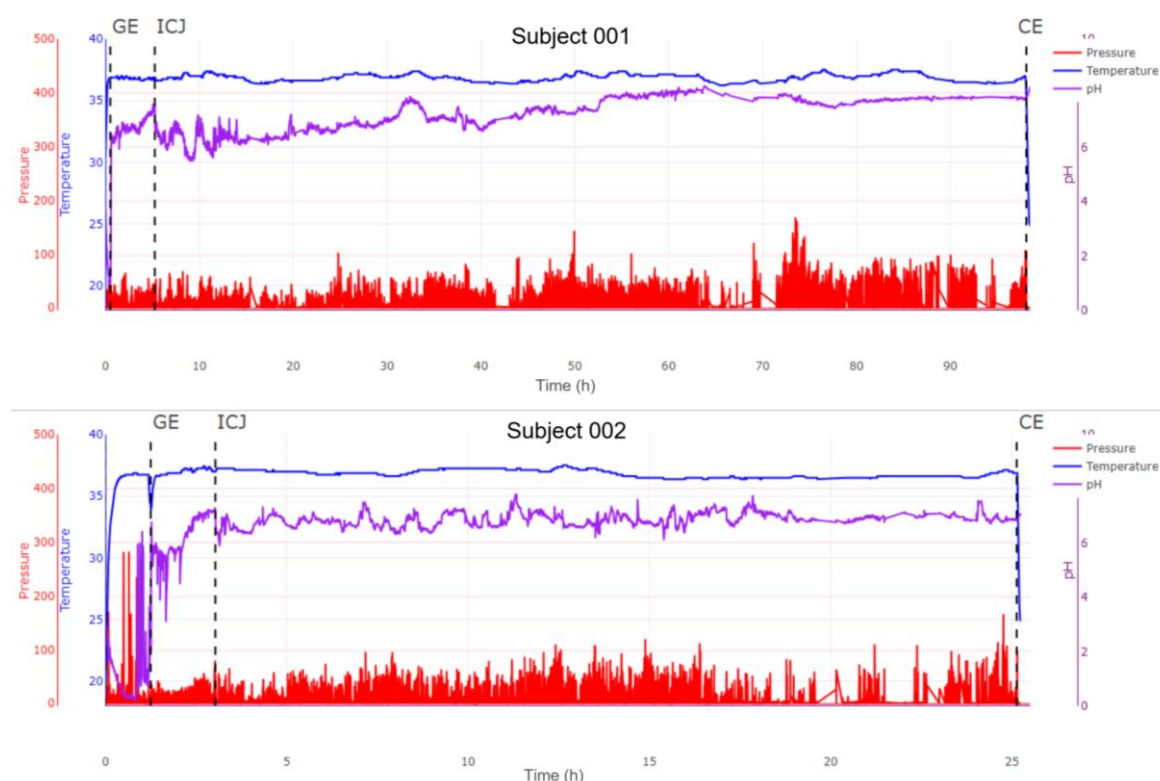


Figure 7. Examples of gastrointestinal pH, pressure, and temperature profiles from WMC readings of two subjects, highlighting various physiological landmarks.

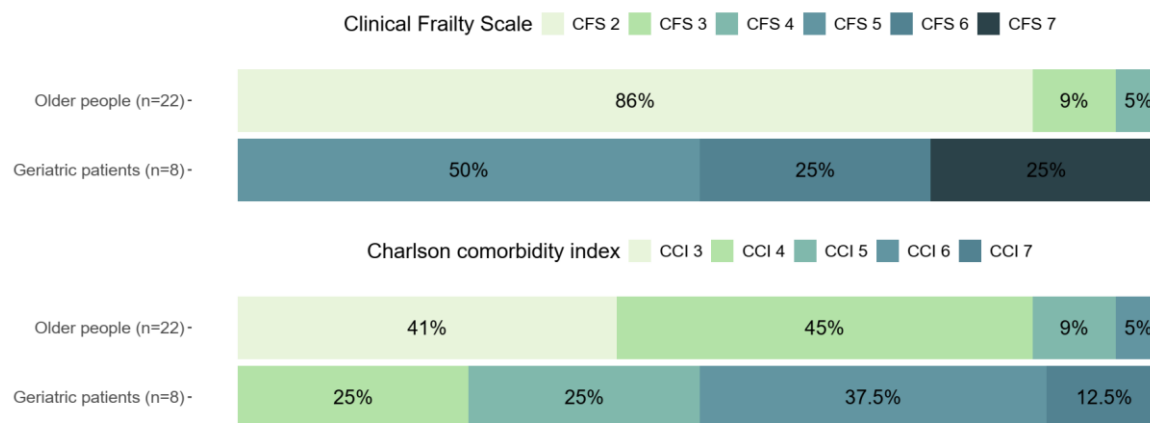
GE: Gastric Emptying; ICJ: Ileocecal Junction; CE: Capsule Excretion

Specific parameters such as the minimum, 25th percentile, median, 75th percentile, and maximum pH values were determined per segment. This enabled us to capture the intraindividual variability given the sheer number of pH measurements recorded during the WMC's transit through the GI tract (17). The highest values of intragastric pressures were extracted. For descriptive statistics, data are presented as median [interquartile range, IQR] unless specified otherwise. The differences in GI physiology parameters, i.e., transit time, pH values and intragastric pressures, between community-dwelling older adults, hospitalized geriatric patients and young volunteers were first tested by using ANOVA for normally distributed data or Kruskal-Wallis for non-normally distributed data. Stepwise multiple linear regression with forward selection was used to assess the association between subject characteristics and GI parameters with a significance level of $p < 0.05$. This allows the inclusion of multiple covariates that may affect GI physiology. Data processing, statistical and graphical analyses were conducted using R in conjunction with RStudio (2023.06.1+524) (19, 20).

Table 1. Subject characteristics of community-dwelling older adults, geriatric patients and young volunteers

Variables	Pooled older cohort (n=30)*			Young volunteers (n=11)*
	Pooled older cohort (n = 30)	Community-dwelling older adults (n=22)	Geriatric patients (n=8)	
<i>Age (y)</i>	80.9 ± 4.3	80.5 ± 4.3	82.0 ± 4.6	29.2 ± 14.3
<i>Sex (% male) (n)</i>	56.7% (n=17)	45.5% (n=10)	87.5% (n=7)	9.1% (n=1)
<i>Body weight (kg)</i>	72.3 ± 15.6	68.9 ± 15.7	81.6 ± 11.5	63.6 ± 7.1
<i>Height (m)</i>	1.68 ± 0.09	1.67 ± 0.09	1.71 ± 0.07	1.71 ± 0.09
<i>BMI (kg/m²)</i>	25.6 ± 4.4	24.7 ± 4.5	27.8 ± 3.5	21.8 ± 1.4
<i>Concomitant medication with:</i>				
<i>PPI (%) (n)</i>	46.7% (n=14)	40.9% (n=9)	62.5% (n=5)	-
<i>Anticholinergic drugs (%) (n)</i>	33.3% (n=10)	31.8% (n=7)	50% (n=4)	-

*Values are expressed as mean ± standard deviation; % (total number)

**Figure 8.** Clinical frailty scale and Charlson comorbidity index in community-dwelling older adults and geriatric patients

3. Results

3.1. Study participants

A total of 33 participants were enrolled in the older cohort. Three were unable to swallow the WMC, resulting in 30 participants who successfully completed the study with a median age of 80 [77–83.75] years. However, data from two participants were incomplete due to a technical failure in the pH detector. The data from one participant could only be used to determine the whole gut transit time, whereas the data from the other subject was limited to gastric emptying time and gastric pH.

Among the participants who completed the study, 22 were community-dwelling older adults with a CFS score of 2 to 4, indicating fit to very mild frail persons. The remaining eight participants were geriatric inpatients with a CFS score of 5 to 7. Community-dwelling older adults generally exhibited a lower CCI than geriatric patients (**Figure 8**). Almost half of the participants routinely took PPIs ($n=14/30$, 46.7%), and one-third used anticholinergic medications ($n=10/30$, 33.3%). More detailed patient characteristics are summarized in **Table 1**.

3.2. Transit time and intragastric pressure

The median gastric transit time in our study cohort was 0.50 [0.3–1.25] h, as shown in **Table 2**. Multivariable linear regression analysis indicated that gastric emptying times in geriatric patients was significantly delayed compared to those of healthy young volunteers (p -value <0.01), while no significant difference was observed between community-dwelling older adults and healthy young volunteers (**Supplementary table 1**). The gastric emptying times for most geriatric inpatients were in the upper quartile of the entire cohort. Three extreme cases with around ≥ 20 times IQR were identified in geriatric patients, with gastric emptying times of approximately 20 h ($n=2$) and 65 h ($n=1$) (**Figure 9**). **Figure 10** shows the intragastric pressure in healthy young volunteers, older adults and geriatric patients. No variables were found to be associated with intragastric pressure values (**Supplementary table 4**).

Small intestine transit time (SITT) exhibited less variability compared to gastric emptying time, with a median of 4.95 [4.58–5.75] h. Six participants had an SITT longer than 6 h, with one participant reaching 13 h. Most participants exhibited a normal colon transit

time (CTT) of less than 59 h (median 43.78 [22.1–70.72] h) (21), although a quarter of the cohort had CTTs ranging from 68 to 120 h.

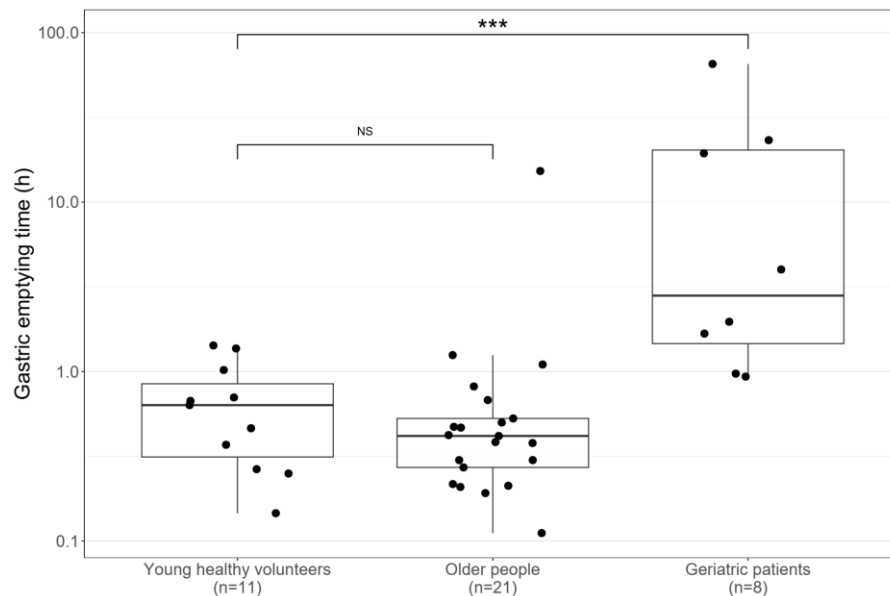


Figure 9. Adjusted differences in gastric emptying times among young healthy volunteers, community-dwelling older adults and geriatric patients. The center lines represent the median values, with the box extending from the 25th percentile to the 75th percentile; the whiskers cover the 1.5-fold interquartile range. Y axis is in logarithmic scale *** P<0.001

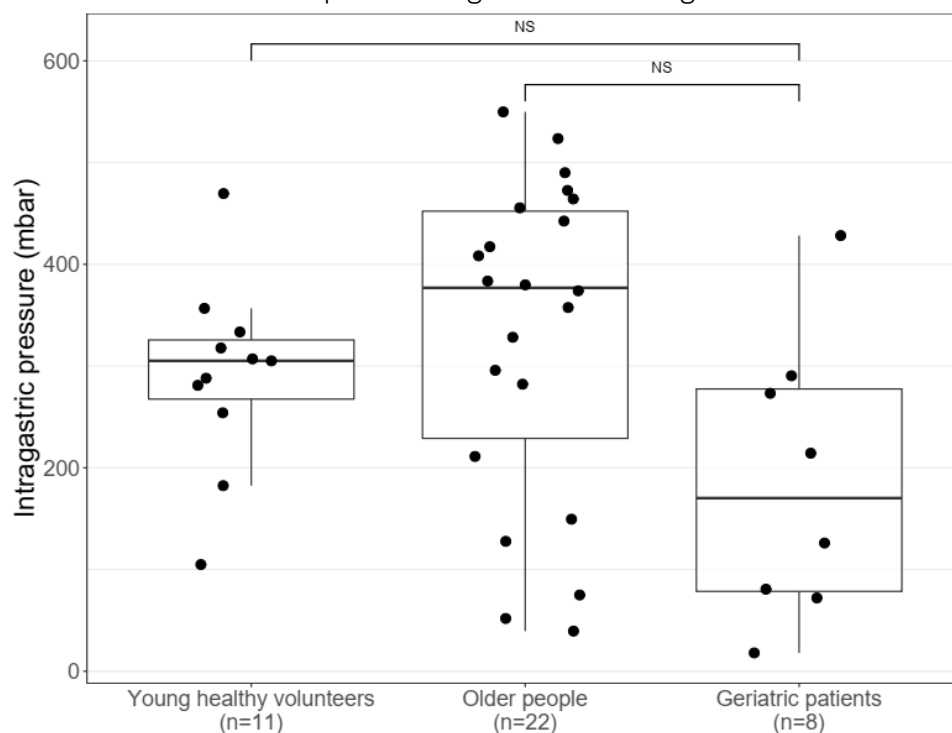


Figure 10. Maximum values of intra-gastric pressures among young healthy volunteers, community-dwelling older adults and geriatric patients. The center lines represent the median values, with the box extending from the 25th percentile to the 75th percentile; the whiskers cover the 1.5-fold interquartile range.

Table 2. Transit times per gastrointestinal segment in community-dwelling older adults, geriatric patients and young volunteers

GI segment	Transit time (h), median [IQR]			
	Older cohort (n=30)			Young volunteers (n=11)
	All older cohort (n = 30)	Community-dwelling older adults (n=22)	Geriatric patients (n=8)	
Gastric	0.50 [0.3–1.25]	0.42 [0.27-0.53]	2.98 [1.5-20.35] ^{a,b}	0.63 [0.32-0.86]
Small intestine	4.92 [4.58–5.75]	4.74 [4.53-5.47]	6.01 [5.13-6.92] ^a	4.17 [3.78-4.86]
Colon	43.78 [22.1–70.72]	41.15 [20.08-68.22]	47.37 [35.73-80.72]	21.87 [17.75-46.24]
Whole gut	50.34 [26.49–78.1]	49.62 [25.1-75.8]	57.83 [52.24-95.3] ^a	26.72 [23.14-51.55]

IQR: Interquartile range, p25: 25th percentile, p75: 75th percentile, SD: Standard deviation

Significant differences are indicated by ^afor comparisons with young volunteers and ^bfor comparisons with community-dwelling older adults

(post-hoc test p-value < 0.05)

Table 3. Intraluminal pH values per gastrointestinal segment in community-dwelling older adults, geriatric patients and young volunteers

GI segment		Intraluminal pH, mean ± SD			
		All older cohort (n = 30)	Community-dwelling older adults (n=22)	Geriatric patients (n=8)	Young volunteers (n=11)
Gastric	p25	2.06 ± 1.93	1.93 ± 1.74	2.39 ± 2.46	1.60 ± 0.35
	median	2.61 ± 2.01	2.54 ± 1.85	2.78 ± 2.51	1.89 ± 0.34
	p75	3.49 ± 2.05	3.51 ± 2	3.43 ± 2.32	2.35 ± 0.35
Small intestine	p25	6.25 ± 0.43	6.14 ± 0.39	6.51 ± 0.41	6.22 ± 0.09
	median	6.77 ± 0.38	6.69 ± 0.39	6.97 ± 0.26	7.15 ± 0.13
	p75	7.11 ± 0.28	7.05 ± 0.29	7.28 ± 0.18	7.43 ± 0.08
Colon	p25	6.1 ± 0.58	6.03 ± 0.45	6.29 ± 0.83	6.16 ± 0.20
	median	6.57 ± 0.61	6.54 ± 0.5	6.63 ± 0.86	6.69 ± 0.21
	p75	6.91 ± 0.57	6.9 ± 0.47	6.96 ± 0.81	7.14 ± 0.19

IQR: Interquartile range, p25: 25th percentile, p75: 75th percentile, SD: Standard deviation

Generally, whole gut transit time was under 48 h with a minimum of 9 h and a maximum of 8 days (median 50.34 [26.49–78.1] h). Geriatric patients had significantly longer whole gut transit time (p-value=0.01). This was due to longer gastric emptying time since SITT and CTT were not affected in geriatric patients. No significant associations were found between other covariates, including sex, body size, comedication with anticholinergics and PPIs, and GI transit time (**Supplementary table 1**).

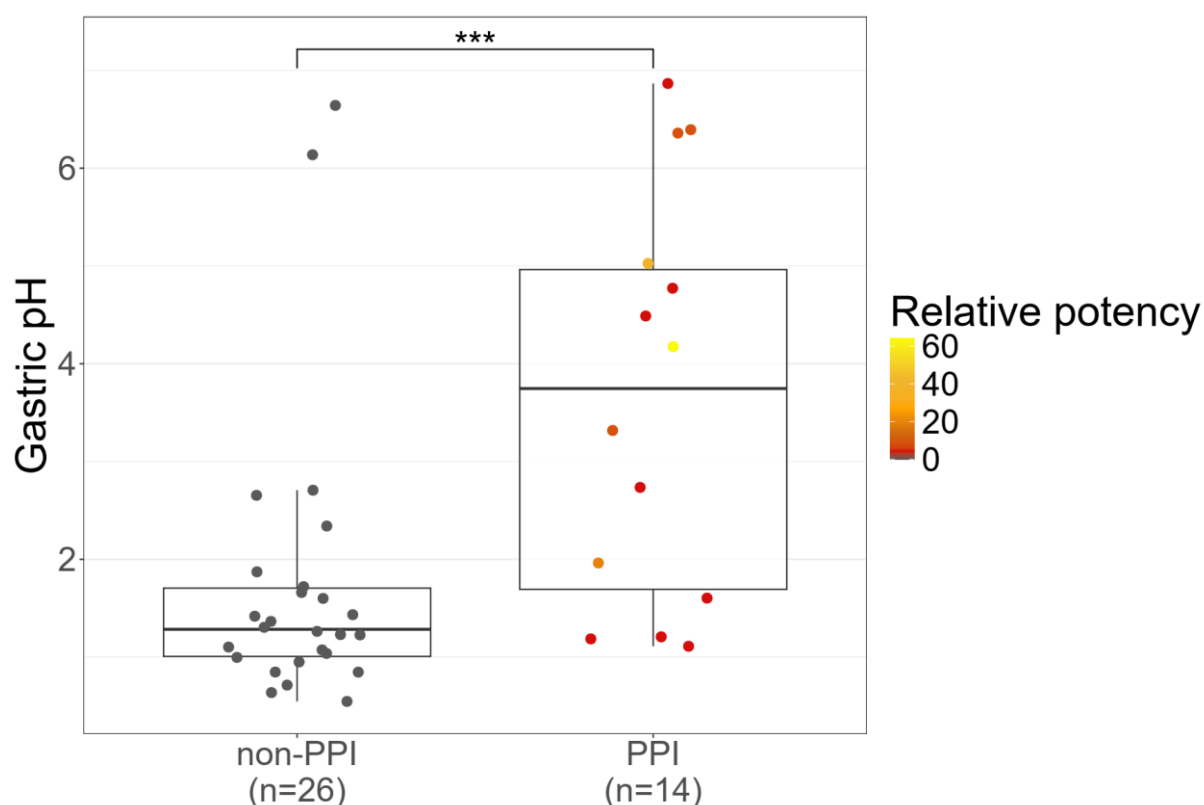


Figure 11. The comparison of gastric pH in participants with PPI and without PPI tested by using linear regression. In the PPI group, colors indicate the relative potency of PPIs as omeprazole equivalents (16), with red representing lower potency and yellow representing higher potency, while dark grey represents non-PPI group. The center lines represent the median values, with the box extending from the 25th percentile to the 75th percentile; the whiskers cover the 1.5-fold interquartile range.

***P<0.001

3.3. Intraluminal pH

Table 3 presents the percentiles of the luminal pH values across different segments in the older cohort. In the pooled multivariable analysis including older and younger cohort (**Supplementary table 2**), only the use of PPIs was significantly associated with increased gastric pH (Beta coefficient: 0.34 [95%CI: 0.15–0.52], p-value<0.01). This association was not observed in other GI segments. The median gastric pH for subjects taking a PPI was 3.32

[1.60–4.77] compared to a median value of 1.28 [1.01–1.71] for subjects not taking PPIs (**Figure 11**). Additionally, small intestinal pH in the combined older cohort was significantly lower than that in young volunteers (6.77 ± 0.38 vs. 7.15 ± 0.13 , **Figure 12**). None of the other covariates was significantly associated with small intestinal pH. Colonic pH values showed no significant relationships with any of the tested covariates (**Supplementary table 2**).

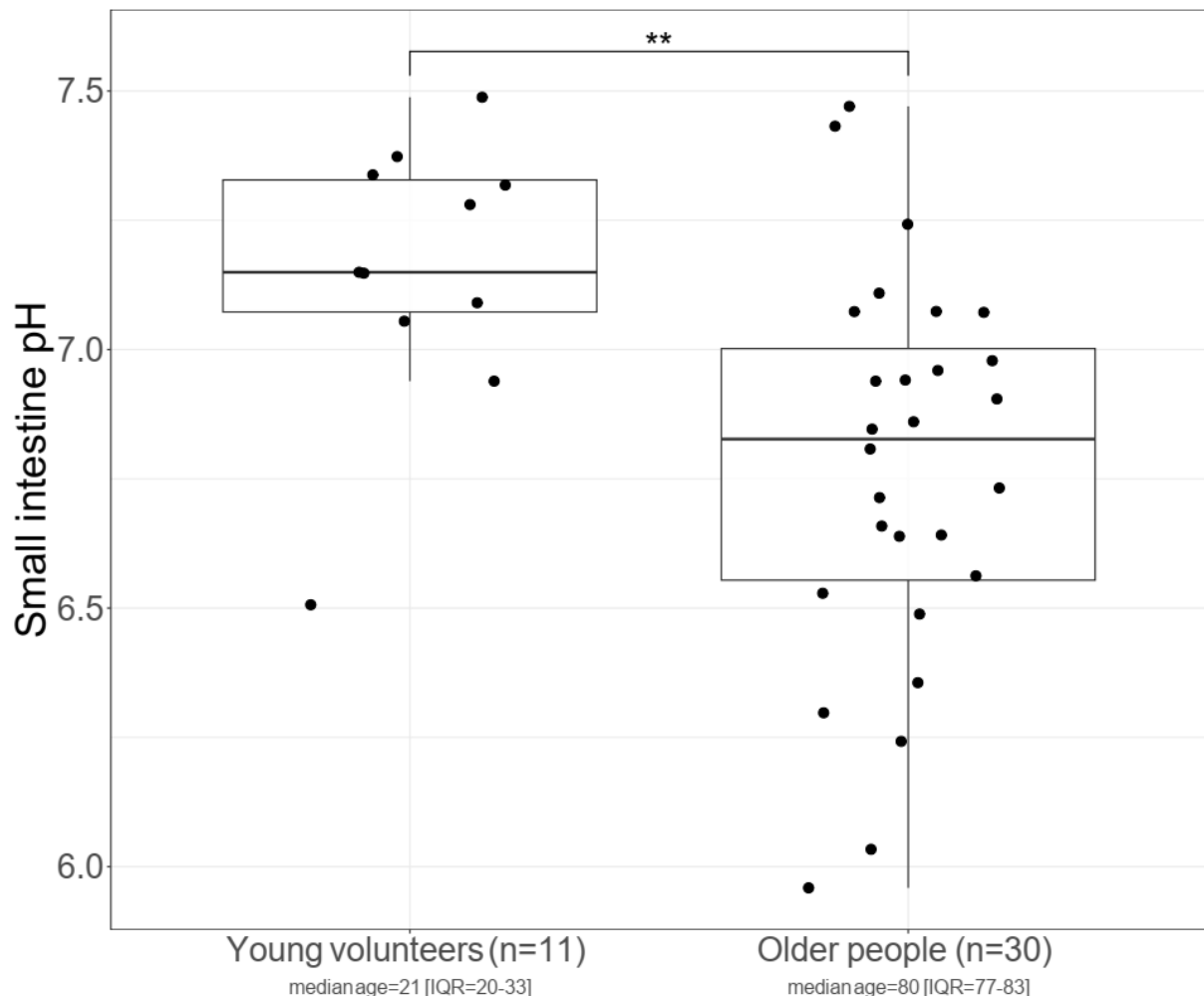


Figure 12. The differences in small intestine pH between young and older cohort tested by using linear regression. The center lines represent the median values, with the box extending from the 25th percentile to the 75th percentile; the whiskers cover the 1.5-fold interquartile range. **P<0.01

4. Discussion

This study investigated GI transit times and intraluminal pH values in community-dwelling older adults and geriatric inpatients, comparing them data from healthy younger adults. In this study, we demonstrated that the gastric emptying time of the WMC was significantly delayed in geriatric inpatients as compared to healthy young adults or community-dwelling

older adults, suggesting that age alone is not the main factor influencing gastric emptying. The wide range of gastric emptying times observed in geriatric patients in our present study, which focuses on monolithic objects, may help explain the high pharmacokinetic variability we previously observed with paracetamol. This variability was noted when paracetamol was administered as immediate-release tablets and granulate, despite its absorption mainly reflecting fluid emptying (12). This variability might have therapeutic consequences as sub-therapeutic exposures lead to insufficient pain relief, and high exposures cause toxicity (22).

The finding of our study highlights the importance of looking beyond chronological age and considering the overall health status of individuals. Geriatric patients often have reduced levels of physical activity, which can slow down gastrointestinal motility. A meta-analysis has shown that shorter gastric emptying times are observed in individuals engaging in lower intensity activities, such as walking (23). This aligns with our data, where most of our healthy older participants were regularly active with a CFS of 2. In contrast, the geriatric cohort, all of whom were hospitalized and characterized by higher comorbidity indices (CCI=4–7) and higher CFS (5–7), spent a majority of their time sedentary or even bedridden. One potential mechanism is vagal nerve stimulation from physical activity, which may promote more stable and powerful gastric myoelectrical activity in active individuals (24, 25). Additionally, although not statistically significant, there is a clear trend indicating that geriatric patients exhibited lower intragastric pressure, which may imply slower gastric emptying.

A previous study using the same WMC technology demonstrated alignment with our young volunteers' data (26). Using this method, another data from Jensen et al. found that individuals who engaged in more light-intensity activity had faster colonic and whole gut transit times. s (27). However, we were unable to confirm the statistical significance of the CTT differences between subgroups; we only observed that geriatric patients had higher median CTT values with a wider distribution compared to the other two groups (Figure S2). The delay in gastric emptying time in frail individuals observed in our study did not extend to colon and whole gut transit times.

Age was not a significant predictor for gastric emptying time in other studies that used alternative methods such as paracetamol absorption as an indirect measure of the rate of gastric emptying (28). The paracetamol absorption technique generally correlates

well with scintigraphy of liquid-phase gastric emptying, and non-caloric liquids are expected to show no difference between age groups due to the *Magenstrasse* pathway, which allows liquids to bypass the slower solid-food movement in the stomach for rapid emptying. While we did not observe a difference between younger and older healthy people, Giezenaar *et al.* demonstrated that gastric emptying of a protein drink was slower in healthy, older men than in younger male participants (29). A previous study showed no difference in chyme gastric emptying time observed between older adults with and without sarcopenia (30). In their studies, gastric motility was measured using scintigraphy, which primarily assesses the emptying of caloric liquids and digestible solids. This difference was anticipated, as older people generally had lower caloric needs leading to slower caloric emptying. In contrast, the WMC tracks the emptying of a large, non-digestible object, which occurs during the fasted state motility, primarily during the phase III of the migrating motor complex (MMC). The difference between our findings and above-mentioned studies may suggest that MMC phase III-driven emptying is less affected by age, but more affected by geriatric profiles, especially (patho-)physiological changes such as muscle mass loss and weakness, morbidities and polypharmacy.

However, predicting the timing of the phase III of the MMC using WMC technology is challenging (31). Previous studies compared gastric emptying measurements using WMC and scintigraphy, the latter being recognized as the gold standard to measure gastric emptying of chyme as it allows precise tracking of radiolabeled solid meals or liquids through the stomach and into the intestines (32, 33). It has been demonstrated that the emptying process measured by the WMC correlates with both the time to the next MMC III and the 90% emptying of a radiolabeled meal (31, 34). However, it is important to note that these studies were conducted in a fed state, and there is lack of research comparing WMC with other techniques in a fasted state.

Geriatric patients represent a complex population characterized by frailty, multiple comorbidities and polypharmacy. Evaluating a single covariate effect on gastrointestinal physiology within this population is nearly impossible due to the presence of confounders. Moreover, focusing on a single factor would not provide a comprehensive understanding of the net effect of the geriatric profile on GI physiology. Hence this study adopted an explorative approach, without excluding any disease or medication that might have

influenced GI physiology. Applying more stringent criteria would have led to a lack of participants, thereby limiting the study's ability to accurately represent the diversity and complexity in the geriatric patient population.

This study also examined the impact of anticholinergic drugs. Theoretically, anticholinergic drugs should influence GI transit by prolonging it, given their mechanism of action (35). However, our study did not observe this anticipated effect. This suggests that the relationship between the use of anticholinergic drug use and GI transit time may be more complex than initially assumed. Moreover, the small sample size for assessing anticholinergic activity may have limited our ability to detect the anticipated effect.

Regarding intraluminal pH, older individuals tended to have higher gastric pH due to PPI use. It is known that PPI are often prescribed to older adults and geriatric patients due to the high prevalence of gastroesophageal reflux disease or to protect GI from ulceration due to comedications (e.g., antiplatelet drugs) (36). In our study, we observed a considerable variability in gastric pH levels within the PPI user group. This variability could be attributed, in part, to differences in PPI potency among the participants (Figure 11). Hypochlorhydria can be another cause of the higher gastric pH in older people (37). One participant in the older group, who was not on PPI and had a gastric pH of 6, was suspected of having hypochlorhydria based on the study results, as no prior indication of impaired gastric function had been reported. Additionally, we noted a significant difference in small intestinal pH levels between the older and young cohort, with the older group exhibiting a lower pH in the small intestine (pH 7.2 vs. pH 6.8). Small intestinal pH levels in the younger volunteers align with previous studies, which report a range of 7.0–7.2 in healthy individuals (38, 39). The lower pH in the older cohort, despite less acidic gastric contents, may suggest reduced bicarbonate secretion in the small intestine. Normally, bicarbonate from the pancreas neutralizes the acidic chyme from the stomach. However, if bicarbonate secretion is insufficient, even the less acidic stomach contents may not be fully neutralized, leading to a lower-than-expected intestinal pH. This decrease in pH could significantly impact drug absorption in some individuals.

An increase in gastric pH and high variability in intragastric pressures can significantly affect drug absorption. In the case of weakly basic drugs, an elevated gastric pH can decrease their dissolution rate, leading to reduced absorption and efficacy (40). This

phenomenon has been demonstrated with ketoconazole and posaconazole, which showed reduced oral bioavailability when co-administered with omeprazole, and with itraconazole, which exhibited a reduced area under the concentration-time curve at 24 h and lower peak plasma concentration when administered alongside omeprazole (41, 42). Although PPIs generally increased gastric pH, some users had levels similar to non-PPI users. Additionally, The intragastric pressures wide variability can lead to inconsistent drug disintegration and dissolution rates. This particularly matters for those with lower pressures as drugs designed for gastric release may dissolve more slowly. Hence, the high variability in gastric pH and pressure suggests the need for more robust formulations, particularly using enteric-release supersaturable formulations with polymers (43). These could not only improve oral absorption but also help reduce variability in absorption caused by changes in gastric pH.

There are some limitations to our study. Firstly, the diversity of diseases and medications in our cohort unable us to test the exact source of delay in gastric emptying time. This limitation prevented us from conducting more detailed analyses, such as evaluating anticholinergic drugs using the anticholinergic burden score (44) and assessing PPIs based on their relative potency as omeprazole equivalents (16). Instead, this comedication use was dichotomized. Secondly, gastric emptying values measured by WMC, which is a relatively large and non-digestible object, may potentially overestimate the actual values of common tablets or capsules. With the exception of modified-release tablets, most oral dosage forms are typically smaller and disintegrate rapidly after oral administration. For this reason, the data should be interpreted carefully. Thirdly, no post-calibration was performed on the data, so pH corrections were not applied. However, even with post-calibration, it would be impossible to precisely determine when the pH drift occurred during transit. Nevertheless, the values obtained from a WMC can still offer valuable insights, particularly when comparing GI physiology across various patient characteristics. WMC remains the safest and easiest method to measure GI transit and pH, although we encountered occasional data loss when patients forgot to wear the recorder, but this did not affect key outcomes. Another limitation to WMC is its relatively large size which restricted its use in many geriatric patients with swallowing difficulties. Future studies could improve compliance by using caregiver support or enhanced WMC technologies with a smaller size. Moreover, our data might be useful to inform *in silico* tools such as

physiologically based pharmacokinetics (PBPK) models and biorelevant *in vitro* tools, which are used to predict drug release and absorption in older adults and geriatric patients in order to guide the development of drug formulations tailored for older adults and geriatric patients.

Given the complexities uncovered in our study and the limitations due to our sample size, further research with larger cohorts of geriatric patients is needed to confirm these findings. This will facilitate a more comprehensive understanding of the complex interplay between GI physiology, associated comorbidities, and polypharmacy in geriatric populations. Investigating the relationship between GI physiology, drug release and absorption processes, as well as pharmacokinetics and pharmacodynamics of medications prevalent in geriatric patients is crucial. We acknowledge that using the WMC in this population carries mild risks, such as capsule obstruction due to the prevalence of diverticula in older adults. However, we did not observe any adverse events in this study. The WMC has less invasive nature and no radiation exposure compared to other methods, especially with close participant monitoring throughout the study. According to Hens *et al.*, there is no single ideal method that gives a complete picture of how GI physiology and oral drug products interact. Therefore, to gain a more comprehensive understanding of this interaction, it is suggested that various methods should be used (45). A crossover study comparing these three methods, e.g., WMC, paracetamol absorption method, and scintigraphy, in both geriatric and non-geriatric populations, might be useful to account for the higher interindividual variability in geriatric patients. However, this approach is challenging due to the significant manpower, time, and funding required to recruit and manage larger cohorts of older adults. An alternative approach would be to conduct a meta-analysis or data pooling from existing studies to incorporate more diverse populations, including different ethnicities and frailty levels. Unfortunately, such efforts have been limited in the geriatric population to date. Nonetheless, our study adds to previous research by utilizing a WMC to investigate this interaction in the older population.

Learning from our data in the aging population that showed a large variability of GI transit times and intraluminal pH, particularly gastric pH, we should develop formulations which are robust against variations in GI transit time and pH values. By this, we can improve the efficacy and safety of oral pharmacotherapy in the aging population.

5. Conclusion

Our study findings suggest that age alone does not independently alter GI physiology. When compared to healthy young participants, geriatric patients exhibited significant delays in gastric emptying, while community-dwelling older adults did not. The delayed gastric emptying observed in geriatric patients in our study, specifically impacting phase III of the MMC, likely results from a combination of frailty, reduced muscle mass and strength, leading to lower physical activity. This might decrease vagal stimulation and weaken gastric myoelectrical activity. Additional factors include high comorbidity and polypharmacy, making it difficult to isolate individual effects. Additionally, PPI use significantly increased gastric pH levels and substantial inter-subject variability in gastric pH was noted in the PPI group, potentially impacting the solubility of pH-sensitive drugs and formulations. Our study provides detailed quantitative data on transit time and intraluminal pH in older adults, essential for creating predictive biopharmaceutics *in vitro* tools and models (e.g., PBPK and PBBM [Physiologically Based Biopharmaceutics Modeling]) used in drug product development. Given the complexity of factors influencing GI physiology, further research with larger cohorts of geriatric patients is necessary to validate these findings. This will help to comprehensively understand the interplay between GI physiology, associated comorbidities, and polypharmacy in geriatric populations.

References

1. Statbel. Bevolkingsvooruitzichten: Synthese van de demografische vooruitzichten 2023-2070 voor België en de gewesten 2023. 2023.
2. Eurostat. Ageing Europe: Looking at the lives of older people in the EU. Luxembourg: Publications Office of the European Union; 2019.
3. Guisado-Clavero M, Roso-Llorach A, López-Jimenez T, Pons-Vigués M, Foguet-Boreu Q, Muñoz MA, et al. Multimorbidity patterns in the elderly: a prospective cohort study with cluster analysis. *BMC Geriatrics* 2018 18:1. 2018;18(1):1-11.
4. Eurostat. Medicine use statistics. 2022.
5. Beers E, Moerkerken DC, Leufkens HG, Egberts TC, Jansen PA. Participation of older people in preauthorization trials of recently approved medicines. *J Am Geriatr Soc.* 2014;62(10):1883-90.
6. van Marum RJ. Underrepresentation of the elderly in clinical trials, time for action. *Br J Clin Pharmacol.* 2020;86(10):2014-6.
7. Agency EM. Reflection paper on the pharmaceutical development of medicines for use in the older population. Amsterdam, the Netherlands; 2020 15 October 2020. Contract No.: EMA/CHMP/QWP/292439/2017.
8. Administration UFaD. Guidance Document: E7 Studies in Support of Special Populations; Geriatrics; Questions and Answers 2012 [Available from: <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/e7-studies-support-special-populations-geriatrics-questions-and-answers>].
9. Talley NJ, Fleming KC, Evans JM, O'Keefe EA, Weaver AL, Zinsmeister AR, et al. Constipation in an elderly community: a study of prevalence and potential risk factors. *Am J Gastroenterol.* 1996;91(1):19-25.
10. O'Mahony D, O'Leary P, Quigley EM. Aging and intestinal motility: a review of factors that affect intestinal motility in the aged. *Drugs Aging.* 2002;19(7):515-27.
11. Soenen S, Rayner CK, Jones KL, Horowitz M. The ageing gastrointestinal tract. *Curr Opin Clin Nutr Metab Care.* 2016;19(1):12-8.
12. Hias J, Van der Linden L, Walgraeve K, Gijssen M, Mian P, Koch BCP, et al. Pharmacokinetics of 2 oral paracetamol formulations in hospitalized octogenarians. *British Journal of Clinical Pharmacology.* 2021.
13. van der Heijden C, Ter Heine R, Kooistra EJ, Brüggemann RJ, Walburgh Schmidt JWJ, de Grouw E, et al. Effects of dalteparin on anti-Xa activities cannot be predicted in critically ill COVID-19 patients. *Br J Clin Pharmacol.* 2022;88(6):2982-7
14. Weitschies W, Muller L, Grimm M, Koziolk M. Ingestible devices for studying the gastrointestinal physiology and their application in oral biopharmaceutics. *Adv Drug Deliv Rev.* 2021;176:113853.
15. Agency EM. Guideline on the Investigation of Bioequivalence. London; 2010.
16. Graham DY, Lu H, Dore MP. Relative potency of proton-pump inhibitors, *Helicobacter pylori* therapy cure rates, and meaning of double-dose PPI. *Helicobacter.* 2019;24(1):e12554.
17. Steenackers N, Wauters L, Van der Schueren B, Augustijns P, Falony G, Koziolk M, et al. Effect of obesity on gastrointestinal transit, pressure and pH using a wireless motility capsule. *Eur J Pharm Biopharm.* 2021;167:1-8.
18. Tran K, Brun R, Kuo B. Evaluation of regional and whole gut motility using the wireless motility capsule: relevance in clinical practice. *Therap Adv Gastroenterol.* 2012;5(4):249-60.
19. Team R. RStudio: Integrated Development for R. Boston, MA: RStudio, PBC; 2020.

20. Team RC. R: A language and environment for statistical computing. Vienna, Austria: R Foundation for Statistical Computing; 2021.
21. Lee YY, Erdogan A, Rao SS. How to assess regional and whole gut transit time with wireless motility capsule. *J Neurogastroenterol Motil.* 2014;20(2):265-70.
22. Cairney DG, Beckwith HK, Al-Hourani K, Eddleston M, Bateman DN, Dear JW. Plasma paracetamol concentration at hospital presentation has a dose-dependent relationship with liver injury despite prompt treatment with intravenous acetylcysteine. *Clin Toxicol (Phila).* 2016;54(5):405-10.
23. Horner KM, Schubert MM, Desbrow B, Byrne NM, King NA. Acute exercise and gastric emptying: a meta-analysis and implications for appetite control. *Sports Med.* 2015;45(5):659-78.
24. Wang Y, Kondo T, Suzukamo Y, Oouchida Y, Izumi S. Vagal nerve regulation is essential for the increase in gastric motility in response to mild exercise. *Tohoku J Exp Med.* 2010;222(2):155-63.
25. Lu CL, Shidler N, Chen JD. Enhanced postprandial gastric myoelectrical activity after moderate-intensity exercise. *Am J Gastroenterol.* 2000;95(2):425-31.
26. Schneider F, Grimm M, Koziolok M, Modess C, Dokter A, Roustom T, et al. Resolving the physiological conditions in bioavailability and bioequivalence studies: Comparison of fasted and fed state. *Eur J Pharm Biopharm.* 2016;108:214-9.
27. Jensen MM, Pedersen HE, Clemmensen KKB, Ekblond TS, Ried-Larsen M, Faerch K, et al. Associations Between Physical Activity and Gastrointestinal Transit Times in People with Normal Weight, Overweight, and Obesity. *J Nutr.* 2024;154(1):41-8.
28. Hellstrom PM, Samuelsson B, Al-Ani AN, Hedstrom M. Normal gastric emptying time of a carbohydrate-rich drink in elderly patients with acute hip fracture: a pilot study. *BMC Anesthesiol.* 2017;17(1):23.
29. Giezenaar C, Oberoi A, Jones KL, Horowitz M, Chapman I, Soenen S. Effects of age on blood pressure and heart rate responses to whey protein in younger and older men. *J Am Geriatr Soc.* 2021;69(5):1291-9.
30. Huang HH, Wang TY, Yao SF, Lin PY, Chang JC, Peng LN, et al. Gastric Mobility and Gastrointestinal Hormones in Older Patients with Sarcopenia. *Nutrients.* 2022;14(9).
31. Cassilly D, Kantor S, Knight LC, Maurer AH, Fisher RS, Semler J, et al. Gastric emptying of a non-digestible solid: assessment with simultaneous SmartPill pH and pressure capsule, antroduodenal manometry, gastric emptying scintigraphy. *Neurogastroenterol Motil.* 2008;20(4):311-9.
32. Lin HC, Prather C, Fisher RS, Meyer JH, Summers RW, Pimentel M, et al. Measurement of gastrointestinal transit. *Dig Dis Sci.* 2005;50(6):989-1004.
33. Abell TL, Camilleri M, Donohoe K, Hasler WL, Lin HC, Maurer AH, et al. Consensus recommendations for gastric emptying scintigraphy: a joint report of the American Neurogastroenterology and Motility Society and the Society of Nuclear Medicine. *Am J Gastroenterol.* 2008;103(3):753-63.
34. Kuo B, McCallum RW, Koch KL, Sitrin MD, Wo JM, Chey WD, et al. Comparison of gastric emptying of a nondigestible capsule to a radio-labelled meal in healthy and gastroparetic subjects. *Aliment Pharmacol Ther.* 2008;27(2):186-96.
35. Wong KKL, Tang LCY, Zhou J, Ho V. Analysis of spatiotemporal pattern and quantification of gastrointestinal slow waves caused by anticholinergic drugs. *Organogenesis.* 2017;13(2):39-62.
36. AlMutairi H, O'Dwyer M, McCarron M, McCallion P, Henman MC. The use of proton pump inhibitors among older adults with intellectual disability: A cross sectional observational study. *Saudi Pharm J.* 2018;26(7):1012-21.

37. Hurwitz A, Brady DA, Schaal SE, Samloff IM, Dedon J, Ruhl CE. Gastric acidity in older adults. *JAMA*. 1997;278(8):659-62.
38. Wang YT, Mohammed SD, Farmer AD, Wang D, Zarate N, Hobson AR, et al. Regional gastrointestinal transit and pH studied in 215 healthy volunteers using the wireless motility capsule: influence of age, gender, study country and testing protocol. *Aliment Pharmacol Ther*. 2015;42(6):761-72.
39. Wegeberg AM, Bertoli D, Ejksjaer N, Brock B, Drewes AM, Brock C. Gastrointestinal function in diabetes is affected regardless of asymptomatic appearance. *J Intern Med*. 2022;291(4):505-12.
40. Lahner E, Annibale B, Delle Fave G. Systematic review: impaired drug absorption related to the co-administration of antisecretory therapy. *Aliment Pharmacol Ther*. 2009;29(12):1219-29.
41. Chin TW, Loeb M, Fong IW. Effects of an acidic beverage (Coca-Cola) on absorption of ketoconazole. *Antimicrob Agents Chemother*. 1995;39(8):1671-5.
42. Jaruratanasirikul S, Sriwiriyan S. Effect of omeprazole on the pharmacokinetics of itraconazole. *Eur J Clin Pharmacol*. 1998;54(2):159-61.
43. Kataoka M, Nakanishi R, Umesaki M, Kobayashi M, Minami K, Higashino H, et al. An enteric polymer mitigates the effects of gastric pH on oral absorption of poorly soluble weak acid drugs from supersaturable formulations: A case study with dantrolene. *Eur J Pharm Biopharm*. 2020;155:29-36.
44. Kiesel EK, Hopf YM, Drey M. An anticholinergic burden score for German prescribers: score development. *BMC Geriatr*. 2018;18(1):239.
45. Hens B, Corsetti M, Spiller R, Marciani L, Vanuytsel T, Tack J, et al. Exploring gastrointestinal variables affecting drug and formulation behavior: Methodologies, challenges and opportunities. *Int J Pharm*. 2017;519(1-2):79-97.

Appendix

Supplementary table 1. Univariable linear regression analysis of transit time in healthy young volunteers, community-dwelling older people and geriatric patients

GI segment	Variables		Beta coefficient	95%CI	p-value (<0.05)
Gastric	Age group	Younger cohort (n=11)	reference		
		Older cohort (n=30)	4.23	-3.53–11.99	0.29
	Population sub-group	Younger cohort (n=11)	reference		
		Community-dwelling older adults (n=22)	0.5	-6.79–7.79	0.89
		Geriatric patients (n=8)	14.02	4.92–23.12	<0.01
	Sex	Female	reference		
		Male	5.8	-1.08–12.67	0.11
	Weight		0.25	-0.11–0.14	0.82*
	BMI		0.73	-0.07–1.54	0.08
	Proton pump inhibitors (n=14)		1.64	-5.72–9	0.66
	Anticholinergics (n=10)		1.41	-6.71–9.52	0.74
Small intestine	Age group	Younger cohort (n=11)	reference		
		Older cohort (n=30)	0.65	-0.71–2	0.36
	Population sub-group	Younger cohort (n=11)	reference		
		Community-dwelling older adults (n=22)	0.1	-1.23–1.42	0.89
		Geriatric patients (n=8)	2.08	0.43–3.74	0.02
	Sex	Female	reference		
		Male	1.05	-0.15–2.24	0.09
	Weight		0.01	-0.03–0.05	0.63
	BMI		-0.02	-0.16–0.13	0.81
	Proton pump inhibitors (n=14)		0.73	-0.53–2	0.26

	Anticholinergics (n=10)		0.41	0.07–2.75	0.05
Colon	Age group	Younger cohort (n=11)	reference		
		Older cohort (n=30)	16.85	-4.43–38.13	0.13
	Population sub-group	Younger cohort (n=11)	reference		
		Community-dwelling older adults (n=22)	13.11	-9.21–35.43	0.26
		Geriatric patients (n=8)	26.67	-1.2–54.54	0.07
	Sex	Female	reference		
		Male	7.31	-12.37–26.99	0.47
	Weight		0.4	-0.27–1.07	0.25
	BMI		2.02	-0.25–4.28	0.08
	Proton pump inhibitors (n=14)		17.84	-1.9–37.58	0.08*
	Anticholinergics (n=10)		20.69	-0.96–42.34	0.07
Whole gut	Age group	Younger cohort (n=11)	reference		
		Older cohort (n=30)	22.4	-1.99–46.78	0.08
	Population sub-group	Younger cohort (n=11)	reference		
		Community-dwelling older adults (n=22)	14.99	-9.66–39.63	0.24
		Geriatric patients (n=8)	42.78	11.77–73.78	0.01
	Sex	Female	reference		
		Male	15.15	-7–37.31	0.19
	Weight		0.67	-0.1–1.44	0.09*
	BMI		1.38	-0.92–3.69	0.23
	Proton pump inhibitors (n=14)		18.98	-3.97–41.93	0.11
	Anticholinergics (n=10)		22.47	-2.74–47.69	0.09

95%CI: 95% Confidence Interval; GI: Gastrointestinal

*Values obtained after sensitivity analysis excluding the influential subject

Supplementary table 2. Univariable analysis of intraluminal pH in healthy young volunteers, community-dwelling older people and geriatric patients

GI segment	Variables	Beta coefficient	95%CI	p-value (<0.05)
Gastric ^a	Age group	Younger cohort (n=11)		Reference
		Older cohort (n=30)	0.11 -0.12–0.34	0.34
	Population subgroup	Younger cohort (n=11)		Reference
		Community-dwelling older adults (n=22)	0.11 -0.13–0.35	0.37
		Geriatric patients (n=8)	0.11 -0.20–0.41	0.48
	Sex	Female		reference
		Male	0.06 -0.15–0.27	0.56
	Weight	0.00	0.00–0.01	0.41
	BMI	0.01	-0.02–0.03	0.47
	Proton pump inhibitors (n=14)	0.34	0.15–0.52	<0.01
	Anticholinergics (n=10)	0.12	-0.11–0.35	0.30
Small intestine	Age group	Younger cohort (n=11)	reference	
		Older cohort (n=30)	-0.38 -0.64– -0.13	<0.01
	Population subgroup	Younger cohort (n=11)	reference	
		Community-dwelling older adults (n=22)	-0.46 -0.68– -0.24	<0.01
		Geriatric patients (n=8)	-0.18 -0.45–0.09	0.26
	Sex	Female		reference
		Male	0.04 -0.183–0.26	0.77
	Weight	-0.002	-0.01–0.01	0.73
	BMI	-0.01	-0.04–0.01	0.47
	Proton pump inhibitors (n=14)	-0.06	-0.29–0.17	0.65
	Anticholinergics (n=10)	-0.14	-0.40–0.11	0.34

Colon	Age group	Younger cohort (n=11)	reference		
		Older cohort (n=30)	-0.12	-0.54–0.30	0.56
	Population subgroup	Younger cohort (n=11)	reference		
		Community-dwelling older adults (n=22)	-0.13	-0.52–0.25	0.56
		Geriatric patients (n=8)	-0.06	-0.53–0.40	0.82
	Sex	Female	reference		
		Male	0.08	-0.26–0.41	0.70
	Weight		-0.001	-0.01–0.01	0.92
	BMI		-0.01	-0.05–0.03	0.70
	Proton pump inhibitors (n=14)		-0.08	-0.42–0.26	0.69
	Anticholinergics (n=10)		-0.07	-0.45–0.31	0.77

95%CI: 95% Confidence Interval; GI: Gastrointestinal

^aValues obtained after box-cox transformation

Supplementary table 4. Univariable analysis of intragastric pressure in healthy young volunteers, community-dwelling older people and geriatric patients

Variables		Beta coefficient	95%CI	p-value (<0.05)
Age group	Younger cohort (n=11)			Reference
	Older cohort (n=30)	1.84	-100.68 - 104.37	0.97
Population subgroup	Younger cohort (n=11)			Reference
	Community-dwelling older adults (n=22)	39.99	-60.94 - 140.91	0.44
	Geriatric patients (n=8)	-103.05	-230.05 - 23.95	0.12
Sex	Female			reference
	Male	-14.61	-106.03 - 76.81	0.76
Weight		-1.09	-4.28 - 2.1	0.51
BMI		-2.68	-13.59 - 8.23	0.63
Proton pump inhibitors (n=14)		15.44	-80.23 - 111.12	0.75
Anticholinergics (n=10)		-88.18	-186.9 - 10.55	0.09

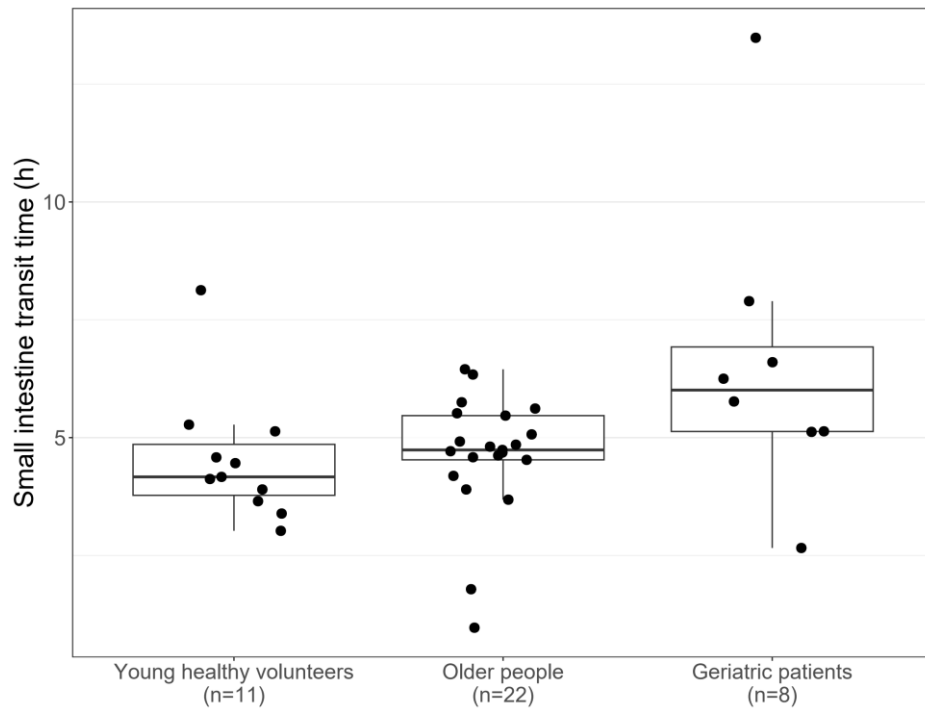


Figure S1. Small intestine transit time among young healthy volunteers, community-dwelling older adults and geriatric patients. The center lines represent the median values, with the box extending from the 25th percentile to the 75th percentile; the whiskers cover the 1.5-fold interquartile range.

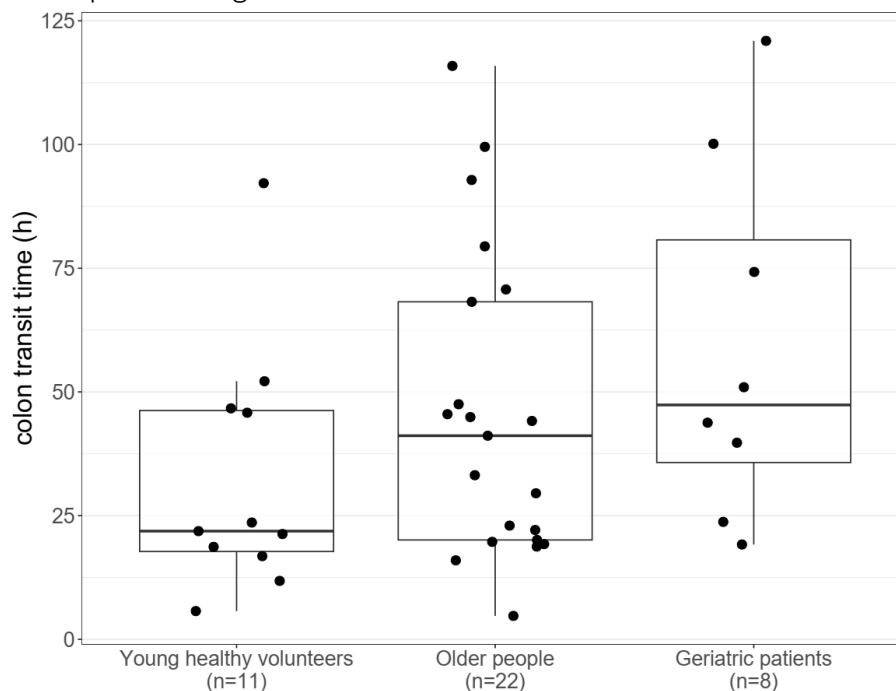


Figure S2. Colon transit time among young healthy volunteers, community-dwelling older adults and geriatric patients. The center lines represent the median values, with the box extending from the 25th percentile to the 75th percentile; the whiskers cover the 1.5-fold interquartile range.

Part 2. Pharmacokinetics of Direct Oral Anticoagulants

Chapter 2. Insights into the Pharmacokinetics and Pharmacodynamics of Direct Oral Anticoagulants in Older Adults with Atrial Fibrillation: A Structured Narrative Review

This chapter is published as a review article

Edwina AE, Dia N, Dreesen E, Vanassche T, Verhamme P, Spriet I, Van der Linden L, Tournoy J. *Insights into the Pharmacokinetics and Pharmacodynamics of Direct Oral Anticoagulants in Older Adults with Atrial Fibrillation: A Structured Narrative Review*. Clin Pharmacokinet. 2023 Mar;62(3):351-373

Abstract

Older adults, the fastest growing population, represent almost 50% of all direct oral anticoagulants (DOACs) users. Unfortunately, we have nearly no relevant pharmacological and clinical data on DOACs especially in older adults with geriatric profile. This is highly relevant as pharmacokinetics and pharmacodynamics (PK/PD) often differ substantially in this population. We hence need to obtain a better understanding of the PK/PD of DOACs in older adults, to ensure appropriate treatment. This review summarises the current insights into PK/PD of DOACs in older adults. A search was undertaken until October 2022 to identify PK/PD studies of apixaban, dabigatran, edoxaban, and rivaroxaban, that included older adults aged ≥ 75 years. This review identified 44 articles. Older age alone did not influence exposure of edoxaban, rivaroxaban and dabigatran, while apixaban peak concentrations were 40% higher in older adults than in young volunteers. Nevertheless, high interindividual variability in DOACs exposure in older adults was noted, which can be explained by distinctive older patient characteristics, such as kidney function, changes in body composition (especially reduced muscle mass), and co-medication with P-gp inhibitors, which is in line with the current dosing reduction criteria of apixaban, edoxaban, and rivaroxaban. Dabigatran had the largest interindividual variability among all DOACs since its dose adjustment criterion is only age, and thus it is not a preferable option. Additionally, DOACs exposure which fell outside of on-therapy ranges was significantly related to stroke and bleeding events. No definite thresholds linked to these outcomes in older adults have been established.

1. Introduction

Atrial fibrillation (AF) affects over 35 million people worldwide (1). Oral anticoagulation is the standard-of-care treatment that has been shown to reduce mortality and disability in patients with AF. Age is one of the most important risk factors for AF; it occurs in about half of geriatric inpatients (2, 3). AF increases the risk of ischemic stroke fivefold, of which two-thirds is reduced by oral anticoagulants (4, 5, 6). Current clinical practice guidelines on the management of AF recommend the use of Direct Oral Anticoagulants (DOACs) (7). Briefly, DOACs are non-inferior to the first generation and highly effective anticoagulants, the vitamin K antagonists of which warfarin is the standard tested compound (see **Table 4**). DOACs are furthermore associated with fewer major and fatal bleeding (8, 9, 10, 11). Unlike warfarin, DOACs do not require routine laboratory monitoring due to their more predictable pharmacokinetic (PK) profile, absence of food interactions, and minimal drug interactions (12, 13). Currently, four agents are marketed in Europe: three direct factor Xa inhibitors (apixaban, edoxaban, rivaroxaban) and one direct thrombin inhibitor (dabigatran).

It remains unclear, however, whether data from pivotal clinical trials can be readily extrapolated to all older adults (e.g., oldest old and geriatric patients) aged ≥ 75 years, as they were not always included into the landmark trials. Since adults over 75 years old represent nearly 50% of all DOACs users in daily clinical practice (14, 15, 16), we are confronted with a proportionally high use of DOACs in those patients for whom we have the least pharmacological and clinical data.

Older adults can exhibit remarkable variability in PK (17). This may be caused by age-related changes and geriatric traits such as reduced organ function, body composition changes, a high comorbidity burden, and the use of multiple other medications. Such factors can lead to supratherapeutic or subtherapeutic exposure which may confer an increased risk of bleeding or breakthrough thrombotic events, respectively. Additionally, the pharmacodynamics (PD) of DOACs can change, as older adults are likely to experience a higher sensitivity to many medications (24). Age-related fragility of tissues and blood vessels increases the risk of bleeding, and vascular changes with age are associated with a higher risk of thrombosis (18). The effect of a given concentration of DOACs on clinical outcomes is therefore also likely to change with age and frailty status.

There is a clear need to better understand the impact of the physiology of aging and the pathophysiology associated with aging on DOACs PK and PD. Therefore, this structured narrative review aims to provide a comprehensive summary of the PK and PD of DOACs in older adults. Considering above-mentioned reasons, this review focuses on articles including old (75-84 years) and oldest-old patients (85 years or older) to capture more subjects with a geriatric profile (19).

2. Methods

A literature search was conducted in EMBASE, PubMed, and the Web of Science Core Collection database in October 2022. A search strategy was developed with the terms specified in Supplementary Material 1. All articles written in English were screened for relevance.

2.1. Eligibility Criteria and Study Selections

Studies on apixaban, edoxaban, rivaroxaban, and dabigatran were included if the following criteria were met. First, they presented an original research study addressing the PK and/or PD of DOACs in patients with AF. Second, they enrolled at least ten older adults, and at least 50% of the study participants were aged ≥ 75 years. Population PK models had to include patients with an upper age range value of > 75 years. Third, PK(/PD) parameters and metrics had to be reported. Fourth, DOACs concentrations were measured by mass spectroscopy or anti-Xa chromogenic assay for apixaban, edoxaban, and rivaroxaban, and direct thrombin inhibitor activity for dabigatran. Studies were excluded if they concerned review papers, meta-analyses, case reports or series, non-English articles, poster presentations, or abstract proceedings.

Literature references were imported into a reference manager (Endnote 20®, Clarivate). Duplicates were removed based on a published guideline (21). Titles and abstracts were screened independently by two reviewers (AE and ND) using a semi-automation screening tool, Rayyan web (Qatar Computing Research Institute, Doha, Qatar) (22). Subsequently, full-text articles were retrieved and reviewed independently by the two reviewers to assess eligibility.

Table 4. DOACs approved dosage regimen for stroke prevention in atrial fibrillation

DOACs	Phase III studies	Standard dosage	Dose reduction	Comparison with warfarin	
				Efficacy	Toxicity
Apixaban (8)	ARISTOTLE, median age 70 [IQR: 63-76] years	5 mg BID	2.5 mg BID if two out of three fulfilled: weight $\leq$ 60 kg, age $\geq$ 80 years, sCr $\geq$ 133 mmol/L (1.5 mg/dL), or single criterion: if CrCl 15–29 mL/min	Superior in preventing stroke or systemic embolism	Lower risks of major and non-major bleeding
Dabigatran (10)	RE-LY, mean age 71.4 (SD: $\pm$ 8.6) years	150 mg BID	110 mg BID if: age $\geq$ 80 years	Superior in preventing stroke or systemic embolism	Similar risk of major haemorrhage
Edoxaban (9)	ENGAGE AF, median age 72 [IQR: 64-78] years	60 mg QD	30 mg QD if: weight $\leq$ 60 kg or CrCl 15–49 mL/min or concomitant medication with strong P-gp inhibitor	Similar in preventing stroke or systemic embolism	Lower risk of bleeding
Rivaroxaban (11)	ROCKET AF, median 73 [IQR: 65-78] years	20 mg QD	15 mg QD if CrCl $\leq$ 15–49 mL/min	Similar in preventing stroke or systemic embolism	Similar risk of major bleeding, lower risks of intracranial and fatal bleeding

BID: twice daily, QD: Once daily, IQR: Interquartile range, sCr: serum creatinine, SD: standard deviation, CrCl: Creatinine clearance, P-gp: p-glycoprotein

2.2. Data extraction

The following information was extracted from the articles: author, year, sample size, median or mean study age, statistical analysis method, endpoints, significant covariates (determined by using statistical analysis), and exposure metrics (trough, peak, and average concentrations, area under the curve (AUC), time to maximum concentration, and half-life). Additionally, typical values of DOACs PK and/or PD parameter estimates, including the absorption rate constant (k_a), lag time (T_{lag}), clearance (CL), intercompartmental clearance (Q), volumes of distribution (Vd), PD surrogate markers, maximum effect (E_{max}), half maximal effective concentration (EC_{50}) and interindividual variabilities (IIVs), were collected from population PK(/PD) studies. When data were only provided in figures, observations were extracted using a digital tool (WebPlot Digitizer, Ankit Rohatgi, available online: <https://automeris.io/WebPlotDigitizer/>). Since data presentation may vary among included studies, median with interquartile range (IQR) or mean with standard deviation (SD) were extracted. One author (AEE) extracted the data and the other author (ND) checked them for accuracy.

3. Results

3.1. Search and included studies

The search strategy yielded 3,836 records, of which 327 full texts were relevant based on the title/abstract as they evaluated the dose-exposure (PK) or exposure-response relationship (PD) of the DOACs of interest. A thorough full-text screening resulted in 44 articles, with 28 studies characterising DOACs PK/PD (23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50). Sixteen publications provided a population PK(/PD) model of DOACs (51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66). One of those 16 publications is a population PK/PD model paper (66) as a continuation of a previously developed population PK model (62). Included studies were published between 2011 and 2022. Some studies analysed more than one DOACs. Twenty-two studies assessed apixaban, eleven rivaroxaban, eight dabigatran, and five focused on edoxaban. The flowchart in **Figure 13** shows the study selection.

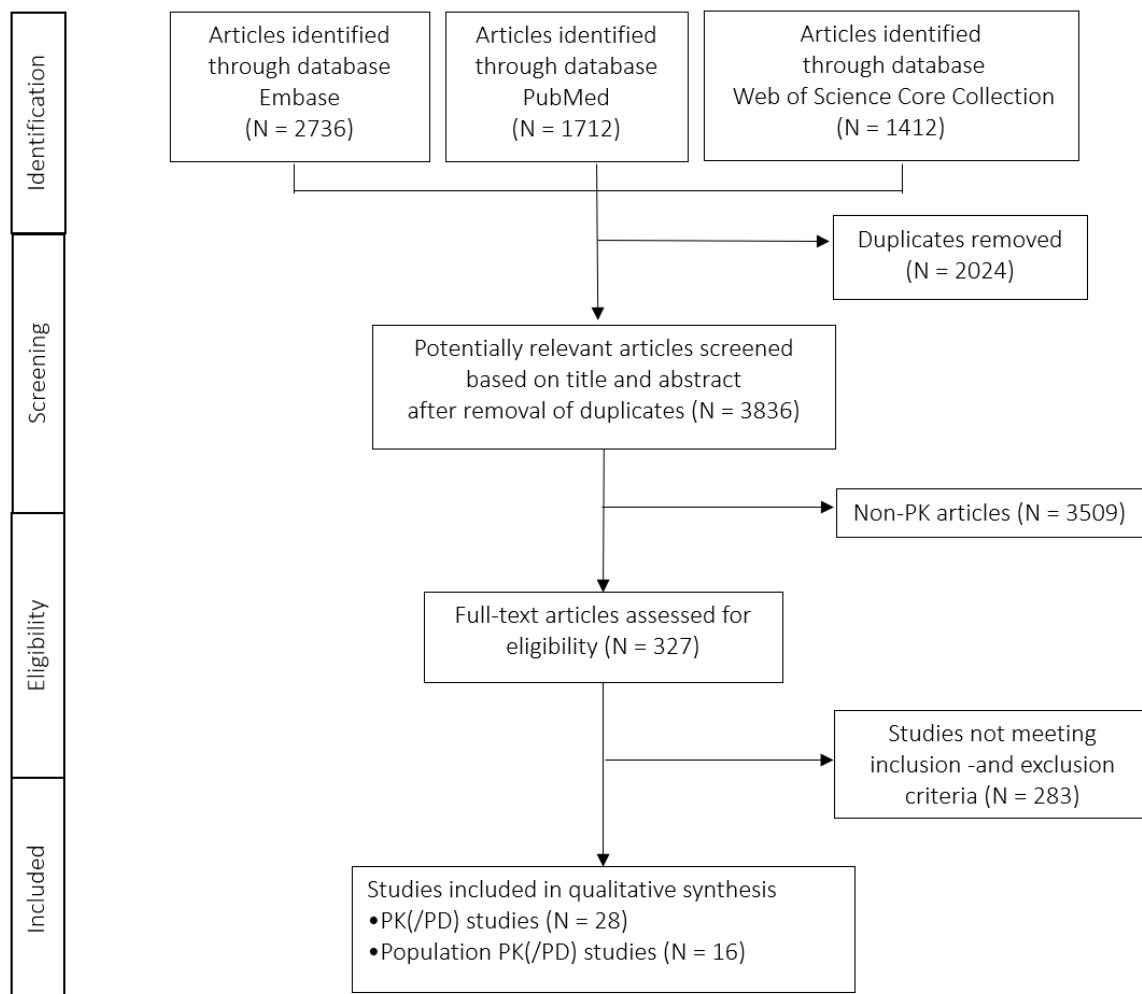


Figure 13. Flowchart of the selection process

3.2. Exposure metrics in studies including older adults

There was heterogeneity in reporting DOACs concentrations in the included studies. Most of the studies stratified DOACs concentrations according to the dosing regimen (n=17) (23, 25, 27, 32, 34, 35, 36, 37, 38, 39, 44, 45, 46, 47, 48, 49), as summarised in **Figure 14**. Additionally, DOACs concentrations were compared in 11 studies across various categories, such as on-label versus (vs.) off-label dose, different age ranges, numerous haemoglobin levels, genotypes, time-after-dose, with vs. without P-glycoprotein (P-gp) inhibitors, acute ischemic stroke vs. intracerebral haemorrhage group (24, 25, 26, 27, 29, 33, 37, 38, 39, 42, 50). The details of exposure metrics of DOACs are provided in **Supplementary Table 1**.

When compared with the phase 1 study, the exposures of edoxaban, dabigatran and rivaroxaban seem to be similar in older adults (67, 68, 69) (see **Figure 14**). Older adults did exhibit much larger between-subject variability however compared to younger subjects, especially in the case of dabigatran which showed the largest variability among all DOACs. In contrast, all mean or median apixaban concentrations were higher than those in the phase 1 study (see **Figure 14**) (70).

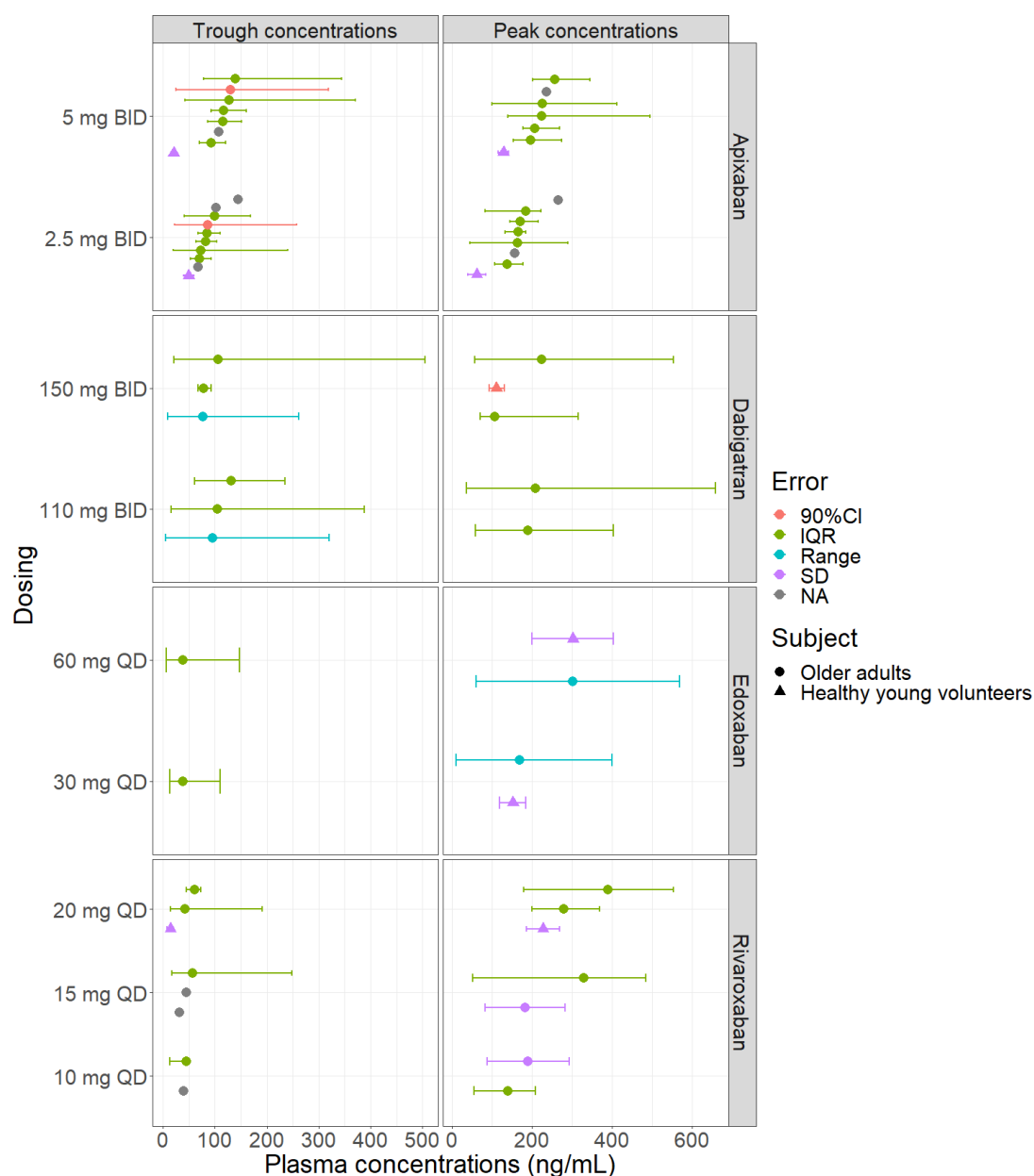


Figure 14. The mean and median of DOACs concentrations summarised from the included studies that classified data based on dosing regimen (N=17). BID: twice daily, QD: once daily, IQR: interquartile range, SD: standard deviation, NA: not available, 90%CI: 90% confidence interval

3.3. PK parameters of DOACs

PK parameters of DOACs were estimated using non-compartmental analysis in 2 studies and using compartmental analysis employing a population PK approach in 15 studies. We provide an overview of the PK parameters of DOACs in **Table 5**, and more details about parameterisation are summarised in the **Supplementary Table 2**.

No population PK study estimated PK parameters solely based on data from older adults aged >75 years. The included population PK studies built structural PK models by using rich sampling data from phase 1 studies or using published models, and covariate models by using sparse sampling data (i.e., only peak and trough concentrations) from phase 2 and phase 3 studies. The age range of included population PK study populations was 18 – 105 years. Most of the included population PK studies tested age as a covariate and are described in the Section 3.4. In addition, we provide the overview of PK parameters estimated by population PK models across different age ranges in **Figure 15**.

Table 5. PK parameters of DOACs estimated in population PK studies including older adults

DOACs	Absorption	Distribution	Elimination
Apixaban	$K_a=0.42-0.47\text{ h}^{-1}$	$V_c=V_p=24-30\text{ L}$	CL in Japanese= 1.53 L/h CL= 3.59 L/h
Edoxaban	$K_a=1.08-5.87\text{ h}^{-1}$ Tlag= $0.25-0.42\text{ h}$	$V_c=151-244\text{ L}$ $V_p=82.2-134\text{ L}$	CL= $11.4-36\text{ L/h}$
Rivaroxaban	$K_a=0.6-1.16\text{ h}^{-1}$	$V_d=40.3-79\text{ L}$	CL= $4.72-6.58\text{ L/h}$
Dabigatran	$K_a=0.754\text{ h}^{-1}$ Tlag= 0.634 h	$V_d=673-728\text{ L}$	CL= $111-124\text{ L/h}$

K_a : absorption rate constant, Tlag: lag time, V_d : volume of distribution, V_c : central volume of distribution, V_p : peripheral volume of distribution, CL: Clearance

3.4. Characteristics related to pharmacokinetics of DOACs in older adults

The effect of patient characteristics on DOACs exposure metrics, as well as the involved statistics and effect size, if available, are described in **Table 6**. **Figure 15** summarises the significant as well as the tested covariates in population PK models with the correlation directions. The relationships between PK parameters and covariates are specified in **Supplementary Table 2**. Below we have provided a detailed description of the six most significant covariate effects.

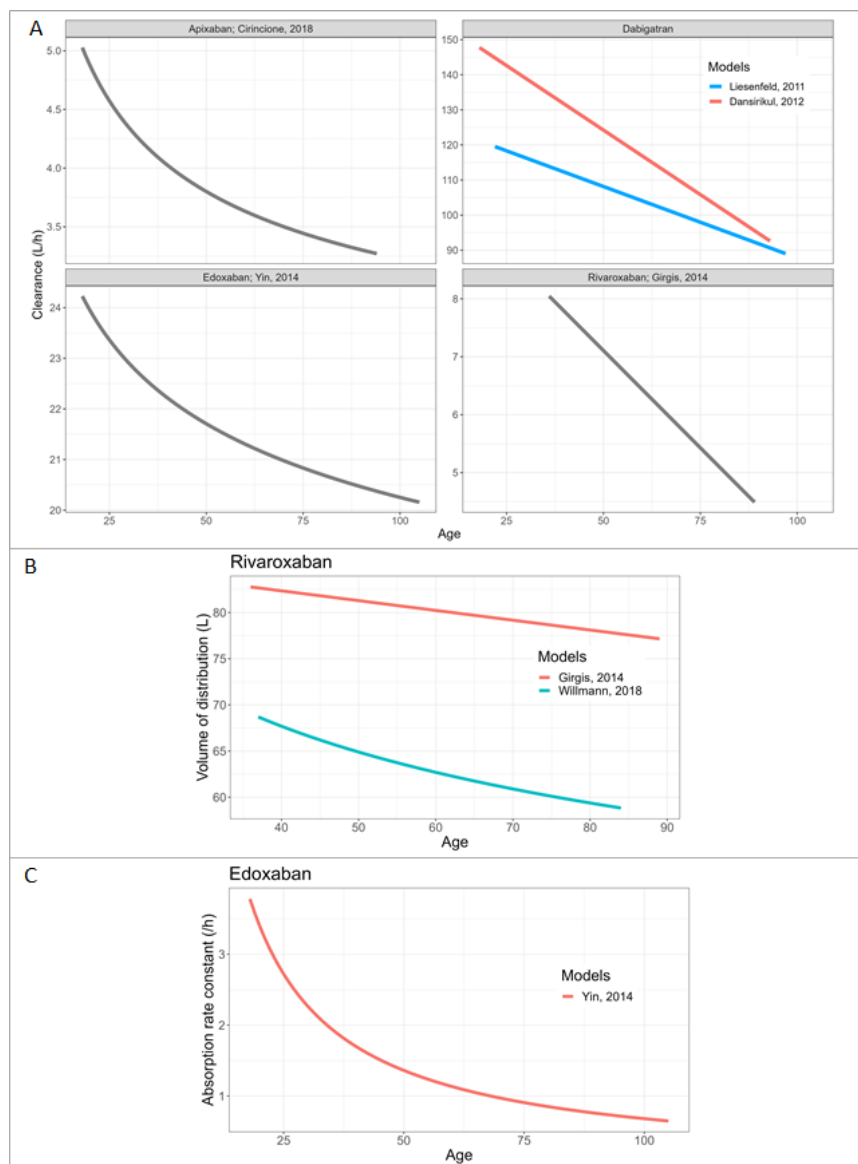


Figure 15. The changes of PK parameters of DOACs estimated by using population PK models across adult lifespan. The age range in every model was based on subjects' age included in the model. (A) The effect of age on the clearance of four DOACs as described by Liesenfeld and Dansirikul model for dabigatran, Cirincione model for apixaban, Yin model for edoxaban, Girgis model for rivaroxaban (51, 52, 53, 56, 64). (B) The effect of age on the volume of distribution of rivaroxaban as described by Girgis and Willmann model (53, 63). For other DOACs, the effect of age on volume of distribution was not significant and hence is not presented in this Figure. (C) The effect of age on the absorption rate constant of edoxaban as described by Yin model (64). For other DOACs, the effect of age on the absorption rate constant was not significant and hence is not presented in this Figure. Note: to accommodate the very different value ranges of PK parameters across DOACs, truncated Y-axes were used. This approach emphasizes age-related trends for each parameter but may visually amplify changes within individual panels. Care should be taken when interpreting the magnitude of these changes.

3.4.1. Impact of age

For apixaban, two out of three clinical criteria are required for dose reduction, and age was one of the three dose adjustment criteria according to the manufacturer (71). Even with the dose reduction there is still a 20-70% higher apixaban concentrations in older adults (39, 47). One apixaban population PK model demonstrated that in the oldest people (age >85 years), non-renal clearance decreased by 10% (51). Similarly, higher dabigatran concentrations were observed in older age groups (26, 29, 44), which might be caused by a 10% reduction of dabigatran clearance as explained in two population PK models (52, 56).

Five out of the six population PK models of rivaroxaban tested the age effect on CL, but no statistical significance was found (54, 57, 60, 63, 65). Only one population PK model of edoxaban and rivaroxaban identified a lower CL due to older age (64). Additionally, two population PK models of rivaroxaban incorporated an age effect on Vd in their final models (53, 63). Other models did not detect any significant finding when they tested the age effect (52, 54, 55, 57, 58, 60, 62, 64, 65, 72).

3.4.2. Body size and composition

Body weight is one of the dose reduction criteria for edoxaban and apixaban. It mainly affected Vd as seven out of 12 models that tested body weight on Vd retained this covariate. Willmann *et al.* and Zhang *et al.* analysed the effect of different body size predictors (e.g., body weight, body mass index, body surface area and lean body mass) on rivaroxaban Vd, and body weight was the most significant predictor (63, 65). Another potentially significant body size predictor on Vd of rivaroxaban was the lean body mass (60).

A PK study by Bendayan *et al.* indicated that age-related loss of muscle mass had an influence on apixaban and rivaroxaban exposure in older adults (23). To determine muscle mass, the appendicular lean mass (ALM, the sum of lean mass in arms and legs) was measured using a bioimpedance device and a dual energy X-ray absorptiometry scanner. ALM was negatively associated with apixaban and rivaroxaban plasma concentrations after the correction for age, total body weight, and renal impairment. Higher concentrations were observed in patients with an ALM less than 20 kg (odds ratio [OR] 0.35, 95% confidence interval [CI]=0.14-0.92). In a univariate analysis, supratherapeutic concentrations were associated with a higher Strength, Ambulation, Rising from a chair, Stair climbing and

history of Falling score (SARC-F) and a reduced handgrip strength (23). These two sarcopenia screening tools could be considered as practical alternatives in the absence of other approaches to measure ALM.

3.4.3. Concomitant medications

P-gp inhibitors such as ketoconazole, diltiazem, erythromycin, azithromycin, quinidine, verapamil, amiodarone, and dronedarone, were shown to influence the PK of DOACs. All edoxaban population PK models included an effect of P-gp inhibitors (55, 58, 64, 72). P-gp inhibitors altered all edoxaban PK parameters (i.e., k_a , bioavailability, V_d and CL), so that edoxaban exposure increased. According to Salazar *et al.*, amiodarone did not increase edoxaban concentration, although it is a P-gp inhibitor (58). As a result, dose adjustment with concomitant amiodarone was not required in the ENGAGE AF-TIMI 48 trial (9). Similar to edoxaban, the effect of amiodarone on apixaban concentrations (30) was inconclusive due to insufficient sample size, and did not appear to influence the CL (51).

A study in patients older than 85 years reported that almost 30% of patients with concomitant intake of P-gp inhibitors had dabigatran concentrations beyond the upper limit of the expected range of 400 ng/mL, which is considerably more than those without P-gp inhibitors (24). This result was in line with that of a population PK model where P-gp inhibitors increased dabigatran bioavailability by 15% (52). Another population PK model indicated that concomitant use of verapamil and amiodarone were associated with a 23% and 12% increase of dabigatran bioavailability, respectively (56). In a rivaroxaban PK study based on seven patients with dronedarone, two of the patients had higher-than-expected concentrations (i.e., >215 ng/mL) (34). Based on a population PK model, rivaroxaban CL decreased slightly by 3.4% with comedications, notably P-gp inhibitors and CYP3A4 inhibitors (63).

Besides P-gp inhibitors, the influence of CYP enzymes inhibitors was studied as well. Concomitant use of CYP3A4 inhibitors resulted in DOACs concentrations above the on-therapy ranges as observed in phase III studies (40). Population PK models showed that lower apixaban and rivaroxaban clearances were associated with use of CYP3A4 inhibitors (51, 63). Similarly, reduced DOACs concentrations were related to the intake of enzyme-inducing anti-seizure medications (e.g., phenytoin, phenobarbital, and carbamazepine)

(40). Additionally, in the population PK models, only dabigatran bioavailability was impacted by proton pump inhibitor (PPI) use, showing a reduction of 12-15% (52, 56).

3.4.4. Decrease of organ function

Adjustment of dose based on kidney function, particularly creatinine clearance (CrCL), is a common criterion shared by all DOACs. Thirteen PK studies found that kidney function predicted DOACs exposure in older adults. Lower CrCL values were associated with higher trough, peak and average concentrations of apixaban, dabigatran and rivaroxaban (28, 31, 32, 44, 47, 50). When compared with normal kidney function patients, older adults with severe renal impairment (CrCL < 30 mL/min) had 63% higher apixaban trough concentrations, albeit after dose reduction (46, 47). This was confirmed by population PK models describing the decrease of apixaban renal CL by 45-63% in severe renally impaired patients (51, 62). For edoxaban and dabigatran, predicted CL using population PK models declined by 32-66% and 25-52%, respectively, in renal impairment when CrCL was <30 mL/min (52, 55, 56, 57, 59, 64). Four of the seven population PK studies retained CrCL in their final models and predicted a decrease in CL of 17-40% in severe renal impairment (54, 60, 63, 65).

In addition to CrCL, other significant kidney function predictors were identified, such as serum creatinine, blood urea nitrogen (BUN) and estimated glomerular filtration rate (eGFR). A model which incorporated the effect of age and serum creatinine on CL of rivaroxaban, predicted that, in renal impairment, CL was around 20% lower in older adults (≥ 75 years) than in young subjects with impaired kidney function (53). Only two rivaroxaban models found that BUN and eGFR were more informative than CrCL and predicted that CL decreased by 30-40% in severe renal impairment (57, 61).

The effect of hepatic function on clearance was demonstrated in a rivaroxaban population PK study that had total bilirubin as covariate (65). It was suggested by Zhang *et al.* model that rivaroxaban clearance was reduced by 10% in the case of elevated total bilirubin (total bilirubin=20 μ mol/L) when compared to a population without reduced liver function (65). Total bilirubin was the only liver-related biomarker tested in the included studies, which limits its ability to capture the multifaceted nature of liver function. Notably,

in three other rivaroxaban population PK models (54, 57, 60), total bilirubin was tested but not retained in the final model.

3.4.5. Comorbidities

The presence of non-AF cardiovascular diseases was a predictor for higher plasma concentrations. Suzuki *et al.* indicated that old patients (mean study age of 82 years) with a history of heart failure or peripheral artery disease or myocardial infarction were almost two times more likely to have higher apixaban concentrations (47). Patients with heart failure have about 7% lower dabigatran clearance than those without this comorbidity in a population PK model (56), however it is not known whether this is clinically relevant or not as the relationship was not found in other studies (60, 61, 65). A higher rivaroxaban concentration was associated with diabetes mellitus in a linear mixed-effects regression test (38).

3.4.6. Dose

The association between DOACs dosing and plasma concentrations has been studied in observational PK studies. Although the intention of dose reduction is to reach the same exposure in more vulnerable patients, there were significant differences in plasma concentrations between patients who received a full or a reduced dose (30, 32, 36, 37, 39, 40, 45). The administration of higher doses of rivaroxaban and dabigatran likely led to higher bioavailability as described by using two population PK models (63, 64).

Five PK studies with median population age of 80 years evaluated the variability of DOACs exposure in standard, reduced, and off-label doses. Two studies indicated a more than 30% decrease in trough concentrations when the dose was reduced by half (standard dose of 5 mg BID) vs reduced dose of 2.5 mg BID) of apixaban (36, 45). Higher than 95th percentile concentrations were observed in older adults who received standard doses of apixaban, rivaroxaban and dabigatran (37). This is in an agreement with a study showing that 75–80% of geriatric older adults with an off-label low dose of apixaban exhibited trough and peak concentrations within the observed on-therapy range derived from clinical trials (27). On the contrary, Rizos *et al.* demonstrated that the inappropriately low dose of four DOACs resulted in the significantly lower area under the curve (AUC) with an OR of 2.125 (95%CI 1.039–4.56) (41).

Table 6. Covariates affecting DOACs concentrations analysed in included dose-exposure studies

Author, year	Population, sample size, mean age $\pm$ SD, median age (minimum-maximum) [IQR]	DOACs	Statistical test	Endpoints	Endpoint cut-off values (ng/mL, unless stated otherwise)	Significant covariates
Siedler, 2022 (43)	Patients with ischemic stroke and atrial fibrillation, • Apixaban, N=251, mean age=80.6 (48–100) • Edoxaban, N=43, mean age=79 (56–93) • Rivaroxaban, N=20, mean age=80.3 (54–91) • Dabigatran, N=83, mean age=72 (32–94)	Apixaban, dabigatran, edoxaban, rivaroxaban	Multivariable linear regression	Higher concentrations	-	Lower GFR (coefficient=-0.336, SE=0.137)
			Propensity score matching	Recurrent stroke	-	Lower apixaban concentrations
Rizos, 2022 (41)	Patients with acute ischemic stroke or transient ischemic attacks, N=211, median age=80 [75-86]	Apixaban, rivaroxaban, edoxaban, dabigatran	Multivariable logistic regression	Lower area under the curve	-	Inappropriate underdose (OR 2.125, 95%CI 1.039–4.56)
Lin, 2021 (34)	Patients with atrial fibrillation, N=511, mean age= 75.3 $\pm$ 8.9	Apixaban, rivaroxaban, dabigatran	Multivariable logistic regression	Higher-than-expected concentrations	Expected trough range = 34-230	Male (OR 17.77, 95%CI 1.87-168.81)
Mavri, 2021 (36)	Patients with atrial fibrillation, N=62, mean age = 78 $\pm$ 8	Apixaban	Mann–Whitney U-test or X ² test	Higher trough and peak concentrations	-	Dose 5 BID
Bendayan, 2021 (23)	Patients with atrial fibrillation, n=62, mean age=77.0 $\pm$ 6.1	Apixaban, rivaroxaban	Multivariable logistic regression	Supratherapeutic trough and peak concentrations	Apixaban 2.5 mg > 162, Apixaban 5 mg > 230, Rivaroxaban 20 mg > 96	• Lower appendicular lean mass (ALM), per 5 kg (OR 0.35, 95%CI=0.14-0.92) • Higher SARC-F score, per point increase (OR 1.68, 95%CI=1.10–2.57) (As an alternative to ALM)

Comans, 2021 (27)	Patients with atrial fibrillation, N=80, median (IQR)=79.5 (77–85)	Apixaban, rivaroxaban, dabigatran	Linear mixed model	Higher peak concentrations	-	Appropriate dose
Suzuki, 2020 and 2021 (46, 47)	Patients with atrial fibrillation, N=943, mean age 81.7 ± 4.6	Apixaban	Multivariable logistic regression	High trough concentrations	≥ 86 ng/dL	In patients with reduced dose apixaban • Female (OR 1.46, 95%CI 1.00-2.14) • Age ≥85 years old (OR 1.73, 95%CI 1.18-2.54) • CrCL <30 mL/min (OR 1.63, 95%CI 1.03-2.58) • Heart failure (OR 1.73, 95%CI 1.18-2.53) • History of peripheral artery disease or myocardial infarction (OR 1.84, 95%CI 1.00-3.36)
Kampouraki, 2021 (32)	Older adults stabilized on an anticoagulant for prevention and treatment of thrombosis, N=151, Median (IQR) = 84 (78–89)	Apixaban, rivaroxaban, dabigatran	Multivariable regression	Higher trough concentrations	-	• Female • Lower eGFR • Dose
				Higher peak concentrations		• Female • Lower CrCL • Dose
Perlman, 2021 (40)	Patients with atrial fibrillation and venous thromboembolic disease, N=131, median age (IQR)=80 (70–85)	Apixaban, rivaroxaban, dabigatran	Mixed-effects Linear - regression	Concentrations	Apixaban and rivaroxaban range: the 5th and 95 th percentiles Dabigatran: the 10th and 90th percentile As reported by analyses of phase III (93, 94)	• Co-medication with enzyme-inducing antiseizure medications (coefficient -0.42, 95%CI -0.64--0.21) • Co-medication with moderate CYP3A4 inhibitors (coefficient 0.26, 95%CI 0.06 –0.45) • Lower dose (coefficient 0.28, 95%CI 0.18–0.37)

			Mixed-effects logistic regression	Lower-than-expected concentrations		Co-medication with enzyme-inducing antiseizure medications (OR 6.26, 95%CI 2.19–17.9)
Gommans, 2021 (29)	Patients with atrial fibrillation, Median age (min-max) per group - <75 years (n=25): 68 (55–74) - ≥75–84 years (n=25): 77 (75–83) - ≥85 years (n=25): 88 (85–94)	Dabigatran	Kruskal-Wallis test	Higher trough concentrations	-	Older age
			Regression analysis			Lower value of eGFR
Gendron 2020 (28)	Patients who had received idarucizumab for dabigatran reversal, N=47, - mean age dabigatran < 264 ng/mL=80.4 ± 12.5 (n=26) - mean age dabigatran ≥ 264 ng/mL=83.6 ± 8.8 (n=21)	Dabigatran	Multivariable logistic regression	Higher-than-threshold concentrations	≥ 264	Lower CrCL Lower haemoglobin
Gulilat, 2019 (30)	Patients with atrial fibrillation, N=358, mean age= 77.2±10.5 (41-97)	Apixaban	Multivariable linear regression	Higher plasma concentrations	-	• Dose 5 mg BID ($\beta=0.1$, 95%CI= 0.081 - 0.119) • Female ($\beta=0.08$, 95%CI=0.035 - 0.124) • Older age ($\beta=0.006$, 95%CI=0.004 - 0.008) • Increased serum creatinine ($\beta=0.002$, 95%CI=0.001-0.002) • Amiodarone ($\beta=0.068$, 95%CI=0.005-0.131) • ABCG2 c.421C > A, per allele ($\beta=0.048$, 95%CI=0.002 - 0.093)
Lin, 2019 (35)	Asian patients with AF, N=73, mean age=74.9 ± 7.5	Rivaroxaban	Multivariate linear regression	Lower than range trough concentrations	-	• Appropriate dose (OR= 4.24, 95%CI 1.02-17.68, p-value = 0.047)

	Asian patients with AF, N=105, mean age=77.3 ± 9.1	Apixaban		Higher than range peak concentrations		Male (OR=3.46, 95%CI 1.19-10.07, p-value=0.023)
Skripka, 2020 (44)	Patients with atrial fibrillation and stage 3 chronic kidney disease, N=60, median age=76 (51–89)	Dabigatran	T-test and Mann-Whitney-U test	Higher dose-normalized plasma concentrations	-	- Lower creatinine clearance - Older age - Higher score of CHA2DS2-VASc and HASBLED
Bhagirath, 2020 (25)	Patients with atrial fibrillation, N=62, Median (IQR) = 87 (84-90)	Apixaban	T-test	Lower median trough concentrations	-	Off-label apixaban 2.5 BID (p-value=0.014)
Bernier, 2019 (24)	Patients with atrial fibrillation, N=58, Mean age= 88 ± 3	Dabigatran	multivariate logistic regression-	Supratherapeutic concentrations	> 400	Co-medications with P-gp inhibitors
Nissan, 2019 (37)	Patients with atrial fibrillation, Mean age per group: - Age < 70 dosage 5 mg BID (n=20): 64.3 ± 3.3 - Age ≥ 80 dosage 5 mg BID (n=20): 83.3 ± 2.7 - Age ≥ 80 dosage 2.5 mg BID, appropriate (n=20): 86.7 ± 3.7 - Age ≥ 80, dosage 2.5 mg BID inappropriate (n=20): 85.6 ± 4.0	Apixaban	ANOVA test	Higher peak concentration	-	5 mg BID
Suwa, 2019 (45)	Patients with atrial fibrillation 5 mg BID (N=58), median age=73±6 2.5 mg BID (N=61), median age=80±6	Apixaban	simple linear regression	Higher peak and trough concentrations	-	5 mg BID
Testa, 2019 (49)	Patients with atrial fibrillation, N=101, median age=80 (61–97)	Edoxaban	Linear regression and correlation test	Concentrations	-	CrCL was tested but it was not significant

Sacco, 2018 (42)	Patients with atrial fibrillation or venous thromboembolism, N=90, median age (IQR)=79 (74–84)	Apixaban	Bi- and multi-variate regression	Higher peak concentrations		Lower haemoglobin level ($\beta = -31.6$, 95%CI: -46.2--17)
Chaussade, 2018 (26)	Patients with atrial fibrillation, N=68, Mean=85.8 $\pm$ 5.1	Dabigatran	Multivariate linear mixed effect regression	Higher trough concentrations	-	Lower albumin level (Estimate -0.18, 95%CI=-0.36 – -0.003) Lower eGFR-MDRD (Estimate -0.16, 95%CI=-0.49 – -0.21)
				Higher peak concentrations	-	Lower eGFR-MDRD (Estimate -0.20, 95%CI=-0.32 – -0.07)
Ikeda 2016 (31)	Patients with atrial fibrillation, N=34, mean=82.9 $\pm$ 10.3	Apixaban	Pearson correlation test	Higher anti-Xa activity	-	Lower CrCL ($r=-0.439$)
Testa, 2016 (48)	Patients with atrial fibrillation, N=330 Apixaban (N=99, mean age= 76.1 $\pm$ 6.5) - Dabigatran (N=160, mean age = 75.4 $\pm$ 8.5) - Rivaroxaban (N= 71, mean age = 74.3 $\pm$ 6.4)	Apixaban, dabigatran, and rivaroxaban	Correlation test and coefficient of determination	Higher trough anti-Xa activity	-	Lower CrCL in patients who received dabigatran 110 mg ($r/r^2=-0.25/0.0625$) and 150 mg group ($r/r^2=-0.32/0.1024$)
Volbers, 2016 (50)	- Patients with acute ischemic stroke (n=15), median age = 80 [71-83] - Patients with intracerebral haemorrhage (n=4), median age = 78 [74-87]	Dabigatran	Linear regression for correlation	Higher dose-normalized concentrations	-	Lower CrCL ($r^2=0.319$)
				Higher concentrations		Lower CrCL ($r^2=0.346$)
Osanai, 2015 (39)	Patients with atrial fibrillation, N=124, mean age=77.5 $\pm$ 8.4	Apixaban	Multivariable regression	Higher trough anti-Xa activity	-	- Higher dose ($\beta = 0.4432$) - Older age ($\beta = 0.2602$) - Higher serum creatinine level ($\beta = 0.3683$)
				Higher peak anti-Xa activity		- Higher dose ($\beta = 0.6686$) - Older age ($\beta = 0.2289$) - Higher serum creatinine level ($\beta = 0.3112$)

Okata, 2014 (38)	Patients with stroke and transient ischemic attack, N=110, mean age=75 ± 9	Rivaroxaban	Linear mixed effect	Higher plasma concentrations	-	• Tablet crushing (Estimate -0.43, 95%CI -0.6 – -0.26) • Diabetes mellitus (Estimate 0.13, 95%CI 0.01-0.25) • Time from stroke/TIA onset to blood sampling (Estimate -0.02, 95%CI -0.03-0.01)
------------------	--	-------------	---------------------	------------------------------	---	---

3.5. The relationship between DOACs concentrations and clinical outcomes

Importantly, age and age-related conditions impact the relation between drug exposure and clinical outcomes. Therefore, information on clinical outcomes is crucial. Eight included studies reported the relationship between DOACs exposure with efficacy outcomes (e.g., stroke and thromboembolic events) and safety outcomes (e.g., bleeding) (n=1,924). This is presented in **Table 7**.

An observational PK study by Rizos *et al.* demonstrated that older adults (N=211) with AUCs below the study average experienced more serious strokes, signified by a higher National Institute of Health (NIHSS) score (41). Rizos *et al.* also suggested that the AUC was a better predictor for stroke severity than a peak or trough concentration at the event. Three other studies also found that DOACs exposure is a reliable predictor for clinical outcomes. Lower trough, peak and 9 hours after-dose rivaroxaban concentrations were significantly linked to a higher NIHSS score (N=110) (38). A lower peak concentration predicted a thromboembolic event in apixaban users and a recurrent stroke for those on dabigatran or apixaban (N=459) (36, 43).

In addition, dabigatran trough concentrations above 200 ng/mL were associated with haemorrhagic events (log-rank tests $p < 0.0003$) (N=68) (26). In Japanese patients with a reduced apixaban dose, plasma concentrations were significantly higher in those with bleeding requiring hospitalization (N=943) (46). Five of the 14 patients experiencing adverse events (i.e., bleeding and thrombosis) had out-of-range plasma apixaban, dabigatran, rivaroxaban concentrations (32). Because of an inadequate sample size, the statistical significance could not be confirmed (32).

Three publications found no significant association between DOACs exposure with bleeding events, stroke, systemic embolism, cardiovascular-related death other than ischaemic stroke, CHA₂DS₂-VASc score for atrial fibrillation stroke risk (36, 43, 46). Suwa *et al.* described that peak apixaban concentrations predicted bleeding, but no statistical significance was reported (45).

3.6. Coagulation biomarkers as PD endpoints

The relationship between DOACs exposure and anticoagulation surrogate markers, including intrinsic factor Xa, prothrombin time (PT), partial thrombin time (aPTT), was characterised by using mathematical equations in the population PK/PD models (52, 54, 57, 61, 66) (59) (Supplementary Table 2). aPTT had a weak correlation with apixaban concentrations (39). This result was supported by several studies showing that PT exhibited a superior correlation with edoxaban and apixaban concentrations compared to aPTT (31, 38, 49). However, both (PT and aPTT) are not acceptable for DOACs measurement and should be carefully interpreted as clinical biomarkers for DOACs anticoagulation activity (31, 38, 49).



Figure 16. Covariates tested using population PK approach in the 15 included studies. The transparent bars and negative numbers indicated covariate not found being significant. The blocked bars and positive numbers indicated covariate found being significant and had positive relationship (blue bars) or negative relationship (yellow bars). The blank rows showed that the covariates were not tested *Subject status: patients with atrial fibrillation, acute coronary syndrome or healthy volunteers. PPI: Proton Pump Inhibitors, P-gp: P-glycoprotein, CrCL: Creatinine clearance, GFR: Glomerular filtration rate, BUN: Blood urea nitrogen, LBM: Lean body mass.

Table 7. Pharmacodynamics of DOACs in older adults

Author, year	DOACs	Statistical test	Endpoints	Significant predictors	Not significant predictors
Rizos, 2022 (41)	Apixaban, rivaroxaban, edoxaban, dabigatran	Logistic regression	Ischemic stroke and TIA	• Lower AUC (OR 2.411, 95%CI 1.254–4.638) • Lower concentration at event onset (OR 4.123, 95%CI 1.834–9.268)	
		Linear regression	Stroke severity (NIHSS score)	Lower AUC (Slope 3.161, 95%CI 0.741 -- 5.581)	
Siedler, 2022 (43)	Apixaban, rivaroxaban, edoxaban	Mann-Whitney test	Bleeding		Plasma trough concentrations
			Ischaemic stroke		
			Composite endpoint of stroke and death		
		Propensity score matching	Recurrent stroke	Lower trough concentrations	
Suzuki, 2020 (46)	Apixaban	Multivariable cox regression	Bleeding requiring hospitalisation	Higher trough concentration (HR 12.12, 95%CI: 1.56-94.22)	
			Stroke and systemic embolism		Concentration
		Correlation test	Prothrombin time on standard dose measured by various assays	Concentrations (r=0.474-0.633)	
			Prothrombin time on reduced dose measured by various assays	Concentrations (r=0.292-0.569)	
Chaussade , 2018 (26)	Dabigatran	Kaplan-Meier curve, Wilcoxon rank sum test or Fisher exact test	Haemorrhage	Higher trough concentrations (>90 th percentiles: ≥ 243.9 ng/mL)	
		Wilcoxon rank sum test or Fisher exact test		Lower serum albumin levels	

Osanai, 2015 (39)	Apixaban	Spearman's correlation coefficient	Prothrombin time	Trough concentrations ($r=0.60$)	
				Peak concentrations ($r=0.69$)	
Suwa, 2019 (45)	Apixaban	Logistic regression	Bleeding		Peak plasma concentrations
Testa, 2019 (49)	Edoxaban	Correlation test and coefficient of determination	Prothrombin time	• Trough concentrations after the administration of edoxaban 30 mg ($r/r^2=0.702/0.493$) and of all patients ($r/r^2=0.386/0.149$) • Peak concentrations after the administration of edoxaban 30 mg ($r/r^2=0.862/0.743$) and 60 mg ($r/r^2=0.911/0.830$) and of all patients ($r/r^2=0.922/0.851$)	
			Partial thromboplastin time	• Trough concentrations after the administration of edoxaban 30 mg ($r/r^2=0.316/0.010$) and of all patients ($r/r^2=0.283/0.080$) • Peak concentrations after the administration of edoxaban 30 mg ($r/r^2=0.545/0.297$) and 60 mg ($r/r^2=0.498/0.248$) and of all patients ($r/r^2=0.567/0.321$)	
Mavri, 2021 (36)	Apixaban	Mann-Whitney U test	Thromboembolic events	Lower peak concentration (average=153 ng/mL (149-169), p -value=0.05)	Trough concentration
			Bleeding		Peak and trough concentration

OR: Odds ratio, CI: confidence interval, AUC: Area Under the Curve, NIHSS: National Institutes of Health Stroke Scale, TIA: Transient Ischemic Attack

4. Discussion: unmet needs and future directions

This review summarises current insights on PK and PD of apixaban, edoxaban, rivaroxaban, and dabigatran in older adults. Older adults exhibited high IIV in DOACs exposure which can be explained by above-mentioned covariates. However, there is often a complex combination of all kinds of variables at the same time, which are not fully explored in terms of their relationships with DOACs PK/PD. In the sub-section below, we focused on specific patient characteristics related to older age.

4.1. Does advanced age influence DOACs PK/PD?

There are some changes in the PK/PD of DOACs in older adults compared with the younger population. However, it is difficult to assess whether age itself influences the PK of DOACs. The conflicting evidence of age influencing PK/PD of DOACs may indicate that most probably age alone does not directly affect PK/PD, since age is associated with many physiological and pathophysiological alterations. Changes of DOACs PK in older adults can be caused by kidney function reduction with aging. This was confirmed in studies comparing healthy elderly subjects to young healthy volunteers, where age had no clinically meaningful effect on DOACs PK (73, 74, 75). An article reviewed dabigatran PK data, and suggested that higher dabigatran exposure or lower clearance in old volunteers could be explained by declined kidney function in older age (76). The effect of age on DOACs Vd is most probably related to the loss of muscle mass or the changes of body mass in older adults since they are frequently burdened with sarcopenia. Moreover, DOACs typically have a relatively high Vd, indicating significant tissue uptake, so the change of body composition, including decreased lean mass and increased fat tissue, will also affect DOACs PK. Likewise, the effect of sex on DOACs PK was also confounded by body size (73, 74, 75). More importantly, age does not seem to influence clinical outcomes of anticoagulated subjects in the included studies. This finding is supported by a large scale observational study, which shows no increased risk of stroke or systemic embolism, bleeding requiring hospitalization, or cardiovascular death in oldest old patients (≥ 85 years) taking on-label doses of apixaban (77). In **Figure 17**, we have categorized the retained covariates, into important and potential covariates. This classification was based on how frequently these covariates were found to

be significant in the included studies. **Figure 17** also presents other relevant geriatric traits that have not been investigated yet on DOACs PK/PD.

4.1.1. Important and potential covariates

Besides reduced kidney function, another important confounding factor between age and DOACs PK/PD is concomitant medication with P-gp inhibitors. There is a higher risk of drug-drug interaction in older adults with geriatric profile as the polypharmacy is common. On average, hospitalized geriatric patients receive 10 medications per day (78). In addition, body size and composition could also be confounding factors as unintentional weight loss is common in older adults (79, 80). Another important covariate affecting DOACs PK is dosing, which is further discussed in the section 4.2. Potentially meaningful characteristics include genotype, race, albumin, bilirubin level, food, haemoglobin, haematocrit, co-medication with PPI and CYP3A4 inhibitors, and certain comorbidities (e.g., diabetes mellitus, heart failure, myocardial infarction and atherosclerosis). Comorbidities, such as diabetes mellitus and heart failure, are likely to influence PK due to prolonged gastric emptying, nephropathy, and changes in blood flow. However, the mechanisms of myocardial infarction and of atherosclerosis potentially impacting the PK of DOACs remain unknown and still need further investigation.

4.1.2. Other relevant geriatric traits not investigated yet on DOACs

Studies evaluating the effect of geriatric traits, such as sarcopenia, frailty, malnutrition, and altered gastrointestinal physiology on PK were limited, while such characteristics may have direct or indirect effects on DOACs PK/PD. So far, only one study evaluating the effect of muscle mass, one of the features of sarcopenia, on PK/PD of DOACs was found (23). No study was found regarding the relation between frailty and DOACs PK, although several evidences demonstrated that frailty is associated with DOACs adverse outcomes, i.e. stroke, bleeding risk and mortality. The effect of frailty on PK was studied in only a few non-DOACs drugs (84). Frailty was defined based on clinical characteristics (e.g., age, mobility, medication use) or on an objective scale (e.g., (reported) Edmonton Frail Scale, (adapted) Fried criteria). Those studies showed the tendency of higher drug exposure and lower clearance in old and frail subjects compared with younger volunteers. Considering that frailty seems to affect other drugs PK and DOACs clinical outcomes, a

study scrutinising the relation between frailty - DOACs PK - clinical outcomes should be performed.

Based on a systematic review, malnutrition might affect PK mediated by impaired absorption related to loss of mucosal surface area and inflammation of the gastrointestinal tract (85). The Vd of several drugs tended to increase in malnutrition due to decreased protein binding. Reduced hepatic oxidative drug biotransformation and conjugation are present in malnutrition and can reduce drug elimination. However, most of the studies included in the systematic review were done in children and a young adult population (85). Additionally, gastrointestinal physiology such as gastric emptying, intestinal motility, gut microbiome, bile solubilization are critical since DOACs are classified as biopharmaceutical classification system class II-IV where solubility and permeability across the gastrointestinal tract are considered as rate-limiting steps. Taking all of this into consideration, several PK studies, especially for high-risk medications such as DOACs, are needed to investigate the extent to which potential geriatric traits might further explain PK variability.

4.2. Is DOACs dosing optimal in this special population with AF?

Studies included in this review indicated less predictable DOACs concentrations in advanced age and high IIV. In some studies, dosing adjustment criteria did not result in precise DOACs concentrations. Of all DOACs, dabigatran exhibited the largest IIV. This is because dabigatran dose adjustment is based solely on age, and as indicated above, age alone is not a reliable dosing adjustment criterion. Because of the large IIV, dabigatran is less preferable than other DOACs. Additionally, drug concentrations after on-label standard and reduced doses were statistically different, although the dose was adjusted to tailor the dose to the important patient characteristics. Paradoxically, some findings showed that on-label doses resulted in supratherapeutic concentrations, while off-label reduced doses led to on-therapy concentrations in older adults. Considering the afore-mentioned finding, standard dosing may still fail to catch the full spectrum of variability and heterogeneity in this unique subpopulation, sticking to the concept of a one-dose to size or age-fits-all. Therefore, it should be questioned whether current dose adjustment criteria are optimal for all older adults with geriatric profiles or whether there is a need to further improve them by

scrutinising important, potential, and relevant geriatric patient characteristics as summarised in this review. We hypothesize that frailty deserves more attention in this regard.

More importantly, those observed on-therapy concentrations should be carefully interpreted. Although PK/PD studies revealed a significant correlation between DOACs exposure and clinical outcomes (e.g., stroke and bleeding), it is difficult to conclude whether certain target concentrations are linked to these outcomes in older adults due to the absence of robust trial evidence. In the case the correlation between supratherapeutic exposure and bleeding, conflicting evidence emerged. A real-life dabigatran study reported that a high concentration higher than 243.9 ng/mL was related to an elevated risk of haemorrhage event (26). This value is just slightly above the upper limit of the expected trough concentration range determined in a pivotal randomized controlled trial according to European Heart Rhythm Association (EHRA) (28-215 ng/mL) (86). Several included studies in older adults revealed stroke-associated subtherapeutic exposure and bleeding-associated supratherapeutic exposure (36, 41, 43, 46). EHRA also provides expected plasma concentrations for apixaban, edoxaban and rivaroxaban, which were also derived from DOAC PK studies (70, 87, 88). However, the references to determine target ranges were in healthy volunteers and younger patient populations. It remains unknown whether the same exposure in older adults will correspond with the same outcomes as in a younger population. No target concentration of apixaban, edoxaban and rivaroxaban was studied specifically in old patients, and the expected ranges were mainly observed in fixed dosing regimen.

DOACs PK/PD data in older adults is lacking. The included population PK studies utilized phase 1 or 2 trial to build structural models and integrated data including a wide age range for covariate model building, and thus the typical PK parameters may not represent older adults appropriately. Moreover, model-based dosing combined with therapeutic drug monitoring have been used to further individualise drug treatment (89, 90). For this vulnerable patient group, this strategy can be useful to explain the complex interplay between PK/PD, covariates, and clinical outcomes.

4.3. The potential role of therapeutic drug monitoring as tool to optimize DOACs therapy in older adults

Therapeutic drug monitoring (TDM) cannot currently not be recommended for DOACs, due to lack of clinical evidence regarding its clinical benefits and the uncertain impact on the therapeutic range in older adults (91). A fixed dose approach has clearly shown its merit in multiple clinical trials, where TDM was done. Conversely, there is obviously a high variability in DOACs PK/PD, which is compounded in older adults. This might imply one of two things. First, the benefits of DOAC (vs warfarin) are apparent on a background of wide variability, signifying the lack of a narrow therapeutic index. Second, the risk/benefit ratio of DOACs might theoretically be enhanced for individual patients by personalising the dose to target a predefined range. Such benefits might be even more apparent in older adults suffering from moderate to severe frailty, which compounds PK variability and hence impacts bleeding/stroke risk in atrial fibrillation.

Importantly, as stated above, there is no robust therapeutic range for DOACs in older adults, which is also the reason why therapeutic drug monitoring is not recommended at this moment. Thus, more efforts should be devoted to ensuring DOACs safety and efficacy by performing an analysis to define the therapeutic target in frail older adults. Without solid evidence, the use of TDM should be limited to highly selected clinical scenarios and/or subgroups of patients (e.g., to exclude residual anticoagulation prior to administering fibrinolysis in acute ischemic stroke) (91).

4.4. Future direction

Considering the unmet needs above, we propose a framework for future research direction of DOACs in adults ≥ 75 years. More PK studies should be performed to better understand the relationship between dose and exposure incorporating important as well as potential and relevant covariates in the older adults with geriatric syndromes as specified in **Figure 17**. Optimal sampling needs to be developed to reduce patient burden considering that this special population live with multimorbidity and polypharmacy since this strategy can identify the most informative sampling times. Future research on the PK/PD of DOACs in older adults should also focus on unraveling the mechanisms and population variability that influence their efficacy and safety. Given the challenges of PK sampling in older adults

and geriatric patients approaches such as in silico modeling especially physiologically-based PK(/PD) or quantitative system pharmacology models are essential while minimizing the need for extensive sampling. This does not imply that monitoring is warranted per se; it might well be the case that the current fixed-dosing approach is found to be the best approach or that additional criteria should be included in the dose reduction criteria (e.g., frailty, sarcopenia, altered GI function).

4.5. Limitations

This narrative review has some limitations. Although we implemented a search strategy to reduce selection bias, it was difficult to compare the exposure of four DOACs, considering the heterogeneity in the DOACs dosing, the way exposure metrics were reported (peak or trough concentration, average concentration, dose-normalised concentration, or AUC, classification based on different factors), time when concentration was measured, the use of different assays with differing sensitivities (see **Supplementary Table 1**). While most samples in the included studies were above the LLOQ, assays with higher LLOQs, such as anti-Xa activity assays, can potentially introduce bias by overestimating median concentrations. However, in our analysis, studies utilizing anti-Xa activity assays did not demonstrate higher concentrations, suggesting that such bias was not evident in these cases. Hence, we can only compare exposures from studies that clearly mentioned the dose group. Although **Figure 14** gives us an overview of DOACs exposure metrics in older adults across studies, it is still difficult to compare due to different effect size. Two articles reported DOACs exposure in a different unit (IU/mL) (31, 39), and dose modifications were not captured, which rendered it difficult to compare the findings to those from other studies. Additionally, no statistical analysis was done when comparing DOACs exposure in older and younger subjects.

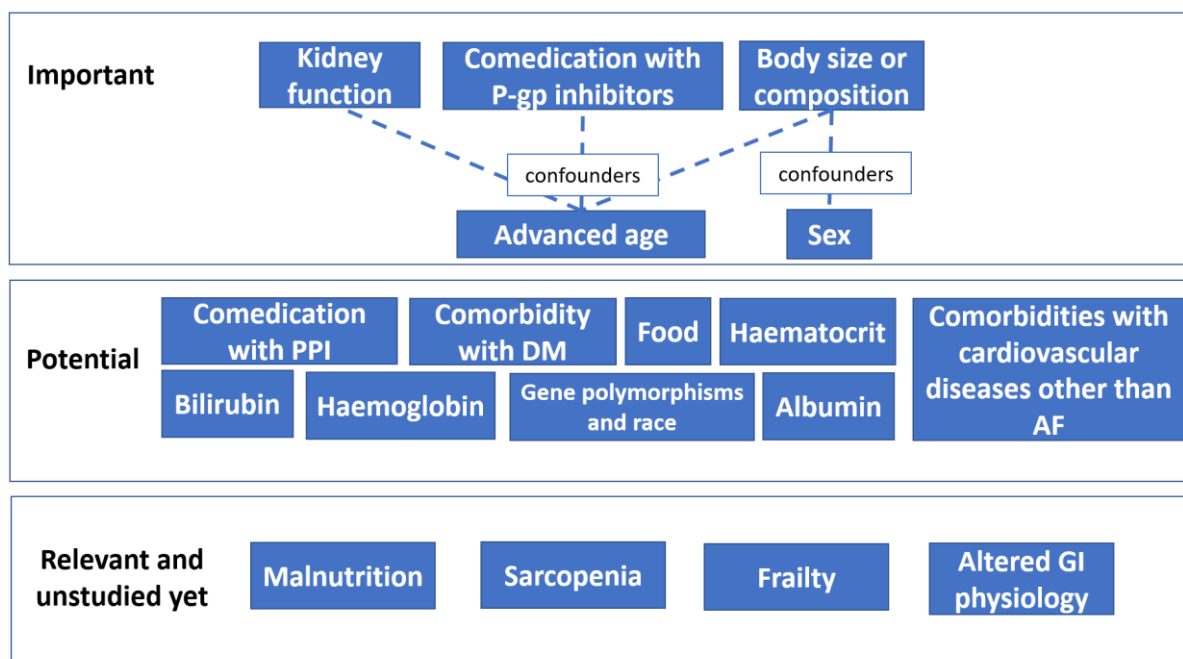


Figure 17. List of identified important, or potential, covariates, and relevant covariates for older adults affecting PK/PD of DOAC. The effect of age on PK/PD of DOAC is more likely confounded by other covariates, including kidney function, comedication with P-gp inhibitors and body size or composition. Likewise, the effect on sex on PK/PD of DOAC can be confounded by body size. P-gp: P-glycoprotein, PPI: proton pump inhibitors, AF: Atrial Fibrillation, GI: Gastrointestinal, DM: diabetes mellitus

5. Conclusion

Older adults exhibit a remarkable interindividual variability in DOACs PK. Advanced age did not directly influence DOACs PK/PD. The effect of age on DOACs PK/PD was confounded by important covariates, including kidney function, concomitant use of P-gp inhibitors and body size or composition. This is in line with the current dosing guideline of apixaban, edoxaban and rivaroxaban, recommending that age alone should not be used as a dose adjustment criterion. For dabigatran, where dose reduction only relies on age, the highest IIV was observed. For that reason, dabigatran is not a preferable option among all DOACs. Data on other potential covariates (e.g., genotype, race, albumin, bilirubin level, food, haemoglobin, haematocrit, co-medication with PPI and CYP3A4 inhibitors, and comorbidities with diabetes mellitus and non-AF cardiovascular diseases) are limited and, should be a topic for further investigation. At this point, there is insufficient evidence to support personalising DOACs dosing in older adults with AF other than current used dosing regimens. There is room for improving individualized treatment of older adults with DOACs

to maximally optimize appropriate prescribing by using a simulation study with a good model.

References

1. Vos T, Abajobir AA, Abate KH, Abbafati C, Abbas KM, Abd-Allah F, et al. Global, regional, and national incidence, prevalence, and years lived with disability for 328 diseases and injuries for 195 countries, 1990–2016: a systematic analysis for the Global Burden of Disease Study 2016. *The Lancet*. 2017;390(10100):1211-59.
2. Wilke T, Groth A, Mueller S, Pfannkuche M, Verheyen F, Linder R, et al. Incidence and prevalence of atrial fibrillation: an analysis based on 8.3 million patients. *Europace*. 2013;15(4):486-93.
3. Wilkinson C, Clegg A, Todd O, Rockwood K, Yadegarfar ME, Gale CP, et al. Atrial fibrillation and oral anticoagulation in older people with frailty: a nationwide primary care electronic health records cohort study. *Age Ageing*. 2021;50(3):772-9.
4. Risk factors for stroke and efficacy of antithrombotic therapy in atrial fibrillation. Analysis of pooled data from five randomized controlled trials. *Arch Intern Med*. 1994;154(13):1449-57.
5. Hart RG, Pearce LA, Aguilar MI. Meta-analysis: antithrombotic therapy to prevent stroke in patients who have nonvalvular atrial fibrillation. *Ann Intern Med*. 2007;146(12):857-67.
6. Singer DE, Chang Y, Fang MC, Borowsky LH, Pomernacki NK, Udaltsova N, et al. The net clinical benefit of warfarin anticoagulation in atrial fibrillation. *Ann Intern Med*. 2009;151(5):297-305.
7. Steffel J, Collins R, Antz M, Cornu P, Desteghe L, Haeusler KG, et al. 2021 European Heart Rhythm Association Practical Guide on the Use of Non-Vitamin K Antagonist Oral Anticoagulants in Patients with Atrial Fibrillation. *EP Europace*. 2021;23(10):1612-76.
8. Granger CB, Alexander JH, McMurray JJV, Lopes RD, Hylek EM, Hanna M, et al. Apixaban versus Warfarin in Patients with Atrial Fibrillation. *New England Journal of Medicine*. 2011;365(11):981-92.
9. Giugliano RP, Ruff CT, Braunwald E, Murphy SA, Wiviott SD, Halperin JL, et al. Edoxaban versus Warfarin in Patients with Atrial Fibrillation. *New England Journal of Medicine*. 2013;369(22):2093-104.
10. Connolly SJ, Ezekowitz MD, Yusuf S, Eikelboom J, Oldgren J, Parekh A, et al. Dabigatran versus Warfarin in Patients with Atrial Fibrillation. *New England Journal of Medicine*. 2009;361(12):1139-51.
11. Patel MR, Mahaffey KW, Garg J, Pan G, Singer DE, Hacke W, et al. Rivaroxaban versus Warfarin in Nonvalvular Atrial Fibrillation. *New England Journal of Medicine*. 2011;365(10):883-91.
12. Agno W, Gallus AS, Wittkowsky A, Crowther M, Hylek EM, Palareti G. Oral Anticoagulant Therapy. *Chest*. 2012;141(2):e44S-e88S.
13. Witt DM, Clark NP, Kaatz S, Schnurr T, Ansell JE. Guidance for the practical management of warfarin therapy in the treatment of venous thromboembolism. *Journal of Thrombosis and Thrombolysis*. 2016;41(1):187-205.
14. De Groot JR, Weiss TW, Kelly P, Monteiro P, Deharo JC, De Asmundis C, et al. Edoxaban for stroke prevention in atrial fibrillation in routine clinical care: 1-year follow-up of the prospective observational ETNA-AF-Europe study. *European Heart Journal - Cardiovascular Pharmacotherapy*. 2021;7(F11):f30-f9.
15. Proietti M, Romanazzi I, Romiti GF, Farcomeni A, Lip GYH. Real-World Use of Apixaban for Stroke Prevention in Atrial Fibrillation. *Stroke*. 2018;49(1):98-106.
16. Ibáñez L, Sabaté M, Vidal X, Ballarin E, Rottenkolber M, Schmiedl S, et al. Incidence of direct oral anticoagulant use in patients with nonvalvular atrial fibrillation and characteristics of users in 6 European countries (2008–2015): A cross-national drug utilization study. *British Journal of Clinical Pharmacology*. 2019;85(11):2524-39.

17. Hias J, Van der Linden L, Walgraeve K, Gijssen M, Mian P, Koch BCP, et al. Pharmacokinetics of 2 oral paracetamol formulations in hospitalized octogenarians. *British Journal of Clinical Pharmacology*. 2021.
18. Lloyd-Jones D, Adams R, Carnethon M, De Simone G, Ferguson TB, Flegal K, et al. Heart Disease and Stroke Statistics—2009 Update. *Circulation*. 2009;119(3):e21-e181.
19. Eurostat. Ageing Europe: Looking at the lives of older people in the EU. Luxembourg: Publications Office of the European Union; 2019.
20. Lip GY, Nieuwlaat R, Pisters R, Lane DA, Crijns HJ. Refining clinical risk stratification for predicting stroke and thromboembolism in atrial fibrillation using a novel risk factor-based approach: the euro heart survey on atrial fibrillation. *Chest*. 2010;137(2):263-72.
21. Bramer WM, Giustini D, De Jonge GB, Holland L, Bekhuis T. De-duplication of database search results for systematic reviews in EndNote. *Journal of the Medical Library Association : JMLA*. 2016;104(3):240-3.
22. Ouzzani M, Hammady H, Fedorowicz Z, Elmagarmid A. Rayyan—a web and mobile app for systematic reviews. *Systematic Reviews*. 2016;5(1).
23. Bendayan M, Mardigyan V, Williamson D, Chen-Tournoux A, Eintracht S, Rudski L, et al. Muscle Mass and Direct Oral Anticoagulant Activity in Older Adults With Atrial Fibrillation. *Journal of the American Geriatrics Society*. 2021;69(4):1012-8.
24. Bernier M, Lancrerot SL, Rocher F, Van-Obberghen EK, Olivier P, Lavrut T, et al. Major bleeding events in octagenarians associated with drug interactions between dabigatran and P-gp inhibitors. *Journal of Geriatric Cardiology*. 2019;16(11):806-11.
25. Bhagirath VC, Chan N, Hirsh J, Ginsberg J, de Vries TAC, Eikelboom J. Plasma Apixaban Levels in Patients Treated Off Label With the Lower Dose. *J Am Coll Cardiol*. 2020;76(24):2906-7.
26. Chaussade E, Hanon O, Bouilly C, Labourée F, Caillard L, Gerotziakas G, et al. Real-Life Peak and Trough Dabigatran Plasma Measurements Over Time in Hospitalized Geriatric Patients with Atrial Fibrillation. *Journal of Nutrition, Health and Aging*. 2018;22(1):165-73.
27. Comans AL, Sennesael AL, Bihin B, Regnier M, Mullier F, de Saint-Hubert M. Inappropriate low dosing of direct oral anticoagulants in older patients with non-valvular atrial fibrillation: Impact on plasma drug levels. *Thrombosis Research*. 2021;201:139-42.
28. Gendron N, Chocron R, Billoir P, Brunier J, Camoin-Jau L, Tuffigo M, et al. Dabigatran Level Before Reversal Can Predict Hemostatic Effectiveness of Idarucizumab in a Real-World Setting. *Frontiers in Medicine*. 2020;7.
29. Gommans E, Grouls RJE, Kerkhof D, Houterman S, Simmers T, Van Der Linden C. Dabigatran trough concentrations in very elderly patients. *European Journal of Hospital Pharmacy*. 2021;28(4):231-3.
30. Gulilat M, Keller D, Linton B, Pananos AD, Lizotte D, Dresser GK, et al. Drug interactions and pharmacogenetic factors contribute to variation in apixaban concentration in atrial fibrillation patients in routine care. *Journal of Thrombosis and Thrombolysis*. 2020;49(2):294-303.
31. Ikeda K, Tachibana H. Clinical implication of monitoring rivaroxaban and apixaban by using anti-factor Xa assay in patients with non-valvular atrial fibrillation. *J Arrhythm*. 2016;32(1):42-50.
32. Kampouraki E, Avery P, Biss T, Wynne H, Kamali F. Assessment of exposure to direct oral anticoagulants in elderly hospitalised patients. *British Journal of Haematology*. 2021;195(5):790-801.
33. Kryukov AV, Sychev DA, Andreev DA, Ryzhikova KA, Grishina EA, Ryabova AV, et al. Influence of ABCB1 and CYP3A5 gene polymorphisms on pharmacokinetics of apixaban in patients with atrial fibrillation and acute stroke. *Pharmacogenomics and Personalized Medicine*. 2018;11:43-9.

34. Lin SY, Kuo CH, Huang TM, Peng YF, Huang CF, Tang SC, et al. Impact of different renal function equations on direct oral anticoagulant concentrations. *Scientific reports*. 2021;11(1):23833.
35. Lin SY, Kuo CH, Yeh SJ, Tsai LK, Liu YB, Huang CF, et al. Real-World Rivaroxaban and Apixaban Levels in Asian Patients With Atrial Fibrillation. *Clinical Pharmacology and Therapeutics*. 2020;107(1):278-86.
36. Mavri A, Vene N, Božič-Mijovski M, Miklič M, Söderblom L, Pohanka A, et al. Apixaban concentration variability and relation to clinical outcomes in real-life patients with atrial fibrillation. *Sci Rep*. 2021;11(1):13908.
37. Nissan R, Spectre G, HersHKovitz A, Green H, Shimony S, Cooper L, et al. Apixaban Levels in Octogenarian Patients with Non-valvular Atrial Fibrillation. *Drugs and Aging*. 2019;36(2):165-77.
38. Okata T, Toyoda K, Okamoto A, Miyata T, Nagatsuka K, Minematsu K. Anticoagulation intensity of rivaroxaban for stroke patients at a special low dosage in Japan. *PLoS ONE*. 2014;9(11).
39. Osanai H, Ajioka M, Masutomi T, Kuwayama T, Ishihama S, Sakamoto Y, et al. Measurement of anti-factor xa activity in patients on apixaban for non-valvular atrial fibrillation. *Circulation Journal*. 2015;79(12):2584-90.
40. Perlman A, Goldstein R, Choshen Cohen L, Hirsh-Racchah B, Hakimian D, Matok I, et al. Effect of Enzyme-Inducing Antiseizure Medications on the Risk of Sub-Therapeutic Concentrations of Direct Oral Anticoagulants: A Retrospective Cohort Study. *CNS Drugs*. 2021;35(3):305-16.
41. Rizos T, Meid AD, Huppertz A, Dumschat C, Purrucker J, Foerster KI, et al. Low Exposure to Direct Oral Anticoagulants Is Associated with Ischemic Stroke and Its Severity. *J Stroke*. 2022;24(1):88-97.
42. Sacco M, Lancellotti S, Berruti F, Arcovito A, Bellelli A, Ricciardelli T, et al. Apixaban Interacts with Haemoglobin: Effects on Its Plasma Levels. *Thrombosis and Haemostasis*. 2018;118(10):1701-12.
43. Siedler G, Macha K, Stoll S, Plechschmidt J, Wang R, Gerner ST, et al. Monitoring of direct oral anticoagulants plasma levels for secondary stroke prevention. *Journal of Thrombosis and Haemostasis*. 2022.
44. Skripka A, Sychev D, Bochkov P, Shevchenko R, Krupenin P, Kogay V, et al. Factors Affecting Trough Plasma Dabigatran Concentrations in Patients with Atrial Fibrillation and Chronic Kidney Disease. *High Blood Pressure and Cardiovascular Prevention*. 2020;27(2):151-6.
45. Suwa M, Morii I, Kino M. Rivaroxaban or Apixaban for Non-Valvular Atrial Fibrillation - Efficacy and Safety of Off-Label Under-Dosing According to Plasma Concentration. *Circ J*. 2019;83(5):991-9.
46. Suzuki S, Yamashita T, Akao M, Okumura K. Clinical implications of assessment of apixaban levels in elderly atrial fibrillation patients: J-ELD AF registry sub-cohort analysis. *European Journal of Clinical Pharmacology*. 2020;76(8):1111-24.
47. Suzuki S, Yamashita T, Akao M, Okumura K. Predictors for a high apixaban level in elderly patients with atrial fibrillation prescribed reduced dose of apixaban. *European Journal of Clinical Pharmacology*. 2021;77(11):1757-8.
48. Testa S, Tripodi A, Legnani C, Pengo V, Abbate R, Dellanoce C, et al. Plasma levels of direct oral anticoagulants in real life patients with atrial fibrillation: Results observed in four anticoagulation clinics. *Thrombosis Research*. 2016;137:178-83.
49. Testa S, Dellanoce C, Paoletti O, Cancellieri E, Morandini R, Tala M, et al. Edoxaban plasma levels in patients with non-valvular atrial fibrillation: Inter and intra-individual variability, correlation with coagulation screening test and renal function. *Thrombosis Research*. 2019;175:61-7.
50. Volbers B, Köhrmann M, Kallmünzer B, Kurka N, Breuer L, Ringwald J, et al. Dabigatran Plasma Levels in Acute Cerebrovascular Events. *Journal of Stroke and Cerebrovascular Diseases*. 2016;25(4):877-82.

51. Cirincione B, Kowalski K, Nielsen J, Roy A, Thanneer N, Byon W, et al. Population Pharmacokinetics of Apixaban in Subjects With Nonvalvular Atrial Fibrillation. *CPT: Pharmacometrics and Systems Pharmacology*. 2018;7(11):728-38.
52. Dansirikul C, Lehr T, Liesenfeld KH, Haertter S, Staab A. A combined pharmacometric analysis of dabigatran etexilate in healthy volunteers and patients with atrial fibrillation or undergoing orthopaedic surgery. *Thrombosis and Haemostasis*. 2012;107(4):775-85.
53. Girgis IG, Patel MR, Peters GR, Moore KT, Mahaffey KW, Nessel CC, et al. Population pharmacokinetics and pharmacodynamics of rivaroxaban in patients with non-valvular atrial fibrillation: results from ROCKET AF. *J Clin Pharmacol*. 2014;54(8):917-27.
54. Kaneko M, Tanigawa T, Hashizume K, Kajikawa M, Tajiri M, Mueck W. Confirmation of model-based dose selection for a Japanese phase III study of rivaroxaban in non-valvular atrial fibrillation patients. *Drug Metabolism and Pharmacokinetics*. 2013;28(4):321-31.
55. Krekels EH, Niebecker R, Karlsson MO, Miller R, Shimizu T, Karlsson KE, et al. Population Pharmacokinetics of Edoxaban in Patients with Non-Valvular Atrial Fibrillation in the ENGAGE AF-TIMI 48 Study, a Phase III Clinical Trial. *Clin Pharmacokinet*. 2016;55(9):1079-90.
56. Liesenfeld KH, Lehr T, Dansirikul C, Reilly PA, Connolly SJ, Ezekowitz MD, et al. Population pharmacokinetic analysis of the oral thrombin inhibitor dabigatran etexilate in patients with non-valvular atrial fibrillation from the RE-LY trial. *Journal of Thrombosis and Haemostasis*. 2011;9(11):2168-75.
57. Liu XQ, Zhang YF, Ding HY, Yan MM, Jiao Z, Zhong MK, et al. Population pharmacokinetic and pharmacodynamic analysis of rivaroxaban in Chinese patients with non-valvular atrial fibrillation. *Acta Pharmacologica Sinica*. 2022.
58. Salazar DE, Mendell J, Kastrissios H, Green M, Carrothers TJ, Song S, et al. Modeling and simulation of edoxaban exposure and response relationships in patients with atrial fibrillation. *Thrombosis and Haemostasis*. 2012;107(5):925-34.
59. Song S, Kang D, Halim AB, Miller R. Population pharmacokinetic-pharmacodynamic modeling analysis of intrinsic FXa and bleeding from edoxaban treatment. *Journal of Clinical Pharmacology*. 2014;54(8):910-6.
60. Speed V, Green B, Roberts LN, Woolcombe S, Bartoli-Abdou J, Barsam S, et al. Fixed dose rivaroxaban can be used in extremes of bodyweight: A population pharmacokinetic analysis. *Journal of Thrombosis and Haemostasis*. 2020;18(9):2296-307.
61. Tanigawa T, Kaneko M, Hashizume K, Kajikawa M, Ueda H, Tajiri M, et al. Model-based dose selection for phase III rivaroxaban study in Japanese patients with non-valvular atrial fibrillation. *Drug Metabolism and Pharmacokinetics*. 2013;28(1):59-70.
62. Ueshima S, Hira D, Kimura Y, Fujii R, Tomitsuka C, Yamane T, et al. Population pharmacokinetics and pharmacogenomics of apixaban in Japanese adult patients with atrial fibrillation. *British Journal of Clinical Pharmacology*. 2018;84(6):1301-12.
63. Willmann S, Zhang L, Frede M, Kubitzka D, Mueck W, Schmidt S, et al. Integrated population pharmacokinetic analysis of rivaroxaban across multiple patient populations. *CPT: Pharmacometrics and Systems Pharmacology*. 2018;7(5):309-20.
64. Yin OQ, Tetsuya K, Miller R. Edoxaban population pharmacokinetics and exposure-response analysis in patients with non-valvular atrial fibrillation. *Eur J Clin Pharmacol*. 2014;70(11):1339-51.
65. Zhang F, Chen X, Wu T, Huang N, Li L, Yuan D, et al. Population Pharmacokinetics of Rivaroxaban in Chinese Patients with Non-Valvular Atrial Fibrillation: A Prospective Multicenter Study. *Clinical Pharmacokinetics*. 2022.
66. Ueshima S, Hira D, Tomitsuka C, Nomura M, Kimura Y, Yamane T, et al. Population Pharmacokinetics and Pharmacodynamics of Apixaban Linking Its Plasma Concentration to

- Intrinsic Activated Coagulation Factor X Activity in Japanese Patients with Atrial Fibrillation. *AAPS Journal*. 2019;21(5).
67. Ogata K, Mendell-Harary J, Tachibana M, Masumoto H, Oguma T, Kojima M, et al. Clinical Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of the Novel Factor Xa Inhibitor Edoxaban in Healthy Volunteers. *The Journal of Clinical Pharmacology*. 2010;50(7):743-53.
 68. Kreutz R, Persson PB, Kubitza D, Thelen K, Heitmeier S, Schwes S, et al. Dissociation between the pharmacokinetics and pharmacodynamics of once-daily rivaroxaban and twice-daily apixaban: a randomized crossover study. *Journal of Thrombosis and Haemostasis*. 2017;15(10):2017-28.
 69. Stangier J, Eriksson BI, Dahl OE, Ahnfelt L, Nehmiz G, Stähle H, et al. Pharmacokinetic profile of the oral direct thrombin inhibitor dabigatran etexilate in healthy volunteers and patients undergoing total hip replacement. *J Clin Pharmacol*. 2005;45(5):555-63.
 70. Frost C, Nepal S, Wang J, Schuster A, Byon W, Boyd RA, et al. Safety, pharmacokinetics and pharmacodynamics of multiple oral doses of apixaban, a factor Xa inhibitor, in healthy subjects. *Br J Clin Pharmacol*. 2013;76(5):776-86.
 71. Summary of Product Characteristics: Eliquis 2.5 mg film-coated tablets [Available from: https://www.ema.europa.eu/en/documents/product-information/eliquis-epar-product-information_en.pdf.
 72. Shimizu T, Tachibana M, Kimura T, Kumakura T, Yoshihara K. Population Pharmacokinetics of Edoxaban in Japanese Atrial Fibrillation Patients With Severe Renal Impairment. *Clinical Pharmacology in Drug Development*. 2017;6(5):484-91.
 73. Mendell J, Shi M. Safety, tolerability, pharmacokinetic (PK) and pharmacodynamic (PD) profiles of edoxaban in healthy post-menopausal or surgically sterile females, and healthy elderly males. *European Heart Journal*. 2011;32:461.
 74. Frost CE, Song Y, Shenker A, Wang J, Barrett YC, Schuster A, et al. Effects of Age and Sex on the Single-Dose Pharmacokinetics and Pharmacodynamics of Apixaban. *Clinical Pharmacokinetics*. 2015;54(6):651-62.
 75. Kubitza D, Becka M, Roth A, Mueck W. The Influence of Age and Gender on the Pharmacokinetics and Pharmacodynamics of Rivaroxaban-An Oral, Direct Factor Xa Inhibitor. *The Journal of Clinical Pharmacology*. 2013;53(3):249-55.
 76. Stangier J. Clinical Pharmacokinetics and Pharmacodynamics of the Oral Direct Thrombin Inhibitor Dabigatran Etexilate. *Clinical Pharmacokinetics*. 2008;47(5):285-95.
 77. Okada M, Inoue K, Tanaka N, Sakata Y, Akao M, Yamashita T, et al. Clinical Outcomes of Very Elderly Patients With Atrial Fibrillation Receiving On-label Doses of Apixaban: J-ELD AF Registry Subanalysis. *Journal of the American Heart Association*. 2021;10(15).
 78. Van Der Linden L, Decoutere L, Walgraeve K, Milisen K, Flamaing J, Spriet I, et al. Combined Use of the Rationalization of Home Medication by an Adjusted STOPP in Older Patients (RASP) List and a Pharmacist-Led Medication Review in Very Old Inpatients: Impact on Quality of Prescribing and Clinical Outcome. *Drugs & Aging*. 2017;34(2):123-33.
 79. de Groot CP, Perdigo AL, Deurenberg P. Longitudinal changes in anthropometric characteristics of elderly Europeans. *SENECA Investigators. Eur J Clin Nutr*. 1996;50 Suppl 2:S9-15.
 80. Newman AB, Yanez D, Harris T, Duxbury A, Enright PL, Fried LP, et al. Weight change in old age and its association with mortality. *J Am Geriatr Soc*. 2001;49(10):1309-18.
 81. Candeloro M, Di Nisio M, Potere N, Di Pizio L, Secinaro E, De Flaviis C, et al. Frailty phenotype as a predictor of bleeding and mortality in ambulatory patients receiving direct oral anticoagulants. *J Am Geriatr Soc*. 2022;70(12):3503-12.

82. Yamamoto T, Yamashita K, Miyamae K, Koyama Y, Izumimoto M, Kamimura Y, et al. The influence of frailty under direct oral anticoagulant use in patients with atrial fibrillation. *Heart Asia*. 2019;11(2):e011212.
83. De Simone V, Mugnolo A, Zanutto G, Morando G. Direct oral anticoagulants for patients aged over 80 years in nonvalvular atrial fibrillation: The impact of frailty. *Journal of Cardiovascular Medicine*. 2020;21(8):562-9.
84. Hilmer SN, Kirkpatrick CMJ. New Horizons in the impact of frailty on pharmacokinetics: latest developments. *Age Ageing*. 2021;50(4):1054-63.
85. Verrest L, Wilthagen EA, Beijnen JH, Huitema ADR, Dorlo TPC. Influence of Malnutrition on the Pharmacokinetics of Drugs Used in the Treatment of Poverty-Related Diseases: A Systematic Review. *Clinical Pharmacokinetics*. 2021;60(9):1149-69.
86. Reilly PA, Lehr T, Haertter S, Connolly SJ, Yusuf S, Eikelboom JW, et al. The effect of dabigatran plasma concentrations and patient characteristics on the frequency of ischemic stroke and major bleeding in atrial fibrillation patients: The RE-LY trial (Randomized Evaluation of Long-Term Anticoagulation Therapy). *Journal of the American College of Cardiology*. 2014;63(4):321-8.
87. Ruff CT, Giugliano RP, Braunwald E, Morrow DA, Murphy SA, Kuder JF, et al. Association between edoxaban dose, concentration, anti-Factor Xa activity, and outcomes: an analysis of data from the randomised, double-blind ENGAGE AF-TIMI 48 trial. *Lancet*. 2015;385(9984):2288-95.
88. Kubitz D, Becka M, Wensing G, Voith B, Zuehlsdorf M. Safety, pharmacodynamics, and pharmacokinetics of BAY 59-7939—an oral, direct Factor Xa inhibitor—after multiple dosing in healthy male subjects. *European Journal of Clinical Pharmacology*. 2005;61(12):873-80.
89. Sturkenboom MGG, Märtson A-G, Svensson EM, Sloan DJ, Dooley KE, Van Den Elsen SHJ, et al. Population Pharmacokinetics and Bayesian Dose Adjustment to Advance TDM of Anti-TB Drugs. *Clinical Pharmacokinetics*. 2021;60(6):685-710.
90. Brooks E, Tett SE, Isbel NM, Staatz CE. Population Pharmacokinetic Modelling and Bayesian Estimation of Tacrolimus Exposure: Is this Clinically Useful for Dosage Prediction Yet? *Clinical Pharmacokinetics*. 2016;55(11):1295-335.
91. Chan N, Sager PT, Lawrence J, Ortel T, Reilly P, Berkowitz S, et al. Is there a role for pharmacokinetic/pharmacodynamic-guided dosing for novel oral anticoagulants? *Am Heart J*. 2018;199:59-67.
92. Stocker SL, Carland JE, Reuter SE, Stacy AE, Schaffer AL, Stefani M, et al. Evaluation of a Pilot Vancomycin Precision Dosing Advisory Service on Target Exposure Attainment Using an Interrupted Time Series Analysis. *Clin Pharmacol Ther*. 2021;109(1):212-21.
93. Eikelboom JW, Quinlan DJ, Hirsh J, Connolly SJ, Weitz JI. Laboratory Monitoring of Non-Vitamin K Antagonist Oral Anticoagulant Use in Patients With Atrial Fibrillation. *JAMA Cardiology*. 2017;2(5):566.
94. Gosselin R, Adcock D, Bates S, Douxfils J, Favaloro E, Gouin-Thibault I, et al. International Council for Standardization in Haematology (ICSH) Recommendations for Laboratory Measurement of Direct Oral Anticoagulants. *Thrombosis and Haemostasis*. 2018;118(03):437-50.

Appendix

Supplementary material 1: Search strategy

PUBMED

Concept 1 = elderly, older people, older patients and geriatric patients OR models

“Aged”[MeSH] OR frail*[tiab] OR aged[tiab] OR “aged people”[tiab] OR “aged adult”[tiab] OR “aged patient”[tiab] OR “aged person”[tiab] OR “aged citizen”[tiab] OR elder*[tiab] OR “older adult”[tiab] OR “older people”[tiab] OR “older patient”[tiab] OR “older person”[tiab] OR “older citizen”[tiab] OR “geriatr”[tiab] OR “old patient”[tiab] OR “old adult”[tiab] OR “old people”[tiab] OR “old person”[tiab] OR “old citizen”[tiab] OR senior*[tiab] OR senil*[tiab] OR “advanced age”[tiab] OR agedness[tiab] OR ageing[tiab] OR aging[tiab] OR senium[tiab] OR septuagenarian*[tiab] OR octogenarian*[tiab] OR nonagenarian*[tiab] OR centenarian*[tiab] OR “Oldest Old”[tiab] OR ((60[tiab] OR 65[tiab] OR 70[tiab] OR 75[tiab] OR 80[tiab] OR 85[tiab] OR 90[tiab] OR 95[tiab] OR 100[tiab]) AND (age[tiab] OR ages[tiab] OR year*[tiab])) OR "Models, Statistical"[Mesh:NoExp] OR model*[tiab] OR "Models, Theoretical"[Mesh:NoExp] OR "Models, Biological"[Mesh:NoExp]

Concept 2 = direct oral anticoagulants (apixaban, dabigatran, edoxaban, rivaroxaban)

“Factor Xa Inhibitors”[MeSH] OR “Factor Xa inhibitor”[tiab] OR “factor 10a inhibitor”[tiab] OR “direct oral anticoagulant”[tiab] OR “Direct-Acting Oral Anticoagulant”[tiab] OR "apixaban" [Supplementary Concept] OR apixaban[tiab] OR eliquis[tiab] OR eliques[tiab] OR “BMS 562247”[tiab] OR BMS562247[tiab] OR BMS-562247-01[tiab] OR “non-vitamin K antagonist oral anticoagulant”[tiab] OR "edoxaban" [Supplementary Concept] OR edoxaban[tiab] OR lixiana[tiab] OR savaysa[tiab] OR DU-176[tiab] OR DU-176b[tiab] OR "Rivaroxaban"[Mesh] OR rivaroxaban[tiab] OR xarelto[tiab] OR kriva[tiab] OR roteas[tiab] OR throsaben[tiab] OR “BAY 59-7939”[tiab] OR “BAY 597939”[tiab] OR "Dabigatran"[Mesh] OR dabigatran[tiab] OR xanirva[tiab] OR pradax*[tiab] OR prazaxa[tiab] OR rendix[tiab] OR “BIBR 1048”[tiab] OR DOAC[tiab] OR DOACs[tiab] OR NOAC[tiab] OR NOACs[tiab] OR “novel oral anticoagulant”[tiab]

Concept 3 = pharmacokinetics and pharmacodynamics endpoints

"Pharmacokinetics"[Mesh] OR pharmacokinetic*[tiab] OR pharmacodynamic*[tiab] OR disposition[tiab] OR ADME[tiab] OR ADMET[tiab] OR ADME-Tox[tiab] OR LADMER[tiab] OR AUC[tiab] OR bioavailability[tiab] OR “dose response”[tiab] OR “concentration dependenc”[tiab] OR “concentration response”[tiab] OR ((drug*[tiab] OR plasma[tiab] OR “anti-Xa”) AND (concentration*[tiab] OR level*[tiab] OR exposure*[tiab] OR absorption*[tiab] OR

accumulation*[tiab] OR distribution*[tiab] OR metabolism*[tiab] OR biotransformation*[tiab] OR “protein binding”[tiab] OR elimination*[tiab] OR clearance*[tiab] OR excretion*[tiab] OR “half-life”[tiab] OR “t1/2”[tiab] OR kinetic*[tiab] OR activit*[tiab] OR response*[tiab] OR interaction*[tiab] OR tolerability[tiab]))

Exclusion

NOT (review[pt] OR review[ti])

EMBASE

Concept 1 = elderly, older people, older patients and geriatric patients

'aged'/exp OR (frail* OR aged OR 'aged people' OR 'aged adult*' OR 'aged patient*' OR 'aged person*' OR 'aged citizen*' OR elder* OR 'older adult*' OR 'older people' OR 'older patient*' OR 'older person*' OR 'older citizen*' OR 'geriatr*' OR 'old patient*' OR 'old adult*' OR 'old people' OR 'old person*' OR 'old citizen*' OR senior* OR senil* OR 'advanced age*' OR agedness OR ageing OR aging OR senium OR septuagenarian* OR octogenarian* OR nonagenarian* OR centenarian* OR 'Oldest Old' OR ((60 OR 65 OR 70 OR 75 OR 80 OR 85 OR 90 OR 95 OR 100) NEAR/3 (age OR ages OR year*))) :ti,ab,kw OR 'statistical model'/de OR 'biological model'/de OR 'theoretical model'/de OR model*:ti,ab,kw OR 'model'/de

Concept 2 = direct oral anticoagulants (apixaban, dabigatran, edoxaban, rivaroxaban)

'blood clotting factor 10a inhibitor'/de OR 'apixaban'/exp OR 'dabigatran'/exp OR 'edoxaban'/exp OR 'rivaroxaban'/exp OR (apixaban OR dabigatran OR edoxaban OR rivaroxaban OR 'direct oral anticoagulant*' OR 'Factor Xa inhibitor*' OR 'factor 10a inhibitor*' OR 'Direct-Acting Oral Anticoagulant*' OR Eliquis OR eliques OR 'BMS 562247' OR BMS562247 OR BMS-562247-01 OR 'non-vitamin K antagonist oral anticoagulant*' OR DU-176 OR DU-176b OR lixiana OR savaysa OR roteas OR xarelto OR kriva OR throsaben OR xanirva OR pradax* OR prazaxa OR rendix OR 'BAY 59-7939' OR 'BAY 597939' OR 'BIBR 1048' OR DOAC OR DOACs OR NOAC OR NOACs OR 'novel oral anticoagulant*') :ti,ab,kw

Concept 3 = pharmacokinetics and pharmacodynamics endpoints

'pharmacokinetics'/de OR 'pharmacodynamics'/de OR 'drug absorption'/de OR 'drug accumulation'/de OR 'drug bioavailability'/exp OR 'drug clearance'/exp OR 'drug disposition'/de OR 'drug distribution'/de OR 'drug elimination'/de OR 'drug excretion'/de OR 'drug half life'/exp OR 'drug metabolism'/de OR 'plasma concentration-time curve'/de OR 'drug interaction'/de OR 'drug response'/de OR 'concentration response'/de OR 'dose response'/de OR 'drug tolerability'/de OR (pharmacokinetic* OR pharmacodynamic* OR disposition OR ADME OR ADMET OR ADME-Tox OR

AUC OR bioavailability OR 'concentration dependenc*' OR 'concentration response*' OR 'dose response*' OR ((drug* OR plasma OR 'anti-Xa') NEAR/3 (concentration* OR level* OR exposure* OR kinetic* OR absorption* OR accumulation* OR distribution* OR 'protein binding' OR metabolism* OR elimination* OR clearance OR 'half life' OR excretion OR half-life OR 't1/2' OR activit* OR response* OR interaction* OR tolerability)))):ab,ti,kw

Exclusion

NOT (review:ti OR review:it OR 'conference abstract':it)

WoS Core Collection

Concept 1 = elderly, older people, older patients and geriatric patients OR models

TS=(frail* OR aged OR "aged people" OR "aged adult*" OR "aged patient*" OR "aged person*" OR "aged citizen*" OR elder* OR "older adult*" OR "older people" OR "older patient*" OR "older person*" OR "older citizen*" OR "geriatr*" OR "old patient*" OR "old adult*" OR "old people" OR "old person*" OR "old citizen*" OR senior* OR senil* OR "advanced age*" OR agedness OR ageing OR aging OR senium OR septuagenarian* OR octogenarian* OR nonagenarian* OR centenarian* OR "Oldest Old" OR ((60 OR 65 OR 70 OR 75 OR 80 OR 85 OR 90 OR 95 OR 100) AND (age OR ages OR year*)) OR model*)

Concept 2 = direct oral anticoagulants (apixaban, dabigatran, edoxaban, rivaroxaban)

TS=("Factor Xa inhibitor*" OR "factor 10a inhibitor*" OR "direct oral anticoagulant*" OR "Direct-Acting Oral Anticoagulant*" OR apixaban OR eliquis OR eliques OR "BMS 562247" OR BMS562247 OR BMS-562247-01 OR "non-vitamin K antagonist oral anticoagulant*" OR edoxaban OR lixiana OR savaysa OR DU-176 OR DU-176b OR rivaroxaban OR xarelto OR kriva OR roteas OR throsaben OR "BAY 59-7939" OR "BAY 597939" OR dabigatran OR xanirva OR pradax* OR prazaxa OR rendix OR "BIBR 1048" OR DOAC OR DOACs OR NOAC OR NOACs OR "novel oral anticoagulant*")

Concept 3 = pharmacokinetics and pharmacodynamics endpoints

TS=(pharmacokinetic* OR pharmacodynamic* OR disposition OR ADME OR ADMET OR ADME-Tox OR LADMER OR AUC OR bioavailability OR "dose response*" OR "concentration dependenc*" OR "concentration response*" OR ((drug* OR plasma OR "anti-Xa") AND (concentration* OR level* OR exposure* OR absorption* OR accumulation* OR distribution* OR metabolism* OR biotransformation* OR "protein binding" OR elimination* OR clearance* OR excretion* OR "half-life" OR "t1/2" OR kinetic* OR activit* OR response* OR interaction* OR tolerability))))

Exclusion

NOT (DT=("meeting abstract" OR "review" OR "editorial material") OR TI=("review"))

Supplementary table 1

DOACs concentrations and PK metrics analysed in included dose-exposure studies

Author, year	Assay	Groups	PK metrics (ng/mL, unless stated otherwise), Mean ± SD, Median [IQR], Median (range)			
			C _{trough}	C _{peak}	C _{avg}	Others
Apixaban						
Rizos, 2022	UHPLC-MS/MS	Concentrations at admission	-	-	123.0 (59.6–210.0)	
Lin, 2021	UHPLC-MS/MS		96.5 ± 55.2			
Mavri, 2021	LC–MS/MS	5 mg BID	117 [93–161]	256 [202–345]		
		2.5 mg BID	85 [68–110]	170 [145–216]		
	Berichrom Heparin anti-Xa with the STA-Apixaban calibrator	5 mg BID	115 [87–152]	206 [178–269]		
		2.5 mg BID	83 [64–104]	165 [133–184]		
	STA–Liquid anti-Xa with the STA-Apixaban calibrator	5 mg BID	93 [71–121]	197 [154–274]		
		2.5 mg BID	70 [53–93]	138 [107–177]		
Bendayan, 2021	Anti-Xa chromogenic assay (HemosIL® Liquid Anti-Xa) with BIOPHEN® Calibrator and Control Kit	Apixaban 2.5 and 5 mg	99.1 [55.8-175.0]	210.4 [149.8, 253.8]		
Comans, 2021	Biophen® Direct Factor Xa inhibitors, STA®-Liquid anti-Xa	Appropriate low dose	145.4 (digitized)	266		
		Inappropriate dose	104.4 (digitized)	169		
Suzuki, 2020 and 2021	STA®-Liquid Anti-Xa with STA®-Apixaban Calibrator and Control on the STA®-R analyzer	2.5 mg BID	86 ng/dL (90% CI = <23-258)			
		5 mg BID	130 ng/dL (90% CI = 25-319)			
Kampouraki, 2021	Liquid anti-Fxa assay with HemosIL® reagents and calibrators	2.5 mg BID	100.1 [41.2-168.2]	185 [82,8, 222,9]		
		5 mg BID	140 [78,1 -- 344,5]	224.2 [139-494.8]		
Bhagirath, 2020	STA Liquid Anti-Xa assay with STAr/Evolution calibrators and controls to convert anti-Xa activity to plasma concentrations	Off-label 2.5 mg BID	66			
		On-label 2.5 mg BID	102			
		On-label 5 mg BID	108			
Lin, 2019	UHPLC-MS/MS		105.9 ± 57.2	221.3 ± 116.5		
Nissan, 2019	Liquid anti-Xa, Instrumentation with TECHNOVIEW apixaban Calibrator and Controls	Age <70, 5 mg BID	123.3 ± 129.8	246.1 ± 161.4		
		Age ≥ 80, 5 mg BID	150.3 ± 97.0	306.6 ± 140.4		
		Age ≥80, 2.5 mg BID, appropriate	112.6 ± 75.2	180.8 ± 95.6		
		Age ≥80, 2.5 mg bid, inappropriate	91.8 ± 61.4	163.9 ± 68.6		
Suwa, 2019	Biophen® DiXal kit with a calibrator and control	5 mg BID	108.0	236.6		

		2.5 mg BID	67.5	157.4		
Sacco, 2018	STA-Liquid Anti-Xa assays with STA-Apixaban Calibrator	Haemoglobin 9 g/dL	149.8	253		AUC=2237 ng.h/mL Vd = 20 L Clearance=1.11 L/h
		Haemoglobin 15 g/dL	49.6	161		AUC=1046 ng.h/mL Vd=31 L CL= 3.75 L/h
		Haemoglobin 18 g/dL	1.9	108		AUC=453.4 ng.h/mL Vd=46 L Cl=6.7 L/h
Kryukov, 2018	LC-MS/MS	CYP3A5 (rs776746) GG genotype	-	116.5	-	AUC _{0-T} =940.1 AUC _{0-inf} = 2745.8 Tmax (h)=3.5 T1/2 (h)= 18.4
		CYP3A5 (rs776746) AG genotype	-	170	-	AUC _{0-T} =1395.9 AUC _{0-inf} = 2775.3 Tmax (h)=2 T1/2 (h)= 11.1
Gulilat, 2019	LC-MS/ MS				208.3 (32.3 -- 1329.0)	
Osanai, 2015	Anti-Xa chromogenic assay, HemosIL Liquid Heparin Kit with ACL TOP Hemostasis Testing System	5 mg BID	1.00 ± 0.66 IU/ml	2.05 ± 0.73 IU/ml		
		2.5 mg BID, appropriate	0.84 ± 0.47 IU/ml	1.51 ± 0.65 IU/ml		
		2.5 mg BID, inappropriate	0.59 ± 0.38 IU/ml	1.23 ± 0.46 IU/ml		
Ikeda, 2016	HemosIL® Liquid Heparin kit	2.5 and 5 mg BID	0.22–2.86 IU/mL	0.83–2.77 IU/mL		
Testa, 2016 (Digitized)	Calibrated anti-Xa assays calibrated	5 mg BID	127.53 [42.83-370.02]	224.73 [100-411.8]		
		2.5 mg BID	73.77 [19.91 -- 241]	163.03 [45.4-289.88]		
Edoxaban						
Testa, 2019	STA- liquid anti-Xa with STA-edoxaban calibration and control	Total	38 (7–147)	238 (10-569)		
		60 mg QD	39 (13–110)	301 (60-569)		
		30 mg QD	38 (7–147)	169 (10-400)		
Rivaroxaban						

Rizos, 2022	UHPLC-MS/MS	Concentrations at admission			127.5 (26.4–291.5)	
Lin, 2021	UHPLC-MS/MS		45.2 ± 60.6			
Bendayan, 2021	Chromogenic anti-Xa assay (HemosIL® Liquid Anti-Xa), BIOPHEN® Calibrator and Control Kit was used to convert anti-Xa activity to plasma concentrations	Rivaroxaban 20 mg	60.7 [45.3 -- 72.6]	279.3 [200.2 -- 368.4]		
Comans, 2021	Biophen® Direct Factor Xa inhibitors, STA®-Liquid anti-Xa	Appropriate dose	26.4 (digitized)	345		
		Inappropriate lower dose	43.9 (digitized)	236		
Kampouraki , 2021	Liquid anti-Fxa assay with HemosIL® reagents and calibrators	10 mg OD	45.1 [13.9 -49.4]	139.5 [55.1 -- 209.2]		
		15 mg OD	56.6 [17.7 -- 248.7]	329.4 [51.6 -- 483.9]		
		20 mg OD	42.4 [15 -- 190.8]	389.2 [= 179.5 -- 553.1]		
Lin, 2019	UHPLC-MS/MS	Total	102.88 ± 59.03	221.29 ± 116.45		
		10 mg OD	39.46 ± 74.61	189.80 ± 102.77		
		15 mg OD	31.90 ± 34.65	182.74 ± 100.14		
Okata, 2014	Anti-Xa chromogenic assay, STA-Liquid Anti-Xa with STA-R coagulometer	Time 0 h			11 [5-22]	
		Time 4 h			168 [109-243]	
		Time 9 h			65 [44-103]	
Dabigatran						
Rizos, 2022	UHPLC-MS/MS	Concentrations at admission			83.8 (47.6–200.8)	
Gommans, 2021	UHPLC- MS/MS	<75 years			99 (29–214), Dose-normalized=0.66 ng/mL/mg (0.19–1.43)	
		≥75–84 years			108 (33–327) Dose-normalized=0.83 ng/mL/mg (0.24–2.97)	
		≥85 years			132 (36–324) Dose-normalized=1.20 ng/mL/mg (0.33–2.95)	
Kampouraki 2021	HemosIL® Direct Thrombin Inhibitor assay with Hyphen BioMed calibrators	110 mg BID	131 [61-235]	188.6 [57.9-403.1]		
		150 mg BID	78.3 [67.6-93]	106.8 [70.1-315.6]		

Comans, 2021	Biophen® Direct Factor Xa inhibitors, STA®-Liquid anti-Xa		19-308	44-620		
Skripka, 2020	LC-MS/MS	Total	102.3 (5.2-319.6)			
		110 mg BID	96.3 (5.2-319.6)			
		150 mg BID	77.5 (9.6-262.1)			
Gendron, 2020	Diluted thrombin time, samples from three patients were quantified by using LC/MS				323 ± 477	
Bernier, 2019	LC/MS-MS	With P-gp inhibitors			182.2 ± 147.3	
		Without P-gp inhibitors			93.7 ± 64.9	
Chaussade, 2018	Hemoclot thrombin inhibitor (Hyphen Biomed)	110 mg BID Day 4	97.4 ± 94.3	153.4 ± 130.1		
		110 mg BID Day 8	110.7 ± 104.9	177.2 ± 136.4		
		110 mg BID Day 15	103 ± 79.3	158.7 ± 98.0		
		110 mg BID Day 30	120.5 ± 105.5	197.3 ± 149 .0		
		110 mg BID Day 45	113.5 ± 91.9	156.1 ± 136.2		
Testa, 2016 (digitized)	Calibrated diluted thrombin time	110 mg BID	104.51 [16.22-388.52]	208.31 [36.65-657.49]		
		150 mg BID	106.09 [21.29-504.57]	224.16 [57.54-552.66]		
Volbers, 2016	Diluted thrombin time (Hemoclot)	Acute ischemic stroke			69.1 [20.6-85.0]	
		Intracerebral haemorrhage			135 [79-218]	
Unstratified DOAC concentrations analysis						
Perlman, 2021 (digitized)	Calibrated anti-Xa-based assay (rivaroxaban and apixaban) and a thrombin time (TT)-based assay (dabigatran)	Rivaroxaban, apixaban, dabigatran			25.67-542.91	
Siedler, 2022	STA-Liquid Anti-Xa with anticoagulant- specific calibration (Diagnostica STAGO)	Apixaban, edoxaban, rivaroxaban and dabigatran (Dabigatran mean age was too young)	95.9 ± 66.9			

Supplementary table 2

PK(/PD) parameters of DOAC estimated by using population PK(/PD) models including older adults

Author, year	Population, age (y)	Parameter s	Fixed effects parameters	Interindividual variability	Significant covariates	Non-significant covariates
Apixaban						
Cirincione, 2018	Phase I, Phase II (Japanese patients with acute coronary syndrome or atrial fibrillation), Phase III (ARISTOTLE trial), n=4,385, median age = 68.0, age range=18-94	Ka (/h)	$= 0.473 \times (1 + time \times -0.433)$ If evening dose, $time=2$, otherwise 0	0.263 (51.3%)	Ka ~Evening dosing CL_R/F ~CrCL CL_{NR} ~Age, female CL/F ~Asian subjects, subject status, Strong/moderate CYP3A4/P-gp inhibitors Vc/F ~body weight, subject status	CL_R ~race, subject status, dosing time CL_{NR} ~body weight, race, subject status, dosing time
		CL/F (L/h)	$= (CL_{NR} + CL_R) \times (1 + race \times -0.119) \times (1 + conmed \times -0.146) \times (1 + AF \times -0.139) \times (1 + ACS \times -0.215)$ - If Asian, $race=1$, otherwise=0 - If concomitant medication with strong/moderate CYP3A4/P-gp inhibitors, $conmed=1$, otherwise=0 - If patient with atrial fibrillation, $AF=1$, otherwise=0 - If patient with acute coronary syndrome, $ACS=1$, otherwise=0	k=0.0954 (30.9%)		
		CL _R /F (L/h)	For patients with CrCL < 150 mL/min: $= 1.57 \times \left(\frac{CrCL}{80}\right)$ For patients with CrCL ≥ 150 mL/min: $= 1.57 \times \left(\frac{150}{80}\right)$ <i>CrCL: creatinine clearance, mL/min: millilitre per minute</i>			
		CL _{NR} /F (L/h)	$CL_{NR} = 2.02 \times \left(\frac{age}{65}\right)^{-0.429} \times (1 - 0.216 \times sex)$ If female, $sex=1$, otherwise=0			
		Q/F (L/h)	=1.91	$k_{21}=0.24$ (49%) $k_{12}=1.55$ (124%)		
		Vc/F (L)	$= 30 L \times \left(\frac{weight}{70 kg}\right)^{0.79} \times (1 + AF \times -0.0405) \times (1 + ACS \times -0.180)$ If patient with atrial fibrillation, $AF=1$, otherwise=0	0.0294 (17.1%)		

			If patient with acute coronary syndrome, ACS=1, otherwise=0			
		Vp/F (L)	=27.0	-		
Ueshima 2018, 2019	Japanese patients with atrial fibrillation, n=81, age range=40.5-84.9	Ka/F (/h)	=0.42/h (fixed)	-	CL~CrCL, genotype	CL~body weight, age, AST, ALT, P-gp inhibitors, co-morbidities Vd~body weight, age, AST, ALT, genotype, genotype, P-gp inhibitors, co-morbidities
		CL/F (L/h)	$= 1.53 \times \left[\left(\frac{CrCL}{70} \right)^{0.7} + 0.312^{CYP3A5} \cdot 0.341^{ABCG2} \right]$ - If patient had CYP3A5*1/*3 or *3/*3 genotype, CYP3A5 = 1, otherwise=0. - If patients had the ABCG2 421A/A genotype, ABCG2=1, otherwise=0. CrCL: Creatinine clearance	26.6%		
		Vc/F (L)	=24.7	56.6%		
		IC ₅₀	$IC_{50} = 45.3 \times 0.723^{HF} \times 1.55^{CI}$ - If patients had heart failure, HF=1, otherwise=0 - If patients with history of cerebral infarction, CI=1, otherwise=0	26.1%		
		FXa	$FXa = 40.2 - \frac{38.4 \times Cp}{IC_{50} + Cp}$ Cp: plasma concentration			
Edoxaban						
Salazar, 2012	Subject with atrial fibrillation included in phase I and II studies, some of which are renally impaired n=1,281, mean age (SD)= 65.2 (8.7) in a study	ka (/h)	$= 5.87 \times e^{x_{food} \times -1.34} \times e^{x_{verap} \times 1.33} \times e^{x_{amio} \times -0.769}$ If indicated covariate presents, x=1, otherwise 0 Verap: verapamil Amio: amiodarone	127.7%	Ka~food, P-gp inhibitors F~food, P-gp inhibitors Vc~weight, subject status CL~P-gp inhibitors	CL~age, sex, race, subject status, and body size, hepatic function Vc~ age, sex, race, hepatic function, CrCL
		F	$= 1 \times e^{x_{food} \times 0.14} \times e^{x_{keto} \times 0.792} \times e^{x_{ery} \times 0.781} \times e^{x_{Qnd} \times 0.727} \times e^{x_{verap} \times 0.388} \times e^{x_{amio} \times 0.751}$ If indicated covariate presents, x=1, otherwise 0 Keto: ketoconazole Ery: erythromycin Qnd: quinidine Verap: verapamil Amio: amiodarone			
		Tlag (h)	=0.421 h	11.7%		
		CL/F (L/h)	$= 36 \times \left(\frac{CrCL}{91} \right)^{0.341} \times e^{x_{keto} \times 0.18} \times e^{x_{ery} \times 0.199} \times e^{x_{Qnd} \times 0.212} \times e^{x_{amio} \times 0.436}$ If concomitant medication presents, x=1, otherwise 0	18.1%		

			CrCL: creatinine clearance Keto: ketoconazole Ery: erythromycin Qnd: quinidine Amio: amiodarone			
		Q/F (L/h)	=6.42		16.8%	
		Vc/F (L)	$= 244 \times \left(\frac{BW}{76}\right)^{0.688} \times e^{NVA \times 0.34}$ NVA: nonvalvular atrial fibrillation BW: Body weight		31.0%	
		Vp/F (L)	=90.3		34.9%	
Krekels, 2016	Healthy volunteers included in phase I studies and patient, renal impairments with atrial fibrillation included in phase III ENGAGE trial, n= 10,432, median age 71 years range=27-95	Ka (/h)	= $2.16 \times -(1 + 0.528)$			F~P-gp inhibitor, Asian race, subject status CL~CrCL, P-gp inhibitor, Asian race, Subject status (CL _{NR}) Q~subject status Vc~Asian race
		Tlag (h)	0.25 (fixed)		59.8%	
		F	$F_{with\ p-gp\ inhibitor} = F \times (1 + 1.20 \times AF)$ • If patients with AF, AF=0.134, otherwise=1 $F_{asian} = F \times (1 + 0.417)$			
		CL/F (L/h)	$= \frac{CL_R}{F} + \frac{CL_{NR}}{F}$ $CL/F_{with\ p-gp\ inhibitor} = CL/F \times (1 + 0.316)$ $CL/F_{asian} = CL/F \times (1 + 0.480)$ $Q/F_{NVA\ patients} = 5.75 \times (1 + 2.98)$ NVA: nonvalvular atrial fibrillation		13.6%	
		CL _R /F (L/h)	For CrCL ≤ 90 $CL_R = 0.197 \times CrCL$ For CrCL > 90 $CL_R = (0.197 - 0.0503a) \times 90 + 0.0503a \times CrCL$ If AF patients, a=2.32, otherwise=1 CrCL: Creatinine clearance			
		CL _{NR} /F (L/h)	NVA patients = 13.8 × 0.851 Study in patients with renal impairment=16.7			
		Q/F	5.75		47.8%	
		Vc/F (L)	= 194 × b		21.5%	

			If Asian, b= 1+0.740, otherwise =1			
		Vp/F (L)	88.6	47.8%		
Song, 2014	Healthy volunteers and patients with either (prevention or treatment) venous thromboembolism, or atrial fibrillation, n=1624 for PK model and 585 for PD model, age range 18-88	Ka (/h)	=3.54/h	102%	Covariate testing was not pursued	
		Tlag (h)	=0.391 h	56.7%		
		CL/F (L/h)	$= 32.1 \times \left(\frac{CrCL}{100}\right)^{0.604}$ CrCL: creatinine clearance	29.1%		
		Q/F (L/h)	=8.85	73.2%		
		Vc/F (L)	$= 214 \times \left(\frac{BW}{80}\right)^{0.918}$ BW: Body weight	36.1%		
		Vp/F (L)	$= 134 \times \left(\frac{BW}{80}\right)^{0.918}$ BW: Body weight	52.5%		
		FXa	$\frac{dFX_a(t)}{dt} = -k_{on}FX_a(t) \cdot Cp + k_{off}FX_a^*(t)$ $\frac{dFX_a^*(t)}{dt} = k_{on}FX_a(t) \cdot Cp - k_{off}FX_a^*(t)$ $k_{on} = k_{off}/K_D$ K _D (equilibrium constant) = 9.04 K _{off} (dissociation rate constant) = 0.220 FXa: free intrinsic anti-Xa activity (active) FX*a: bound intrinsic anti-Xa activity (inactive) Cp: plasma concentration	K _D =73.9%		
Yin, 2014	Healthy volunteers, patients with renal impairment, subjects with end-stage renal disease,	F _{absolute}	$F = 0.583 \times e^{0.57 \times x_{keto}} \times e^{0.382 \times x_{ver}} \times e^{0.553 \times x_{ery}} \times e^{0.513 \times x_{qnd}} \times e^{0.3 \times x_{amio}} \times e^{0.712 \times x_{dose}} \times e^{0.271 \times x_{sst}}$ If concomitant medication presents, x=1, otherwise 0 Ver=verapamil Ery=erythromycin Qnd= quinidine Amio=amiodarone SST= study-specific effect	19.5%	F~P-gp inhibitors, dose, study specific effect Ka~P-gp inhibitors CL~body weight (CL _{NR}), age (CL _{NR}), CrCL (CL _R), IV quinidine	CL~sex, race, dose, Pgp-inhibitors, study specific effect Vc~sex, race, age, study specific effect F~age, body weight, sex, race

(Japanese) patients with either deep vein thrombosis or atrial fibrillation, n=1134, age range=18-105	K _a (/h)	$k_a = 1.08 \times \left(\frac{Age}{63}\right)^{-1} \times e^{1.4 \times x_{ver}} \times e^{-0.895 \times x_{amio}}$ If concomitant medication presents, x=1, otherwise 0 Ver: verapamil Amio: amiodarone	79.4%	Vc ~P-gp inhibitors, weight, dose Bleeds ~country, trough concentrations Ka ~body weight, sex, dose, race, study specific effect Bleeds ~age, sex, race	
	Tlag (h)	= 0.250 (fixed)	50.2%		
	CL (L/h)	$= \left[11.4 \times \left(\frac{BW}{81}\right)^{0.528} \times \left(\frac{Age}{63}\right)^{-0.188} + 0.0822 \times CrCL \right] \times e^{0.233 \times x_{qnd, IV}}$ If CrCL > 115 mL/min, CrCL = 115 mL/min <i>BW: body weight</i> <i>CrCL: creatinine clearance</i>	10.1%		
	Q (L/h)	2.73	16.6%		
	Vc/F (L)	$= TV_{Vc} \times \left(\frac{BW}{81}\right)^{0.712} \times e^{-0.181 \times x_{keto}} \times e^{-0.164 \times x_{ery}} \times e^{a \times x_{QND}} \times e^{0.555 \times x_{amio}}$ - If oral dose, TV_Vc= 151, if IV dose, TV_Vc= 82.2 - If concomitant medication presents, x=1, otherwise=0 - If oral dose, a=-0.218, if IV dose a=-0.0794 <i>BW: body weight</i> <i>Keto: ketoconazole</i> <i>Ery: erythromycin</i> <i>QND: quinidine</i> <i>Amio: amiodarone</i>	18.6%		
	Vp/F (L)	=42.9 L	40.0%		
	Minor bleeds	$= -4.19 + \frac{Slope \times C_{trough}}{14.3 + C_{trough}}$			
	Moderate bleeds	$= -4.81 + \frac{Slope \times C_{trough}}{14.3 + C_{trough}}$			
	Major bleeds	$= -6.44 + \frac{Slope \times C_{trough}}{14.3 + C_{trough}}$ <i>Slope</i> = 4.69 + a - If Eastern European countries, a=-1.89, otherwise=0			

		All bleeds	$= -4.18 + \frac{Slope \times C_{trough}}{14.6 + C_{trough}}$ $Slope = 4.9 + a$ - If Eastern European countries, a=-2.11, otherwise=0 <i>Ctrough: trough concentration</i>			
Rivaroxaban						
Girgis 2014	Patients with atrial fibrillation, n=161, age range=36 -- 89	Ka (/h)	=1.16		CL ~Age, creatinine serum V ~Age, lean body mass PT, PiCT ~CrCL, rivaroxaban concentration	Covariate testing was not pursued. The structural model and covariates included in the final model were based on previous models (Mueck 2012 and Xu 2012)
		CL/F (L/h)	$= 6.10 \times (1 - 0.011 \times (age - 65) - 0.194 \times (sCr - 1.05))$ <i>sCr: serum creatinine</i>	35.2%		
		Vc/F (L)	$= 79.7 \times (1 - 0.00133 \times (Age - 65) + 0.0118 \times (LBM - 56.62))$ <i>LBM: Lean body mass</i>	17.6%		
		PT	$= 11.40 \times (1 - 0.000192 \times (CrCl - 76)) + 0.0426 \times (Cp(1 - 0.0000551 \times (1 + 0.0174 \times (CrCl - 76)) \times Cp))$ <i>Cp: plasma concentration</i> <i>CrCL: creatinine clearance</i>			
		PiCT (prothrombin induced clotting time)	$= 7.97 \times (1 - 0.0016 \times (CrCl - 76)) + 0.0954 \times (Cp(1 - 0.000263 \times (1 + 0.00293 \times (CrCl - 76)) \times Cp))$ <i>Cp: plasma concentration</i> <i>CrCL: creatinine clearance</i>			
Liu, 2022	Chinese patients with atrial fibrillation, n=195, age range= 28–96	Ka (/h)	= 0.617 (fixed)		CL ~eGFR PT ~total bilirubin, GFR, rivaroxaban plasma concentration	CL ~age, sex, body weight, body surface area, lean body mass, white blood cell count, red blood cell count, haemoglobin, haematocrit, platelet count, aspartate aminotransferase (AST),
		CL/F (L/h)	$= 5.03 \times \left(\frac{eGFR}{80}\right)^{0.53}$ <i>eGFR: estimated glomerulus filtration rate</i>	35.4%		
		V/F (L)	= 40.3			
		PT	$= 13.9 \times e^{0.00574 \times (TBIL-12)} \times \left(1 - 0.0872 \times \left(\frac{eGFR}{80}\right)\right) + 0.0133 \times Cp$			

			<i>TBIL</i> =total bilirubin <i>eGFR</i> : estimated glomerulus filtration rate <i>Cp</i> : plasma concentration			alanine aminotransferase (ALT), alkaline phosphatase (ALP), total bilirubin (TBIL), direct bilirubin, albumin, gamma-glutamyl transpeptidase, urea nitrogen, urea acid, serum creatinine, CrCl Ka, V ~age, sex, body weight, body surface area, lean body mass, white blood cell count, red blood cell count, haemoglobin, haematocrit, platelet count, AST, ALT, ALP, total bilirubin (TBIL), direct bilirubin, albumin, gamma-glutamyl transpeptidase, urea nitrogen, urea acid, serum creatinine, CrCl, GFR PT ~age, sex, body weight, body surface area, lean body mass, white blood cell count, red blood cell count, haemoglobin, haemato crit, platelet count, AST, ALT, ALP, direct bilirubin, albumin, gamma-glutamyl transpeptidase, urea nitrogen, urea acid, serum creatinine, CrCl
Kaneko, 2013	Japanese patients with	Ka (/h)	=0.617	58.2% F1=37.7%		CL ~ following covariates measured at baseline: age,

	atrial fibrillation phase III, n=597, age range=34-89	CL/F (L/h)	$= 4.73 \times \left(\frac{CrCL}{67.11} \right)^{0.159} \times (1 - 0.0132 \times (HCT - 42.14))$ <i>CrCL: creatinine clearance</i> <i>HCT: haematocrit</i>	41.0%	CL~ CrCL at week 24, haematocrit at week 24 PT~ age at baseline, lean body mass at baseline, albumin at baseline, haemoglobin at week 24	body weight, body surface area, lean body mass, body fat at baseline, sex and the effect of time, and following covariates measured at baseline and week 24: serum creatinine, albumin, blood urea nitrogen, AST, ALT, total bilirubin and haemoglobin K_a, V~ following covariates measured at baseline: age, body weight, body surface area, lean body mass, body fat at baseline, sex and the effect of time, and following covariates measured at baseline and week 24: serum creatinine, CrCL, albumin, blood urea nitrogen, AST, ALT, total bilirubin, haematocrit and haemoglobin PT~ following covariates measured at baseline: body weight, body surface area, body fat at baseline, sex and the effect of time; and following covariates measured at baseline and week 24: $K_a = 0.617 \times e^{58.2 \text{ to be calculated}}$ serum creatinine, CrCL, blood urea nitrogen, AST, ALT, total bilirubin, hematocrit
		V/F (L)	= 43.8	63.6%		
		PT	$= 11.4 + 0.0467 \times Cp^{FACT \times Cp}$ $Base = 11.4 \times (1 + 0.00335 \times (age - 70.98)) \times (1 + 0.00242 \times (LBM - 49.69)) \times [1 - 0.065 \times (Alb - 4.22)] \times [1 - 0.015 \times (Hb - 14.06)]$ $FACT = 1 - \left(\frac{Tbil}{0.82} \right)^{-1.11} \times 0.000155$ <i>LBM=Lean body mass</i> <i>Alb=albumin at baseline</i> <i>Hb=Hemoglobin at week 24</i> <i>Tbil= Total bilirubin at baseline</i>			

						and hemoglobin; and albumin at week 24
Speed 2020	Patients on anticoagulants for prevention of stroke resulting from atrial fibrillation or secondary prevention from venous thromboembolism and other reasons, weighing between 39 kg and 172 kg, n=913, age mean (SD)= 67.03 (15.00), range = 19-96	Ka (/h)	=0.707		CL~CrCl calculated using lean body weight Vd~lean body weight	CL~sex, age, body weight, lean body weight, albumin, bilirubin, creatinine, CrCl calculated using LBW, the diagnosis of atrial fibrillation or a documented diagnosis of heart failure Vd~ sex, age, TBW, albumin, and the diagnosis of atrial fibrillation or a documented diagnosis of heart failure
		CL/F (L/h)	$= 5.57 \times \left(\frac{CrCLLBW}{55} \right)^{0.446} \times e^{23.02}$ <i>CrCLLBW: creatinine clearance calculated using lean body weight</i>	23.02%		
		Vc/F (L)	$= 59.4 \times \left(\frac{LBW}{55} \right)^{0.519}$ LBW: Lean body weight			
Tanigawa 2013	Japanese patients with atrial fibrillation phase III, n=182, age range= 30–92	Ka (/h)	=0.6	0.463 (68%)	CL~blood urea nitrogen Factor Xa baseline~age HepTest EC50~albumin on day 28	CL~body surface area V~lean body mass k _a ~lean body mass HepTest EC50~study number and albumin at screening Factor Xa baseline~age and serum glutamic oxaloacetic trans-aminase at screening time Factor Xa Emax~study number
		CL/F (L/h)	$= 4.72 \times (1 - 0.0165 \times (BUN - 16.73))$ <i>BUN: blood urea nitrogen</i>	0.0452 (21.3%)		
		Vc/F (L)	= 42.9	0.0596 (24.4%)		
		PT	$= 13.7 + 0.0227 \times Cp^{1.1}$ <i>Cp: plasma concentration</i>			
		aPTT	$= 32.6 + 0.0658 \times Cp^{1-0.000156 \times Cp}$ <i>Cp: plasma concentration</i>			
		HepTest	$= 17.9 + \left(\frac{43.2 \times Cp^{1.18}}{EC_{50}^{1.18} + Cp^{1.18}} \right)$ • $EC_{50} = 240 \times (1 + 0.147 \times (ALB - 4.28))$			

			<i>EC50: Half maximal effective concentration</i> <i>Cp: plasma concentration</i> <i>ALB: albumin</i>			
		FXa	$= base \times \left(1 - \frac{0.928 \times Cp^{1.16}}{221^{1.16} + Cp^{1.16}}\right)$ $base = 0.803 \times (1 - 0.00656 \times (Age - 65.59))$ <i>Cp: plasma concentration</i>			
Willmann 2018	Multiple patient populations with all approved rivaroxaban indication (venous thromboembolism prevention and treatment, acute coronary syndrome, atrial fibrillation, Phase II and III), n= 4,918, mean age (SD)=60.53 (11.82)	Ka (/h)	=0.821	0.628	F ~dose CL ~CrCL, body weight, concomitant medication (P-glycoprotein-inhibitors, weak/medium/strong CYP3A4 inhibitors, CYP3A4 inducer), diagnosis (AF, VTE, ACS) Vd ~body weight, age, sex	CL ~age, sex, serum creatinine, BMI, BSA, lean body mass Vd ~BMI, BSA, lean body mass
		F	$= 0.59 + (0.66) \times e^{-\frac{\ln 2}{14.4} \times dose \times 1000}$			
		CL/F (L/h)	$= 6.58 \times \left(\frac{CrCL}{93}\right)^{0.406} \times \left(\frac{BW}{81}\right)^{-0.278} \times comed \times diag$ - Concomitant medication (<i>comed</i>): - If with P-glycoprotein inhibitors, <i>comed</i>=0.966 - If with strong CYP3A4 inhibitors, <i>comed</i>=0.978 - If with medium CYP3A4 inhibitor, <i>comed</i>=0.863 - If with weak CYP3A4 inhibitor, <i>comed</i>=0.939 - if with CYP3A4 inducer, <i>comed</i>=1.3 - otherwise=1 - Diagnosis (<i>diag</i>): - If with atrial fibrillation (AF), <i>diag</i>=0.849 - If with ACS, <i>diag</i>=1.14 - If for venous thromboembolism (VTE) prevention ≤ 72h, <i>diag</i> = 1.04 - If for VTE prevention > 72h, <i>diag</i> = 1.29 Otherwise=1 <i>BW: body weight</i> <i>CrCL: creatinine clearance</i>	0.167		
		Vd/F (L)	$= 62.5 \times \left(\frac{BW}{81}\right)^{0.216} \times \left(\frac{age}{61}\right)^{-0.189} \times sex^{0.889}$ $\times e^{0.0391 \text{ to be calculated}}$ If female, <i>sex</i>=0.889, otherwise=1 <i>BW: body weight</i>	0.0391		

Zhang, 2022	Chinese patients with atrial fibrillation, n=150, age range= 23–90	Ka (/h)	= 0.821 (fixed)		CL~CrCL, total bilirubin, ABCB1 gene polymorphism (rs4728709) Vd~ body weight	CL, Vd~age, body weight (CL only), body surface area, body mass index, CHA2DS2-VASc scores, HAS-BLED scores, albumin, serum creatinine, GFR, haemoglobin, platelet, ALT, AST, and blood urea nitrogen, sex, smoking, drinking, hypertension, diabetes, congestive heart failure, vascular disease, antiplatelet drugs, amiodarone, statins, proton pump inhibitors (PPIs), and CYP3A4/CYP3A5/ABCG2 gene polymorphisms Vd~total bilirubin, creatinine clearance, ABCB1 gene polymorphism
		CL/F (L/h)	$= 5.79 \times e^{(CrCL-76.1) \times 0.00586} \times e^{-(TBIL-14) \times 0.0144} \times (1 + rs09 \times 0.476)$ <i>CrCL: creatinine clearance</i> <i>TBIL: total bilirubin</i> <i>Rs099=rs4728909 expressed as AA or GA = 1</i>	38.3%		
		Vd/F (L)	$= 51.5 \times e^{(BW-66) \times 0.00873}$ <i>BW: body weight</i>	18.5%		
Dabigatran						
Liesenfeld, 2011	Patients with atrial fibrillation, n=9552, mean age (SD)=71.2 (8.7), age range=22-97	Ka (/h)	=0.754 (fixed)		F~ proton pump inhibitor, amiodarone, verapamil CL~CrCL, age, race, patient with heart failure, sex Vc~body weight, haemoglobin	CL~left ventricular dysfunction Vc~hemoglobin F~P-glycoprotein inhibitors, H ₂ receptor antagonist,
		Tlag (h)	=0.634 (fixed)			
		F	$= 1 \times PPI \times Amio \times Vera \times e^{44.3 \text{ to be calculated}}$ • If patient with proton pump inhibitor, <i>PPI</i>=0.875, otherwise=1 • If patient with amiodarone, <i>Amio</i>=1.12, otherwise=1 • if patient with verapamil, <i>Vera</i>=1.23, otherwise=1	44.3%		
		CL/F (L/h)	$= 124 \times \frac{CRCL^{1.29}}{56.7^{1.29} + CRCL^{1.29}} \times (1 + (-0.0041 \times (age - 72)) \times race \times HF \times sex)$ • If South Asian patients, <i>race</i>=0.797, otherwise 1			

			If patients with heart failure, $HF=0.933$, otherwise 1 if female, $sex=0.917$, otherwise 1			
		Q/F (L/h)	=35.5 (fixed)			
		Vc/F (L)	$= 673 \times (1 + (0.77 \times (BW - 80.3)) \times (1 + (-3.99 \times (Hb - 14.3) \times e^{20.5 \text{ to be calculated}}))$ BW = body weight, Hb = hemoglobin	20.5%		
		Vp/F (L)	=345 (fixed)			
Dansirikul, 2012	Healthy volunteers and patients with atrial fibrillation or undergoing orthopaedic surgery, n=2807, age range=18 - 93	Ka (/h)	= 0.754	95.3%	F ~co-medications with P-gp inhibitors and proton pump inhibitors CL ~CrCL, age, non-healthy volunteer, atrial fibrillation, sex Vc ~body weight	CL ~weight, race, smoking status, alcohol history, sCr, left ventricular dysfunction, Hb, P-gp inhibitors, PPI, H2RA, verapamil, amiodarone, antacids Vc ~ age, subject status, sex, race, smoking status, alcohol history, CrCL, sCr, left ventricular dysfunction, Hb, P-gp inhibitors, PPI, H2RA, verapamil, amiodarone, antacids Vp ~ age, weight, subject status, sex, race, smoking status, alcohol history, CrCL, sCr, left ventricular dysfunction, Hb, P-gp inhibitors, PPI, H2RA, verapamil, amiodarone, antacids F ~ age, weight, subject status, sex, race, smoking status, alcohol history, CrCL, sCr, left ventricular dysfunction, Hb, H2RA,
		Tlag (h)	= 0.634			
		F	= 1 (fixed) If with co-medications $F = 1 \times Pgp \times PPI$ - if patient with P-glycoprotein inhibitors, $Pgp= 1.15$ otherwise 1 - If patient with proton pump inhibitor, $PPI=0.854$, otherwise 1	44.7%		
		CL/F (L/h)	If not a healthy volunteer + CrCL < 120: $\frac{CL}{F} = 111 \times (1 + 0.00644 \times (CLCr - 120)) \times (1 - 0.00662 \times (age - 68)) \times AF \times sex$ If not a healthy volunteer + CrCL > 120 $\frac{CL}{F} = 111 \times (1 - 0.00662 \times (age - 68)) \times AF \times sex$ - If patients with atrial fibrillation, $AF=0.939$, otherwise=1 - If female, $sex=0.875$, otherwise=1			
		Q/F (L/h)	= 35.5 (fixed)			
		Vc/F (L)	= $728 \times (1 + 0.011 \times (WT - 80))$	26.1%		
		Vp/F (L)	= 345 (fixed)			
		aPTT	$aPTT = Base + EFF + Slope \times Cp$ $Base = 30.426$ $Slope = 0.574$ $EFF = \frac{Emax \times Cp}{EC_{50} + Cp}$			

			• $E_{max}=25.56$ • $EC_{50} = 184 \times EC_{50}$ If AF patients, $EC_{50} = 0.713$, if healthy volunteers, $EC_{50} = 0.433$			verapamil, amiodarone, antacids
--	--	--	---	--	--	------------------------------------

Ka/F: apparent absorption rate constant, F: bioavailability, Tlag: lag time, CL/F: apparent clearance, CLR/F: apparent renal clearance; CLNR/F : apparent non-renal clearance; Q/F: apparent intercompartmental clearance, Vd/F: apparent volume of distribution; Vc/F: apparent central volume of distribution, Vp/F: apparent peripheral volume of distribution, aPTT: Partial thromboplastin time, PT: prothrombin time, FXa: factor Xa activity, IC50: half maximal inhibitory concentration

Chapter 3. Fixed-Dose Edoxaban in Frail Geriatric Patients: A Population Pharmacokinetics Study

This chapter is based on:

Edwina AE*, Van der Linden L*, Debruyne L, Foerster K, Burhenne J, Verhamme P, Vanassche T, Spriet I, Dreesen E†, Tournoy J†. (*shared first author, †shared last author). *Fixed-dose edoxaban in frail geriatric patients: a population pharmacokinetics study*. Submitted

Abstract

Background: Atrial fibrillation affects nearly half of geriatric inpatients. Edoxaban is a factor Xa-inhibitor used for stroke prevention. However, geriatric patients are underrepresented in trials. The limited data and high pharmacokinetic variability result in using this high-risk medication without a full understanding of its exposure in this population.

Objectives: To characterize the dose–exposure relationship of edoxaban and predict the adequacy of current fixed-dose guidelines applied to geriatric patients.

Methods: A single-center prospective PK study was performed, including geriatric inpatients with AF who received on-label edoxaban doses. Blood samples were obtained at steady state, with exposure assessed by mass spectrometry and chromogenic assays. A population pharmacokinetics (popPK) model was built and various dosing strategies (n=1000) were simulated. The observed trough concentrations ($C_{\min}$) range in ENGAGE AF TIMI 48 (12–43 ng/mL) was set as the target range for simulation.

Results: Seventeen frail patients (median age 87 years, IQR 83–90) contributed to 85 steady-state PK samples. A two-compartment popPK model best described the edoxaban and M4 metabolite concentrations, with proton pump inhibitors (PPI) reducing relative bioavailability by 24%. Key parameters included absorption rate constant 0.104 h^{-1} , clearance 9.02 L/h, and central and peripheral volumes 47.4 L and 42.7 L. Edoxaban concentrations correlated with anti-factor Xa activity ($p=0.966$). Nearly half of the patients had higher exposures ($C_{\min} > 43\text{ ng/mL}$). Simulations showed a 6.75% probability of higher exposures with 15 mg, 42.4% with 30 mg, and 84.25% with 60 mg.

Conclusion: Estimated PK parameters of edoxaban in geriatric patients differed from those reported in pivotal trial populations, which might be a reason for the increased exposure, thereby potentially increasing the bleeding risk. Further investigation is needed to assess the benefits of alternative dosing in this population.

1. Introduction

Atrial fibrillation (AF) is associated with a fivefold risk of stroke and is present in nearly half of geriatric inpatients (1, 2). Anticoagulants reduce the stroke risk by two-thirds (3, 4). Direct oral anticoagulants (DOACs) are recommended over vitamin K antagonists (VKA) for stroke prevention in AF, with edoxaban being a commonly used option (5, 6). Their predictable pharmacokinetics (PK) allows flat dosing of DOACs without the need for dose individualization (7, 8).

Today, geriatric patients represent about 50% of edoxaban users in Europe (9). However, they were underrepresented in the edoxaban pivotal trials with only 17.8% classified as mild to moderately frail and just 1.7% as severely frail (10). This has led to a knowledge gap regarding the safety and effectiveness of standard dosing in this vulnerable population. Geriatric patients were shown to be more prone to adverse events of DOACs (11).

The lack of clinical data on edoxaban PK in geriatric patients is further compounded by a remarkable PK variability in geriatric patients for many drugs (12), increasing the risk of both over- and underexposure. Previous studies have established a relation between plasma concentrations of DOACs and the hazard risk of stroke and bleeding, both for which risks significantly increase with age (13-15). Taken together, this means that we are administering a high-risk medication to a vulnerable population without having insight into the drug exposure, and thus, safety and effectiveness. Additionally, while the edoxaban M4 metabolite is expected to be minor in the general population ($\leq 10\%$) and has a half-maximal inhibitory concentration on factor Xa of 1.8 nmol/L (compared to 3 nmol/L for edoxaban) (16), patients with reduced glomerular filtration rate, liver insufficiency, or drug-drug interactions may exhibit an altered parent-to-metabolite ratio (17). Hence, characterizing the PK of edoxaban in geriatric patients with AF is desperately needed.

The primary purpose of this study was to exploit population PK (popPK) modeling to describe the dose–exposure relationship of edoxaban in frail geriatric patients and explore patient characteristics effect on PK. We performed popPK simulations to predict whether the current fixed-dose guideline for edoxaban offers safe and effective stroke prevention for geriatric patients. Additionally, the link between anti-factor Xa activity and edoxaban plasma concentrations was evaluated.

2. Methods

2.1. Study Design and Population

We performed a prospective PK study on the geriatric wards of the University Hospitals Leuven, Belgium. This study was conducted according to the ethical guidelines of the Declaration of Helsinki and approved by the Ethics Committee Research UZ/KU Leuven (EC reference number S67063) and competent authority The Federal Agency for Medicines and Health Products (EudraCT no: 2022-004136-26). All volunteers and/or legal representatives were informed about the details of this study and provided written informed consent.

All patients aged 75 years and above, admitted to the geriatric wards of the University Hospitals Leuven, Belgium, who received stable edoxaban doses for at least four consecutive days to prevent stroke in AF were eligible for inclusion. Blood samples were collected via a peripheral venous catheter on a single occasion at pre-dose (i.e., 24 hours after intake of the previous dose), and 0.5, 1, 1.5, 2, 4, 5, and 8 hours after administration.

On the day of blood collection, we collected demographic and biochemical data from the electronic patient record, including age, sex assigned at birth, total body weight, height, serum creatinine, estimated creatinine clearance according to the Cockcroft and Gault (CrCl) formula, blood urea nitrogen, serum albumin, total protein levels, and liver function biomarkers such as total bilirubin, aspartate aminotransferase (AST), alanine aminotransferase (ALT), γ -glutamyltransferase (GGT), and alkaline phosphatase concentrations (ALP). Concomitant therapy with proton pump inhibitors and moderate or strong P-gp inhibitors was identified from the medication list. Multimorbidity was quantified by using the Charlson Comorbidity Index. Other patient characteristics such as lean mass, frailty degree, and malnutrition screening using mini-nutritional assessment were also collected on the same day. Lean mass was measured using bioelectrical impedance analysis (Bodystat Quadscan 4000, Bodystat, Isle of Man, UK). Three different frailty scales were used including (i) Fried criteria to test physical frailty phenotype identified a physical phenotype of frailty based on five criteria (unintentional weight loss, self-reported exhaustion, low physical activity, slow walking speed, and weakness/grip strength) (18); (ii) cumulative frailty using Edmonton frailty scale (19); and (iii) routinely assessed clinical frailty scale (20).

2.2. Edoxaban and Factor Xa Bioassay

Plasma samples were obtained from the supernatant fraction of blood tubes after centrifugating for 5 minutes at $2636 \times g$ at 19°C . The plasma samples were stored at -80°C until bioanalysis, with a maximum storage duration of 8 months. Concentrations of edoxaban and its main metabolite, M4, were quantified by using ultra-performance liquid chromatography–tandem mass spectrometry (UPLC–MS/MS) as previously described (21). For edoxaban, the lower limit of quantification (LLOQ) was 0.1 ng/mL and the upper limit of quantification (ULOQ) was 500 ng/mL . For the M4 metabolite, the LLOQ was 0.1 ng/mL and the ULOQ was 100 ng/mL . The accuracy and precision of edoxaban and its M4 metabolite were evaluated across concentrations of 0.15 , 2.5 , and 25 ng/mL , with an additional concentration of 250 ng/mL assessed only for edoxaban. For edoxaban, accuracy ranged from 90.6% to 107% , with precision improving from 13.5% at 0.15 ng/mL to 2.4% at 250 ng/mL . For the M4 metabolite, accuracy ranged from 90.2% to 104% , while precision ranged from 0.1% at 1 ng/mL to 1.7% at 25 ng/mL .

Anti-factor Xa activity in plasma samples was measured using an ACL-TOP 700 LAS (Werfen, Brussels, Belgium), with the HemosIL® Liquid Anti-Xa kit (Werfen) according to the manufacturer's protocol. Calibration curves were generated with Biophen® Edoxaban Calibrator (Nodia, Boom, Belgium). The lower limit of quantification of the anti-factor Xa assay was 30 ng/mL .

2.3. Population PK Modelling and Simulation

A base popPK model was fitted to the edoxaban plasma concentration–time data. A one-compartment model with first-order absorption and linear elimination was used as a starting point. From there, two-, and three-compartment models and the addition of an absorption lag time were explored. Interindividual variability (IIV) was tested on all PK parameters. Additive, proportional, and combined error models were evaluated for describing residual unexplained variability. Individual PK parameters were assumed to be log-normally distributed, which was achieved using an exponential function. Data below the limit of quantification were discarded if they constituted $\leq 10\%$ of the elimination phase data (22).

Parameter estimation was performed using first-order conditional estimation with interaction and differential equation solver ADVAN 13. Base model selection was guided by objective function value (OFV; calculated as $-2 \times \log\text{-likelihood}$) comparisons (difference

≥ 3.84 points; $P \leq 0.050$), plausibility and precision of parameter estimates, and goodness-of-fit (GOF) plots. All covariates, including collected demographic, biochemical, and frailty data, were first screened using plots of the empirical Bayes estimates of the parameters versus covariates and generalized additive modeling. Subsequently, potential covariates were tested through stepwise forward inclusion and backward deletion of covariates with an OFV change of ≥ 3.84 ($P \leq 0.050$) and ≥ 6.63 ($P \leq 0.01$), respectively. The final model was evaluated using the GOF plots and prediction-corrected visual predictive checks (pcVPCs; $n=1000$ simulated replicates of the original dataset). The parameter uncertainty was evaluated using sample importance resampling (SIR) with 2000 samples and 1000 resamples (23).

Monte Carlo simulations ($n=2000$) were performed using the final popPK model to evaluate once-daily edoxaban 30 mg, 60 mg, as well as a 15 mg off-label dose. Target attainment was defined based on the trough concentrations target range from 12 – 43 ng/mL as recommended by the European Heart Rhythm Association (13, 24).

2.4. Establishing The Relationship Between Edoxaban Concentrations and Anti-Factor Xa Activity

The correlation between anti-factor Xa activity and edoxaban concentrations was tested using the Concordance Correlation Coefficient (CCC). The agreement was visually presented using Bland–Altman plot and Passing–Bablok regression to capture systematic error (25, 26). The conversion factor, defined as the ratio between edoxaban concentrations and anti-Xa activity levels, was estimated after fixing all other parameters in the final population popPK model. In addition, error models for anti-Xa activity levels were estimated.

2.5. Software

Dataset formatting, exploration, and statistical analysis were performed using R (v4.2.1; R Core Team, Vienna, Austria) using custom scripts based on packages including *dplyr*, *ggplot2*, and *Xpose*. The popPK modeling and simulations were performed using NONMEM (v7.5; ICON Development Solutions, Gaithersburg, MD, USA), with a GNU Fortran 95 compiler and the Perl-speaks-NONMEM (PsN; v5.2.6.) toolkit on the interface software Pirana (v2.9.7; Certara, Inc., Princeton, NJ, USA).

Table 8. Baseline Characteristics of Study Participants

Characteristic	Value
Number of participants	17
<i>Demographics</i>	
Age, years, median [IQR]	87 [83–90]
Sex, male, n (%)	10 (58.8)
Weight, kg, median [IQR]	67 [60–78]
Height, cm, median [IQR]	162 [154–171]
BMI, kg/m ² , median [IQR]	25.2 [24–27.8]
Lean mass*, kg, median [IQR]	41 [32–50]
<i>Serology</i>	
Serum albumin, g/L, median [IQR]	35.6 [34.8–40.5]
Total protein†, g/L, median [IQR]	68.5 [66.3–76.0]
BUN, mg/dL, median [IQR]	61.0 [56.0–77.8]
Serum creatinine, mg/dL, median [IQR]	1.33 [1.07–1.70]
CrCL _{Cockcroft-Gault} , mL/min, median [IQR]	32 [24–41]
Bilirubin‡, mg/dL, median [IQR]	0.6 [0.5–0.9]
Aspartate aminotransferase‡, IU/L, median [IQR]	21 [17–23.3]
Alanine transaminase‡, IU/L, median [IQR]	12.0 [10.5–16.5]
Alkaline phosphatase‡, IU/L, median [IQR]	89.5 [64.8–116.3]
Gamma-glutamyl transferase‡, IU/L, median [IQR]	37.5 [19.0–69.5]
<i>Concomitant medication</i>	
PPI, n (%)	10 (58.8)
Charlson Comorbidity Index, n (%)	
4–6	7 (41.2)
6–9	8 (47.1)
≥10	2 (11.8)
<i>Frailty</i>	
Fried criteria, n (%)	
Pre-frail	4 (23.5)
Frail	13 (76.5)
Clinical Frailty Scale	
5	3 (17.6)
6	10 (58.8)
7	4 (23.5)
Edmonton Frailty Scale, n (%)	
Vulnerable	5 (29.4)
Mild frailty	3 (17.6)
Moderate frailty	8 (47.1)
Severe frailty	1 (5.9)

Abbreviations: BMI, body mass index; BUN, blood urea nitrogen; CrCL, creatinine clearance; IQR, interquartile range; PPI, proton pump inhibitor.

* Lean mass was missing for 4 participants.

† Total protein was missing for 7 participants.

‡ Bilirubin, AST, ALT, GGT, ALP were missing for 1 participant

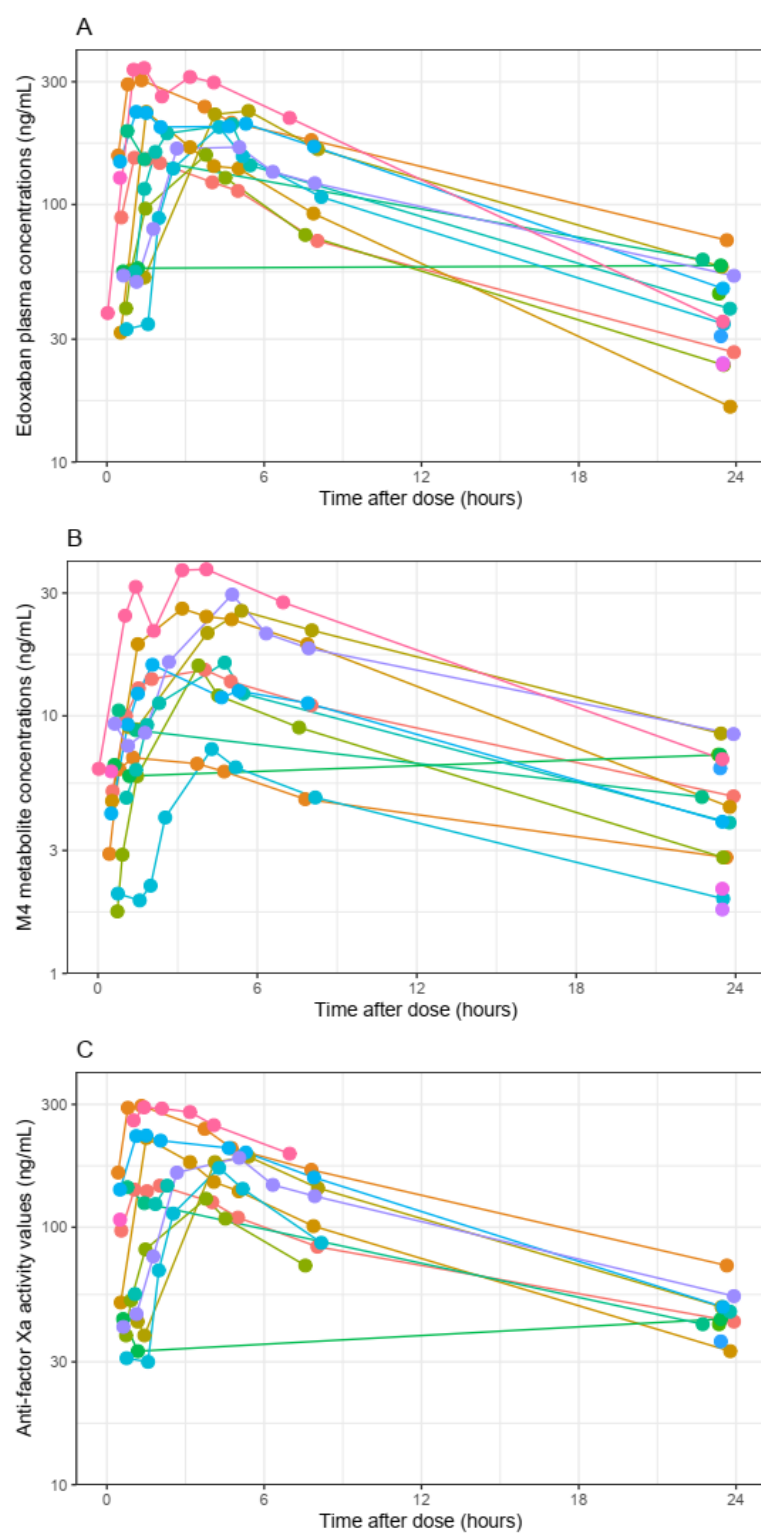


Figure 18. (A) Observed edoxaban plasma concentration–time profile. (B) Observed M4 concentration–time profile. (C) Observed anti-factor Xa activity–time profile. Observations (dots) of the same participant are connected with lines.

3. Results

3.1. Study Population

Seventeen geriatric participants contributed 85 plasma samples to the PK study. In each sample, the edoxaban and M4 concentrations and anti-factor Xa activity were measured (**Figure 18**). All measured edoxaban and edoxaban M4 concentrations were above the lower limit of quantification. Six anti-factor Xa activities (7%) were below the limit of quantification and were therefore excluded from the analysis.

Fourteen patients received once-daily 30 mg edoxaban and three patients received once-daily 60 mg edoxaban. Patient characteristics are presented in **Table 8**. Most of the participants were in the oldest-old age group (≥ 85 years), with a median age of 87 years (interquartile range [IQR]: 83–90). They exhibited a high degree of frailty, as indicated by a CFS score of 5–7, an EFS score of 7–12, and a Fried criteria score of 2–5.

The median edoxaban trough concentration in the study cohort was 37.3 (range=16.4–72.7) ng/mL. None of these trough concentrations were under the lower target range of 12 ng/mL, while 7 geriatric patients (44%) reached steady-state trough concentrations above the upper target range of 43 ng/mL.

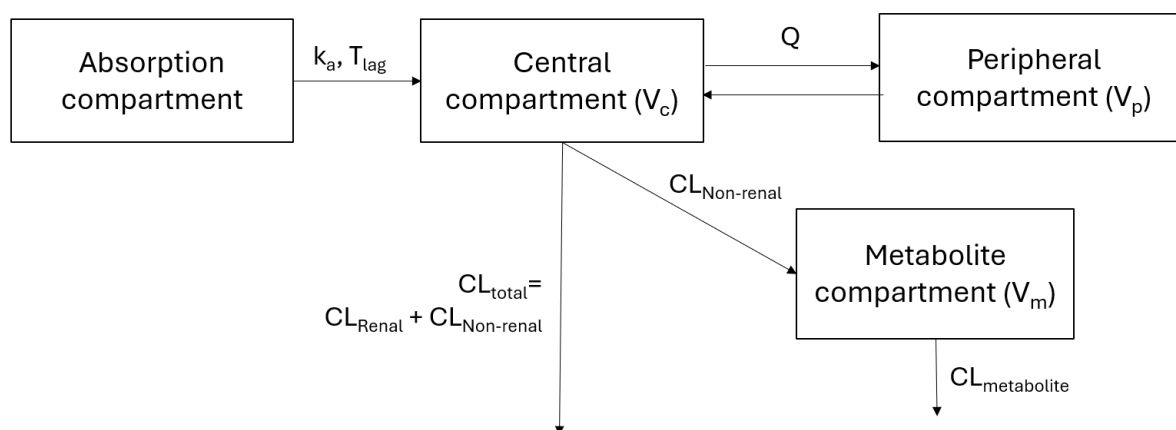


Figure 19. Schematic representation of population PK model for edoxaban

Abbreviations. $CL_{\text{metabolite}}$: M4 metabolite clearance, $CL_{\text{non-renal}}$: non-renal clearance, CL_{renal} : renal clearance, CL_{total} : total clearance, k_a : absorption rate constant, Q : intercompartmental clearance, T_{lag} : lag time, V_c : central volume of distribution, V_m : metabolite M4 volume of distribution, V_p : peripheral volume of distribution

Table 9. Parameters of the Base and Final Population Pharmacokinetics Model of Edoxaban

Parameter Description	Estimate (%RSE) [%Shrinkage]	SIR median (95% CI)
Typical values		
k_a (h ⁻¹)	0.104 (9)	0.104 (0.094–0.114)
Lag time (h)	0.25 (FIXED)	-
CL/F (L/h)	9.02 (1)	9.010 (8.880–9.157)
V_c/F (L)	47.4 (10.3)	47.645 (40.260–55.197)
Q/F (L/h)	3.27 (15.9)	3.312 (2.636–4.070)
V_p/F (L)	42.7 (10)	42.043 (35.870–48.418)
F	1 (FIXED)	–
F with PPI comedication	0.764 (0.7)	0.763 (0.756– 0.771)
CL-M4	37.3 (14.1)	38.618 (31.922–46.541)
V-M4	35.3 (8.1)	34.902 (30.844–39.275)
Interindividual variability*		
on CL/F (%CV)	34.5 (4.1) [2]	34.6 (32.2–36.7)
on V_c/F (%CV)	149 (2.6) [24]	148.3 (142.7–154.2)
Correlation CL~ V_c	40%	0.206 (0.182–0.232)
on CL-M4 (%CV)	46.7 (3.3) [5]	46.47 (44.16–48.78)
Residual variability		
Proportional variability on edoxaban concentration measured by using LC-MS/MS (%CV)	25.7 (0) [7]	25.7 (25.67–25.68)
Proportional variability on edoxaban M4 metabolite concentration (%CV)	21.4 (0) [10]	21.3 (21.33–21.35)

Abbreviations: CI: Confidence Interval, CL/F: apparent clearance, CL-M4: clearance of M4 metabolite, CV: coefficient of variations, K_a : absorption rate constant, LC-MS/MS: Liquid Chromatography Tandem Mass Spectrometry, Q/F: apparent intercompartmental clearance, V_c/F : apparent central volume of distribution, V-M4: Volume of distribution of M4 metabolite, V_p/F : apparent peripheral volume of distribution.

*Interindividual variabilities were calculated as:

$$Parameter = Typical\ parameter \times e^{\eta_i} \text{ with } \eta_i \propto N(0, \omega^2)$$

Table 10. Parameter estimates comparison with other popPK studies using our study's median patient characteristics (CrCL = 32 mL/min, weight = 67 kg, and age = 87 years)

Parameter Description	Parameter estimates			
	Salazar, 2012 (27)	Krekels, 2016 (28)	Song, 2014 (29)	Yin, 2014 (30)
Subjects	Subject with atrial fibrillation included in phase I and II studies, n=1,281, mean age (SD)= 65.2 (8.7)	ENGAGE-AF study population, n= 10,432, median age 71 years range=27-95	Healthy volunteers and patients with either (prevention or treatment) venous thromboembolism, or atrial fibrillation, n=1624 for PK model and 585 for PD model, age range 18-88	Healthy volunteers, patients with renal impairment, subjects with end-stage renal disease, (Japanese) patients with either deep vein thrombosis or atrial fibrillation, n=1134, age range=18-105
k_a (h ⁻¹)	5.87	2.16	3.54	0.78
Lag time (h)	0.421	0.25 (fixed)	0.391	0.25 (fixed)
CL/F (L/h)	25.2	23	16.1	12.32
V_d/F (L)	315.5	194	182	132
Q/F (L/h)	6.42	5.75	8.85	2.73
V_p/F (L)	90.3	88.6	114	42.9

3.2. Population Pharmacokinetics

The edoxaban concentration–time course was best described by a linear two-compartmental popPK model with first-order absorption, a lag time, and linear elimination (Table 9, Figure 19 and 20, Supplementary Figure 1, 2 and 3). Comedication with PPIs reduced the relative bioavailability by 24%. The NONMEM control stream is available in the Supplementary material.

3.3. The correlation between edoxaban concentrations and anti-Xa activity values

Edoxaban concentrations and anti-Xa activity values are well correlated and the time-course profiles of anti-Xa activity mirrored their plasma concentration–time profiles (Figure 18). The ratio between concentrations and anti-Xa values was estimated to be 1.07 with residual proportional error 25.7%. Passing-Bablok regression showed neither systematic nor proportional bias as the 95% confidence interval (CI) of the calculated intercept (-0.06 [95%CI, -0.35 to -0.29] mg/L) and slope (0.999 [95%CI, 0.93–1.06]) enclosed 0 and 1, respectively (Supplementary Figure 4). In addition, the Bland–Altman plot showed an average bias of -1.16% with a 95% lower and upper limit of agreement of -10.61% and 8.29%, respectively (Supplementary Figure 5).

3.4. Dosing simulation

Stochastic simulations with once-daily 15 mg edoxaban predict an overexposure attainment of 10.9% for non-PPI group and 2.6% for PPI group, which is lower than that of once-daily 30 mg edoxaban (53% in non-PPI and 31.8% in PPI group) and 60 mg edoxaban (89.7% in non-PPI and 78.8% in PPI group). More details about the probability of geriatric patients attaining underexposure and therapeutic range were presented in Figure 21.

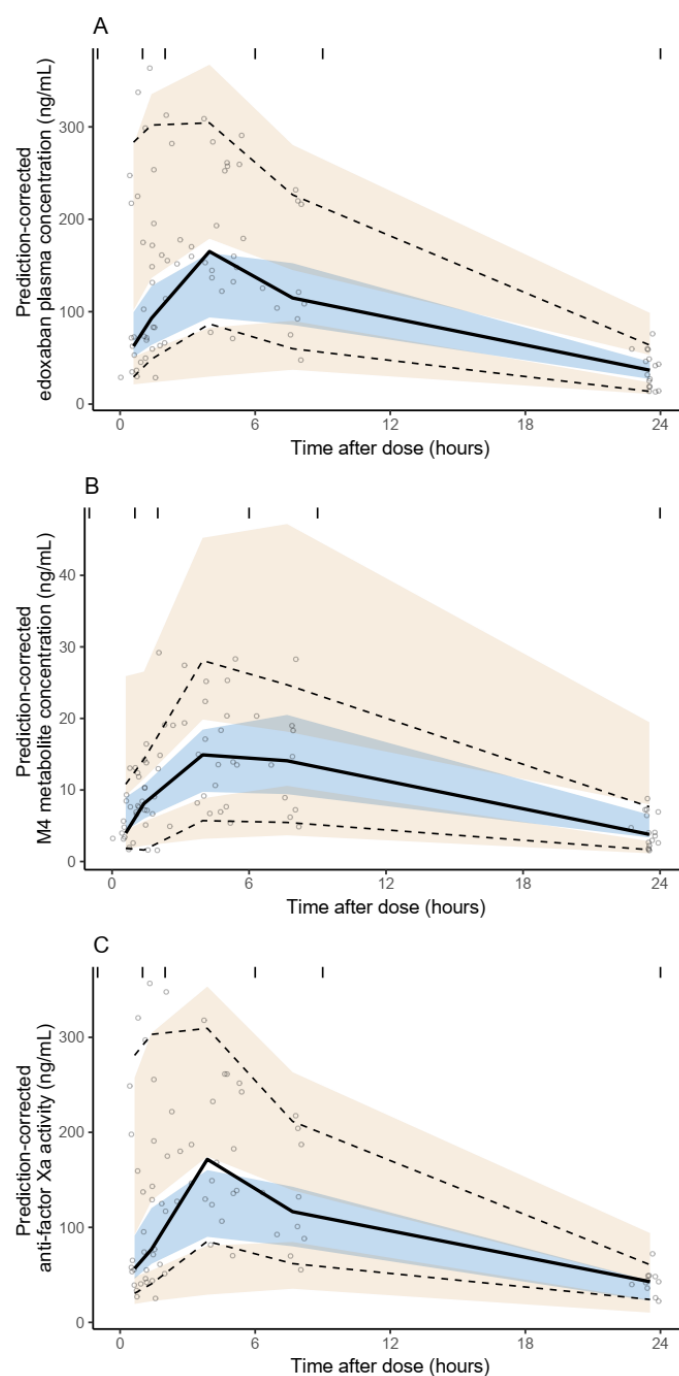


Figure 20. Prediction-corrected visual predictive checks of edoxaban (A), M4 (B), and anti-factor Xa activity (C). Open circles are the individual observations. The solid black line represents the median observation. The dashed black lines represent the observed 5th and 95th percentiles. The shaded blue area is the simulation-based 95% prediction interval of the median. The shaded beige areas are the simulated 95% prediction intervals of the 5th and 95th percentiles. Vertical lines at the top are the bin separators.

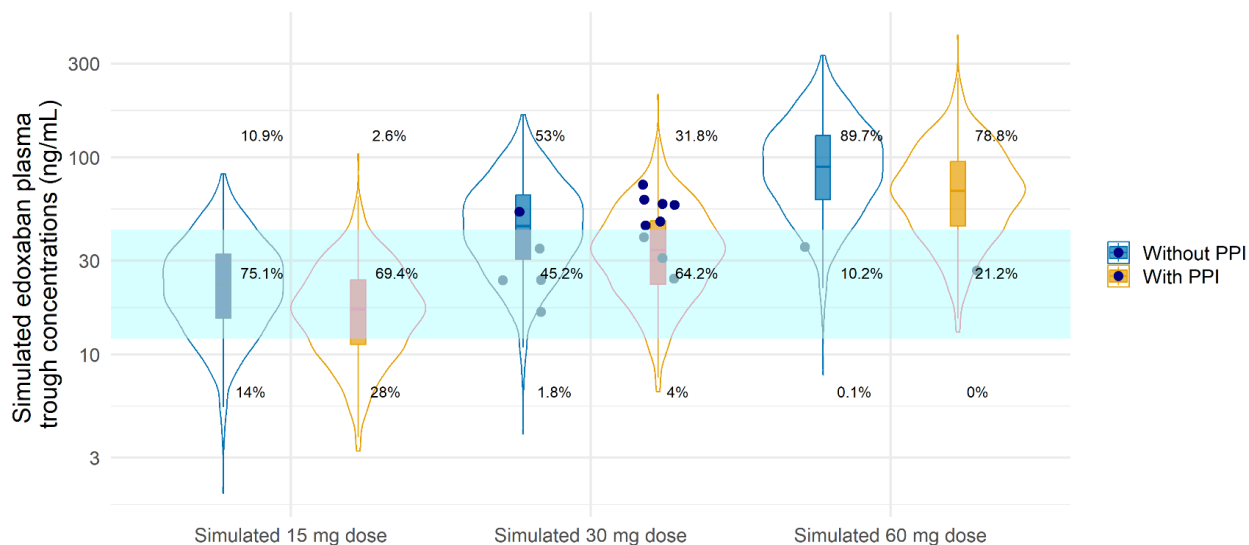


Figure 21. Target attainment of steady-state trough concentrations in simulated virtual patients (n=1000) after administration of 15 mg, 30 mg and 60 mg edoxaban dose. Blue area is the therapeutic range of 12–43 ng/mL (31). Dark blue dots are the observed edoxaban trough plasma concentrations in 16 patients (a trough concentration in one patient was not obtained). The numbers are percentages of the probability of attaining the target range, over-and underexposure.

4. Discussion

This study is the first real-world characterization of edoxaban PK for stroke prevention in frail geriatric patients with AF. Utilizing a popPK modeling approach with rich sampling, our study shows differences in edoxaban PK compared to younger and non-frail populations studied in the previous PK studies. Specifically, total clearance was reduced by 0.3–3-fold, volumes of distribution decreased by 3–6-fold, and the absorption rate constant was significantly lower—being 5–50 times lower—compared to typical values reported in other popPK studies as detailed in the **Table 10**, including a model based on ENGAGE AF-TIMI 48 data (27-30).

Several factors may explain the observed PK changes in geriatric patients. In our popPK modeling, co-medication with PPI significantly reduced relative bioavailability by 24% and improved model fit. However, this finding may not be clinically relevant, as it did not impact the attainment of the trough concentration target range. In **Table 10**, although decreased kidney function was already factored, edoxaban clearance was still markedly lower in geriatric patients, suggesting that additional age-related changes beyond CrCL reductions contribute to the observed differences. The influence of frailty, although

potentially important, could not be demonstrated in this study. Frailty is often associated with sarcopenia—a condition characterized by a loss of lean muscle mass. As demonstrated by Bendayan et al. studying DOAC exposures, older adults with sarcopenia had significantly higher anti-Xa activity levels than those with greater lean mass, attributable to a reduced volume of distribution (32). Given that the drug is eliminated via the kidneys and liver, decreased liver function common in older age, may also substantially contribute to lower edoxaban clearance (16). Additionally, creatinine-based equations may overestimate true kidney function in geriatric patients due to lower muscle mass and altered creatinine production (33). Thus, even if the model corrects for impaired kidney function via CrCL, the actual renal capacity may still be overestimated, leading to less clearance than expected. A decrease in liver size and function associated with older age may also contribute to reduced edoxaban clearance (34). Because 49.1% of the administered dose was recovered in feces, surpassing the theoretical unabsorbed fraction of 38.2%, it has been inferred that edoxaban undergoes active intestinal secretion (35). In older adults, slower splanchnic blood flow may further diminish this secretory process, thereby reducing edoxaban clearance (36). Additionally, gastrointestinal physiological changes, such as slower gastric emptying times, could be responsible for the reduced absorption rate constant observed in geriatric patients (37). These factors become particularly apparent when compared to the ENGAGE AF-TIMI 48 data, where the oldest old patient group with a frailty profile was underrepresented. To accommodate these geriatric-specific physiological changes, future research should develop physiologically based pharmacokinetic (PBPK) models that integrate altered gastrointestinal parameters, renal and liver function, and intestinal secretion in geriatric patients.

Observed PK alterations resulted in nearly half of the steady-state trough concentrations in our cohort exceeding the range recommended by the European Heart Rhythm Association (>43 ng/mL), and none were below the target range (24). This finding warrants attention because, as demonstrated by Ruff et al., the risk of major bleeding rose steeply after edoxaban concentrations exceed 50 ng/mL, especially beyond 100 ng/mL, while the risk of stroke or systemic embolism declined more gradually with increasing concentrations in the trial population (13). Major bleeding events like intracranial hemorrhage (ICH) are among the most feared and devastating complications of anticoagulant therapy. ICH not only has the highest disability rate among survivors but also

carries a 40% to 50% mortality rate within 30 days, which is twice that of ischemic stroke, with only 27% of patients achieving functional independence at 90 days (38).

Our study findings, which reveal changes in PK leading to higher-than-expected edoxaban concentrations, warrant further investigation of lower dosing of edoxaban (15 mg) in the frail oldest old patient groups. The ELDERCARE-AF study evaluated the efficacy and safety of edoxaban 15 mg once daily versus placebo in Japanese patients with AF aged over 80 years who were ineligible for standard oral anticoagulant therapy (39). The results demonstrated that a once-daily 15 mg dose of edoxaban was superior to placebo in preventing stroke or systemic embolism consistently in patients older than 80 years, as well as those aged 90 years or older, with a numerically yet non-statistically significant higher incidence of bleedings. To date, there is no study in older Caucasian patients. Nevertheless, Comans et al. found that for other DOACs (i.e., dabigatran, apixaban, and rivaroxaban), although geriatric patients often received doses that did not align with guideline recommendations, most had plasma levels within the expected range (40). These findings necessitate the need for further studies to define the DOAC levels associated with toxic effects in geriatric patients, as the current target ranges were based on non-frail population.

In Europe, edoxaban 15 mg is not currently indicated as monotherapy due to concerns about decreased efficacy (41). It is only recommended during the transition from edoxaban 30 mg, for patients with one or more clinical factors leading to increased exposure, to a VKA, or is used in conjunction with an appropriate VKA dose. Our simulation results also indicate that the probability of individuals receiving a 15 mg dose falling below the therapeutic range is relatively high at 21%, which reduces our confidence in prescribing edoxaban 15 mg alone for geriatric patients and warrants further research on variable dosing for this patient population. Moreover, although the ELDERCARE-AF study demonstrated the superiority of edoxaban 15 mg compared to placebo, no trials have yet compared the safety and efficacy of the 30 mg and 15 mg doses specifically in frail geriatric patients.

Unfortunately, in this pilot study, determining the covariates driving the PK in geriatric patients was challenging due to the small sample size and high variability observed. Collecting blood samples from geriatric patients was difficult due to the high prevalence of 'difficult intravenous access' (42), resulting in a sampling scheme efficiency of only 62.5% (85 out of 136 expected samples, with 8 samples per patient). However, the missing data

appeared to be completely random, so no bias was expected. Nonetheless, the small sample size limits our ability to further stratify and predict geriatric patients who might benefit from an edoxaban 15 mg dose or who would still benefit from edoxaban 30 mg. For example, we could not ascertain whether patients with low body weight (common among geriatric patients) or those with reduced renal function would benefit more from the 15 mg dose. To confirm this hypothesis, larger, confirmatory safety and efficacy studies are needed. Our popPK modeling and simulation can serve as valuable tools in designing future clinical trials to address this gap.

Building on the findings of our study, we propose that reduced edoxaban dosing—30 mg instead of 60 mg, and 15 mg for patients meeting guideline criteria for dose reduction—may be more appropriate for geriatric patients. In cases of uncertainty, therapeutic drug monitoring (TDM) or even better model-informed precision dosing (MIPD) could be employed to individualize therapy, though these approaches may pose challenges due to increased costs and workload in geriatric wards. To facilitate this, MIPD tools should be integrated into electronic health record systems with user-friendly interfaces tailored to the needs of clinicians. Highlighting the importance of MIPD, the MAS study demonstrated that despite low dosing, 19.0% of patients had high trough concentrations of DOACs and experienced a high rate of bleeding events (43). Furthermore, high correlation between edoxaban concentrations and anti-factor Xa activity facilitates the implementation of edoxaban TDM, potentially making it a more feasible option for optimizing therapy in this population. While the anti-factor Xa assay is less sensitive than the LC-MS/MS assay (as demonstrated in Supplementary Figure 4, where a "resistance" is observed at lower concentrations), it remains a valuable tool for routine TDM or MIPD due to its practicality and accessibility in clinical settings.

Compared to other DOACs, edoxaban offers the advantage of once-daily dosing, making it more practical for geriatric patients who are often on multiple medications and may struggle with adherence due to forgetfulness. While rivaroxaban also shares the benefit of once-daily administration, its higher protein binding (95% vs. 50% for edoxaban) may pose a concern in geriatric populations where malnutrition is prevalent. Malnourished patients often have lower protein levels, which can affect the unbound fraction of highly protein-bound drugs like rivaroxaban. For instance, a 1% increase in the unbound fraction of rivaroxaban can result in a 17% increase in plasma concentrations, potentially

heightening the risk of bleeding. Further research is needed to explore the impact of frailty-related factors, such as malnutrition and low protein levels, on the pharmacokinetics and safety profiles of all DOACs in geriatric patients.

The results should be interpreted with caution, as this is a pilot study with a small number of patients. In addition, while the PPI use was statistically significant with F, the current data does not allow for mechanistic conclusions. Hence, future research could explore the physiological basis of the association, including the impact of PPI-induced pH changes on drug solubility or absorption, especially using tool such as PBPK modeling. Despite these limitations, we successfully developed a popPK model of edoxaban using novel data from a highly burdened and often overlooked population, which differs significantly from those studied in large trials yet aligns closely with the real-world treatment population. Our model can be used to address clinically relevant questions. Through simulations, we can predict the potentially safer dose for geriatric patients (15 mg over 30 mg). Other utilities of our popPK model are predicting when to stop edoxaban in periprocedural situations or when switching to or from other anticoagulant agents, specifically in this geriatric population.

In summary, frail geriatric patients experience significant changes in edoxaban PK, leading to higher exposure and potentially an increased risk of bleeding. This study suggests that further investigation into the benefits of a 15 mg edoxaban dose for frail geriatric patients is needed.

References

1. Wilke T, Groth A, Mueller S, Pfannkuche M, Verheyen F, Linder R, et al. Incidence and prevalence of atrial fibrillation: an analysis based on 8.3 million patients. *Europace*. 2013;15(4):486-93.
2. Wilkinson C, Clegg A, Todd O, Rockwood K, Yadegarfar ME, Gale CP, et al. Atrial fibrillation and oral anticoagulation in older people with frailty: a nationwide primary care electronic health records cohort study. *Age Ageing*. 2021;50(3):772-9.
3. Hart RG, Pearce LA, Aguilar MI. Meta-analysis: antithrombotic therapy to prevent stroke in patients who have nonvalvular atrial fibrillation. *Ann Intern Med*. 2007;146(12):857-67.
4. Singer DE, Chang Y, Fang MC, Borowsky LH, Pomernacki NK, Udaltsova N, et al. The net clinical benefit of warfarin anticoagulation in atrial fibrillation. *Ann Intern Med*. 2009;151(5):297-305.
5. Giugliano RP, Ruff CT, Braunwald E, Murphy SA, Wiviott SD, Halperin JL, et al. Edoxaban versus Warfarin in Patients with Atrial Fibrillation. *New England Journal of Medicine*. 2013;369(22):2093-104.
6. Eisen A, Giugliano RP, Ruff CT, Nordio F, Gogia HS, Awasty VR, et al. Edoxaban vs warfarin in patients with nonvalvular atrial fibrillation in the US Food and Drug Administration approval population: An analysis from the Effective Anticoagulation with Factor Xa Next Generation in Atrial Fibrillation-Thrombolysis in Myocardial Infarction 48 (ENGAGE AF-TIMI 48) trial. *Am Heart J*. 2016;172:144-51.
7. Ageno W, Gallus AS, Wittkowsky A, Crowther M, Hylek EM, Palareti G. Oral Anticoagulant Therapy. *Chest*. 2012;141(2):e44S-e88S.
8. Witt DM, Clark NP, Kaatz S, Schnurr T, Ansell JE. Guidance for the practical management of warfarin therapy in the treatment of venous thromboembolism. *Journal of Thrombosis and Thrombolysis*. 2016;41(1):187-205.
9. De Groot JR, Weiss TW, Kelly P, Monteiro P, Deharo JC, De Asmundis C, et al. Edoxaban for stroke prevention in atrial fibrillation in routine clinical care: 1-year follow-up of the prospective observational ETNA-AF-Europe study. *European Heart Journal - Cardiovascular Pharmacotherapy*. 2021;7(F11):f30-f9.
10. Wilkinson C, Wu J, Searle SD, Todd O, Hall M, Kunadian V, et al. Clinical outcomes in patients with atrial fibrillation and frailty: insights from the ENGAGE AF-TIMI 48 trial. *BMC Medicine*. 2020;18(1).
11. Kampouraki E, Abohelaika S, Avery P, Biss T, Murphy P, Wynne H, et al. Elderly people are inherently sensitive to the pharmacological activity of rivaroxaban: implications for DOAC prescribing. *Journal of Thrombosis and Thrombolysis*. 2021;52(1):170-8.
12. Hias J, Van der Linden L, Walgraeve K, Gijzen M, Mian P, Koch BCP, et al. Pharmacokinetics of 2 oral paracetamol formulations in hospitalized octogenarians. *Br J Clin Pharmacol*. 2022;88(3):1020-30.
13. Ruff CT, Giugliano RP, Braunwald E, Morrow DA, Murphy SA, Kuder JF, et al. Association between edoxaban dose, concentration, anti-Factor Xa activity, and outcomes: an analysis of data from the randomised, double-blind ENGAGE AF-TIMI 48 trial. *Lancet*. 2015;385(9984):2288-95.
14. Lip GY, Nieuwlaat R, Pisters R, Lane DA, Crijns HJ. Refining clinical risk stratification for predicting stroke and thromboembolism in atrial fibrillation using a novel risk factor-based approach: the euro heart survey on atrial fibrillation. *Chest*. 2010;137(2):263-72.
15. Pisters R, Lane DA, Nieuwlaat R, de Vos CB, Crijns HJ, Lip GY. A novel user-friendly score (HAS-BLED) to assess 1-year risk of major bleeding in patients with atrial fibrillation: the Euro Heart Survey. *Chest*. 2010;138(5):1093-100.
16. Parasrampur DA, Truitt KE. Pharmacokinetics and Pharmacodynamics of Edoxaban, a Non-Vitamin K Antagonist Oral Anticoagulant that Inhibits Clotting Factor Xa. *Clin Pharmacokinet*. 2016;55(6):641-55.

17. Siriez R, Yildiz H, Bouvy C, Haguet H, Maloteau V, Hardy M, et al. The edoxaban-M4 metabolite and measurement of edoxaban by chromogenic assays in human plasma. *Research and Practice in Thrombosis and Haemostasis*. 2022;6(3).
18. Fried LP, Tangen CM, Walston J, Newman AB, Hirsch C, Gottdiener J, et al. Frailty in older adults: evidence for a phenotype. *J Gerontol A Biol Sci Med Sci*. 2001;56(3):M146-56.
19. Rolfson DB, Majumdar SR, Tsuyuki RT, Tahir A, Rockwood K. Validity and reliability of the Edmonton Frail Scale. *Age Ageing*. 2006;35(5):526-9.
20. Rockwood K. A global clinical measure of fitness and frailty in elderly people. *Canadian Medical Association Journal*. 2005;173(5):489-95.
21. Foerster KI, Huppertz A, Muller OJ, Rizos T, Tilemann L, Haefeli WE, et al. Simultaneous quantification of direct oral anticoagulants currently used in anticoagulation therapy. *J Pharm Biomed Anal*. 2018;148:238-44.
22. Xu XS, Dunne A, Kimko H, Nandy P, Vermeulen A. Impact of low percentage of data below the quantification limit on parameter estimates of pharmacokinetic models. *J Pharmacokinet Pharmacodyn*. 2011;38(4):423-32.
23. Dosne AG, Bergstrand M, Harling K, Karlsson MO. Improving the estimation of parameter uncertainty distributions in nonlinear mixed effects models using sampling importance resampling. *J Pharmacokinet Pharmacodyn*. 2016;43(6):583-96.
24. Steffel J, Verhamme P, Potpara TS, Albaladejo P, Antz M, Desteghe L, et al. The 2018 European Heart Rhythm Association Practical Guide on the use of non-vitamin K antagonist oral anticoagulants in patients with atrial fibrillation. *Eur Heart J*. 2018;39(16):1330-93.
25. Martin Bland J, Altman DG. Statistical Methods for Assessing Agreement between Two Methods of Clinical Measurement. *The Lancet*. 1986;327(8476):307-10.
26. Passing H, Bablok W. A new biometrical procedure for testing the equality of measurements from two different analytical methods. Application of linear regression procedures for method comparison studies in clinical chemistry, Part I. *Journal of clinical chemistry and clinical biochemistry*. 1983;21(11):709-20.
27. Salazar DE, Mendell J, Kastrissios H, Green M, Carrothers TJ, Song S, et al. Modeling and simulation of edoxaban exposure and response relationships in patients with atrial fibrillation. *Thrombosis and Haemostasis*. 2012;107(5):925-34.
28. Krekels EH, Niebecker R, Karlsson MO, Miller R, Shimizu T, Karlsson KE, et al. Population Pharmacokinetics of Edoxaban in Patients with Non-Valvular Atrial Fibrillation in the ENGAGE AF-TIMI 48 Study, a Phase III Clinical Trial. *Clin Pharmacokinet*. 2016;55(9):1079-90.
29. Song S, Kang D, Halim AB, Miller R. Population pharmacokinetic-pharmacodynamic modeling analysis of intrinsic FXa and bleeding from edoxaban treatment. *Journal of Clinical Pharmacology*. 2014;54(8):910-6.
30. Yin OQ, Tetsuya K, Miller R. Edoxaban population pharmacokinetics and exposure-response analysis in patients with non-valvular atrial fibrillation. *Eur J Clin Pharmacol*. 2014;70(11):1339-51.
31. Steffel J, Collins R, Antz M, Cornu P, Desteghe L, Haeusler KG, et al. 2021 European Heart Rhythm Association Practical Guide on the Use of Non-Vitamin K Antagonist Oral Anticoagulants in Patients with Atrial Fibrillation. *EP Europace*. 2021;23(10):1612-76.
32. Bendayan M, Mardigyan V, Williamson D, Chen-Tournoux A, Eintracht S, Rudski L, et al. Muscle Mass and Direct Oral Anticoagulant Activity in Older Adults With Atrial Fibrillation. *Journal of the American Geriatrics Society*. 2021;69(4):1012-8.
33. Raman M, Middleton RJ, Kalra PA, Green D. Estimating renal function in old people: an in-depth review. *Int Urol Nephrol*. 2017;49(11):1979-88.
34. Tan JL, Eastment JG, Poudel A, Hubbard RE. Age-Related Changes in Hepatic Function: An Update on Implications for Drug Therapy. *Drugs Aging*. 2015;32(12):999-1008.
35. Kato T, Mikkaichi T, Yoshigae Y, Okudaira N, Shimizu T, Izumi T, et al. Quantitative analysis of an impact of P-glycoprotein on edoxaban's disposition using a human physiologically based pharmacokinetic (PBPK) model. *International Journal of Pharmaceutics*. 2021;597.

36. Bender AD. The Effect of Increasing Age on the Distribution of Peripheral Blood Flow in Man. *J Am Geriatr Soc.* 1965;13:192-8.
37. Edwina AE, Koziolk M, Steenackers N, Vanuytsel T, Lopez Marmol A, Matthys C, et al. Tracking the journey: Gastrointestinal transit and intraluminal pH dynamics in aging populations using wireless motility capsule. *Int J Pharm.* 2024;669:125016.
38. Woo D, Comeau ME, Venema SU, Anderson CD, Flaherty M, Testai F, et al. Risk Factors Associated With Mortality and Neurologic Disability After Intracerebral Hemorrhage in a Racially and Ethnically Diverse Cohort. *JAMA Netw Open.* 2022;5(3):e221103.
39. Okumura K, Akao M, Yoshida T, Kawata M, Okazaki O, Akashi S, et al. Low-Dose Edoxaban in Very Elderly Patients with Atrial Fibrillation. *N Engl J Med.* 2020;383(18):1735-45.
40. Comans AL, Sennesael AL, Bihin B, Regnier M, Mullier F, de Saint-Hubert M. Inappropriate low dosing of direct oral anticoagulants in older patients with non-valvular atrial fibrillation: Impact on plasma drug levels. *Thrombosis Research.* 2021;201:139-42.
41. Summary of product characteristics: Lixiana (Edoxaban). European Medicines Agency; 2024.
42. Santos-Costa P, Paiva-Santos F, Bernardes RA, Sousa LB, Ventura F, Faria J, et al. Difficult Intravenous Access in Older Adults with Cancer: Can Vein-Locating Technology Be Key for Vessel Health? *International Workshop on Gerontechnology; Utrecht, the Netherlands* 2023. p. 241–50.
43. Palareti G, Testa S, Legnani C, Paoletti O, Cini M, Antonucci E, et al. The use of reduced DOAC doses in atrial fibrillation patients does not always lead to good anticoagulation levels and avoid adverse events. *Int J Cardiol.* 2024;416:132484.

Appendix

Supplementary material 1 NONMEM control stream

```
$PROBLEM          Edoxaban and metabolite

$INPUT            ID TIME TAD AMT RATE DV TYPE METHODS MDV EVID
CMT

                    AGE SEX SCR CRCL BMI HT WT LM HB ALBUMIN
                    PROTEIN BILIRUBIN AST ALT GGT ALP FRIED CFS EFS
                    MNA CCI PPI

$DATA             DS.csv IGNORE=@
$SUBROUTINE       ADVAN13 TOL2
$MODEL            COMP=(DEPOT DEFDOSE) COMP=(CENTRAL)
COMP=(PERIPH)
COMP=(MET)

$PK

TVCL                = THETA(1)
TVV2                = THETA(2)
TVV3                = THETA(3)
TVQ3                = THETA(4)
TVKA                = THETA(5)
PPIX                = THETA(11)
TVF1                = THETA(10)
IF(PPI.EQ.1) TVF1   = THETA(10)*PPIX
TVALAG1             = THETA(6)

CL                  = TVCL *EXP(ETA(1))
V2                  = TVV2*EXP(ETA(2))
V3                  = TVV3
Q                   = TVQ3
KA                  = TVKA*EXP(ETA(3))
ALAG1               = TVALAG1
F1                  = TVF1

;M4
TVVM                = THETA(7)
TVCLM               = THETA(8)
V4                  = TVVM
CLM                 = TVCLM*EXP(ETA(4))

;CLR fraction
FR_R                = THETA(9)
CLR                 = CL * FR_R      ;Renal clearance
CLN                 = CL * (1-FR_R)  ;Non-renal
clearance

;Microconstant and scaling
S2                  = V2/1000
```

```

S3      = V3/1000
S4      = V4/1000
K20     = CL/V2
K24     = Q/V2
K42     = Q/V4
K23     = CLN/V2
K30     = CLM/V3

```

\$DES

```

DADT(1)  = -KA*A(1)
DADT(2)  = KA*A(1) - K23*A(2) - K24*A(2) +
K42*A(4)
          - K20*A(2)
DADT(3)  = K23*A(2) - K30*A(3)
DADT(4)  = K24*A(2) - K42*A(4)

```

\$ERROR

```

IF (METHODS.EQ.1) THEN                ;Edoxaban error
model
    IPRED = A(2)/S2
    W=SQRT(IPRED**2*SIGMA(1,1)+SIGMA(2,2))
    Y=IPRED*(1+EPS(1))+EPS(2)
ENDIF

IF (METHODS.EQ.3) THEN                ;M-4 error model
    IPRED = A(3)/S3
    W=SQRT(IPRED**2*SIGMA(3,3)+SIGMA(4,4))
    Y=IPRED*(1+EPS(3))+EPS(4)
ENDIF

IF (W.EQ.0) W=1
IRES  = DV-IPRED
IWRES = IRES/W

```

\$THETA

```

(0, 9.02)      ;CL(L/h)
(0, 47.4)      ;Vc(L)
(0, 42.7)      ;Vp(L)
(0, 3.27)      ;Q(L/h)
(0, 0.104)     ;KA
(0.25) FIX     ;Lag time
(0, 35.3)      ;V4
(0, 37.3)      ;CLM
(0, 0.5) FIX   ;Fraction CLR
(0, 1) FIX     ;F1
(0, 0.764)     ;PPI~F1

```

\$OMEGA BLOCK(2)

```

0.119          ;IIV_CL
0.208 2.22     ;IIV_Vc
$OMEGA         0 FIX      ;IIV_V3

```

```

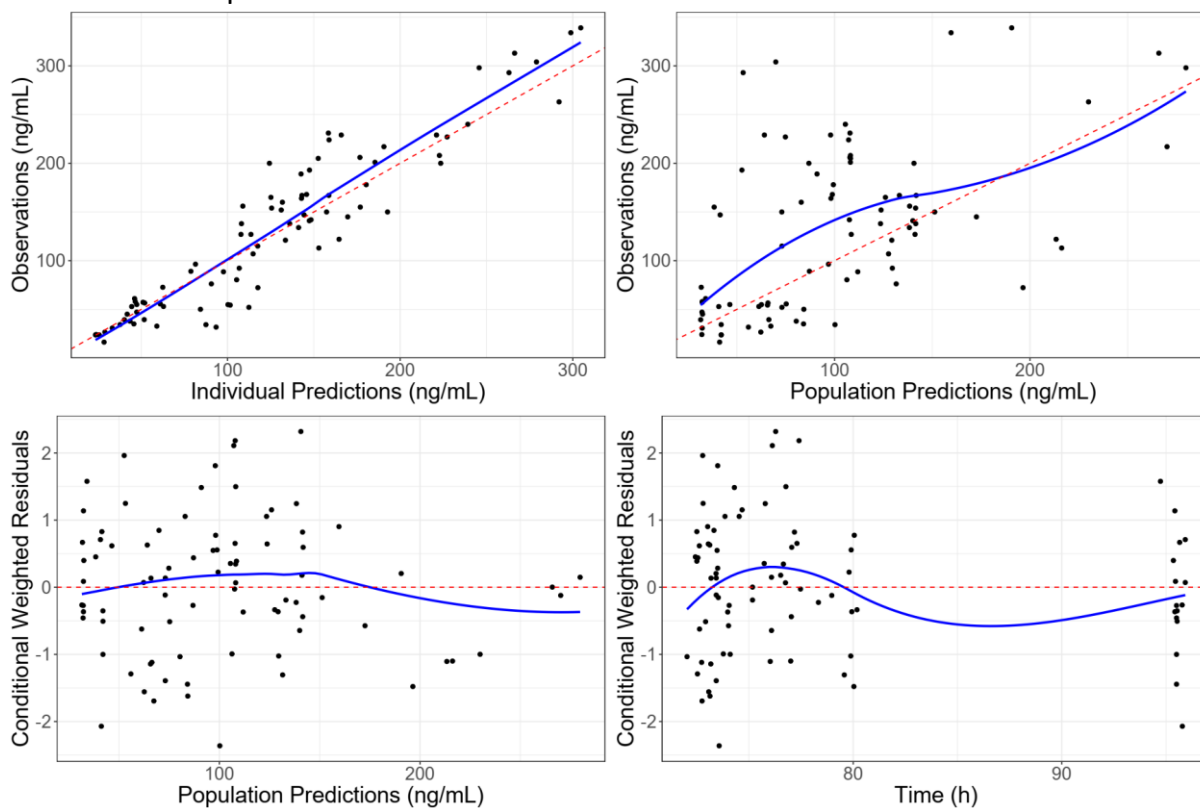
$OMEGA          0 FIX          ;IIV_KA
$OMEGA          0 FIX          ;IIV_Lag time
$OMEGA          0 FIX          ;IIV_V4
$OMEGA          0.218          ;IIV_CLM

$SIGMA
                0.0659
                0 FIX
                0.0456
                0 FIX

$ESTIMATION      METHOD=1 INTER MAXEVAL=9999 PRINT=5 NOABORT
POSTHOC
$COV              PRECOND=2 PFCOND=2

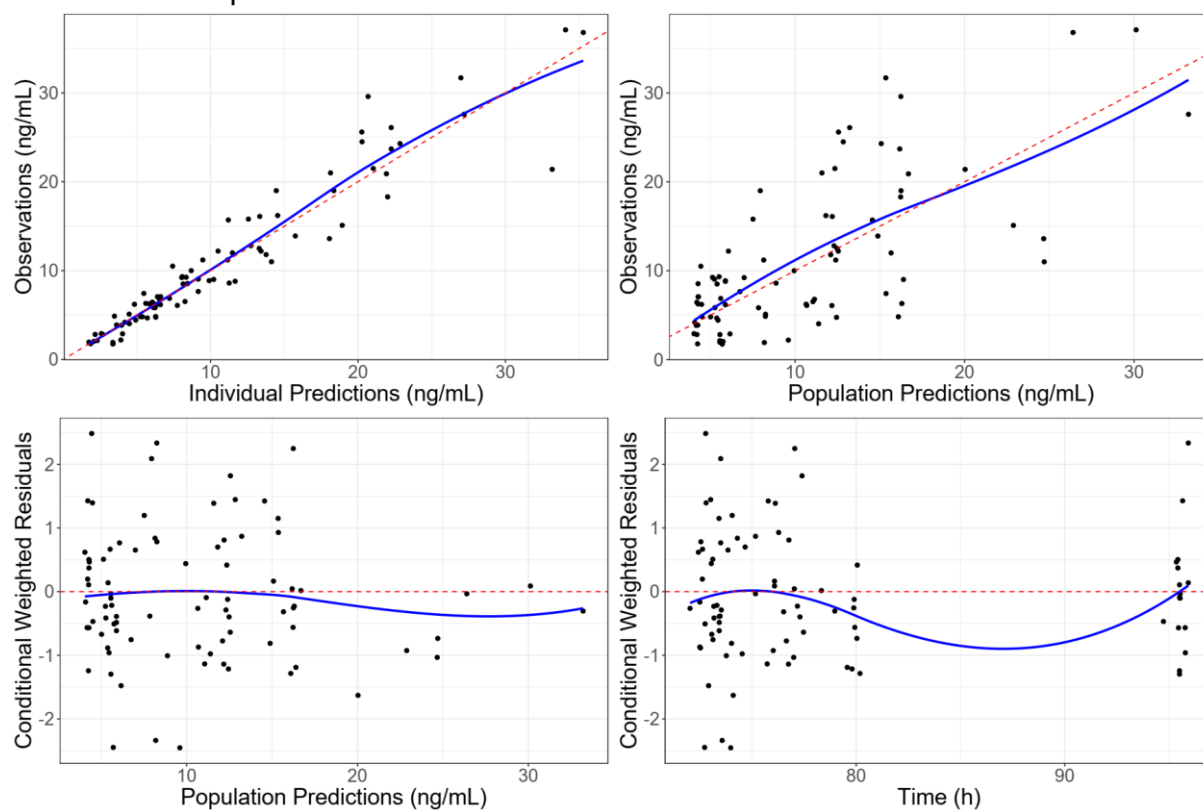
```

Supplementary figure 1
Goodness-of-fit plots of edoxaban concentrations

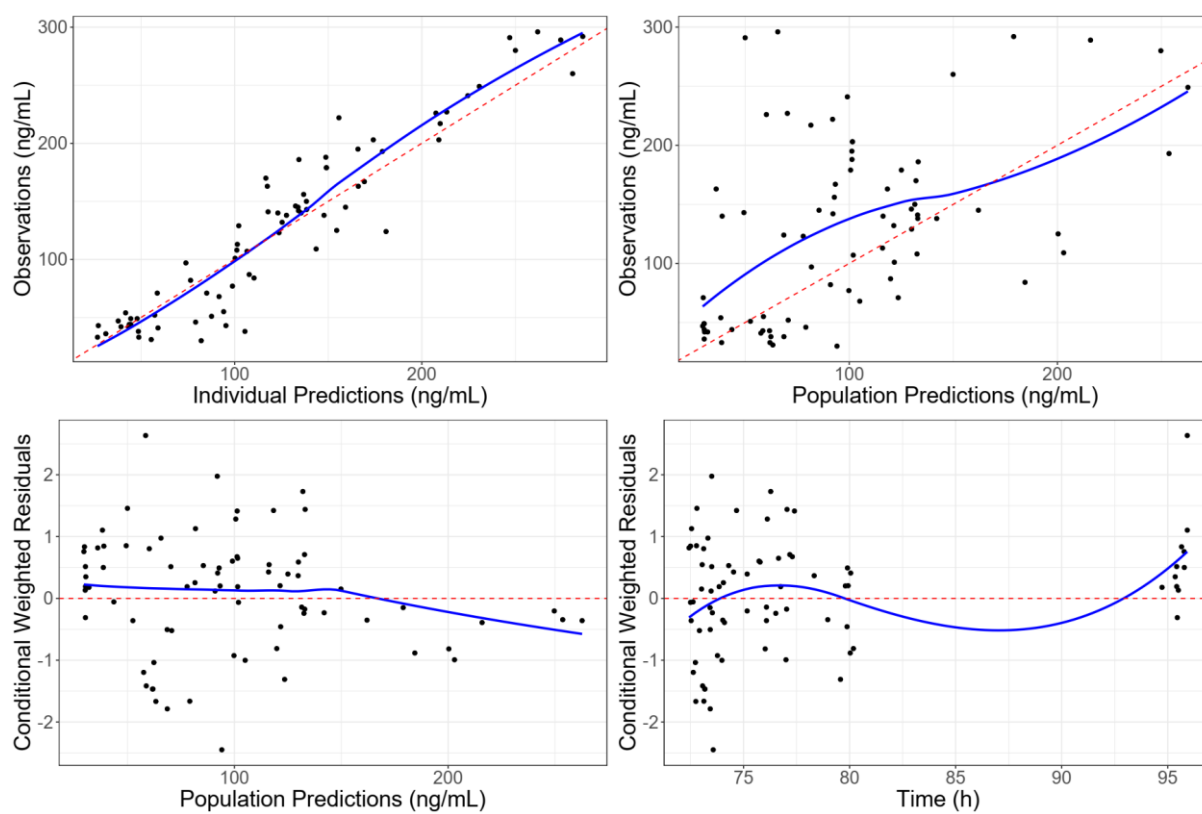


Supplementary figure 2

Goodness-of-fit plots of edoxaban M-4 metabolite concentrations

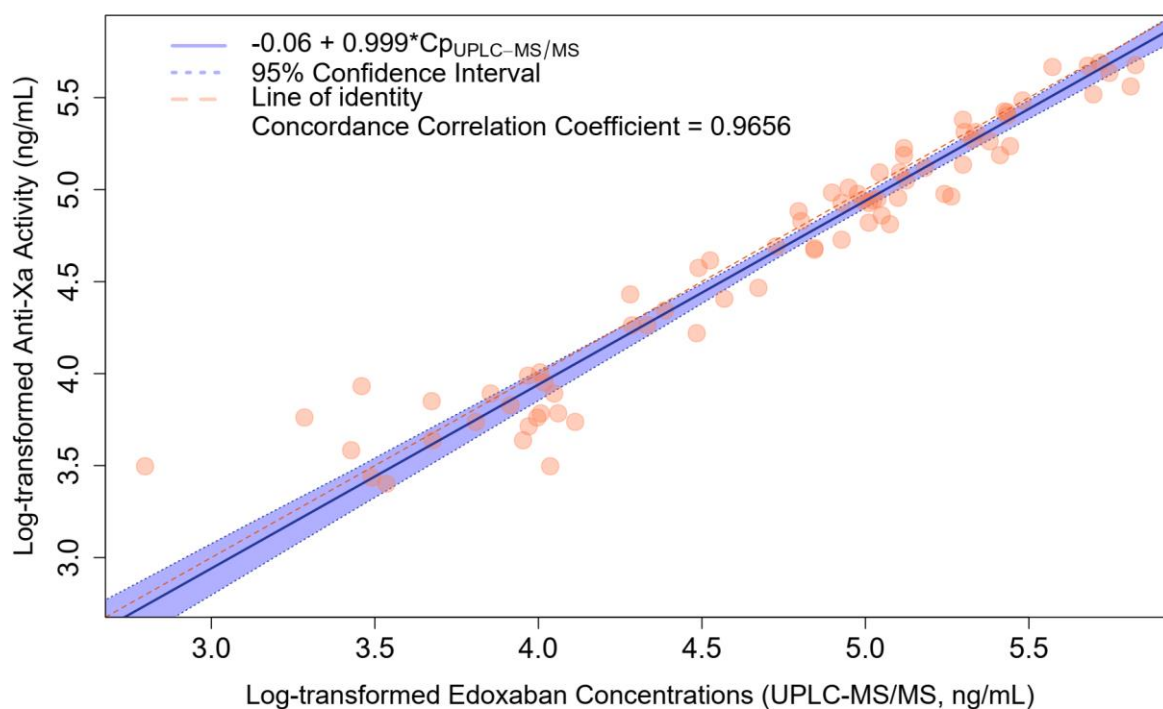


Supplementary figure 3
Goodness-of-fit plots of anti-Xa activity levels



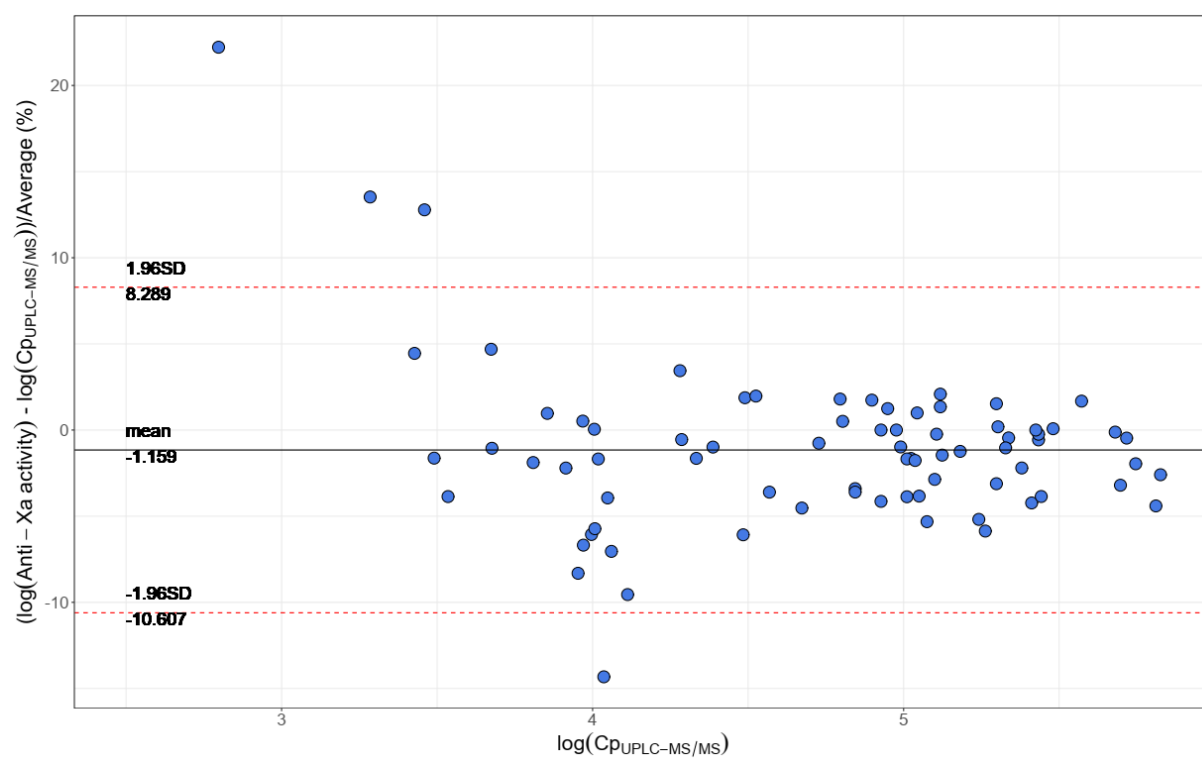
Supplementary figure 4

Passing-Bablok regression for comparison between edoxaban concentrations and anti-Xa activity levels



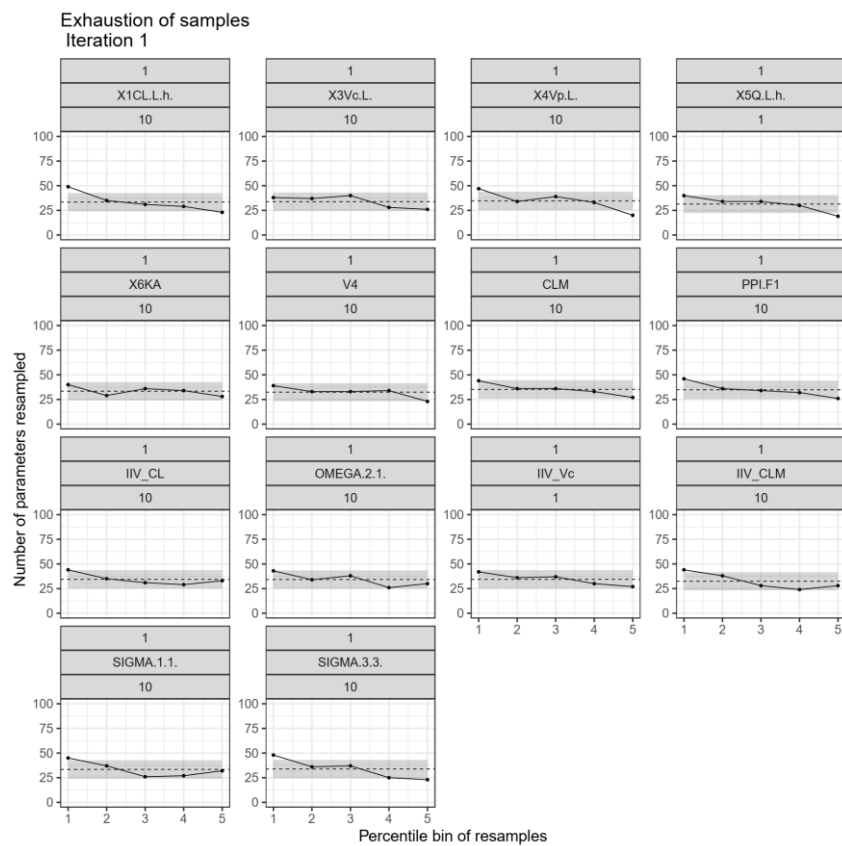
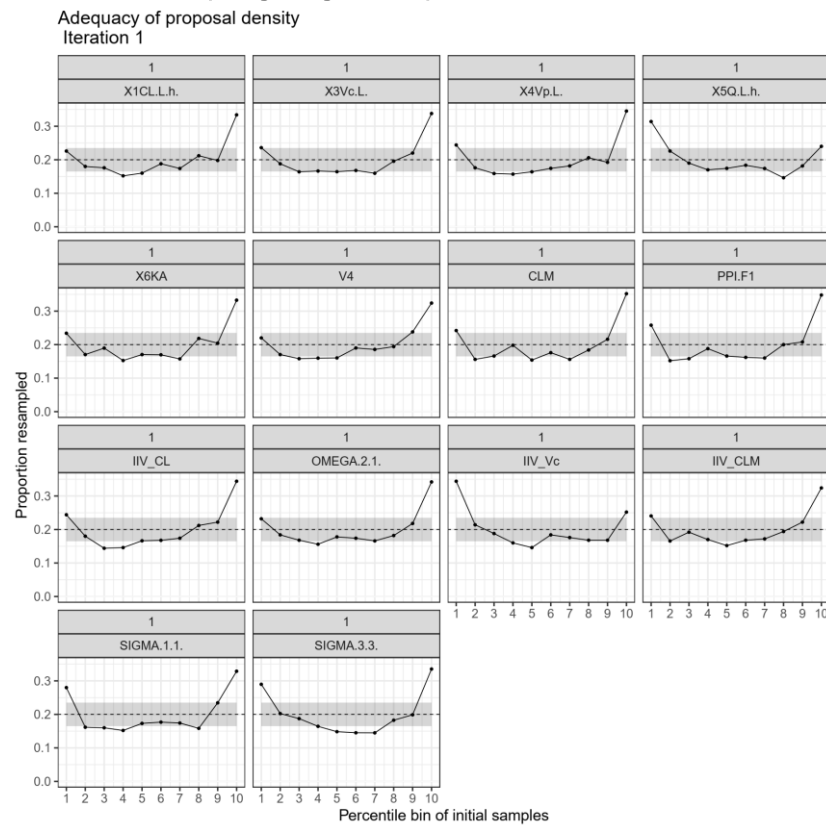
Supplementary figure 5

Bland-Altman plot for comparison between edoxaban concentrations and anti-Xa activity levels



Supplementary Figure 6

Sampling Importance Resampling diagnostic plots



Part 3. Pharmacokinetics of Vancomycin

Chapter 4. Decreased kidney function explains higher vancomycin exposure in older adults

This chapter is published as

Edwina AE, Dreesen E, Gijzen M, van den Hout HC, Desmet S, Flamaing J, Van der Linden L, Spriet I, Tournoy J. *Decreased Kidney Function Explains Higher Vancomycin Exposure in Older Adults*. *Drugs Aging*. 2024 Sep;41(9):753-762

Abstract

Introduction: Older adults face a higher risk of vancomycin-related toxicity given their (patho)-physiological changes, making early management of supratherapeutic exposure crucial. Yet, data on vancomycin exposure in older adults is scarce. This study aimed to compare vancomycin concentrations between older and younger patients, emphasizing supratherapeutic concentrations and the effect of patient characteristics when similar doses were applied.

Methods: This observational retrospective study was conducted in the University Hospital of Leuven (EC Research S65213). We analyzed early (first) vancomycin concentrations between older (≥ 75 years) and younger patients. Multivariable analyses were conducted to evaluate the association between baseline patient characteristics with supratherapeutic exposure (logistic regression), and dose-normalized concentrations (linear regression).

Results: We included 449 patients aged ≥ 75 years (median 80) and 1,609 aged < 75 years (median 61). In univariable analysis, the first-measured vancomycin concentrations were significantly higher in older adults ($p < 0.001$), who more frequently reached supratherapeutic concentrations (30.7% vs. 21%; $p < 0.001$). In multivariable analysis, factors associated with supratherapeutic concentrations were decreased $\text{eGFR}_{\text{CKD-EPI}}$ (Odds Ratio [OR]=0.98, Confidence Interval [CI]= 0.97–0.98). Supratherapeutic concentrations had inverse association with giving lower loading dose (OR=0.59, CI=0.39–0.90), and lower maintenance dose (OR=0.45, CI=0.26–0.77). Factors that predicted increased dose-normalized concentrations included decreased $\text{eGFR}_{\text{CKD-EPI}}$ (Coefficient=-0.05, CI=-0.06– -0.04), lower body weight (Coefficient=-0.04, CI=-0.05– -0.03), increased blood urea nitrogen (Coefficient=0.02, CI=0.01–0.03), and delayed time to therapeutic drug monitoring (TDM) sampling (Coefficient=0.08, CI=0.06–0.09).

Conclusion: The absence of age as a significant factor in the multivariable analysis suggests that $\text{eGFR}_{\text{CKD-EPI}}$ mediated the relationship between age and vancomycin exposure. Older adults may benefit more from vancomycin TDM.

1. Introduction

Infectious diseases are one of the leading causes of hospitalization among older adults (1-5). In addition, this age group is more susceptible to hospital-acquired infections compared to their younger counterparts (6). A particularly concerning pathogen in this regard is the methicillin-resistant *Staphylococcus aureus* (MRSA), given its higher prevalence and lethality among older adults (7-9).

The glycopeptide antibiotic vancomycin is often the preferred agent for treating MRSA and other severe gram-positive infections (10). However, due to its narrow therapeutic window and high pharmacokinetic variability, guidelines recommend therapeutic drug monitoring (TDM) of vancomycin to control the risks of over- and underdosing. The objective herein is to maintain therapeutic efficacy and circumvent renal toxicity by targeting a 24-hour area under the concentration-time curve (AUC_{24h}) between 400–600 mg×h/L (11). However, determining AUCs is laborious and financially demanding (12). As a pragmatic alternative, many clinical settings have adopted single-point concentration measurements as a standard proxy with target trough concentration between 15–20 mg/L for intermittent dosing and 20–25 mg/L for continuous infusion (12-14).

Given the increased vulnerability of older adults to supratherapeutic exposure due to changes such as decreased kidney function and reduced volume of distribution, this population is particularly at risk of vancomycin-related nephrotoxicity (15, 16). The challenge of managing out-of-range exposures, especially the critical issue of supratherapeutic levels, remains unresolved in the initial treatment days due to the lack of TDM data (17). This problem emphasizes the necessity to address supratherapeutic exposure in older adults to mitigate risks from the outset of treatment. Moreover, the scarcity of comprehensive data on vancomycin exposure in older adults adds complexity to determining whether doses based solely on renal function are sufficient for achieving therapeutic targets within this age group.

Therefore, our study specifically aims to assess the achievement of therapeutic targets at the first TDM measurement of vancomycin in older adults compared to younger patients. Additionally, we explored how various patient characteristics influence vancomycin concentrations.

2. Methods

This was a retrospective, monocentric observational study, which was approved by the Ethics Committee Research of UZ/KU Leuven (S65213). All adult patients admitted to the 1754-bed tertiary referral centre in the University Hospital of Leuven between October 2019 and May 2022, and treated with either an intermittent or a continuous vancomycin infusion, were eligible for inclusion. To ensure a comprehensive representation of the hospital's adult patient population, we included inpatients admitted to intensive care units (ICU) and non-ICU wards. Patients without complete vancomycin dosing records within 48 hours prior to the concentration measurement, patients with dialysis, patients with vancomycin concentrations > 50 mg/L as upper limit of quantification were excluded from analysis. To address potential differences in outcomes between steady-state and pre-steady-state concentrations, we also excluded patients whose first measurements were taken after 72 hours. This aligns with our primary focus on identifying potential toxicity as early as possible, thereby facilitating timely intervention.

2.1. Exposures and study endpoints

The first vancomycin concentration measurement available per patient was classified as either subtherapeutic (below the therapeutic range), on target (within the therapeutic range), and supratherapeutic (above the therapeutic range), as defined in the section *Local vancomycin dosing guideline* (see below). The primary endpoint was the occurrence of a supratherapeutic concentration. In addition, the dose-normalized vancomycin concentration and subtherapeutic concentration were also analyzed as secondary endpoints.

2.2. Local vancomycin dosing guideline

The local vancomycin dosing guideline is derived from the Infectious Diseases Society of America guidelines (11), and details are provided in Supplementary Table 1. Before February 15th, 2021, no specific guidance was given regarding the vancomycin loading dose. Most patients received a standard vancomycin maintenance dosing regimen of 1000 mg every twelve hours (q12h). Since February 15th, 2021, the loading dose has been calculated using the patient's actual body weight, whereas the maintenance dose is adjusted according to the estimated glomerular filtration rate according to the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation ($eGFR_{CKD-EPI}$). These dosing standards have

been uniformly applied across all age groups, irrespective of the patient's clinical condition. Subsequent TDM-based dose adjustments aimed at achieving trough concentrations of 12.5 to 17.5 mg/L on intermittent infusion or plateau concentrations of 20 to 25 mg/L on continuous infusion (11, 18). For the trough concentration, the target of 12.5-17.5 mg/L in our study was based on a combination of the IDSA guidelines and considerations for assay variability. According to the IDSA guidelines, the recommended trough target for vancomycin is 15–20 mg/L (11). However, a meta-analysis has shown that concentrations above 15 mg/L were strongly associated with three-fold increased risk of nephrotoxicity (18). To mitigate this risk while ensuring therapeutic efficacy, our hospital guidelines set a target concentration of 15 mg/L. Additionally, we included a margin of 2.5 mg/L to account for an acceptable assay error of less than 20%. This adjustment ensured that minor variations in assay performance did not lead to significant clinical misinterpretations.

2.3. Patient characteristics

Detailed information on baseline patient characteristics, recorded on the day of the initial vancomycin dosage, was extracted from the electronic health records. The following variables were collected: age as continuous variable and as a binary covariate using 75-year age cutoff (as this is a cut-off for admission in the geriatric wards in our hospital), sex assigned at birth, body weight, height, body mass index (BMI), admission to Intensive Care Unit (ICU), and diagnosis of diabetes mellitus. Baseline laboratory results such as serum creatinine, $eGFR_{CKD-EPI}$, blood urea nitrogen (BUN), serum albumin, and total protein were collected as well. Beside $eGFR_{CKD-EPI}$, we also calculated $eGFR$ using the Full Age Spectrum ($eGFR_{FAS}$) and Berlin Initiative Study ($eGFR_{BIS}$) equations that incorporate serum creatinine (19), since these equations showed more accurate estimates of GFR for older adults (20). Other included variables were time to first vancomycin concentration measurement, defined as the interval between the first vancomycin administration and the time of the first TDM blood sample; the appropriateness of the loading dose and the maintenance dose based on the local guideline (Supplementary Table 1); and whether patients received vancomycin before or after implementation of the 2021 guideline.

2.4. Statistical analysis

Categorical data were represented as percentages (%), continuous data as means with standard deviations for normally distributed data and medians with interquartile ranges

(IQR) for non-normally distributed data. The baseline characteristics of patients were compared with those of younger adults using the Student's *t*-test for normally distributed data and the Mann-Whitney U test for non-normally distributed data.

Logistic regression analyses were performed to identify predictors of supratherapeutic and subtherapeutic concentrations. The effect of patient characteristics on dose-normalized vancomycin concentrations was assessed using linear regression. Firstly, these patient characteristics were tested univariably. Characteristics showing a statistically significant association ($p < 0.05$) were further analyzed in multivariable analyses, considering $p < 0.01$ as significant. Models incorporating eGFR calculated by CKD-EPI, FAS, and BIS equations were compared, and the best model was selected based on the objective function value or -2 log likelihood. A threshold value of significant covariate(s) was selected using the Youden J statistic (21). A two-sided $p \leq 0.05$ considered statistical significance. Multicollinearity of final models was evaluated by calculating the Variance Inflation Factor (VIF) (22).

2.5. Software

All statistical and graphical analyses were conducted using R (version 4.3.1, R Core Team, Vienna, Austria) in conjunction with Rstudio (2023.06.1+524) (23, 24). The multivariable analyses were performed using the *glmulti* package, utilizing a genetic algorithm to select the best model with the lowest AIC (25).

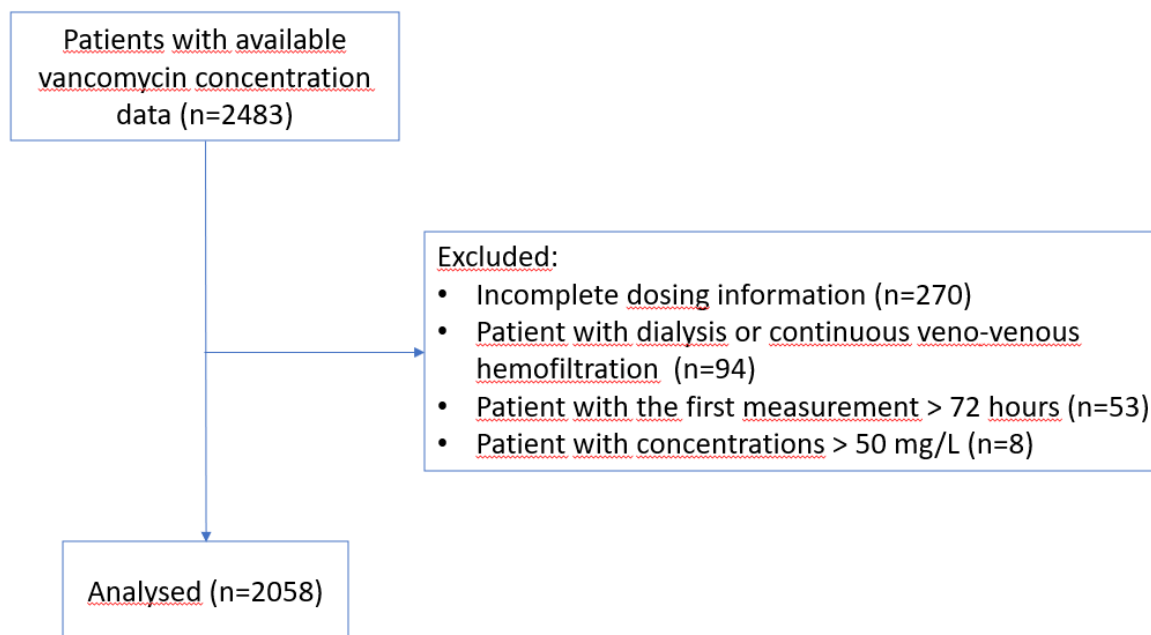


Figure 22. Flowchart for patient inclusion

3. Results

A total of 2,058 patients were included in our study, with 449 aged 75 years or older (21.8%; median: 80 years; range: 75–99 years). The flowchart of patient inclusion process is shown in **Figure 22**. The remaining 1,609 patients were younger than 75 years (78.2%; median 61 years; range 18–74 years). While older adults generally had lower body weight and height compared to their younger counterparts, their BMI remained comparable (**Table 11**). Markers of kidney function—eGFR_{CKD-EPI}, eGFR_{FAS}, eGFR_{BIS}, serum creatinine, and BUN—indicated more pronounced renal impairment in the older group ($p < 0.001$). Additionally, older adults had significantly reduced serum albumin levels (median 31.0 g/L versus 32.3 g/L; $p < 0.001$).

The first vancomycin concentration, taken between day 0 to day 3 (median = 33.2 hours) after start of vancomycin therapy, was significantly higher in older adults (median 16.4 mg/L, IQR 12.4–20.5 mg/L) as compared to younger individuals (median 13.8 mg/L, IQR 9.4–19.3 mg/L) ($p < 0.001$; **Figure 23**). Older individuals achieved more on-target and supratherapeutic concentrations relative to their younger counterparts (33.3% vs. 23.2% and 30.7% vs. 21%, respectively; $p < 0.001$; **Figure 23**). A higher percentage of younger patients fell into the subtherapeutic category compared to older group (55.8% vs. 35%; $p < 0.001$).

Table 11. Patient characteristics

Variables	<75 years (n=1609)	≥75 years or older (n=449)	p-value
Age, years, median [IQR]	61 [51-68]	80 [77-84]	<0.001
Sex, male, n (%)	62	57.5	0.6138
Body weight, kg, median [IQR]	75 [62.575-87]	70 [60.9-81]	<0.001
Height, cm, median [IQR]	172 [165-179]	168 [161-174]	<0.001
BMI, kg/m ² , median [IQR]	25.1 [21.8-29.2]	25 [22.4-28.7]	0.9732
Serum creatinine, mg/dL, median [IQR]	0.82 [0.61-1.13]	0.99 [0.79-1.33]	<0.001
eGFR _{CKD-EPI} , mL/min/1.73m ² , median [IQR]	93.24 [64.04-109.02]	63.16 [43.7-84.3]	<0.001
eGFR _{FAS} , mL/min/1.73m ² , median [IQR]	88 [62.74-121.6]	53.49 [40.5-67.56]	<0.001
eGFR _{BIS} , mL/min/1.73m ² , median [IQR]	86.43 [6.4-60.07]	53.85 [39.2-70.26]	<0.001
BUN, mg/dL, median [IQR]	34 [23-54]	45 [31-67]	<0.001
Serum albumin, g/dL, median [IQR]	32.3 [28.3-37.4]	31 [27.6-34.725]	<0.001
Total protein, g/dL, median [IQR]	60 [54-68]	62 [55-69]	0.3182
Patients with diabetes (%)	21.8	29.3	0.292
ICU admission (%)	35.9	26.5	0.1998

BUN: Blood Urea Nitrogen, eGFR_{CKD-EPI}: estimated glomerular filtration rate calculated by using the Chronic Kidney Disease Epidemiology Collaboration equation, eGFR_{FAS}: estimated glomerular filtration rate calculated by using the Full Age Spectrum equation; eGFR_{BIS}: estimated glomerular filtration rate calculated by using Berlin Initiative Study equation; ICU: Intensive Care Unit; IQR: Interquartile Range

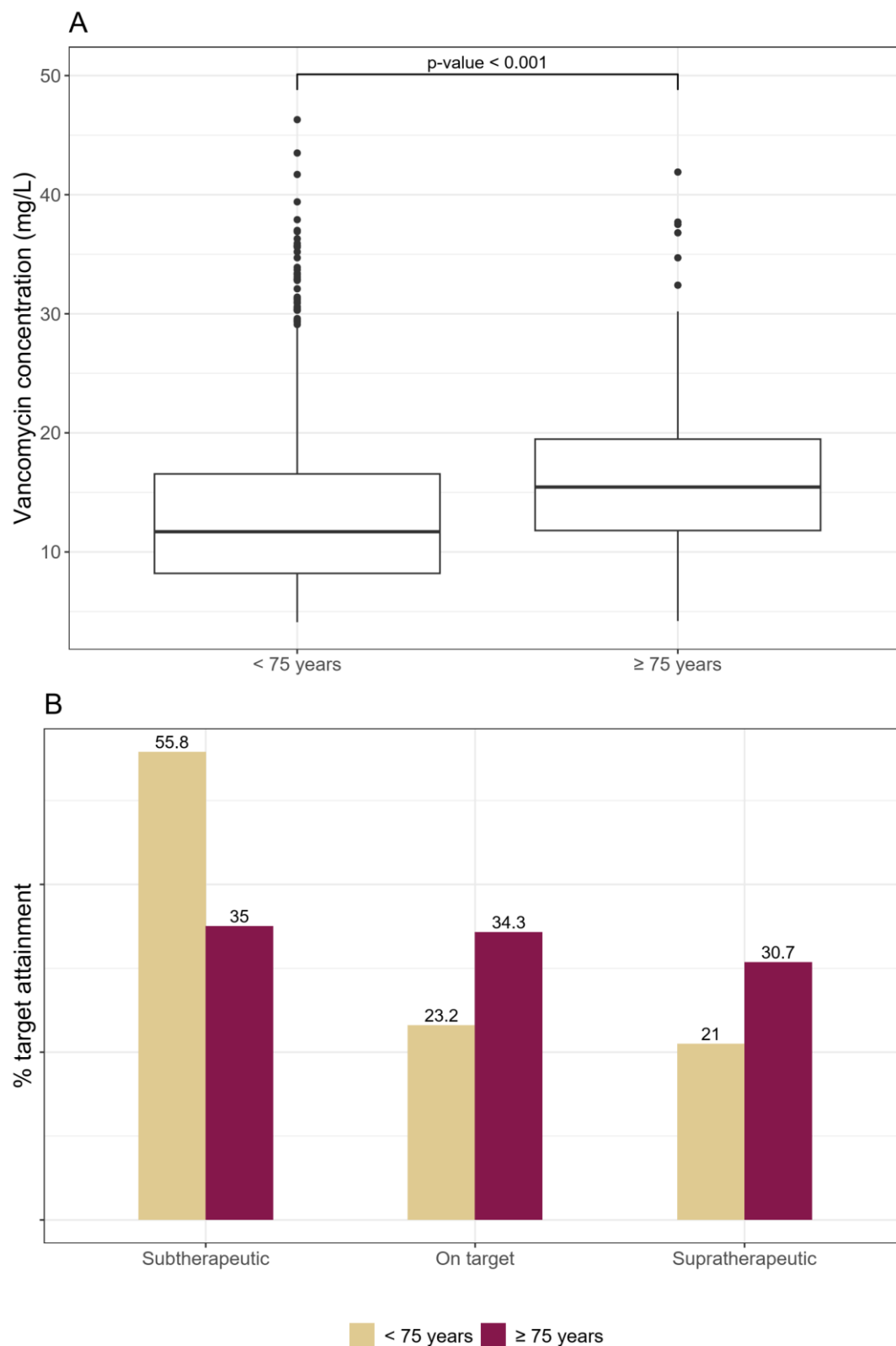


Figure 23. First vancomycin plasma concentrations (A) and the percentage of target attainment (B) in older adults (≥75 years) and younger patients (<75 years).

Multivariable models with $eGFR_{CKD-EPI}$ were chosen since they showed superiority compared to models with $eGFR_{FAS}$ and $eGFR_{BIS}$. Higher $eGFR_{CKD-EPI}$ (OR = 0.979, Confidence Interval [CI] = 0.974–0.984) was associated with reduced risk of supratherapeutic concentrations. Additionally, giving a lower loading dose (in mg/kg) (OR = 0.593, CI = 0.391–0.901) and lower maintenance dose (OR = 0.452, CI = 0.259–0.767) were linked to a lower risk of experiencing supratherapeutic concentrations (**Table 12**). For the risk of having subtherapeutic concentrations, higher $eGFR_{CKD-EPI}$ values (OR = 1.025, CI = 1.019–1.032), no loading dose (OR = 1.518, CI = 1.009–2.283), lower loading doses (OR = 1.526, CI = 1.051–2.214) and lower maintenance doses (OR = 2.151, CI = 1.421–3.255) were statistically significant covariates (**Supplementary table 2**).

Table 12. Multivariable logistic regression model to assess the association between patient characteristics and the risk of supratherapeutic concentrations

Variables		Multivariable analysis using age cut-off 75 years and age group <65, 65–75, >75 years ^a		
		p-value (*<0.01)	OR	99%CI
Sex	Female	0.056	1.382	0.894–2.136
	Male	Reference		
$eGFR_{CKD-EPI}$		<0.001*	0.979	0.973–0.986
Blood urea nitrogen		0.117	1.003	0.998–1.008
Serum albumin		0.08	0.983	0.957–1.008
Loading dose	Appropriate dose	Reference		
	No loading dose	0.011	0.641	0.407–1.008
	Higher dose	0.199	1.501	0.651–3.361
	Lower dose	0.001*	0.593	0.391–0.901
Maintenance dose	Appropriate dose	Reference		
	Higher dose	0.163	1.223	0.845–1.778
	Lower dose	<0.001*	0.452	0.259–0.767
Diabetes	Yes	0.787	0.961	0.654–1.4
	No	Reference		
Weight		0.092	0.993	0.983–1.004
Height		0.014	0.977	0.953–1.001

BMI: Body Mass Index, BUN: Blood Urea Nitrogen, CI: Confidence Interval, $eGFR_{CKD-EPI}$: estimated glomerular filtration rate calculated by using the Chronic Kidney Disease Epidemiology Collaboration equation, OR: Odds Ratio

Table 13. Multivariable linear regression analysis to assess the association between patient characteristics and dose-normalized concentrations

Variables		Multivariable analysis using age cut-off 75 years and age group <65, 65–75, >75 years ^a		
		p-value (*<0.01)	Beta coefficient	99%CI
Weight		<0.001*	-0.039	-0.05– -0.027
eGFR _{CKD-EPI}		<0.001*	-0.051	-0.059– -0.043
Blood Urea Nitrogen		<0.001*	0.019	0.012–0.026
Albumin		0.130	-0.018	-0.049–0.013
Time to TDM		<0.001*	0.077	0.062–0.092
Diabetes	Yes	0.761	-0.056	-0.533–0.421
	No	Reference		

BMI: Body Mass Index, BUN: Blood Urea Nitrogen, CI: Confidence Interval, eGFR_{CKD-EPI}: estimated glomerular filtration rate calculated by using the Chronic Kidney Disease Epidemiology Collaboration equation, TDM: Therapeutic Drug Monitoring

A decrease in eGFR_{CKD-EPI} was associated with higher dose-normalized vancomycin concentrations (Coefficient = -0.051, CI = -0.059– -0.043) (**Table 13**). A higher Blood Urea Nitrogen (BUN) level (Coefficient = 0.019, CI = 0.012–0.026) and a lower body weight (Coefficient = -0.039, CI = -0.05– -0.027) were independently associated with an increase of vancomycin concentrations. Longer time to TDM was associated with higher vancomycin concentrations (Coefficient = 0.077, CI = 0.062–0.092).

The area under the ROC curve (AUROC) of the eGFR_{CKD-EPI} as a predictor for supratherapeutic concentrations was 0.726 (95%CI=0.7–0.752, **Figure 24**) indicating a moderate discrimination of the model. An eGFR_{CKD-EPI} of 73.37 mL/min/1.73m² was identified as a supposedly optimal threshold (0.67 sensitivity and 0.705 specificity). In addition, correlation tests as shown in **Supplementary Figure 1** and **2** showed that eGFR_{CKD-EPI} was moderately correlated with age with correlation coefficient -0.47 (p-value<0.001), while body weight was not correlated with age (correlation coefficient -0.031, p-value=0.16).

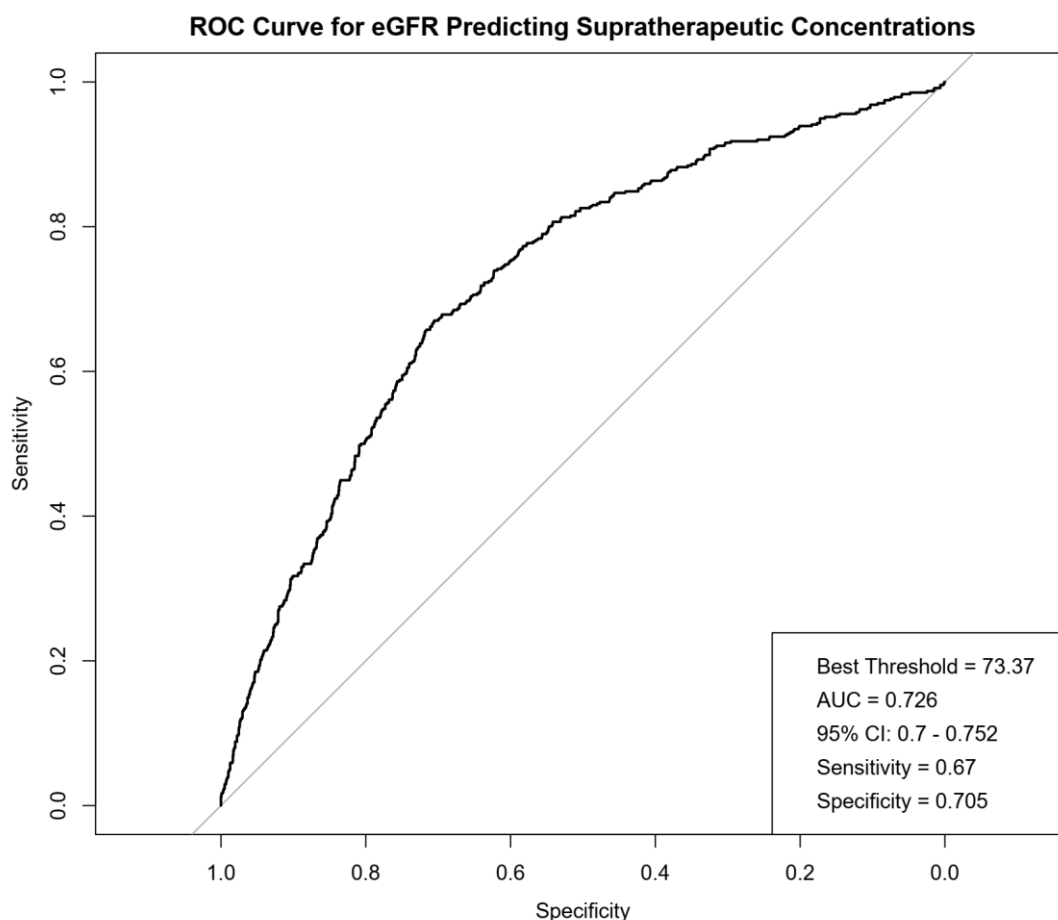


Figure 24. First vancomycin plasma concentrations (A) and the percentage of target attainment (B) in older adults (≥ 75 years) and younger patients (< 75 years).

4. Discussion

We conducted a hospital-wide retrospective analysis to evaluate vancomycin exposure in older adults. We found that adults aged 75 years or older had significantly higher vancomycin concentrations and more supratherapeutic concentrations compared to younger patients when given similar dosing guideline. However, after correcting for the impact of $\text{eGFR}_{\text{CKD-EPI}}$, age was not found to be associated with vancomycin exposure. The relationship between age and vancomycin exposure that was confounded by $\text{eGFR}_{\text{CKD-EPI}}$ was further supported by a correlation test indicating a moderate correlation between age and $\text{eGFR}_{\text{CKD-EPI}}$.

Beyond kidney function, body composition is a relevant factor that might influence vancomycin PK in older adults. It is of importance to understand if differences in body composition between younger and older individuals could affect drug exposure through

alterations in its volume of distribution. This is especially relevant considering that older adults often experience sarcopenia, leading to changes such as reduced muscle mass, higher fat tissue, and decreased hydration (26, 27). In our study, while body weight significantly impacted vancomycin dose-corrected concentrations, we did not find a correlation between body weight and age. This suggests that kidney function is the primary driver of the age-concentration relationship. To better understand drug concentrations in older people, future pharmacokinetic studies should include lean mass measurements, as lean mass may better explain the relationship between aging and sarcopenia than total body weight and therefore provide a more accurate assessment of drug distribution and concentration in the older population.

Our findings are in line with previous studies on vancomycin exposures including older adults population. An earlier univariable analysis comparing patients below and above 65 years of age found that older adults had higher vancomycin trough concentrations than their younger counterparts when given similar renal function-adjusted vancomycin doses (28). Pan et al.'s study further highlighted that while age was associated with decreased renal function influencing vancomycin concentrations, the study also found that vancomycin levels were significantly impacted by BMI and creatinine clearance in both younger and older patients (29). This indicates that age alone does not account for higher vancomycin concentrations. Furthermore, our study extends these observations with a larger sample size, offering more robust evidence.

Contrary to our findings, Gu et al. proposed that in addition to eGFR and body weight, patient age should also be considered when determining the initial maintenance dosing to optimize vancomycin levels. Their analysis showed that older adults exhibited higher first vancomycin concentrations when initial maintenance doses were given twice daily, showing an increase of 1.12 mg/L for every additional decade in age, after adjustments were made for eGFR, sex, and dosing (30). However, the age demographics in Gu et al.'s study were younger and had a narrower range than ours, with a median of 62.5 years and an interquartile range of 48.9–73.2 years.

The difference in results between our study and Gu et al.'s is probably explained by the physiological changes related to aging might be more uniform among participants with narrower age range. Specifically, participants aged 40–70 years might experience similar

rates of metabolism changes, renal function decline, or changes in body composition that impact drug pharmacokinetics. Conversely, our study's wider age range captures a broader variety of (patho)physiological states and complex interactions with other conditions not directly assessed in our study, potentially diluting the observable effect of age. For example, conditions more common in middle-old and oldest-old age group like malnutrition and comorbidities might impact vancomycin pharmacokinetics. A decrease in serum albumin levels due to malnutrition may lead to an increased volume of distribution (31), whereas comorbid conditions like heart failure may reduce it (32). Thus, the complex interplay between these age-related factors could complicate the relationship between age and vancomycin exposure.

Another common age-related condition might affect vancomycin exposure is diabetes mellitus. It was shown in a previous population pharmacokinetics study that comorbidity with diabetes significantly decreased intercompartmental clearance (33). This decrease is likely due to diabetic microangiopathy, which impairs drug distribution into tissues, resulting in higher drug concentrations. However, after adjusting for other covariates, our study found that diabetes was not a significant predictor of higher vancomycin concentrations.

While the maintenance dose according to our hospital guideline is based on $eGFR_{CKD-EPI}$, our findings underscore its persistent significance as a key predictor of vancomycin exposure. This influence on supratherapeutic concentrations persisted even after accounting for dose appropriateness (i.e. lower, appropriate and higher dose) and remained evident when dose-normalized concentration was established as the outcome. Further investigation is needed to review $eGFR_{CKD-EPI}$ categories in dosing guideline specifically for older adults. In this age group, the reliability of $eGFR_{CKD-EPI}$ calculations, which are predominantly creatinine-based, becomes questionable as low serum creatinine levels in this age group may not necessarily signify normal kidney function. Instead, it could reflect reduced muscle mass, commonly observed in older adults, thus potentially leading the $eGFR_{CKD-EPI}$ value to overestimate true kidney function (34-36). It is worth noting that numerous studies have concluded a more robust association between cystatin C and vancomycin concentrations as opposed to serum creatinine, suggesting a potential alternative for more predictive renal elimination (37). Despite its potential advantages, the

routine measurement of cystatin C in clinical practice is not common and is expensive, which limits its utility in optimizing vancomycin dosing strategies.

The findings from this study provide insights into real-world variations in vancomycin concentrations between older and younger patients, even though there is a recent shift in vancomycin TDM from monitoring trough concentrations to focusing on AUC_{24h}/MIC (11). Using trough concentrations to guide vancomycin dose can cause an underestimation of the true AUC_{24h} by nearly 25% (38). The correlation between trough levels and AUCs in older adults remains undefined. Alternative targets may be needed because of physiological changes and complex characteristics that can influence vancomycin pharmacokinetics in this patient group. Confirming this hypothesis by measuring AUC_{24h} has been challenging due to repeated blood sampling which is labor-intensive. A more viable alternative is to employ population pharmacokinetic (popPK) models for prediction and incorporating those models in a Bayesian software tool. Additionally, popPK modeling and simulation should be used to refine vancomycin dosing stratification based on $eGFR_{CKD-EPI}$ value and confirm our finding regarding the $eGFR_{CKD-EPI}$ cut-off value to predict supratherapeutic concentrations.

Although we conducted a large retrospective analysis which resulted in valuable insights, the findings should be considered within the context of the following limitations. Conducted within a single center, the retrospective nature of the study introduces potential biases and may overlook unmeasured confounding factors, thereby affecting the robustness and validity of the findings. Moreover, in this study, clinical outcomes including nephrotoxicity were not evaluated as our study was not designed or powered to detect such associations, although we observed higher exposure in older patients. Despite this, focusing on vancomycin exposures provides a practical approach to understanding its pharmacokinetics in older populations, where such data are less well characterized. Per current guidelines, this approach indirectly indicates potential clinical impact. Overall, further studies are needed, especially prospective external validation to confirm our findings.

5. Conclusion

The primary determinant of vancomycin concentration appears to be $eGFR_{CKD-EPI}$, confounding the impact of age on vancomycin exposure. This also suggests that relying solely on age may not accurately predict vancomycin exposure. Clinicians should consider

comprehensive assessments of renal function when prescribing vancomycin to the older adults who might also experience greater benefits from vancomycin TDM. Further research is needed to prospectively validate these observations.

References

1. Goto T, Yoshida K, Tsugawa Y, Camargo CA, Hasegawa K. Infectious Disease-Related Emergency Department Visits of Elderly Adults in the United States, 2011-2012. *Journal of the American Geriatrics Society*. 2016;64(1):31-6.
2. Buurman BM, Hoogerduijn JG, De Haan RJ, Abu-Hanna A, Lagaay AM, Verhaar HJ, et al. Geriatric Conditions in Acutely Hospitalized Older Patients: Prevalence and One-Year Survival and Functional Decline. *PLoS ONE*. 2011;6(11):e26951.
3. Ukkonen M, Jämsen E, Zeitlin R, Pauniah S-L. Emergency department visits in older patients: a population-based survey. *BMC Emergency Medicine*. 2019;19(1).
4. Gjesten MT, Brønnick K, Testad I. Characteristics and predictors for hospitalizations of home-dwelling older persons receiving community care: a cohort study from Norway. *BMC Geriatrics*. 2018;18(1).
5. Kurtaran B, Kusu F, Korkmaz P, Ozdemir B, Inan D, Oztoprak N, et al. A snapshot of geriatric infections in Turkey: ratio of geriatric inpatients in hospitals and evaluation of their infectious diseases: A multicenter point prevalence study. *International Journal of Infectious Diseases*. 2020;100:337-42.
6. Solis-Hernandez PS, Vidales-Reyes M, Garza-Gonzalez E, Guajardo-Alvarez G, Chavez-Moreno S, Camacho-Ortiz A. Hospital-Acquired Infections in Elderly Versus Younger Patients in an Acute Care Hospital. *International Journal of Infection*. 2015;3(1).
7. Tacconelli E, Pop-Vicas AE, D'Agata EM. Increased mortality among elderly patients with methicillin-resistant *Staphylococcus aureus* bacteraemia. *J Hosp Infect*. 2006;64(3):251-6.
8. Pomorska-Wesołowska M, Róžańska A, Natkaniec J, Gryglewska B, Szczypa A, Dzikowska M, et al. Longevity and gender as the risk factors of methicillin-resistant *Staphylococcus aureus* infections in southern Poland. *BMC Geriatrics*. 2017;17(1).
9. McClelland RS, Fowler VG, Sanders LL, Gottlieb G, Kong LK, Sexton DJ, et al. *Staphylococcus aureus* Bacteremia Among Elderly vs Younger Adult Patients. *Archives of Internal Medicine*. 1999;159(11):1244.
10. Liu C, Bayer A, Cosgrove SE, Daum RS, Fridkin SK, Gorwitz RJ, et al. Clinical practice guidelines by the infectious diseases society of america for the treatment of methicillin-resistant *Staphylococcus aureus* infections in adults and children: executive summary. *Clin Infect Dis*. 2011;52(3):285-92.
11. Rybak MJ, Le J, Lodise TP, Levine DP, Bradley JS, Liu C, et al. Therapeutic monitoring of vancomycin for serious methicillin-resistant *Staphylococcus aureus* infections: A revised consensus guideline and review by the American Society of Health-System Pharmacists, the Infectious Diseases Society of America, the Pediatric Infectious Diseases Society, and the Society of Infectious Diseases Pharmacists. *Am J Health Syst Pharm*. 2020;77(11):835-64.
12. Kufel WD, Seabury RW, Mogle BT, Beccari MV, Probst LA, Steele JM. Readiness to implement vancomycin monitoring based on area under the concentration-time curve: A cross-sectional survey of a national health consortium. *Am J Health Syst Pharm*. 2019;76(12):889-94.
13. Williams PG, Tabah A, Cotta MO, Sandaradura I, Kanji S, Scheetz MH, et al. International survey of antibiotic dosing and monitoring in adult intensive care units. *Critical Care*. 2023;27(1).
14. Flannery AH, Hammond DA, Oyler DR, Li C, Wong A, Smith AP, et al. Vancomycin Dosing Practices among Critical Care Pharmacists: A Survey of Society of Critical Care Medicine Pharmacists. *Infectious Diseases: Research and Treatment*. 2020;13:117863372095207.
15. Cutler NR, Narang PK, Lesko LJ, Ninos M, Power M. Vancomycin disposition: the importance of age. *Clin Pharmacol Ther*. 1984;36(6):803-10.

16. Kan W-C, Chen Y-C, Wu V-C, Shiao C-C. Vancomycin-Associated Acute Kidney Injury: A Narrative Review from Pathophysiology to Clinical Application. *International Journal of Molecular Sciences*. 2022;23(4):2052.
17. Saugel B, Gramm C, Wagner JY, Messer M, Lahmer T, Meidert AS, et al. Evaluation of a dosing regimen for continuous vancomycin infusion in critically ill patients: an observational study in intensive care unit patients. *J Crit Care*. 2014;29(3):351-5.
18. Bosso JA, Nappi J, Rudisill C, Wellein M, Bookstaver PB, Swindler J, et al. Relationship between vancomycin trough concentrations and nephrotoxicity: a prospective multicenter trial. *Antimicrob Agents Chemother*. 2011;55(12):5475-9.
19. Pottel H, Hoste L, Dubourg L, Ebert N, Schaeffner E, Eriksen BO, et al. An estimated glomerular filtration rate equation for the full age spectrum. *Nephrol Dial Transplant*. 2016;31(5):798-806.
20. Ma Y, Shen X, Yong Z, Wei L, Zhao W. Comparison of glomerular filtration rate estimating equations in older adults: A systematic review and meta-analysis. *Arch Gerontol Geriatr*. 2023;114:105107.
21. Robin X, Turck N, Hainard A, Tiberti N, Lisacek F, Sanchez J-C, et al. pROC: an open-source package for R and S+ to analyze and compare ROC curves. *BMC Bioinformatics*. 2011;12(1):77.
22. O'Brien RM. A caution regarding rules of thumb for variance inflation factors. *Qual Quant*. 2007;41(5):673-90.
23. Team R. RStudio: Integrated Development for R. Boston, MA: RStudio, PBC; 2020.
24. Team RC. R: A language and environment for statistical computing. Vienna, Austria: R Foundation for Statistical Computing; 2021.
25. Calcagno V, de Mazancourt C. glmulti: An R Package for Easy Automated Model Selection with (Generalized) Linear Models. *Journal of Statistical Software*. 2010;34(12):1 - 29.
26. Donini LM, Busetto L, Bischoff SC, Cederholm T, Ballesteros-Pomar MD, Batsis JA, et al. Definition and Diagnostic Criteria for Sarcopenic Obesity: ESPEN and EASO Consensus Statement. *Obes Facts*. 2022;15(3):321-35.
27. Olde Rikkert MG, Deurenberg P, Jansen RW, van't Hof MA, Hoefnagels WH. Validation of multi-frequency bioelectrical impedance analysis in detecting changes in fluid balance of geriatric patients. *J Am Geriatr Soc*. 1997;45(11):1345-51.
28. Yahav D, Abbas M, Nassar L, Ghrayeb A, Shepshelovich D, Kurnik D, et al. Attention to age: similar dosing regimens lead to different vancomycin levels among older and younger patients. *Age Ageing*. 2019;49(1):26-31.
29. Pan Y, He X, Yao X, Yang X, Wang F, Ding X, et al. The effect of body mass index and creatinine clearance on serum trough concentration of vancomycin in adult patients. *BMC Infect Dis*. 2020;20(1):341.
30. Gu Q, Jones N, Drennan P, Peto TE, Walker AS, Eyre DW. Assessment of an institutional guideline for vancomycin dosing and identification of predictive factors associated with dose and drug trough levels. *J Infect*. 2022;85(4):382-9.
31. Barber KE, Bell AM, Stover KR, Wagner JL. Intravenous Vancomycin Dosing in the Elderly: A Focus on Clinical Issues and Practical Application. *Drugs Aging*. 2016;33(12):845-54.
32. Shammass FV, Dickstein K. Clinical pharmacokinetics in heart failure. An updated review. *Clin Pharmacokinet*. 1988;15(2):94-113.
33. Mangin O, Urien S, Mainardi JL, Fagon JY, Faisy C. Vancomycin pharmacokinetic and pharmacodynamic models for critically ill patients with post-sternotomy mediastinitis. *Clin Pharmacokinet*. 2014;53(9):849-61.

34. Tap L, Boye NDA, Hartholt KA, van der Cammen TJM, Mattace-Raso FUS. Association of estimated glomerular filtration rate with muscle function in older persons who have fallen. *Age Ageing*. 2018;47(2):269-74.
35. Lees JS, Shlipak MG. Age-adapted eGFR thresholds underestimate the risks beyond kidney failure associated with CKD in older populations. *J Am Geriatr Soc*. 2022;70(12):3660-4.
36. Nankivell BJ, Nankivell LFJ, Elder GJ, Gruenewald SM. How unmeasured muscle mass affects estimated GFR and diagnostic inaccuracy. *EClinicalMedicine*. 2020;29-30:100662.
37. Barreto EF, Rule AD, Murad MH, Kashani KB, Lieske JC, Erwin PJ, et al. Prediction of the Renal Elimination of Drugs With Cystatin C vs Creatinine: A Systematic Review. *Mayo Clin Proc*. 2019;94(3):500-14.
38. Neely MN, Youn G, Jones B, Jelliffe RW, Drusano GL, Rodvold KA, et al. Are vancomycin trough concentrations adequate for optimal dosing? *Antimicrob Agents Chemother*. 2014;58(1):309-16.

Appendix

Supplementary Table 1. Vancomycin local dosing guideline

	Intermittent infusion				Continuous infusion		
	Renal function (ml/min/1.73 m ²)	Dose	Duration	Restriction	Dose	Duration	Restriction
Loading dose	Not dependent on renal function	25 mg/kg, max. 2500 mg	120 min	Max. 40 mg/ml for central administration	25 mg/kg, max. 2500 mg	120 min	Only central administration, maximal concentration 40 mg/ml
	Intermittent hemodialysis (administration during the last hour of dialysis)	15 mg/kg	60-90 min	Max. 5 mg/ml for peripheral administration	15 mg/kg	60-90 min	
Maintenance dose	eGFR _{CKD-EPI} > 100	2 x 1,25 g/ 24 hour	90 min		2.5 g / 24 hour	24 hour	
	eGFR _{CKD-EPI} 50-100	2 x 1 g /24 hour	60-90 min		2 g /24 hour	24 hour	
	eGFR _{CKD-EPI} 10-50	2 x 500 mg /24 hour	60 min		1 g / 24 hour	24 hour	
	Intermittent hemodialysis (administration during the last hour of dialysis)	Based on level 48-68 hours after first dose					

Supplementary Table 2. Uni- and multivariable analyses to assess associations between patient characteristics and the risk of subtherapeutic vancomycin concentrations

Variables		Univariable analysis			Multivariable analysis with age as continuous variable			Multivariable analysis with age cut-off 75 years			Multivariable analysis, age group <65, 65–75, >75 years		
		p-value (*<0.05)	OR	95%CI	p-value (*<0.01)	OR	99%CI	p-value (*<0.01)	OR	99%CI	p-value (*<0.01)	OR	99%CI
Sex	Male	Reference											
	Female	<0.001*	0.592	0.495–0.708	0.027	0.716	0.484–1.058	0.036	0.730	0.496–1.075	0.033	0.726	0.492–1.069
Baseline age		<0.001*	0.970	0.965–0.976	0.149	0.994	0.983–1.005	-	-	-	-	-	-
Older than 75 years	Yes		0.428	0.345–0.532	-	-	-	0.020	0.723	0.505–1.035	-	-	-
	No	Reference											
Age group	< 65 years	Reference											
	65–75 years	<0.001*	0.556	0.455–0.68	-	-	-	-	-	-	0.013	0.675	0.449–1.015
	> 75 years	<0.001*	0.331	0.261–0.421	-	-	-	-	-	-	0.560	0.927	0.661–1.298
Diabetes	Yes	<0.001*	0.637	0.515–0.787	0.568	0.926	0.655–1.309	0.596	0.931	0.659–1.316	0.610	0.934	0.660–1.321
	No	Reference											
Weight		<0.001*	1.010	1.005–1.015	0.012	1.009	0.999–1.018	0.018	1.008	0.999–1.018	0.018	1.008	0.999–1.018
Height		<0.001*	1.040	1.031–1.05	0.092	1.014	0.993–1.036	0.064	1.016	0.994–1.038	0.077	1.015	0.993–1.037
BMI		0.427	1.006	0.991–1.022	-	-	-	-	-	-	-	-	-

Critically ill	Yes	0.644	0.95 8	0.798–1.149	-	-	-	-	-	-	-	-	-
	No	Reference											
sCr		<0.001*	0.38 6	0.324–0.459	ns	-	-	ns	-	-	ns	-	-
eGFR CKD-EPI		<0.001*	1.02 8	1.025–1.031	<0.001*	1.025	1.020–1.031	<0.001*	1.025	1.020–1.031	<0.001*	1.025	1.019–1.030
Blood urea nitrogen		<0.001*	0.98 5	0.982–0.988	ns	-	-	ns	-	-	ns	-	-
Albumin		0.009*	1.01 8	1.004–1.032	0.463	1.00 6	0.984–1.029	0.490	1.006	0.984–1.028	0.503	1.00 6	0.984–1.028
Total protein		0.191	1.00 5	0.996–1.015	-	-	-	-	-	-	-	-	-
Before/after the implementation of the new guideline	Before	0.019	1.23 0	1.035–1.463	-	-	-	-	-	-	-	-	-
	After	Reference											
Loading dose	Appropriate dose	Reference											
	No loading dose	0.054	1.26 9	0.995–1.618	0.008*	1.520	1.010–2.287	0.008*	1.521	1.011–2.286	0.008*	1.518	1.009–2.283
	Higher dose	0.208	0.73 3	0.452–1.189	0.189	0.66 0	0.292–1.490	0.197	0.666	0.295–1.500	0.197	0.66 5	0.295–1.501
	Lower dose	<0.001*	1.471	1.172–1.846	0.004*	1.510	1.041–2.192	0.004*	1.520	1.047–2.205	0.004*	1.526	1.051–2.214
Maintenance dose	Appropriate dose	Reference											
	Higher dose	0.010	0.77 2	0.633–0.941	0.073	0.79 2	0.566–1.107	0.075	0.794	0.569–1.108	0.067	0.78 9	0.564–1.102
	Lower dose	<0.001*	2.05 7	1.589–2.662	<0.001*	2.125	1.406–3.213	<0.001*	2.158	1.426–3.266	<0.001*	2.151	1.421–3.255
Time to TDM		0.271	0.99 6	0.99–1.003	-	-	-	-	-	-	-	-	-

OR: Odds Ratio, BMI: Body Mass Index, sCr: serum creatinine, eGFR: estimated glomerular filtration rate calculated by using the Chronic Kidney Disease Epidemiology Collaboration equation, CVVH: Continuous Veno-Venous Hemofiltration, TDM: Therapeutic Drug Monitoring

Supplementary table 3. Variance Inflation Factors for Predictors in the Subtherapeutic Multivariable Regression Model

Variables	Variance Inflation Factors
Sex	1.3
Age	1.1
Weight	1.2
Height	1.5
eGFR _{CKD-EPI}	1.1
Diabetes	1.0
Albumin	1.0
Loading dose	1.0
Maintenance dose	1.0

Supplementary Table 4. Uni- and multivariable analyses to assess associations between patient characteristics and the risk of supratherapeutic vancomycin concentrations

Variables		Univariable analysis			Multivariable analysis using age as continuous variable			Multivariable analysis using age cut-off 75 years and age group <65, 65–75, >75 years ^a		
		p-value (*<0.05)	OR	95%CI	p-value (*<0.01)	OR	99%CI	p-value (*<0.01)	OR	99%CI
Sex	Female	<0.001*	1.780	1.447–2.189	0.169	1.263	0.815–1.956	0.056	1.382	0.894–2.136
	Male	Reference								
Baseline age		<0.001*	1.015	1.008–1.022	0.013	0.988	0.976–1.000	-	-	-
Older than 75 years	Yes	<0.001*	1.669	1.321–2.108	-	-	-	ns	-	-
	No	Reference								
Age group		Reference								
	65–75 years	0.036*	1.294	1.017–1.646	-	-	-	ns	-	-
	> 75 years	<0.001*	1.899	1.465–2.462	-	-	-	ns	-	-
Diabetes	Yes	0.017*	1.341	1.054–1.707	0.906	0.983	0.671–1.438	0.787	0.961	0.654–1.4
	No	Reference								
Weight		<0.001*	0.989	0.983–0.995	0.127	0.994	0.983–1.004	0.092	0.993	0.983–1.004
Height		<0.001*	0.959	0.948–0.970	*0.004	0.973	0.949–0.997	0.014	0.977	0.953–1.001
BMI		0.563	0.995	0.976–1.013	-	-	-	-	-	-
Critically ill	Yes	0.332	1.112	0.897–1.378	ns	-	-	ns	-	-
	No	Reference								
sCr		<0.001*	2.145	1.863–2.47	-	-	-	-	-	-
eGFR _{CKD-EPI}		<0.001*	0.975	0.972–0.978	<0.001*	0.975	0.970–0.981	<0.001*	0.979	0.973–0.986
Blood urea nitrogen		<0.001*	1.014	1.011–1.017	ns	-	-	0.117	1.003	0.998–1.008

Serum albumin		0.001*	0.972	0.956–0.988	0.030	0.978	0.953–1.004	0.08	0.983	0.957–1.008
Total protein		0.522	0.996	0.986–1.007	ns	-	-	ns	-	-
Before/after the implementation of the new guideline	Before	0.036*	0.802	0.654–0.985	ns	-	-	ns	-	-
	After	Reference								
Loading dose	Appropriate dose	Reference								
	No loading dose	0.011*	0.696	0.526–0.921	0.012	0.641	0.408–1.009	0.011	0.641	0.407–1.008
	Higher dose	0.138	1.454	0.887–2.383	0.153	1.568	0.696–3.533	0.199	1.501	0.651–3.361
	Lower dose	<0.001*	0.643	0.495–0.835	0.002*	0.609	0.401–0.925	0.001*	0.593	0.391–0.901
Maintenance dose	Appropriate dose	Reference								
	Higher dose	0.014	1.335	1.060–1.682	0.179	1.214	0.837–1.762	0.163	1.223	0.845–1.778
	Lower dose	<0.001*	0.511	0.363–0.720	<0.001*	0.465	0.270–0.798	<0.001*	0.452	0.259–0.767
Time to TDM		0.099	1.006	0.999–1.014	-	-	-	-	-	-

OR: Odds Ratio, BMI: Body Mass Index, sCr: serum creatinine, eGFR_{CKD-EPI}: estimated glomerular filtration rate calculated by using the Chronic Kidney Disease Epidemiology Collaboration equation, CVVH: Continuous Veno-Venous Hemofiltration, TDM: Therapeutic Drug Monitoring.

^aTwo multivariable analyses using the age as binary variable with cut-off 75 years and age group <65, 65–75, >75 years resulted in the identical models as these age cut-offs were not retained as significant covariates in the respective models.

Supplementary table 5. Variance Inflation Factors for Predictors in the Supratherapeutic Multivariable Regression Model

Variables	Variance inflation factor
Sex	1.3
Weight	1.1
Height	1.4
Diabetes	1.0
eGFR _{CKD-EPI}	1.3
Albumin	1.0
Blood urea nitrogen	1.3

Loading dose	1.0
Maintenance dose	1.0

Supplementary Table 6. Uni- and multivariable analyses to assess the association between patient characteristics and dose-normalized concentrations

Variables		Univariable analysis			Multivariable analysis using age as continuous variable			Multivariable analysis using age cut-off 75 years and age group <65, 65–75, >75 years ^a		
		p-value (*<0.05)	Beta coefficient	95%CI	p-value (*<0.01)	Beta coefficient	99%CI	p-value (*<0.01)	Beta coefficient	99%CI
Sex	Female	<0.001*	1.438	1.091–1.785	ns	-	-	ns	-	-
	Male	Reference								
Baseline age		<0.001*	0.059	0.049–0.07	0.068	-0.011	-0.026–0.004	-	-	-
Older than 75 years	Yes	<0.001*	1.815	1.406–2.224	-	-	-	ns	-	-
	No	Reference								
Age group	< 65 years	Reference								
	65–75 years	<0.001*	2.320	1.876–2.764				ns	-	-
	> 75 years	<0.001*	1.079	0.694–1.465				ns	-	-
Diabetes	Yes	<0.001*	0.904	0.487–1.322	ns	-	-	0.761	-0.056	-0.533–0.421
	No	Reference								
Weight		<0.001*	-0.034	-0.044– -0.025	<0.001*	-0.039	-0.05– -0.028	<0.001*	-0.039	-0.05– -0.027
Height		<0.001*	-0.105	-0.123– -0.087	-	-	-	-	-	-
BMI		0.009*	-0.032	-0.063– -0.001	-	-	-	-	-	-
Critically ill	Yes	0.089	0.315	-0.048–0.678	ns	-	-	ns	-	-
	No	Reference								
sCr		<0.001*	1.890	1.708–2.071	-	-	-	-	-	-
eGFR _{CKD-EPI}		<0.001*	-0.061	-0.066– -0.057	<0.001*	-0.054	-0.062– -0.045	<0.001*	-0.051	-0.059– -0.043

Blood Urea Nitrogen	<0.001*	0.041	0.037–0.045	<0.001*	0.018	0.011–0.025	<0.001*	0.019	0.012–0.026
Albumin	<0.001*	-0.057	-0.084– -0.029	0.074	-0.022	-0.053–0.01	0.130	-0.018	-0.049–0.013
Total protein	0.829	0.002	-0.017–0.021	ns	-	-	ns	-	-
Time to TDM	<0.001*	0.063	0.05–0.075	<0.001*	0.078	0.063–0.093	<0.001*	0.077	0.062–0.092

BMI: Body Mass Index, sCr: serum creatinine, eGFR: estimated glomerular filtration rate calculated by using the Chronic Kidney Disease Epidemiology Collaboration equation, CVVH: Continuous Veno-Venous Hemofiltration, TDM: Therapeutic Drug Monitoring

^aTwo multivariable analyses using the age as binary variable with cut-off 75 years and age group <65, 65–75, >75 years resulted in the identical models as these age cut-offs were not retained as significant covariates in the respective models.

Supplementary Table 7. Variance Inflation Factors for Predictors in the Dose-normalized Concentrations Multivariable Regression Model

Variables	Variance inflation factor
Weight	1.1
Diabetes	1.1
eGFR _{CKD-EPI}	1.7
Albumin	1.0
Blood urea nitrogen	1.7
Time to TDM	1.0

Supplementary table 8. Comparison of the Akaike Information Criteria between multivariable models for supratherapeutic concentration incorporating three different equations for kidney function predictor

Model	Variables	p-value	OR	CI	Objective function value (-2 Log Likelihood)
eGFR-BIS	Objective function value (-2 Log Likelihood)				1612.757
	eGFR-BIS	<0.001*	0.991	0.986–0.995	
	Albumin	0.093	0.984	0.959–1.009	
	Loading dose	No loading dose	0.014	0.653	0.418–1.02
		Higher dose	0.32	1.368	0.595–3.053
		Lower dose	0.002*	0.611	0.405–0.922
	Maintenance dose	Higher dose	0.083	1.28	0.889–1.851
		Lower dose	<0.001*	0.423	0.244–0.714
	Diabetes	Yes (No diabetes is reference)	0.917	1.015	0.695–1.471
	Sex	Female (male is reference)	0.015	1.502	0.977–2.312
	Height		0.011	0.976	0.953–1
	Blood urea nitrogen		<0.001*	1.009	1.004–1.014
	Weight		0.059	0.992	0.982–1.003
eGFR-FAS	Objective function value (-2 Log Likelihood)				1584.29
	eGFR-FAS	<0.001*	0.984	0.979–0.99	
	Albumin	0.043	0.98	0.955–1.005	
	Loading dose	No loading dose	0.01	0.637	0.406–1
		Higher dose	0.276	1.416	0.611–3.199
		Lower dose	0.001*	0.593	0.392–0.899
	Maintenance dose	Higher dose	0.108	1.259	0.872–1.827
		Lower dose	<0.001*	0.457	0.262–0.776
	Diabetes	Yes (No diabetes is reference)	0.911	0.984	0.672–1.429
	Sex	Female (male is reference)	0.025	1.46	0.946–2.255
	Height		0.023	0.979	0.955–1.003

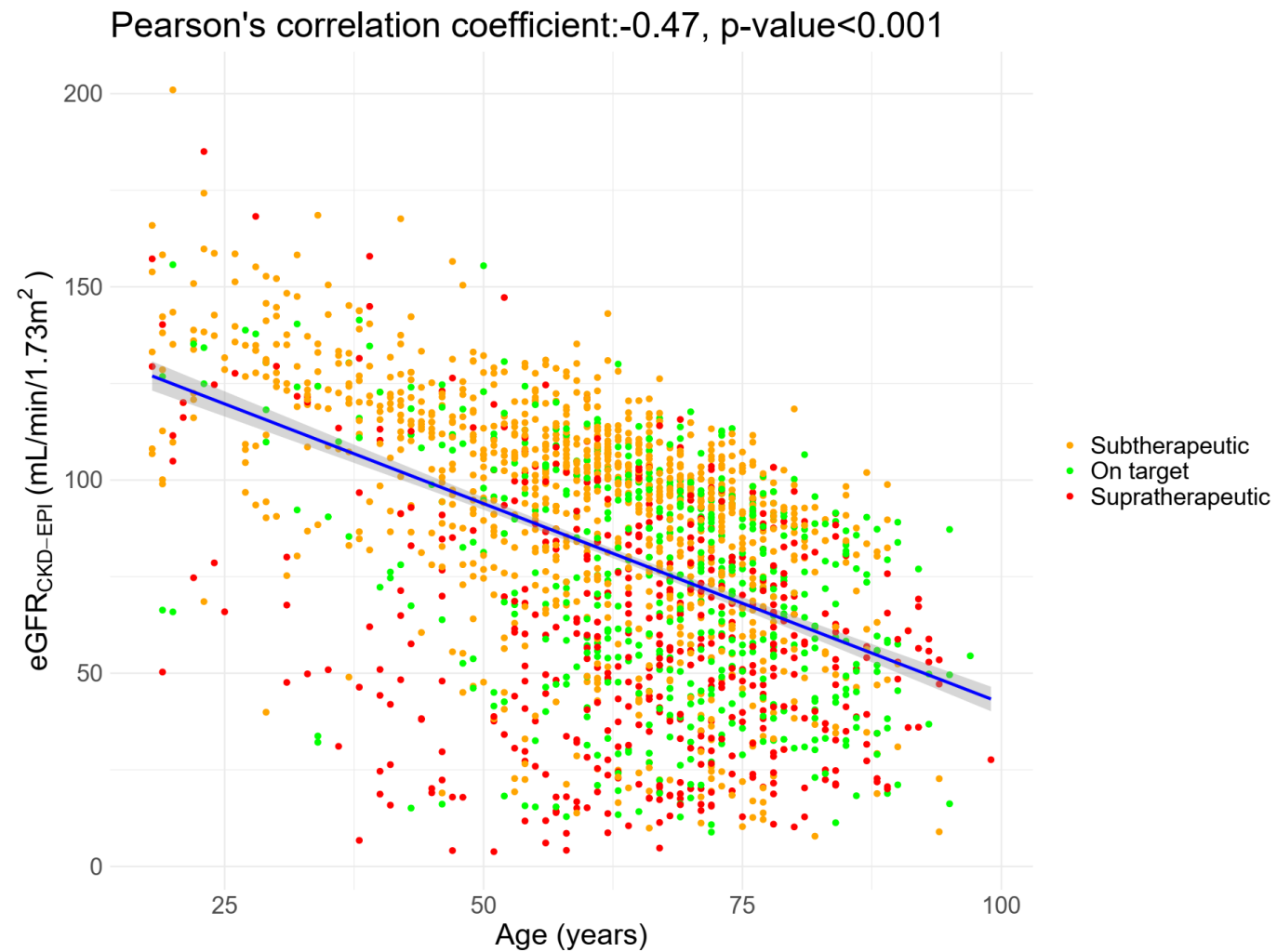
		Blood urea nitrogen	0.003*	1.006	1.001–1.01	
		Weight	0.051	0.992	0.982–1.002	
eGFR-CKD EPI	Objective function value (-2 Log Likelihood)					1579.249
		eGFR-CKD EPI	<0.001*	0.979	0.973–0.986	
		Albumin	0.08	0.983	0.957–1.008	
		Loading dose				
		No loading dose	0.011	0.641	0.407–1.008	
		Higher dose	0.199	1.501	0.651–3.361	
		Lower dose	0.001*	0.593	0.391–0.901	
		Maintenance dose				
		Higher dose	0.163	1.223	0.845–1.778	
		Lower dose	<0.001*	0.452	0.259–0.767	
	Diabetes	Yes (No diabetes is reference)	0.787	0.961	0.654–1.4	
	Sex	Female (male is reference)	0.056	1.382	0.894–2.136	
	Height		0.014	0.977	0.953–1.001	
	Blood urea nitrogen		0.117	1.003	0.998–1.008	
	Weight		0.092	0.993	0.983–1.004	

Supplementary table 9. Comparison of the Akaike Information Criteria between multivariable models for dose corrected concentration incorporating three different equations for kidney function predictor

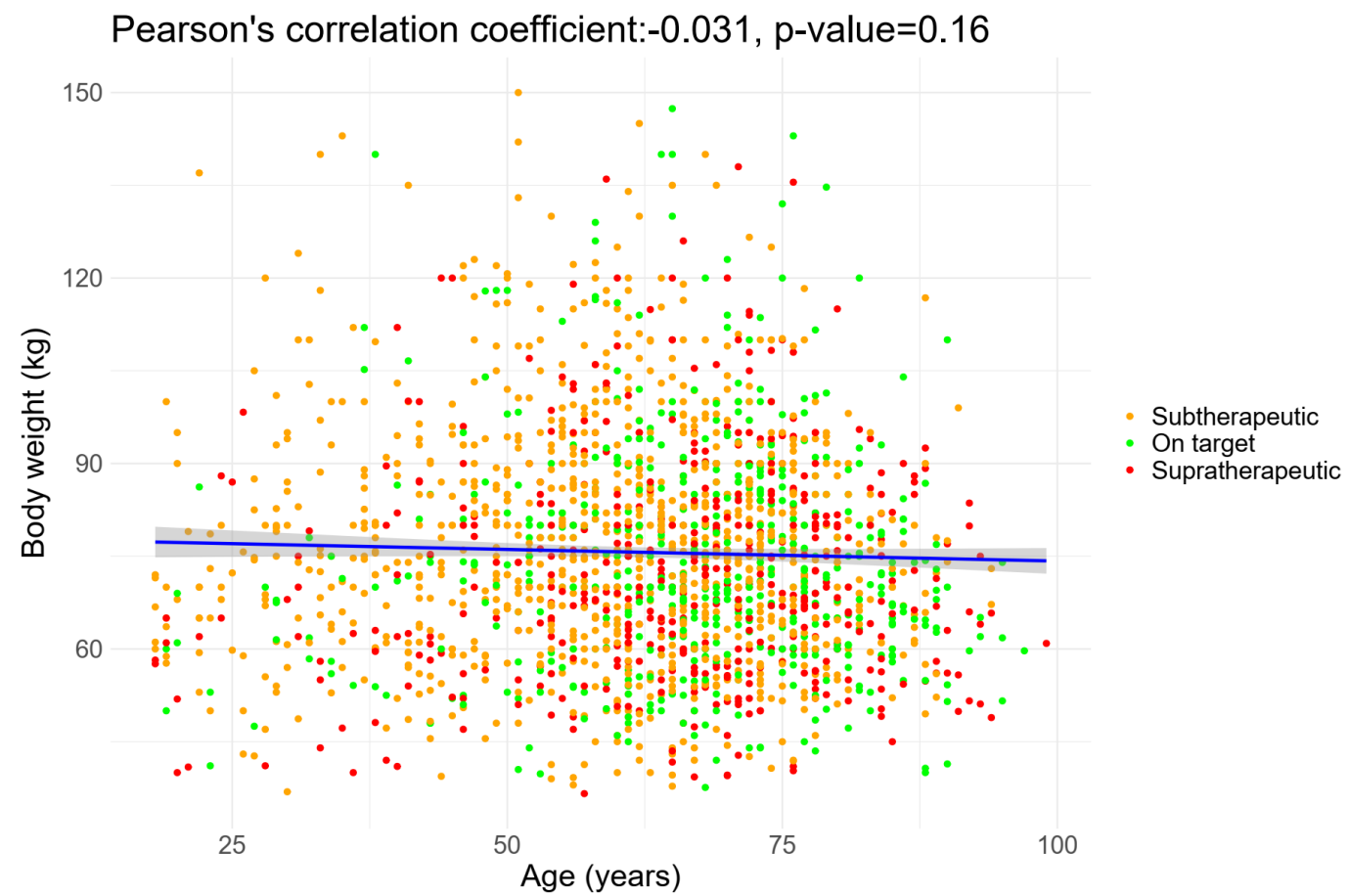
Model	Variables	p-value	Beta coefficients	CI	Objective function value (-2 Log Likelihood)
eGFR-BIS	Objective function value (-2 Log Likelihood)				9370.05
	eGFR-BIS	<0.001*	-0.015	-0.019– -0.011	
	Albumin	0.181	-0.017	-0.049–0.016	
	Diabetes	yes (no diabetes as reference)	0.513	0.127	-0.373–0.627
	Time to TDM	<0.001*	0.077	0.061–0.093	
	Blood urea nitrogen	<0.001*	0.037	0.03–0.043	
	Weight	<0.001*	-0.042	-0.054– -0.03	
eGFR-FAS	Objective function value (-2 Log Likelihood)				9312.395
	eGFR-FAS	<0.001*	-0.024	-0.029– -0.019	
	Albumin	0.039	-0.026	-0.057–0.006	
	Diabetes	yes (no diabetes as reference)	0.677	0.079	-0.412–0.571
	Time to TDM	<0.001*	0.076	0.061–0.092	
	Blood urea nitrogen	<0.001*	0.031	0.025–0.038	
	Weight	<0.001*	-0.042	-0.053– -0.03	
eGFR-CKDEPI	Objective function value (-2 Log Likelihood)				9295.036
	eGFR-CKD EPI	<0.001*	-0.051	-0.059– -0.043	
	Albumin	0.13	-0.018	-0.049–0.013	
	Diabetes	yes (no diabetes as reference)	0.761	-0.056	-0.533–0.421
	Time to TDM	<0.001*	0.077	0.062–0.092	
	Blood urea nitrogen	<0.001*	0.019	0.012–0.026	

Weight	<0.001*	-0.039	-0.05	-0.027
--------	---------	--------	-------	--------

Supplementary figure 1. The correlation between age and eGFR CKD-EPI



Supplementary figure 2. The correlation between age and eGFR CKD-EPI



Chapter 5: Optimizing vancomycin dosing in older patients: predictive performance of pharmacometrics models

This chapter is based on:

Edwina AE*, Dia N*, Gijsen M, Van der Linden L, Spriet I, Tournoy J†, Dreesen E†.

Optimizing vancomycin dosing in older patients: predictive performance of pharmacometrics models. In preparation

*shared first authorship, †shared last authorship

Abstract

Objective: Model-informed precision dosing is increasingly used to guide vancomycin dosing. However, the suitability of published population pharmacokinetics (popPK) models for older patients has not been evaluated. We aimed to assess the predictive performance of these models in the older patient population.

Methods: We encoded nine published popPK models of vancomycin in NONMEM 7.5. We used retrospectively collected data from 255 hospitalized older patients aged 75 years and above who were on vancomycin treatment and contributed three therapeutic drug monitoring (TDM) concentrations. We evaluated the *a priori* (solely based on covariates) and *a posteriori* (based on covariates and vancomycin concentrations) predictive performance (accuracy and precision) of single-model and multi-model selection/averaging algorithms.

Results: *A priori* predictions varied largely (relative bias; rBias -37%–45%, relative root mean squared error; rRMSE 47%–78%). *A posteriori* prediction using the most recent TDM sample (time between TDM sample and predicted occasion: median 36.0 hours, IQR 24.0–48.6 hours) provided the best predictive performance (rBias -18%–24%, rRMSE 25%–45%) with the models of Thomson *et al.* and Ji *et al.* (age median 78 years) performing best. The multi-model algorithm did not improve performance. Adding an earlier sample (T-2) to T-1 resulted in only slight improvement or even worsens performance. Models with less accurate and precise predictive performance produced highly variable and skewed AUC_{0-24h} predictions.

Conclusion: PopPK models developed specifically for older populations and with high interindividual variability, were clinically suitable for guiding vancomycin dosing in older patients, especially when paired with Bayesian forecasting using recent concentration data. The Thomson model showed the best performance, followed by the Ji model, both of which provided more reliable AUC_{0-24h} predictions.

1. Introduction

Infectious diseases are one of the leading causes of hospitalization among older adults (1-5). A particularly concerning pathogen in this regard is methicillin-resistant *Staphylococcus aureus* (MRSA), given its higher prevalence and lethality among older adults (6-8). The glycopeptide antibiotic vancomycin is often the preferred agent for treating MRSA and other severe gram-positive infections (9). Due to its narrow therapeutic window, guidelines recommend therapeutic drug monitoring (TDM) of vancomycin (10). The ratio of the 24-h area under the concentration–time curve (AUC_{24h}) over the minimal inhibitory concentration (MIC) of the infective organism (AUC_{24h}/MIC) of 400–600 mg·h/L has been shown to be the surrogate marker for vancomycin efficacy and safety. However, determining AUCs is laborious and financially demanding (11). As a result, many clinical settings continue to use single-point concentration measurements as a standard alternative, aiming for a target trough concentration of 15–20 mg/L for intermittent dosing and 20–25 mg/L for continuous infusion (12, 13).

Model-informed precision dosing (MIPD) has emerged as a promising strategy to anticipate personalized dosing strategies (14). Typically, MIPD uses population pharmacokinetics (popPK) models to streamline TDM-based dose optimization with fewer TDM samples (15). This means that vancomycin concentrations measured in TDM samples are used for Bayesian forecasting to deduce the maximum *a posteriori* PK parameter estimates of the patient. However, in the absence of TDM samples, patient covariates can be used to deduce *a priori* PK parameter estimates of the patient, making MIPD suitable for first-dosing determination (15).

We previously performed a systematic search and identified 86 popPK models of vancomycin developed in adult patient populations (16), but their predictive performance and thus suitability for MIPD of vancomycin in older patients remains unclear. Selecting the right popPK model for MIPD of vancomycin to older patients is challenging, yet besides manual selection, automated multi-model selection algorithms (MSAs) exist, as well as multi-model averaging algorithms (MAAs) (17, 18). These multi-model algorithms may offer a potential advantage by automating vancomycin MIPD for more accurate and precise predictions (17, 18).

With this study, we aimed to evaluate the predictive performance of both single-model and multi-model approaches using published popPK models to guide vancomycin dosing in older patients.

2. Methods

2.1. Clinical data

Vancomycin TDM data were collected retrospectively from hospitalized older patients aged 75 years and above admitted to any ward at the University Hospitals Leuven between October 2019 and May 2022 and treated with vancomycin infusions. Patients were excluded if they did not provide a minimum of three vancomycin plasma concentrations measured consecutively during one treatment course. Patients without complete vancomycin dosing records within 48 hours prior to the concentration measurement, and patients with vancomycin concentrations above the upper limit of quantification of 50 mg/L were excluded from analysis. The study was approved by the Ethics Committee Research of UZ / KU Leuven (S65213).

2.2. Candidate models

Nine popPK models were selected. Seven of these models were developed in distinct patient populations covering a broad range of settings, including critically ill, extremely obese, dialysis, sepsis, trauma, and post-cardiac surgery patients (19-25). Therefore, these vancomycin models were considered representative of the hospital-wide adult patient population investigated in our study. Additionally, two popPK models developed using data from older patients, with a mean or median age of 75 years or more (26, 27), were selected based on a systematic literature search to ensure adequate representation of older patients (16).

2.3. Model evaluations

The global fit of the clinical data to the candidate models was visually inspected with goodness-of-fit plots, including measured versus individual predicted concentrations and conditional weighted residual versus time plots. In addition, simulation-based evaluations were performed, including prediction-corrected visual predictive checks (pcVPCs) and normalized prediction distribution errors (NPDEs). A normal distribution of NPDEs $N(0, 1)$ was assessed using three tests: the Wilcoxon signed-rank test to compare the NPDE median to zero, the Fisher variance test to compare its variance to one, and the Shapiro–

Wilk test to evaluate whether the NPDEs followed a normal distribution. An adjusted p-value of all three tests (a global test) was calculated to identify the best fitting model (28)

2.4. Multi-model approaches

Two multi-model approaches were assessed using the nine candidate models: an MSA and an MAA (17). Both approaches aimed to predict the vancomycin concentration at relative time +1 (T_{+1}) using earlier vancomycin concentrations at relative times -1 (T_{-1}) and 0 (T_0). First, the individual PK parameters were estimated with each of the nine popPK models based on observed covariate data alone (*a priori* prediction) and covariate data in combination with vancomycin concentrations at T_{-1} and/or T_0 (*a posteriori* prediction). Next, the individually predicted concentration–time data was automatically weighted based on the maximum likelihood estimate (MLE) of each candidate model i in relation to the sum of MLEs of all n candidate models (Equation 1).

$$W_{MLE_i} = \frac{MLE_i}{\sum_{i=1}^n MLE_i} = \frac{e^{(-0.5 \times OFV_i)}}{\sum_{i=1}^n e^{(-0.5 \times OFV_i)}} \quad (1)$$

In the MSA, the model with the highest weight was selected, while the others were excluded. In the MAA, a combined prediction was generated from all models using Equation 2.

$$\text{Prediction} = \sum_{i=1}^n (\text{Prediction}_{model_i} \times W_{model_i}) \quad (2)$$

The theory behind these multi-model approaches was previously published by Uster *et al.* (17).

2.5. Predictive performance evaluations

For both *a priori* and *a posteriori* predictions, the predictive performance was evaluated from the differences between the predicted and the measured concentrations at T_{+1} (C_{+1}). Three *a posteriori* prediction settings were compared: Prediction with C_{T_0} only, $C_{T_{-1}}$ only, and both C_{T_0} and $C_{T_{-1}}$. The model-predicted versus measured C_{+1} in the different single-/multi-model approaches, *a priori/posteriori* prediction modalities, and evaluation settings were compared by calculating the relative bias (rBias; Equation 3) and the relative root mean square error (rRMSE; Equation 4) to determine accuracy and precision, respectively. An rBias between $\pm 20\%$ with a 95% confidence interval (CI) including zero was deemed clinically acceptable (17). No rRMSE threshold for clinical acceptability was prespecified. Lower rRMSE values indicated more precise predictions.

$$rBias = \frac{1}{n} \times \sum_{i=1}^n \left(\frac{predicted_i - observed_i}{observed_i} \right) \times 100\% \quad (3)$$

$$rRMSE = \sqrt{\frac{1}{n} \times \sum_{i=1}^n \frac{(predicted_i - observed_i)^2}{(observed_i)^2}} \times 100\% \quad (4)$$

with n the total number of patients, and i each individual patient.

2.6. Robustness analysis, sensitivity analysis and Bland–Altman analysis

The predictive performance and the computation time of the multi-model approaches was assessed with a subset of popPK models compared to the nine popPK models in a robustness analysis. The predictive performance of the single and the multi-model approaches in the sensitivity analysis was assessed by 1) changing the covariate data using a single imputation with the model's median covariate value when centered, and otherwise the clinical dataset's median, and 2) modifying the models by using flattened priors for all the pharmacokinetic parameters (29). The agreement between the predicted and measured C_{T1} values over the entire range of measured C_{T1} was evaluated in a Bland–Altman analysis. In addition, the analysis was conducted over the time intervals from C_{T-1} or C_{T0} to C_{T1} to determine if the difference between predicted and measured C_{+1} values changed over time after TDM (30).

2.7. Software

All models were coded in NONMEM (version 7.5; Icon plc) and are available in the *Supplementary Material*. Predictions were executed using NONMEM with a GNU Fortran 95 compiler. Dataset formatting, data analysis, and visualization were conducted in R (version 4.3.1, R Core Team, Vienna, Austria) in conjunction with RStudio (2023.06.1+524).

Table 14. Patient characteristics

Characteristic	Values
Number of patients	255
<i>Demographics</i>	
Age, years , median [IQR]	79 [77–84]
Sex, male, n (%)	147 (57.6)
Body weight, kg , median [IQR]	68.0 [60.8–80.9]
Height, cm , median [IQR]	167 [161–174]
Body mass index, kg/m² , median [IQR]	24.7 [22.4–28.6]
<i>Serology</i>	
Serum creatinine, mg/dL , median [IQR]	1.00 [0.78–1.44]
eCrCL _{CG} , mL/min , median [IQR]	54.0 [36.5–73.8]
Blood urea nitrogen, mg/dL , median [IQR]	49.0 [31.5–67.5]
Serum albumin, g/dL , median [IQR]	31.0 [27.2–34.8]
Total protein, g/dL , median [IQR]	62.0 [54.5–68.0]
<i>Comorbidities, comedication, dialysis, admission to the intensive care unit</i>	
Patients with diabetes*, n (%)	65 (25.5)
Comedication with furosemide, n (%)	166 (65.1)
Intensive care unit patients, n (%)	72 (28.2)
Patients with dialysis, n (%)	3 (1.2)
Patients with CVVH, n (%)	7 (2.7)

Abbreviations: CVVH, continuous veno-venous hemofiltration; eCrCL_{CG}, estimated creatinine clearance using Cockcroft–Gault equation; IQR, interquartile range

* Diabetes status was missing for 12 patients.

3. Results

3.1. Clinical data

Data were collected from 255 older patients, each contributing three consecutive TDM samples, resulting in a total of 765 vancomycin concentrations. The median age was 79 years, with an interquartile (IQR) of 77 to 84 years. Detailed patient characteristics are presented in **Table 14**. The median (IQR) vancomycin concentration C_{T-1} , C_{T0} , and C_{T+1} were 17.0 mg/L (13.0–21.0), 19.9 mg/L (15.7–23.9), and 19.4 mg/L (15.9–23.6), respectively.

3.2. Population pharmacokinetics models of vancomycin

The nine popPK models, including two elderly models by Ji *et al.* (2018) and Zhou *et al.* (2019), varied in structure as five are one-compartment models and four are two-compartment models (**Supplementary Table S1**). Differences were also observed in the covariates included, as detailed in **Supplementary Table S2**. Eight models included estimated creatinine clearance calculated by using Cockcroft and Gault formula ($eCrCL_{CG}$) to estimate the vancomycin clearance, and one model incorporated serum creatinine, body weight, and sex as covariates on vancomycin clearance. Seven models included body weight for estimating the volume of distribution. Age was included as a covariate in three models, including Ji model, affecting clearance in two models and the volume of distribution in one.

3.3. Model evaluations

The individual predicted concentrations from each model generally aligned closely with the observed concentrations, although not all models capture the full variability of the data well (**Figure S1–S3**). From visual inspections of these plots, we concluded that the models by Goti *et al.* and Ji *et al.* fitted the clinical data best. The Goti *et al.*, Ji *et al.*, and Thomson *et al.* models were identified as the best-fitting models in terms of the NPDEs (**Figure S4**).

3.4. Influence of the models in the multi model approaches

When calculating model weights for multi-model approaches using single sample C_{T0} or C_{T-1} , the Zhou *et al.* model was the most frequently selected in both intermittent and continuous infusion administration. When both C_{T-1} and C_{T0} were included following intermittent infusion administration, the distribution among models was more balanced, with the Zhou model still held the highest weight at 26% ($n=66/255$) and the Thomson model close behind at 25% ($n=63/255$). In continuous infusion administration, the Thomson model

was selected slightly more often than the Zhou model (28% vs. 26%, respectively) (Figure 25). Detailed model weights for each patient are presented in Figure S6.

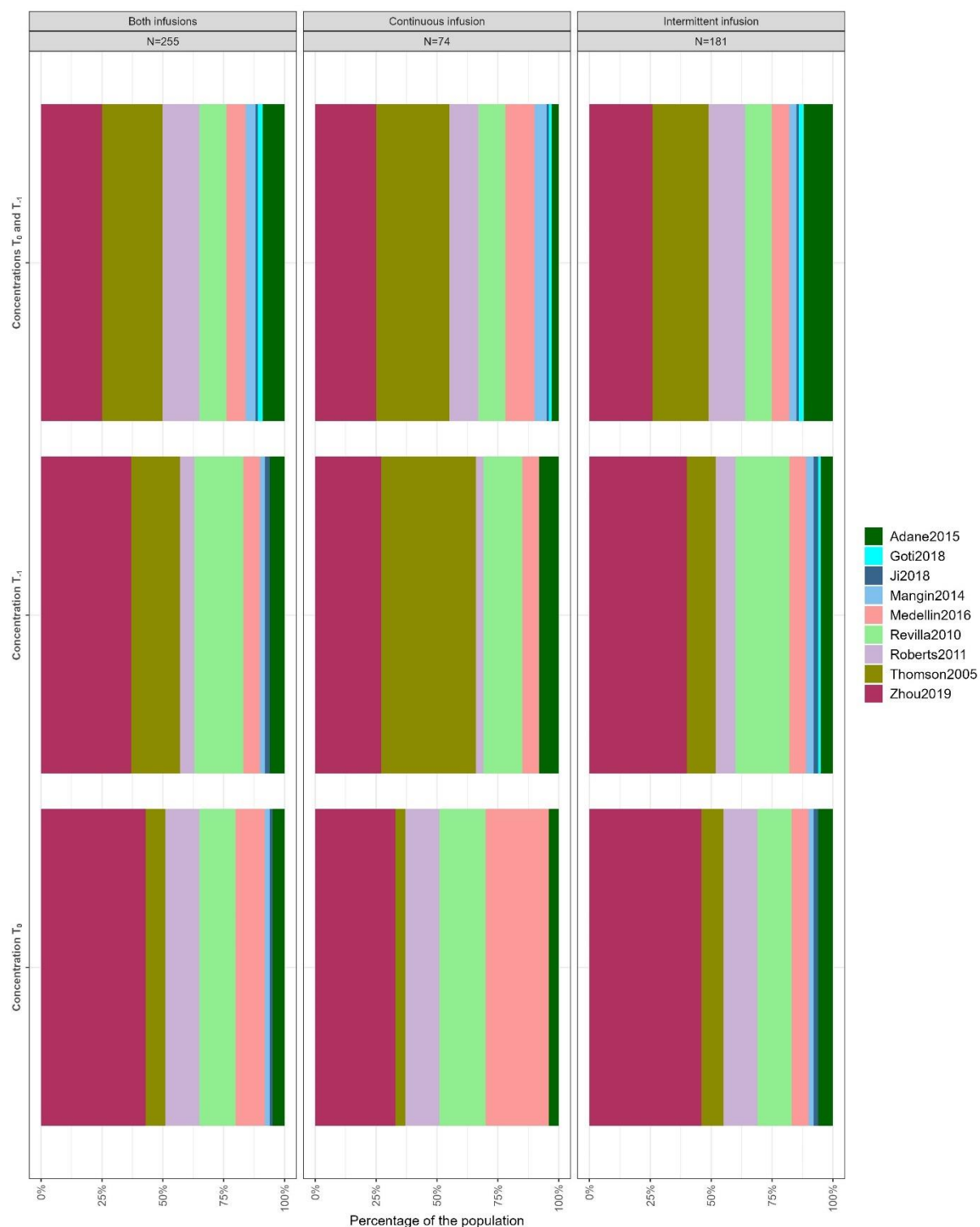


Figure 25. Weight distribution of the population pharmacokinetic models in a three *a posteriori* prediction settings. Zhou model was selected the most whereas Goti model was selected the least. T, relative time; N, total number of patients in the population

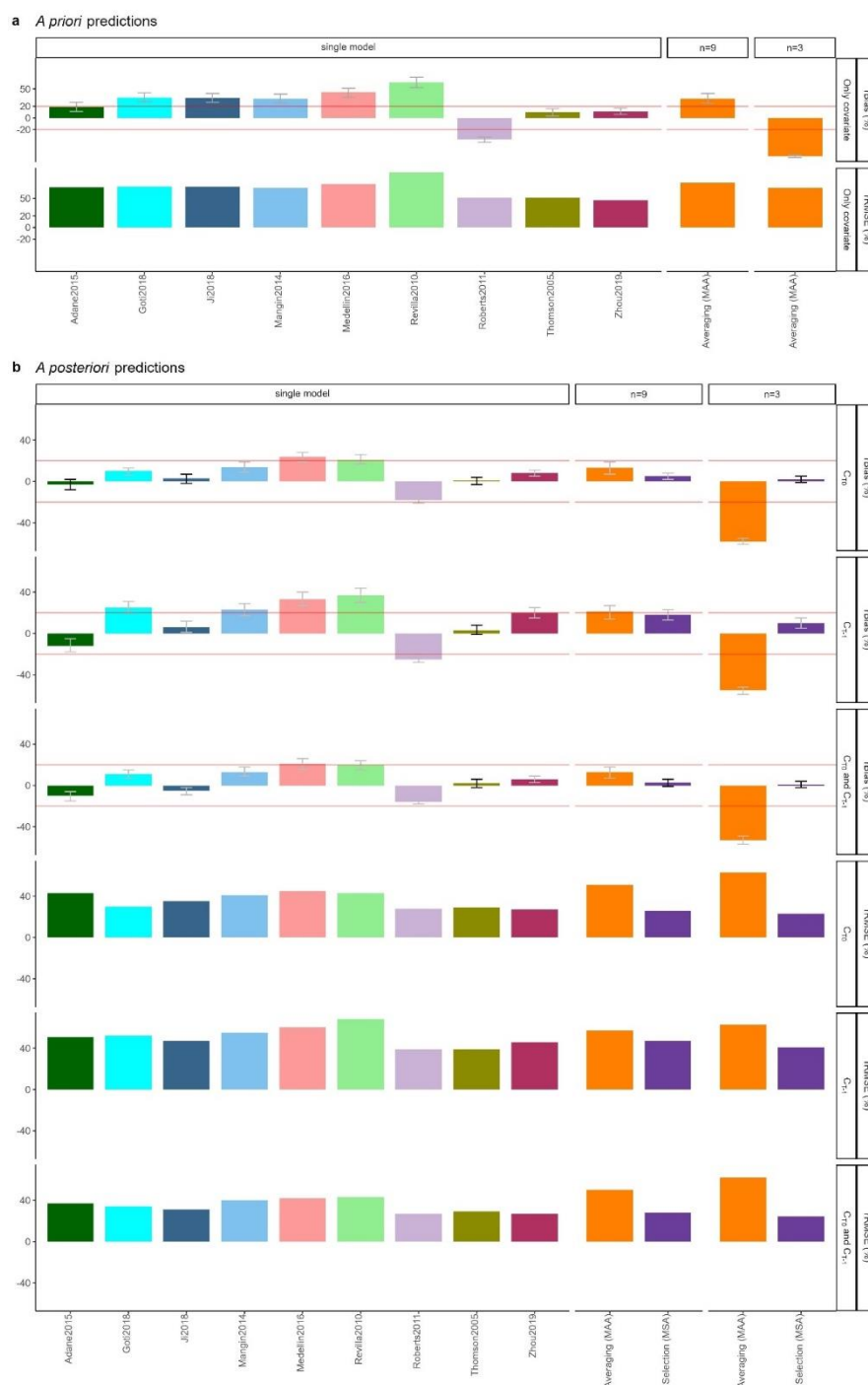


Figure 26. The predictive performance of the single and multi-model approaches using all the 9 models versus the three models to predict vancomycin concentration (C) at a time +1 (T+1). **(a)** *A priori* predictions using the clinical dataset with covariate data without previous concentrations, and **(b)** *a posteriori* prediction settings using the clinical dataset with covariate data and previous concentrations (CT_{-1} , CT_0 , CT_{-1} and CT_0). Whiskers cover the 95% confidence interval of the relative bias calculated via the standard error (black whisker indicated 95% CI including 0). Model weighing in *a priori* predictions correspond to 1/number of models, disabling model selection. The horizontal lines in relative bias panels indicate $\pm 20\%$ of the relative bias. rBias, relative bias; rRMSE, relative root mean square error; C, concentration; T, relative time

3.5. Predictive performances evaluations

A priori prediction of C_{+1} was clinically unacceptable with the single and the multi-model approaches (rBias -37% to +45%, rRMSE 47% to 78%; **Figure 26a**). The predictive performance improved when at least one concentration was accounted for in all the popPK models (**Figure 26b**). The greatest improvement in predictive performance was observed when the most recent concentration (C_0) was included (rBias -18% to +24%, rRMSE 25% to 45%). The least improvement in predictive performance was observed when an earlier concentration (C_{-1}) was included (rBias -25% to +33%, rRMSE 39% to 60%). When the two concentrations were considered (C_{T-1} and C_{T0}), the predictive performance improved slightly or worsened depending on the model/approach (rBias -16% to +21%, rRMSE 27% to 42%). Overall, the Thomson *et al.* model had the best predictive performance, followed by the Ji *et al.* model, while for continuous infusion Adane and Ji models showed the best predictive performance. The multi-model approaches did not perform better than the single-model approach (**Figure 26b**).

3.6. Robustness analysis, sensitivity analysis and Bland–Altman analysis

Three popPK models, Roberts, Thomson, and Zhou models, were selected based on their predictive performance in the *a posteriori* prediction settings. The predictive performance of the multi-model approaches with the subset models showed more bias compared to approaches with all 9 models, especially for MAA (**Figure 26**). The average computation time with the subset versus all the models decreased by 68% from 0.52 to 0.16 seconds per patient. Removing the effect of covariates makes the *a posteriori* prediction of all single models not clinically acceptable, except for MAA and MSA (**Figure S7**). Flattening priors resulted in rBias closer to 0 and smaller rRMSE (**Figure S8**). The prediction bias was not different across time after TDM (**Figure 27**), and was the least across a measured concentration range of 12.5 mg/L to 17.5 mg/L when C_0 was included in the *a posteriori* predictions (**Figure S9**).

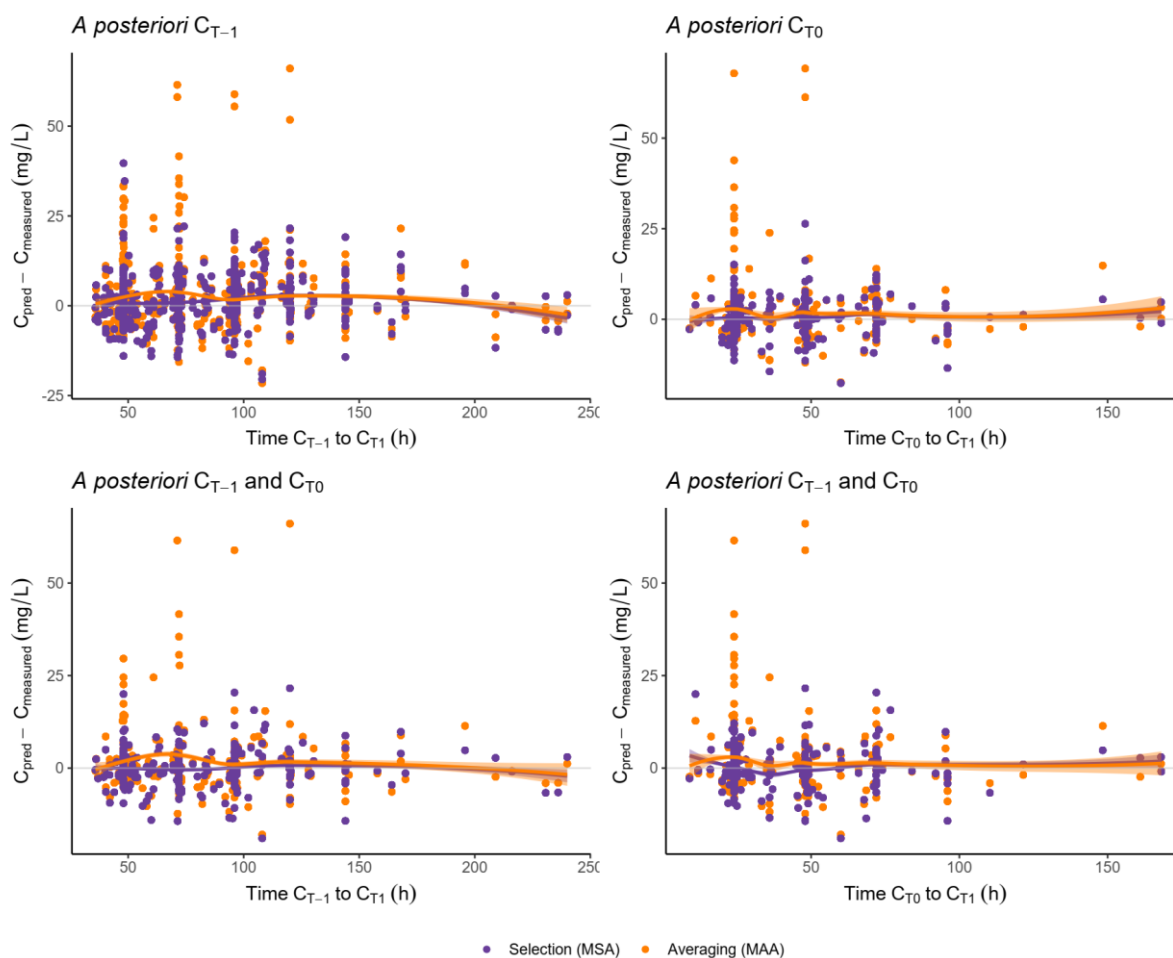


Figure 27. Bland–Altman plots where an agreement between the measured and the predicted vancomycin concentrations was observed in the *a posteriori* predictions using the model selection algorithm (MSA; purple) and the model averaging algorithm (MAA; orange) over time after therapeutic drug monitoring. C, concentration; C_{pred} , predicted concentration; T, relative time.

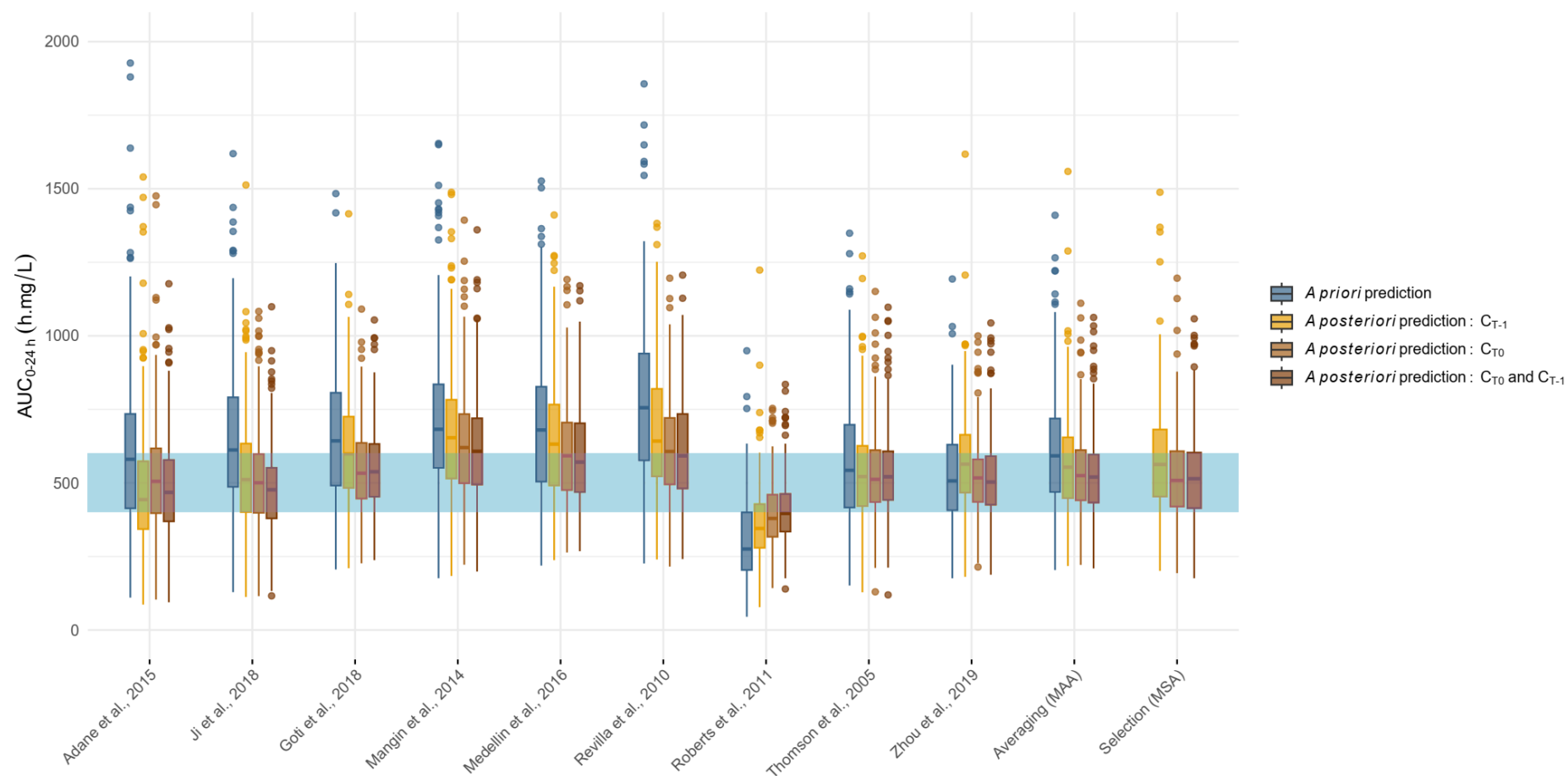


Figure 28. A priori predicted and Bayesian estimated vancomycin AUC_{0-24h} in 255 older inpatients. Box plots represent the 25th, 50th and 75th percentiles; the whiskers cover the 1.5-fold interquartile range and outliers are marked as dots. The blue area indicates the AUC_{0-24h} target of 400–600 $h \cdot mg/L$

3.7. Evaluation of the predicted AUC_{0-24h}

The AUC_{0-24h} of a course of therapy was predicted *a priori* (covariates only) and estimated from the observed vancomycin concentrations (Bayesian) with each model. The distribution of the predicted vancomycin drug exposures was heterogeneous (Figure 28). The *a priori* predicted AUC_{0-24h} ranged from 279–757 mg·h/L showing a wider range compared to the *a posteriori* predictions, which ranged from 348 to 656 mg·h/L. Models such as that of Thomson *et al.* which displayed a comparably low bias and precision in the previous analyses demonstrated similar ranges for the *a priori* and Bayesian estimated AUC_{0-24h} .

4. Discussion

When applying MIPD in special patient populations, such as older hospitalized patients, careful selection or development of the algorithm is crucial to ensure accurate guidance. In our external validation of nine vancomycin popPK models, we found that the superior predictive performance of the Thomson model is likely due to its weaker priors, characterized by higher interindividual variability compared to other models. Our analysis showed that when the priors were flattened, predictive performance improved, suggesting that models with less restrictive assumptions about variability can better capture the individual differences in this population. Additionally, models developed using data from older patients, such as the Ji and Zhou models (median age 78 years), also resulted in superior predictions for our older patient cohort. This demonstrates the importance of using models with suitable variability assumptions and aligning them with the target population, as models that accounted for high interindividual variability and were developed for similar age groups tend to predict outcomes more accurately in older adults.

These single models were more accurate and precise compared to multi-model algorithms, which contrasts with previous findings by Uster *et al.*, where MAA and MSA were reported to result in unbiased and precise predictions (17). The observed underperformance of the multi-model algorithms in our study is likely due to potentially PK alterations during sampling and the non-uniform sampling times, with half of the samples not being trough concentrations. This variability may have complicated the weight calculation process in the multi-model approach.

The top-performing models identified in previous studies evaluating the predictive performance in adults are different from that we identified. In the analysis by Broeker *et al.*,

the Goti model demonstrated the best predictive performance among the models evaluated (22, 31). Broeker *et al.* included a broad hospital-wide population without restrictions, and the Goti model was developed using the largest dataset of all the investigated models (2,765 vancomycin concentrations from 1,812 hospitalized patients). In contrast, our study focused solely on older patients. Although the Goti model showed good general performance in our study as well, it was surpassed by models built on older population. On the other hand, the Roberts model did not perform as well in our study, particularly in terms of general fit. This contrasts with the findings of Cunio *et al.*, who analyzed critically ill patients and found that the Roberts model, which was developed for this population, provided clinically acceptable bias in both overall model evaluation and Bayesian forecasting analyses (32). The difference in results can be explained by the fact that our dataset included a smaller proportion of critically ill patients.

Removing the effect of covariates from *a posteriori* predictions resulted in different results, indicating that including covariates is likely crucial, particularly for older hospitalized patients. This finding aligns with the argument by Broeker *et al.*, who emphasized the importance of body weight and creatinine clearance in vancomycin dosing (31). Our analysis suggests that, beside weight and creatinine clearance, age is also an important covariate, as models developed from older patient populations performed better for the older patients in our dataset, especially the Ji model, which includes age as a covariate. Age may serve as a proxy for various (patho-)physiological changes that accompany aging, such as changes in body composition and comorbidities, which influence both drug distribution and clearance.

The predicted AUC_{0-24h} varied significantly between models, which has important clinical implications since dosing decisions based on these predictions would differ. For example, the supratherapeutic AUCs predicted by the Revilla, Medellin, and Mangin models would suggest a dose reduction, while the subtherapeutic AUCs from the Roberts model would indicate a need for a dose increase. The latter is particularly concerning due to the notable negative bias in the Roberts model, which could lead to overdosing. Although our study lacked the rich sampling necessary to compare predicted and true AUCs, an integrated evaluation of predictive performance metrics and models evaluations using VPC, NPDE and goodness-of-fit plots provides evidence that models such as those from Thomson and Ji are more reliable for unbiased AUC predictions.

This study has several limitations that should be considered. First, the analysis was restricted to a limited number of popPK models, which may have excluded potentially relevant models that could offer further insights or improved predictions. Second, as a single-center, retrospective study, the generalizability of the findings may be constrained. Third, while all models use accessible covariates, there is one model that included the SOFA score, restricting its use to only ICU patients. Furthermore, none of the models accounted for geriatric patients with a frailty profile, instead focusing on older adults without explicitly addressing frailty. While our dataset represents the largest cohort of older patients analyzed to date, addressing the notable underrepresentation of this population in pharmacometrics studies, we acknowledge that larger datasets alone do not necessarily enhance generalizability if they primarily consist of similar patient profiles. However, our cohort includes diverse patient characteristics as it is a hospital-wide study, which we believe enhances the applicability of our findings to a broader geriatric population and strengthens the relevance of our conclusions. In addition, much can be learned from data in other populations. Age alone is not the primary determinant of PK alterations; instead, it is the (patho)physiological changes associated with aging. For instance, studies in non-geriatric populations with a wide range of renal function can provide valuable insights into PK changes, which are particularly relevant for geriatric patients who often experience age-related declines in renal function, a key factor in vancomycin clearance. Moreover, in the case of either observed or predicted concentration is uncertain, future evaluations of vancomycin models could benefit from incorporating modified error structures, such as the one demonstrated by Schatz et al. (30).

In conclusion, popPK models developed from older populations and with high interindividual variabilities were found to be clinically acceptable for guiding vancomycin dosing in older patients, particularly when using Bayesian forecasting with the most recent concentration. The Thomson model demonstrated the best performance, followed by the Ji model.

References

1. Goto T, Yoshida K, Tsugawa Y, Camargo CA, Hasegawa K. Infectious Disease-Related Emergency Department Visits of Elderly Adults in the United States, 2011-2012. *Journal of the American Geriatrics Society*. 2016;64(1):31-6.
2. Buurman BM, Hoogerduijn JG, De Haan RJ, Abu-Hanna A, Lagaay AM, Verhaar HJ, et al. Geriatric Conditions in Acutely Hospitalized Older Patients: Prevalence and One-Year Survival and Functional Decline. *PLoS ONE*. 2011;6(11):e26951.
3. Ukkonen M, Jämsen E, Zeitlin R, Pauniah S-L. Emergency department visits in older patients: a population-based survey. *BMC Emergency Medicine*. 2019;19(1).
4. Gjesten MT, Brønnick K, Testad I. Characteristics and predictors for hospitalizations of home-dwelling older persons receiving community care: a cohort study from Norway. *BMC Geriatrics*. 2018;18(1).
5. Kurtaran B, Kuscu F, Korkmaz P, Ozdemir B, Inan D, Oztoprak N, et al. A snapshot of geriatric infections in Turkey: ratio of geriatric inpatients in hospitals and evaluation of their infectious diseases: A multicenter point prevalence study. *International Journal of Infectious Diseases*. 2020;100:337-42.
6. Tacconelli E, Pop-Vicas AE, D'Agata EM. Increased mortality among elderly patients with methicillin-resistant *Staphylococcus aureus* bacteraemia. *J Hosp Infect*. 2006;64(3):251-6.
7. Pomorska-Wesołowska M, Różańska A, Natkaniec J, Gryglewska B, Szczypka A, Dzikowska M, et al. Longevity and gender as the risk factors of methicillin-resistant *Staphylococcus aureus* infections in southern Poland. *BMC Geriatrics*. 2017;17(1).
8. McClelland RS, Fowler VG, Sanders LL, Gottlieb G, Kong LK, Sexton DJ, et al. *Staphylococcus aureus* Bacteremia Among Elderly vs Younger Adult Patients. *Archives of Internal Medicine*. 1999;159(11):1244.
9. Liu C, Bayer A, Cosgrove SE, Daum RS, Fridkin SK, Gorwitz RJ, et al. Clinical practice guidelines by the infectious diseases society of america for the treatment of methicillin-resistant *Staphylococcus aureus* infections in adults and children: executive summary. *Clin Infect Dis*. 2011;52(3):285-92.
10. Rybak MJ, Le J, Lodise TP, Levine DP, Bradley JS, Liu C, et al. Therapeutic monitoring of vancomycin for serious methicillin-resistant *Staphylococcus aureus* infections: A revised consensus guideline and review by the American Society of Health-System Pharmacists, the Infectious Diseases Society of America, the Pediatric Infectious Diseases Society, and the Society of Infectious Diseases Pharmacists. *Am J Health Syst Pharm*. 2020;77(11):835-64.
11. Kufel WD, Seabury RW, Mogle BT, Beccari MV, Probst LA, Steele JM. Readiness to implement vancomycin monitoring based on area under the concentration-time curve: A cross-sectional survey of a national health consortium. *Am J Health Syst Pharm*. 2019;76(12):889-94.
12. Williams PG, Tabah A, Cotta MO, Sandaradura I, Kanji S, Scheetz MH, et al. International survey of antibiotic dosing and monitoring in adult intensive care units. *Critical Care*. 2023;27(1).
13. Flannery AH, Hammond DA, Oyler DR, Li C, Wong A, Smith AP, et al. Vancomycin Dosing Practices among Critical Care Pharmacists: A Survey of Society of Critical Care Medicine Pharmacists. *Infectious Diseases: Research and Treatment*. 2020;13:117863372095207.
14. Minichmayr IK, Dreesen E, Centanni M, Wang Z, Hoffert Y, Friberg LE, et al. Model-informed precision dosing: State of the art and future perspectives. *Adv Drug Deliv Rev*. 2024:115421.
15. Wicha SG, Martson AG, Nielsen EI, Koch BCP, Friberg LE, Alffenaar JW, et al. From Therapeutic Drug Monitoring to Model-Informed Precision Dosing for Antibiotics. *Clin Pharmacol Ther*. 2021;109(4):928-41.
16. Dia N, Hoffert Y, Edwina AE, Spriet I, Dreesen E. Model-informed Precision Dosing of Vancomycin: A Systematic Review of Population Pharmacokinetics Models and Benchmark Study of Software Tools.

17. Uster DW, Stocker SL, Carland JE, Brett J, Marriott DJE, Day RO, et al. A Model Averaging/Selection Approach Improves the Predictive Performance of Model-Informed Precision Dosing: Vancomycin as a Case Study. *Clin Pharmacol Ther.* 2021;109(1):175-83.
18. Kantasiripitak W, Outtier A, Wicha SG, Kensert A, Wang Z, Sabino J, et al. Multi-model averaging improves the performance of model-guided infliximab dosing in patients with inflammatory bowel diseases. *CPT Pharmacometrics Syst Pharmacol.* 2022;11(8):1045-59.
19. Adane ED, Herald M, Koura F. Pharmacokinetics of vancomycin in extremely obese patients with suspected or confirmed *Staphylococcus aureus* infections. *Pharmacotherapy.* 2015;35(2):127-39.
20. Revilla N, Martin-Suarez A, Perez MP, Gonzalez FM, Fernandez de Gatta Mdel M. Vancomycin dosing assessment in intensive care unit patients based on a population pharmacokinetic/pharmacodynamic simulation. *Br J Clin Pharmacol.* 2010;70(2):201-12.
21. Thomson AH, Staatz CE, Tobin CM, Gall M, Lovering AM. Development and evaluation of vancomycin dosage guidelines designed to achieve new target concentrations. *J Antimicrob Chemother.* 2009;63(5):1050-7.
22. Goti V, Chaturvedula A, Fossler MJ, Mok S, Jacob JT. Hospitalized Patients With and Without Hemodialysis Have Markedly Different Vancomycin Pharmacokinetics: A Population Pharmacokinetic Model-Based Analysis. *Ther Drug Monit.* 2018;40(2):212-21.
23. Medellin-Garibay SE, Ortiz-Martin B, Rueda-Naharro A, Garcia B, Romano-Moreno S, Barcia E. Pharmacokinetics of vancomycin and dosing recommendations for trauma patients. *J Antimicrob Chemother.* 2016;71(2):471-9.
24. Mangin O, Urien S, Mainardi JL, Fagon JY, Faisy C. Vancomycin pharmacokinetic and pharmacodynamic models for critically ill patients with post-sternotomy mediastinitis. *Clin Pharmacokinet.* 2014;53(9):849-61.
25. Roberts JA, Taccone FS, Udy AA, Vincent JL, Jacobs F, Lipman J. Vancomycin dosing in critically ill patients: robust methods for improved continuous-infusion regimens. *Antimicrob Agents Chemother.* 2011;55(6):2704-9.
26. Ji XW, Ji SM, He XR, Zhu X, Chen R, Lu W. Influences of renal function descriptors on population pharmacokinetic modeling of vancomycin in Chinese adult patients. *Acta Pharmacol Sin.* 2018;39(2):286-93.
27. Zhou Y, Gao F, Chen C, Ma L, Yang T, Liu X, et al. Development of a Population Pharmacokinetic Model of Vancomycin and its Application in Chinese Geriatric Patients with Pulmonary Infections. *Eur J Drug Metab Pharmacokinet.* 2019;44(3):361-70.
28. Brendel K, Comets E, Laffont C, Laveille C, Mentre F. Metrics for external model evaluation with an application to the population pharmacokinetics of gliclazide. *Pharm Res.* 2006;23(9):2036-49.
29. Hughes JH, Keizer RJ. A hybrid machine learning/pharmacokinetic approach outperforms maximum a posteriori Bayesian estimation by selectively flattening model priors. *CPT Pharmacometrics Syst Pharmacol.* 2021;10(10):1150-60.
30. Schatz LM, Greppmair S, Kunzelmann AK, Starp J, Brinkmann A, Roehr A, et al. Predictive performance of multi-model approaches for model-informed precision dosing of piperacillin in critically ill patients. *Int J Antimicrob Agents.* 2024;64(4):107305.
31. Broeker A, Nardecchia M, Klinker KP, Derendorf H, Day RO, Marriott DJ, et al. Towards precision dosing of vancomycin: a systematic evaluation of pharmacometric models for Bayesian forecasting. *Clin Microbiol Infect.* 2019;25(10):1286 e1- e7.
32. Cunio CB, Uster DW, Carland JE, Buscher H, Liu Z, Brett J, et al. Towards precision dosing of vancomycin in critically ill patients: an evaluation of the predictive performance of pharmacometric models in ICU patients. *Clin Microbiol Infect.* 2020.

Appendix

Table S1. Patient characteristics in the included studies

Authors. year of publication Journal	Patient population	Age (years), mean $\pm$ SD, median (range) [IQR]	Body weight (kg), mean $\pm$ SD, median (range) [IQR]	BMI (kg/m ²), mean $\pm$ SD, median (range) [IQR]	Serum creatinine, mean $\pm$ SD, median (range) [IQR]	CrCL, mean $\pm$ SD, median (range) [IQR]
Adane <i>et al.</i> , 2015	Extremely obese (BMI $\geq$ 40 kg/m ²) adults with suspected or confirmed <i>Staphylococcus aureus</i> infection requiring treatment for $\geq$ 3 days in USA	43 [38.5–53.0]	147.9 [142.8–178.3]	49.5 [44.3–54.8]	–	124.8 [106.0–133.9] (mL/min/1.73 m ²)
Goti <i>et al.</i> , 2018	Adult patients hospitalized at a unit with high prevalence of end-stage renal diseases	57 (17–101)	79 [33–255]	–	1.3 [0.2–16.6] mg/dL	62 (4–150) mL/min
Ji <i>et al.</i> , 2018	Hospitalized adult patients on continuous infusion of vancomycin, and not in need for renal replacement therapy	78 (42–95)	65 (38–90)	22.31 (12.85–36.89)	75 (24–893) μ mol/L	58.02 (5.45–224.0) mL/min
Mangin <i>et al.</i> , 2014	Intensive care unit adult patients with post-sternotomy mediastinitis during their primary closed Redon catheter drainage	63 (35–81)	82 (62–104)	28 (22–36)	138 (32–606) μ mol/L	–
Medellin–Garibay <i>et al.</i> , 2016	Traumatic adult patients with proven or suspected infection	74.3 $\pm$ 14, 77 (37–100)	72.0 $\pm$ 15	27.5 $\pm$ 5	–	90.5 $\pm$ 51.67 mL/min
Revilla <i>et al.</i> , 2010	Intensive care unit adult patients	61.1 $\pm$ 16.3 (18–85)	73.0 $\pm$ 13.3	26.2 $\pm$ 4.1	–	86.1 $\pm$ 55.1 mL/min
Roberts <i>et al.</i> , 2011	Septic critically ill adults in the intensive care unit	58.1 $\pm$ 14.8	74.8 $\pm$ 15.8	25.9 $\pm$ 5.4	–	90.7 $\pm$ 60.4 (mL/min/1.73 m ²)

Authors. year of publication Journal	Patient population	Age (years), mean $\pm$ SD, median (range) [IQR]	Body weight (kg), mean $\pm$ SD, median (range) [IQR]	BMI (kg/m ²), mean $\pm$ SD, median (range) [IQR]	Serum creatinine, mean $\pm$ SD, median (range) [IQR]	CrCL, mean $\pm$ SD, median (range) [IQR]
Thomson <i>et al.</i> , 2009	Hospitalized adult patients during routine clinical care at the hospital	66 (16–97)	72 (40–159)	–	98 (30–573) μ mol/L	64 (12–216) mL/min
Zhou <i>et al.</i> , 2019	Hospitalized geriatric patients ($\geq$ 65 years) with a pulmonary infection (a hospital– or a community–acquired pneumonia)	78.3 $\pm$ 6.96	60.7 $\pm$ 10.2	–	90.6 $\pm$ 31 μ mol/L	56.3 $\pm$ 22.1 mL/min

BMI, body mass index; CrCL, creatinine clearance; SD, standard deviation; IQR, interquartile range

Table S2. Properties and standardized pharmacokinetic parameters (standard patient, 50 years old, male, body weight: 75 kg, body height: 1.7 m, serum creatinine of 85 µmol/L) of the evaluated pharmacometrics models

Authors. year of publication Journal	Vancomycin administration	No. patients	Nr. samples	Covariates	CL	Vc	Vp	Q	IIV	RUV
Adane <i>et al.</i> , 2015	Intermittent infusion	29	93	TVCL=CL*(CrCL/125) TVV =Vc* Weight	4.74	38.3	–	–	CL: 26.7% Vp: 23.9%	P: 18.9% A: –
Goti <i>et al.</i> , 2018	n.r.	1812	2765	TVCL = CL*((CrCL/120)**0.8) *0.7 (if dialysis) TVVc = Vc*(Weight/70)*0.5 (if dialysis) TVVp = Vp*(Weight/70)	3.82	62.6	38.4	6.5	CL: 38.4% Vc: 81.9%, Vp: 57.1%	P: 22.7% A: 3.4 mg/L
Ji <i>et al.</i> , 2018	Continuous infusion	160	251	TVCL = CL*(1+0.0084*(CrCL-80)) *(75/Age)**0.8143	4.52	52.1	–	–	CL: 32.4% Vc: 28.9%	P: 26.8% A: 2.7 mg/L
Mangin <i>et al.</i> , 2014	Intermittent infusion	30	359	TVCL=CL*0.66 (if female) *((Weight/70)**0.75) *((SAPSII/50)**(-0.5)) *(((SCr/88.42)/100)**(-0.9)) TVVc=Vc*(Weight/70) TVVp=Vc*(Weight/70) TVQ= Q*0.3(if diabetes) *((Weight/70)**0.75)	2.60	23.5	72.9	6.01	CL: 29% Vc: 53% Q: 101%	P: – A: 7.3 mg/L
Medellín–Garibay <i>et al.</i> , 2016	Intermittent infusion	118	392	TVCL = CL*CrCL TVCL = CL _{furosemide} *CrCL	2.87	55.5	442.5	0.81	CL: 36.7% Vc: 40.0%	P: 19.2% A: 3.5 mg/L

				TVVc = Vc*Weight TVVp = Vp*Weight						
Revilla <i>et al.</i> , 2010	Intermittent and continuous infusion	191	569	TVCL=CL*CrCL+(Age**(-0.24))*Weight) TVVc= Vc*2.49 (if sCr>1)*Weight	4.84	61.5	–	–	CL: 30.1% Vc: 22.8%	P: – A: 4.2 mg/L
Roberts <i>et al.</i> , 2011	Continuous infusion	206	579	TVCL = CL*(UCrCL/100) TVVc = Vc*Weight	4.15	114.8	–	–	CL: 38.9% Vc: 37.4%	P: 19.9% A: 2.4 mg/L
Thomson <i>et al.</i> , 2009	Intermittent infusion	398	1557	TVCL = CL*CrCL*0.0154 TVVc = Vc*Weight TVVp = Vp*Weight	4.45	50.6	54.9	2.28	CL: 27%, Q: 49% Vc: 15%, Vp: 120%	P: 15% A: 1.6 mg/L
Zhou <i>et al.</i> , 2019	Intermittent infusion	70	125	TVCL = CL*(CrCL/56.28)**0.542	3.30	154	–	–	CL: 17.4% Vc: 33.9%	P: 6.6%

A, additive; Vc, central volume of distribution; CrCL, creatinine clearance; CL, clearance; Q, intercompartmental clearance; IIV, interindividual variability; n.r., not reported; Vp, peripheral volume of distribution; P, proportional; TBW, total body weight; RUV, residual unexplained variability; SCr, serum creatinine; UCrCL, urinary creatinine clearance

Table S3. Statistical tests to assess normality of the normalized prediction distribution errors (NPDE) of the population pharmacokinetic models.

Model	Wilcoxon test ¹	Fisher test ²	Shapiro–Wilks test ³	Global test ⁴	Ranking ⁵
Adane <i>et al.</i> , 2015	1.5e-15	0e+00	2.8e-01 (.)	0	2
Goti <i>et al.</i> , 2018	9.3e-01 (.)	0e+00	8.2e-11	0	3
Ji <i>et al.</i> , 2018	0e+00	2.9e-12	8.6e-12	0	4
Mangin <i>et al.</i> , 2014	9e-09	0e+00	3e-20	0	5
Medellín-Garibay <i>et al.</i> , 2016	0e+00	5.2e-06	1.3e-12	0	6
Revilla <i>et al.</i> , 2010	0e+00	0e+00	3.7e-06	0	7
Roberts <i>et al.</i> , 2011	0e+00	3.1e-05	9.9e-07	0	8
Thomson <i>et al.</i> , 2009	1e-02 (***)	9.3e-11	2.1e-08	2.8 e-10	1
Zhou <i>et al.</i> , 2019	9e-04	0e+00	2.8e-10	0	9

¹Wilcoxon test compared the mean of the NPDE to 0.

²Fisher test compared the variance of the NPDE to 1.

³Shapiro–Wilks test compared the NPDE to the normal distribution.

⁴P-value of the global test was equal to the minimum of the adjusted p-values. The adjusted p-values are the raw p-value of the three individual tests multiplied by 3.

⁵Ranking based on p-value of the global test (highest value to lowest value).

‘.’ >0.1; ‘*’ >0.05; ‘**’ >0.01; ‘***’ >0.001, Significant levels indicate that the NPDEs are different from N(0,1) distribution for the specified tests.

Table S4. Classification accuracy is defined as no change in exposure category (i.e., subtherapeutic, supratherapeutic or therapeutic) according to the predicted vancomycin concentration compared to the observed concentration.

	Model	Classification accuracy, n (%)
1	Adane, 2015	150 (59%)
2	Goti, 2018	177 (69%)
3	Ji, 2018	168 (66%)
4	Mangin, 2014	166 (65%)
5	Medellin, 2016	167 (65%)
6	Revilla, 2010	168 (66%)
7	Roberts, 2011	134 (53%)
8	Thomson, 2005	183 (72%)
9	Zhou, 2019	182 (71%)
10	MAA	177 (69%)
11	MSA	171 (67%)

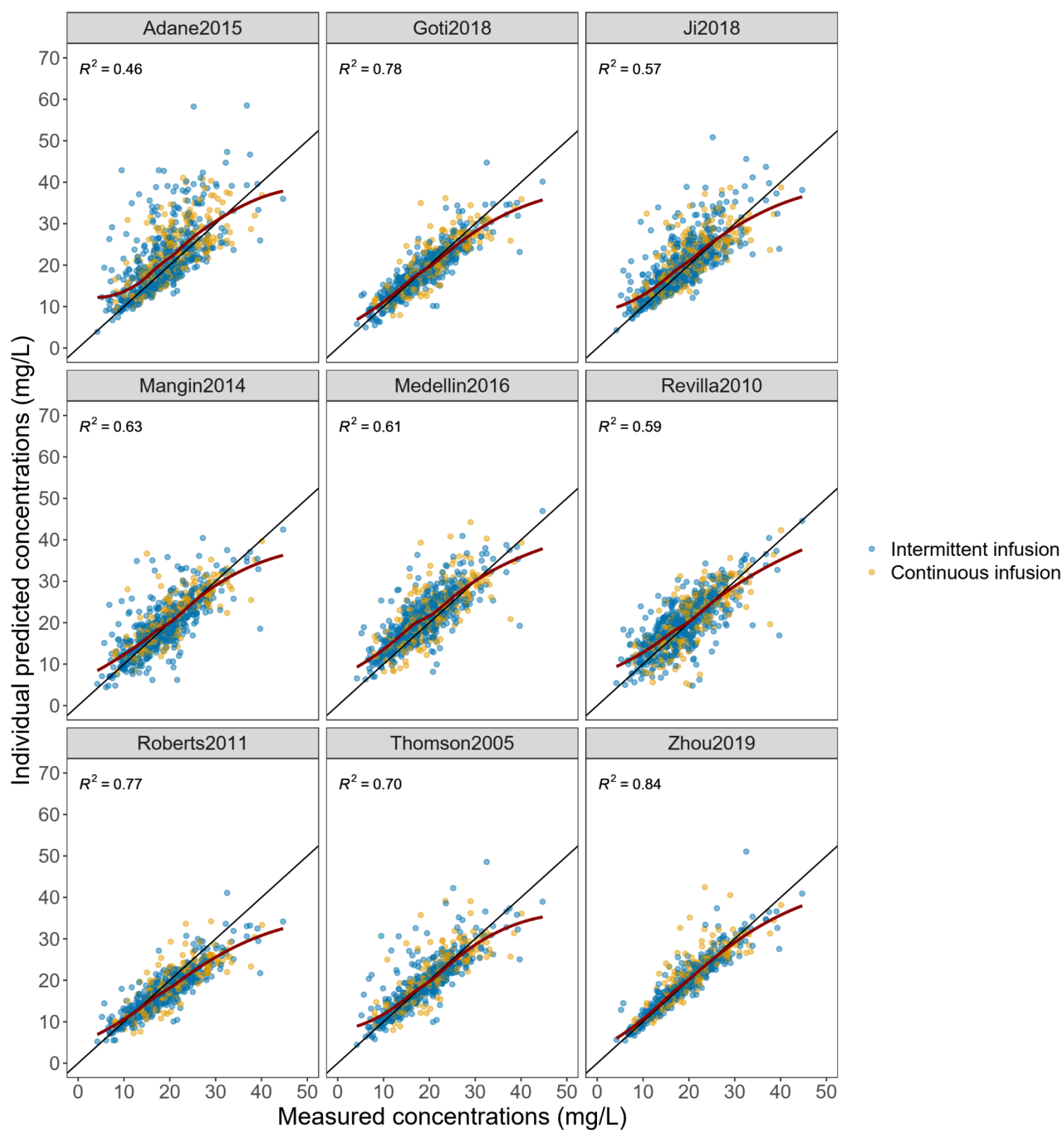


Figure S1. Goodness-of-fit plots of the measured concentration versus individual predicted concentration. The solid black and red lines represent the line of identity, and a locally weighted smoother, respectively.

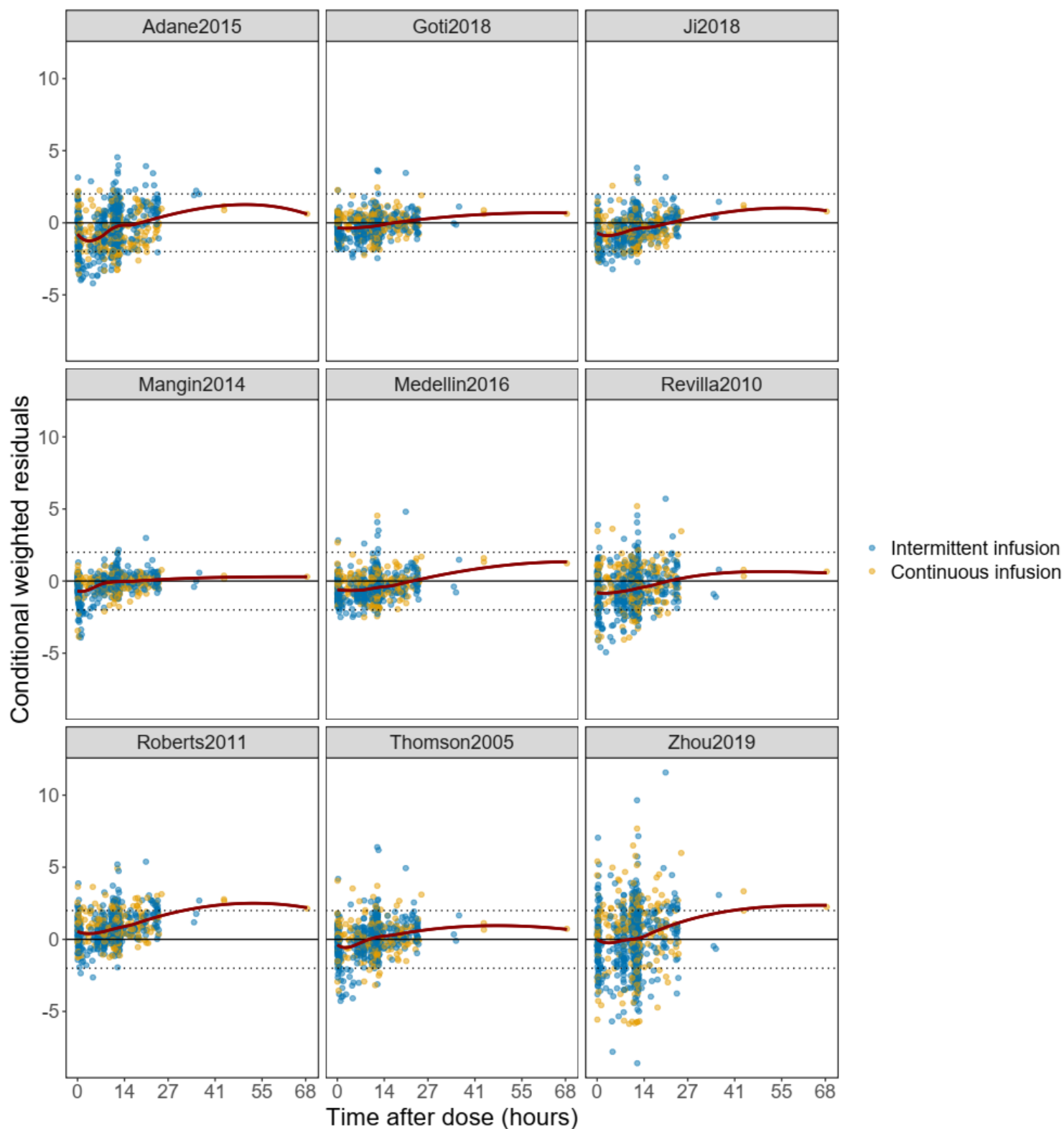


Figure S2. Goodness-of-fit plots of the time after dose (hours) versus conditional weighted residuals. The solid black and red lines represent the line of identity, and a locally weighted smoother, respectively. The dashed lines represent a conditional weighted residuals of ± 3 .

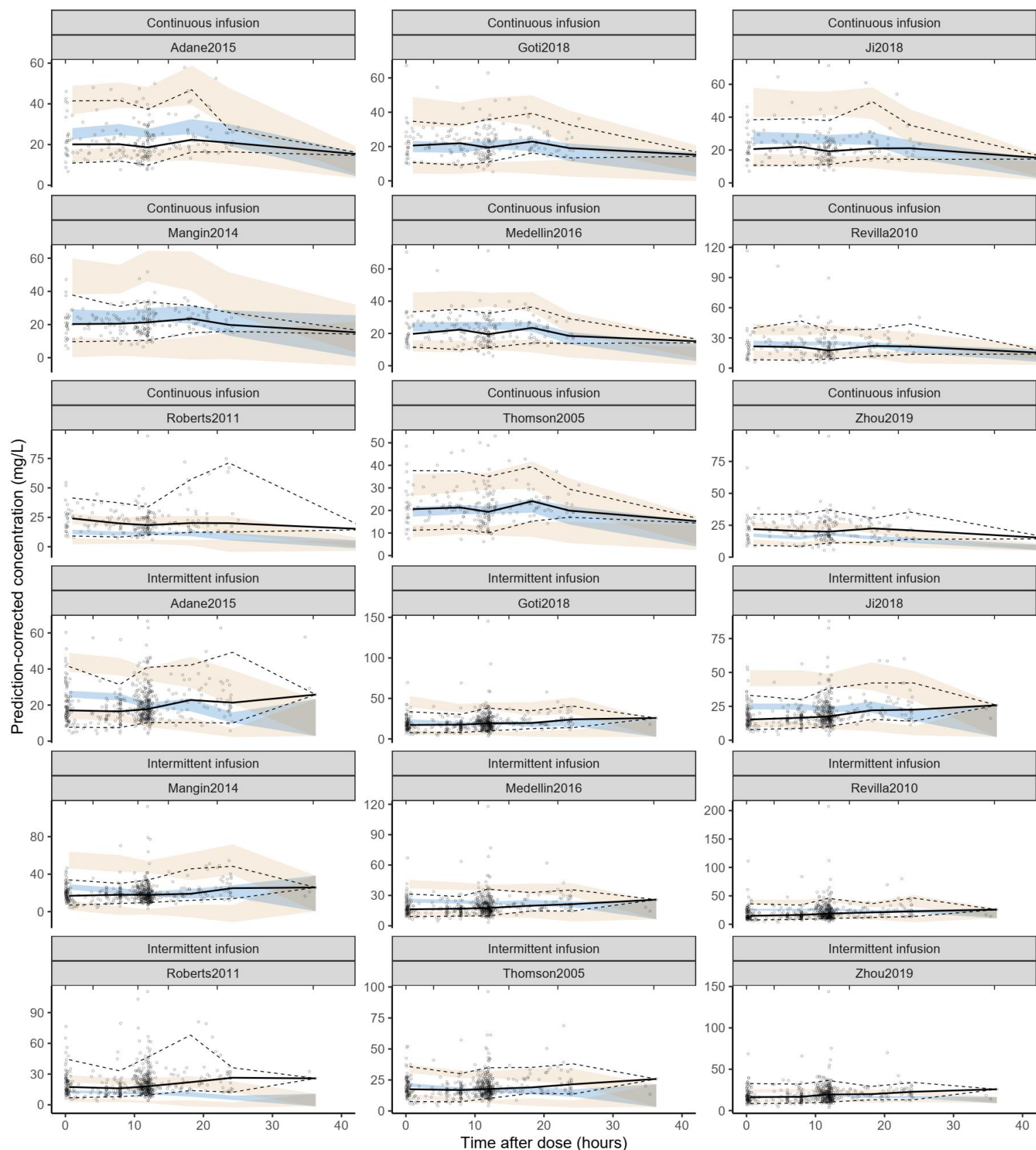


Figure S3. Prediction-corrected visual predictive checks of the prediction-corrected vancomycin concentration versus time after dose (hours) in 255 patients for each of the 9 population pharmacokinetics models. The solid black lines represent the median of the data. The dashed lines represent the 5th and 95th percentile of the data. The shaded areas represent the 90% confidence intervals of the predictions. The vertical dashes at the top of each plot represent the binning intervals.

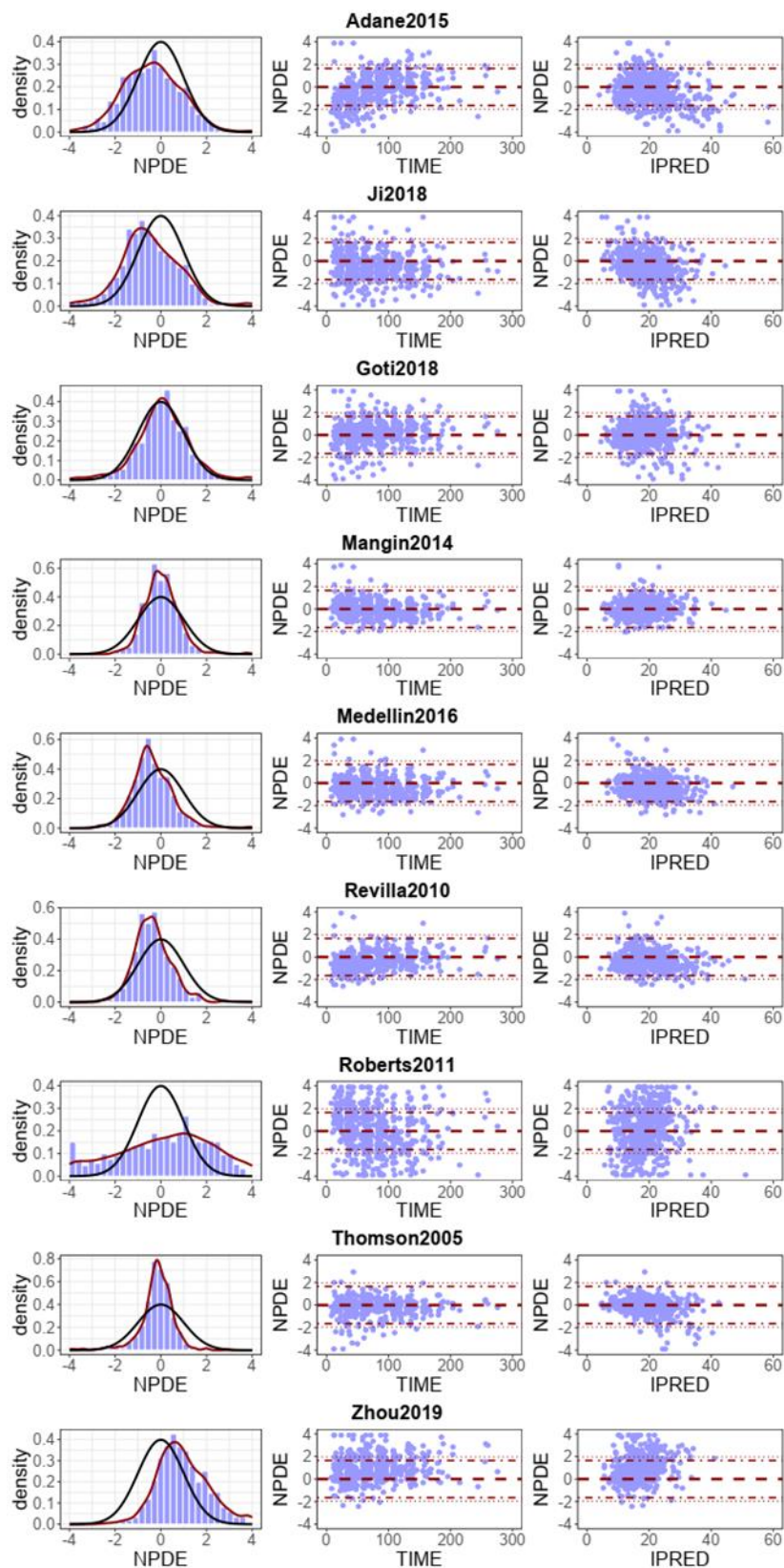


Figure S4. Plots illustrating normalized prediction distribution errors (NPDE) of the 9 population pharmacokinetic models following **intermittent infusion** administration. *The first plot:* Histogram and density (solid red line) of the distribution of NPDE overlaid with theoretical $\mathcal{N}(0,1)$ distribution (solid black line). *The second plot:* Scatterplot of NPDE versus Time (hours). *The third plot:* Scatterplot of NPDE versus individual predicted concentrations (IPRED). The dashed horizontal lines are the null line (NPDE = 0). The dot-dashed horizontal lines indicate the 90% range (NPDE = -1.645, 1.645). The dotted lines indicate the 95% range (NPDE = -1.96, 1.96). ESAMPLE was set at 10,000

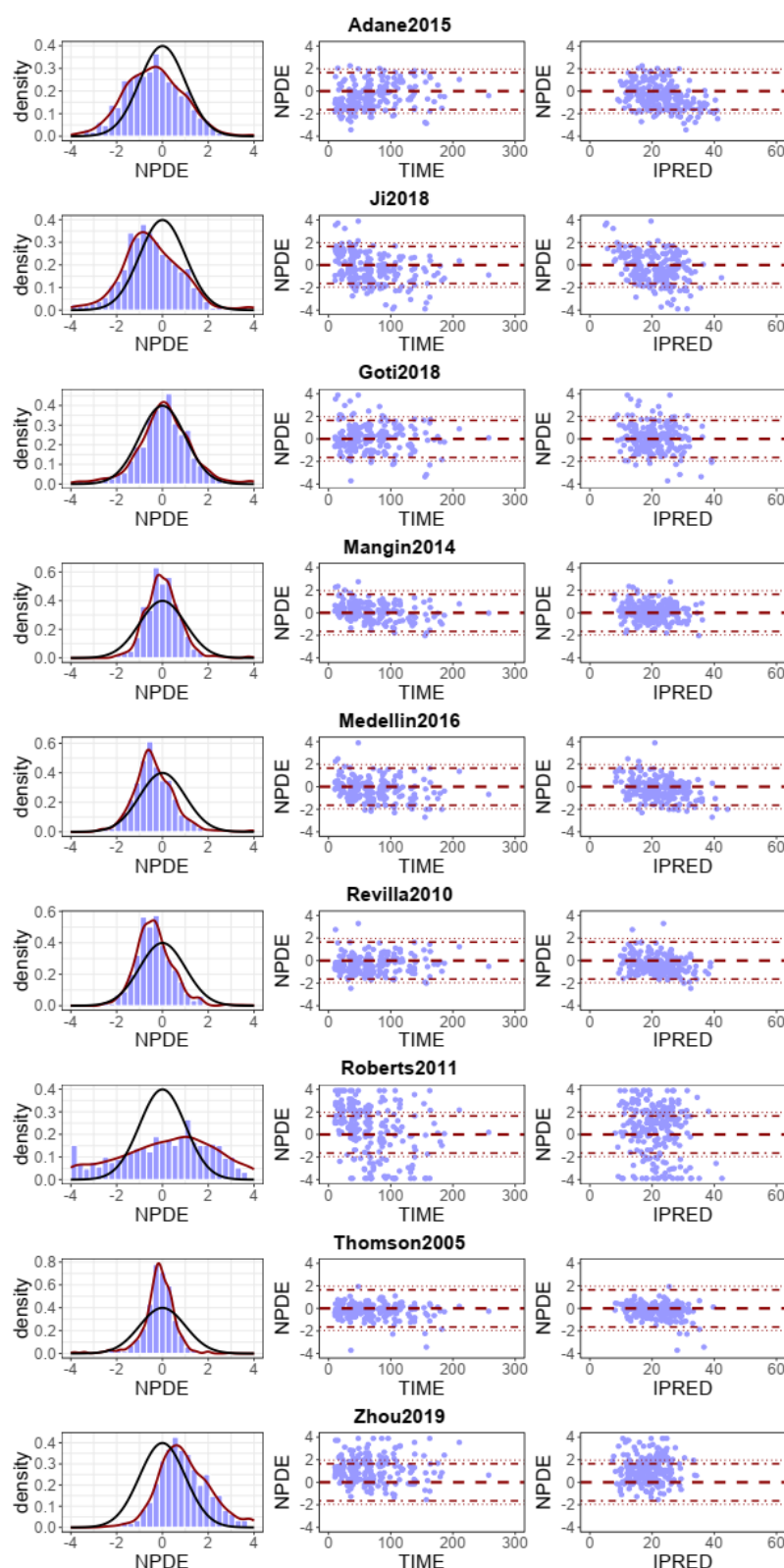


Figure S5. Plots illustrating normalized prediction distribution errors (NPDE) of the 9 population pharmacokinetic models following **continuous infusion** administration. *The first plot:* Histogram and density (solid red line) of the distribution of NPDE overlaid with theoretical $\mathcal{N}(0,1)$ distribution (solid black line). *The second plot:* Scatterplot of NPDE versus Time (hours). *The third plot:* Scatterplot of NPDE versus individual predicted concentrations (IPRED). The dashed horizontal lines are the null line (NPDE = 0). The dot-dashed horizontal lines indicate the 90% range (NPDE = -1.645, 1.645). The dotted lines indicate the 95% range (NPDE = -1.96, 1.96). ESAMPLE was set at 10,000

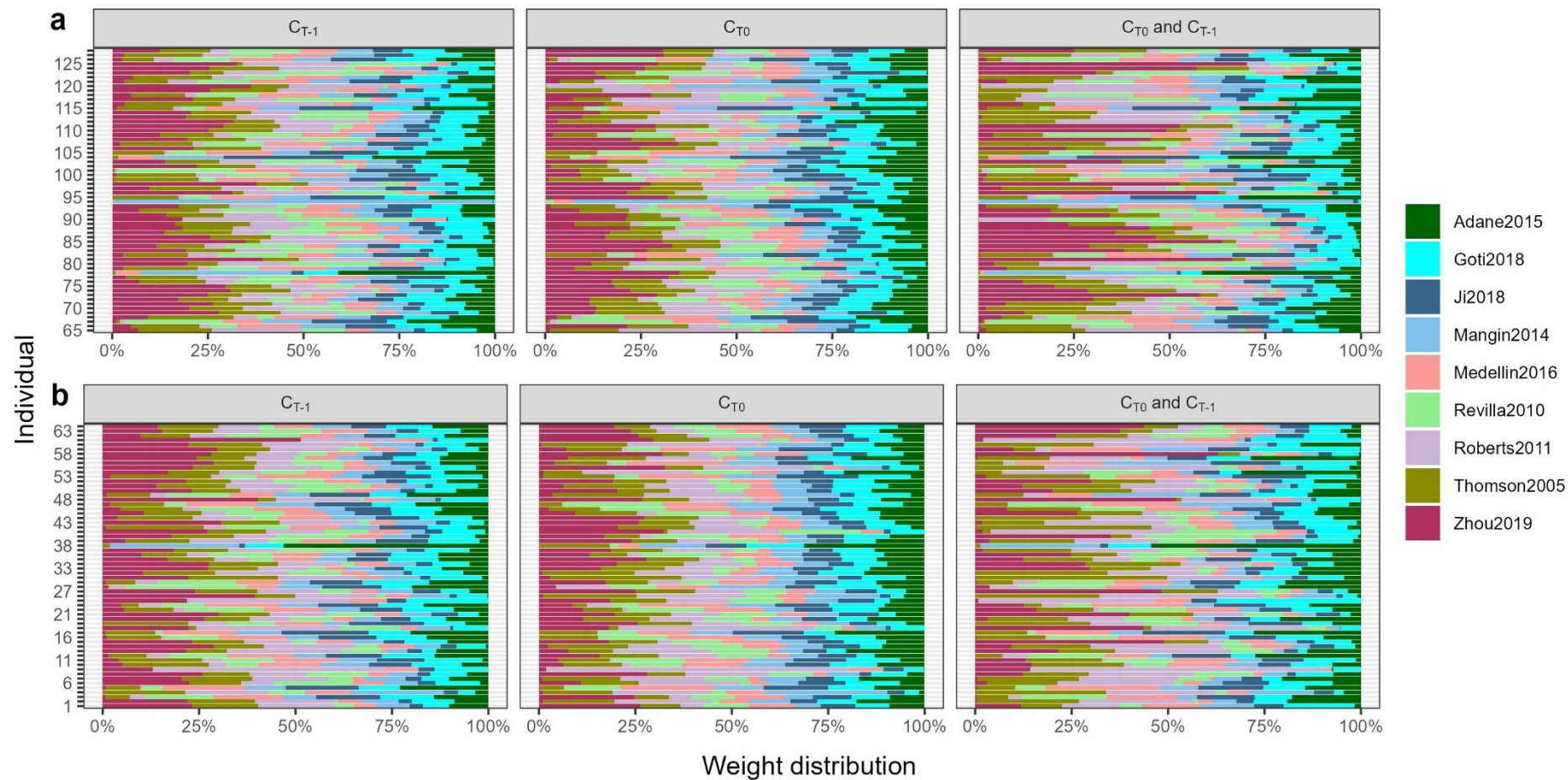


Figure S6 (continue 1/2). Weight contribution of each model per individual in the *a posteriori* predictions with a single most concentration (C_{T0}), single earlier concentration (C_{T-1}), and both concentrations (C_{T0} and C_{T-1}). C, concentration; T, relative time

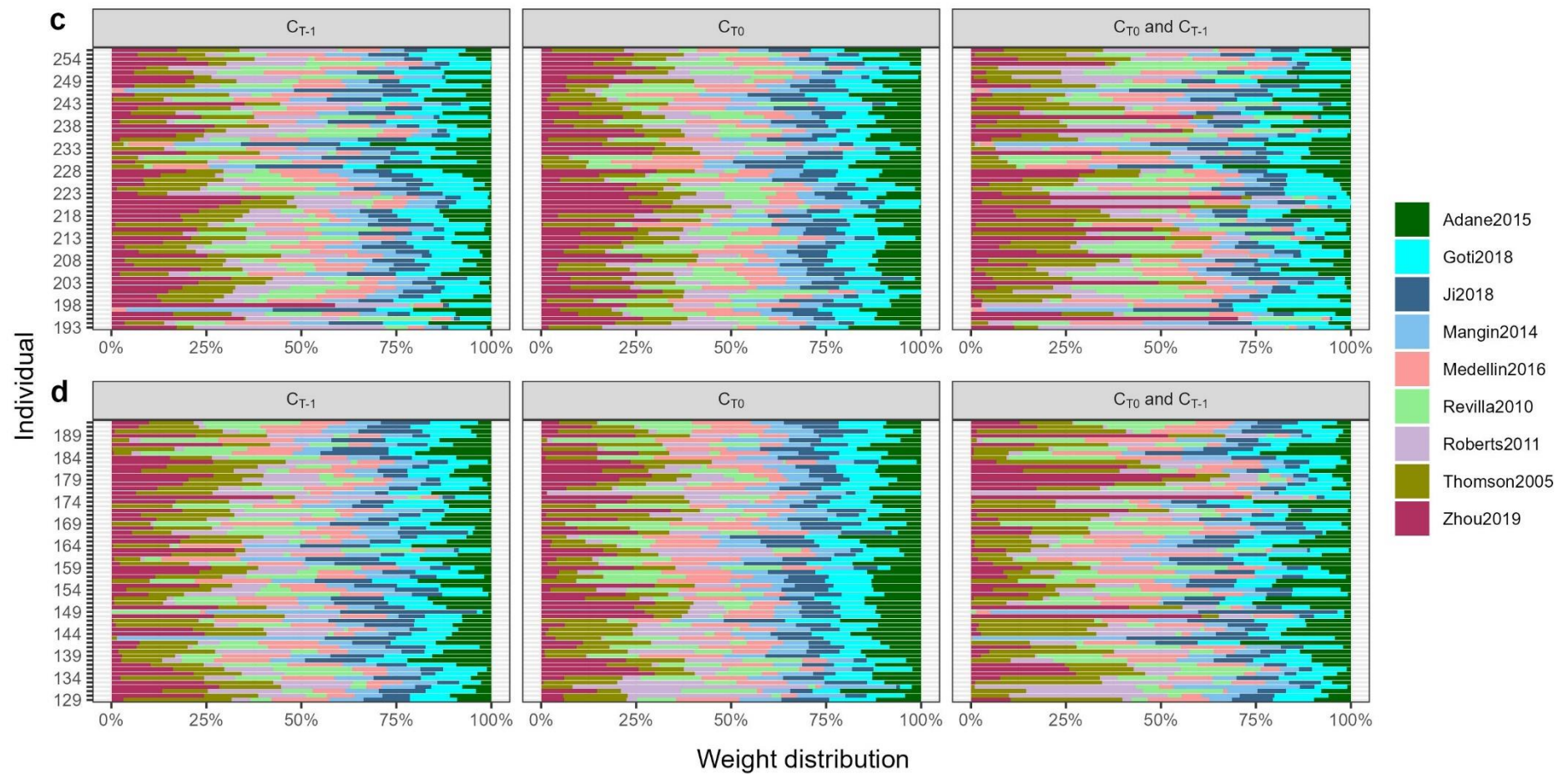


Figure S6 (continue 2/2). Weight contribution of each model per individual in the *a posteriori* predictions with a single most concentration (C_{T0}), single earlier concentration (C_{T-1}), and both concentrations (C_{T0} and C_{T-1}). C, concentration; T, relative time

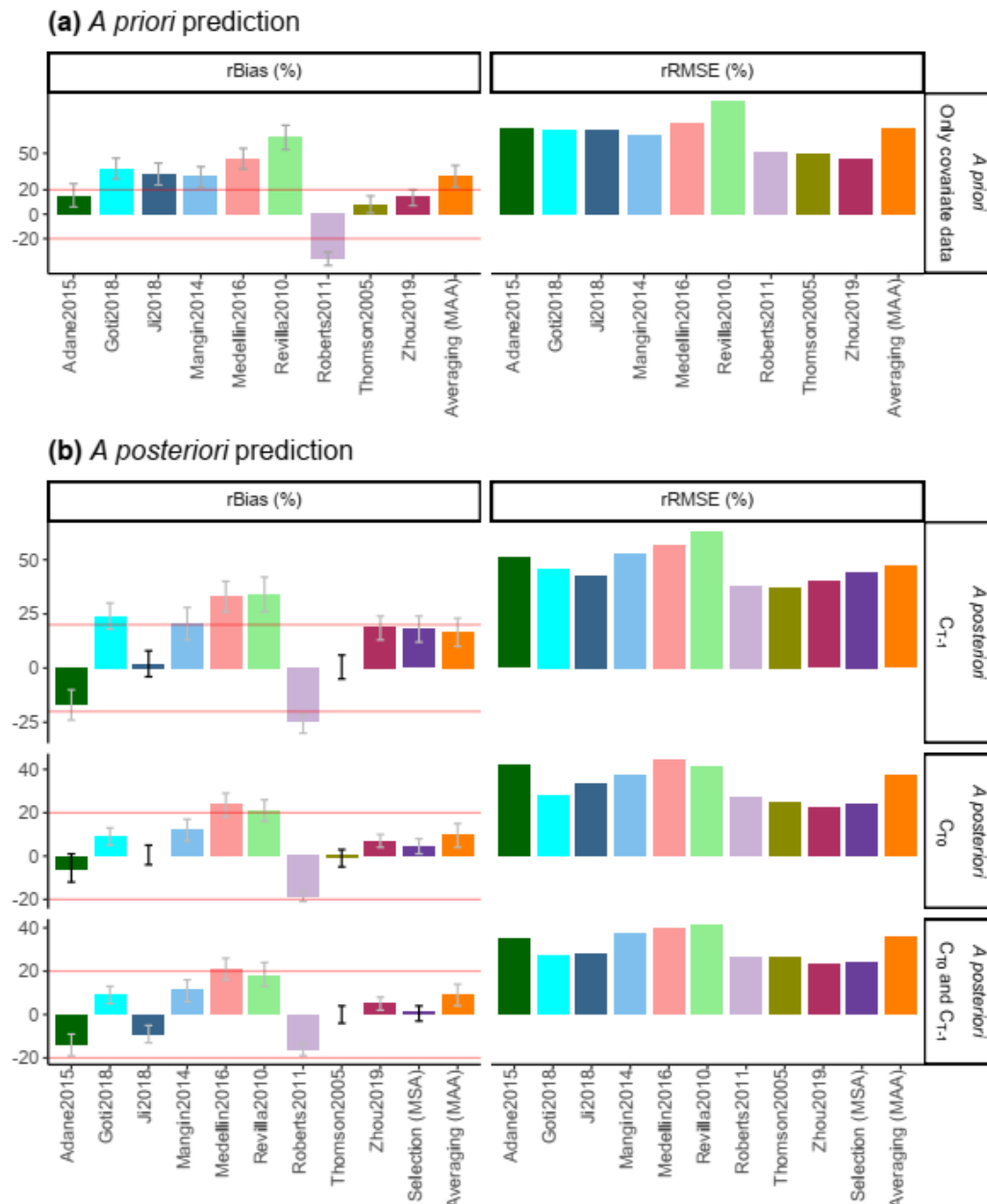


Figure S6. The predictive performance of the single and multi-model approaches using all the 9 models versus the three models to predict vancomycin concentration (C) at a time +1 (T+1) following **intermittent infusion** administration. **(a)** *A priori* predictions using the clinical dataset with covariate data without previous concentrations, and **(b)** *a posteriori* prediction settings using the clinical dataset with covariate data and previous concentrations (CT₋₁, CT₀, CT₋₁ and CT₀). Whiskers cover the 95% confidence interval of the relative bias calculated via the standard error (black whisker indicated 95% CI including 0). Model weighing in *a priori* predictions correspond to 1/number of models, disabling model selection. The horizontal lines in relative bias panels indicate $\pm 20\%$ of the relative bias. rBias, relative bias; rRMSE, relative root mean square error; C, concentration; T, relative time

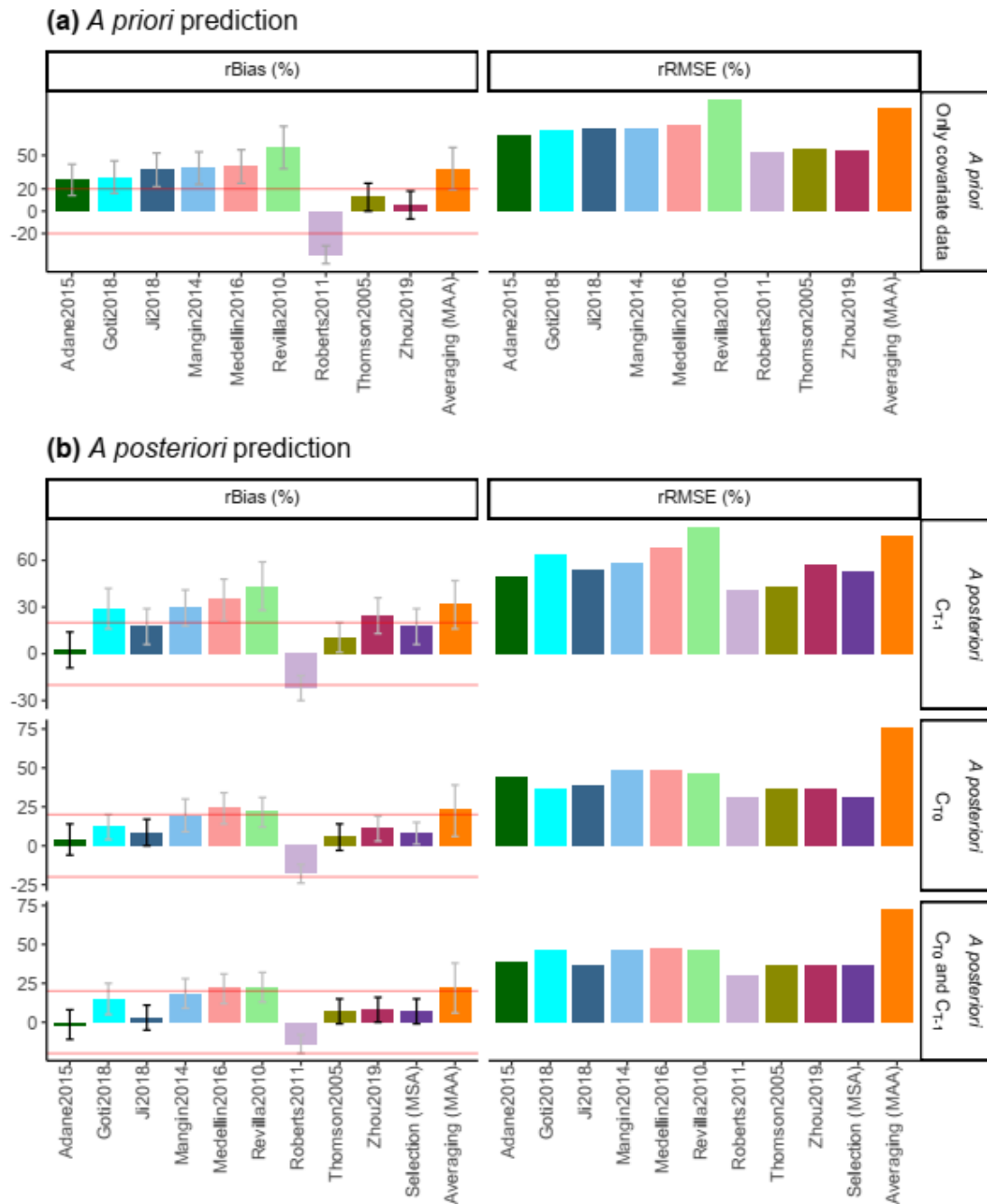


Figure S7. The predictive performance of the single and multi-model approaches using all the 9 models versus the three models to predict vancomycin concentration (C) at a time +1 (T+1) following **continuous infusion** administration. **(a)** *A priori* predictions using the clinical dataset with covariate data without previous concentrations, and **(b)** *a posteriori* prediction settings using the clinical dataset with covariate data and previous concentrations (CT₋₁, CT₀, CT₋₁ and CT₀). Whiskers cover the 95% confidence interval of the relative bias calculated via the standard error (black whisker indicated 95% CI including 0). Model weighing in *a priori* predictions correspond to 1/number of models, disabling model selection. The horizontal lines in relative bias panels indicate $\pm 20\%$ of the relative bias. rBias, relative bias; rRMSE, relative root mean square error; C, concentration; T, relative time

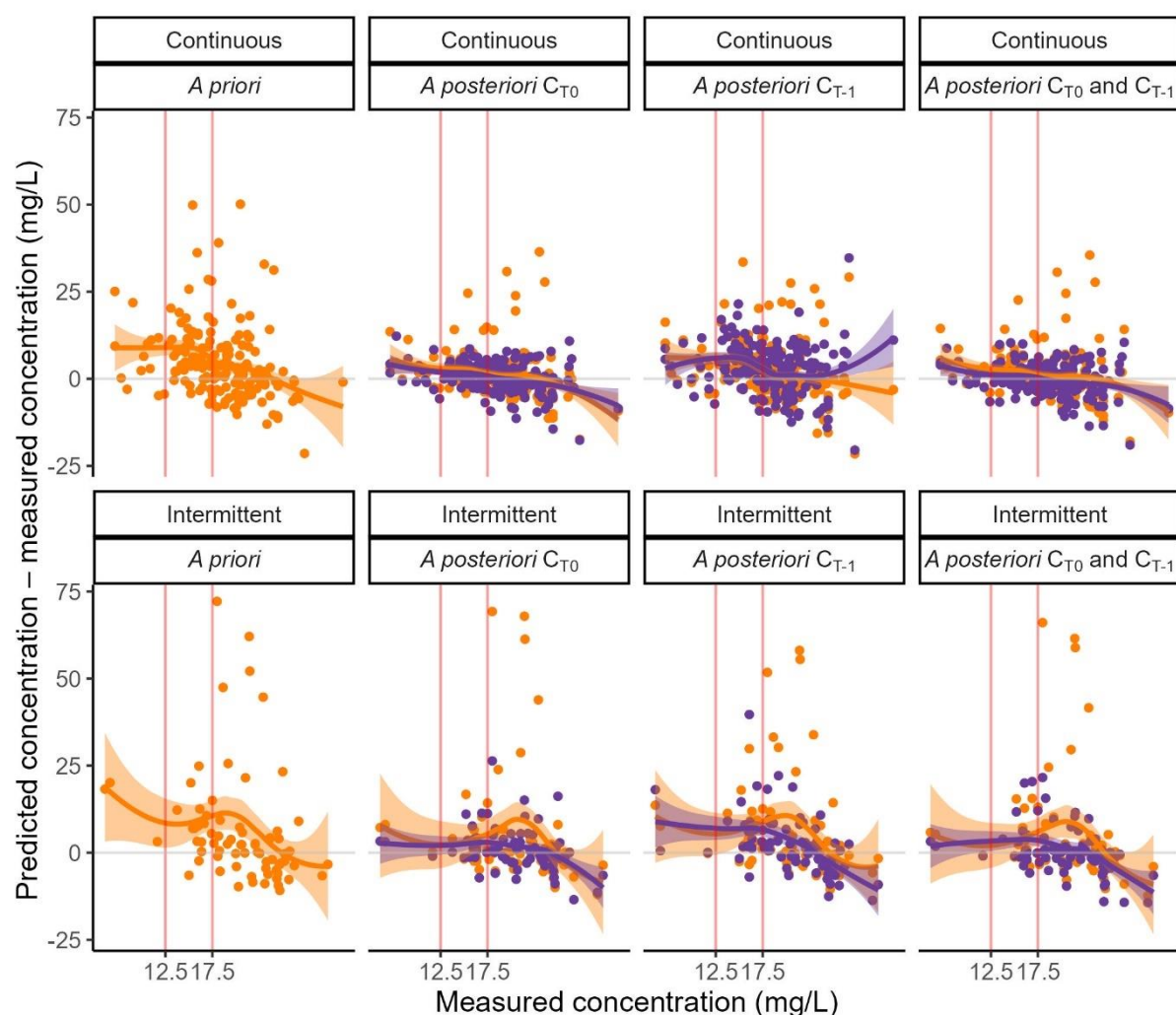


Figure S9. Bland–Altman plots stratified by type of infusion where an agreement between the measured and the predicted vancomycin concentrations was observed in the *a priori* and *a posteriori* predictions using the model selection algorithm (MSA; purple) and the model averaging algorithm (MAA; orange). The vertical red lines indicates the 12.5 and 17.5 mg/L concentration target range. The solid line with shaded area is a locally smoother with 95% confidence interval based on the data (MSA in purple and MAA in orange). C, concentration; T, relative time

Supplementary material
NONMEM control streams

```

$PROBLEM      Adane et al. 2015 Pharmacotherapy – PMID: 25644478
$INPUT        ID TIME TAD AMT RATE DV MDV EVID CMT SEX AGE DIABETES WEIGHT
              HEIGHT BMI ICU SCr EGFR BUN ALBUMIN TOTAL_PROTEIN DIALYSIS
              CVVH FUROSEMIDE DOSE ETH
$DATA         data.csv IGNORE=C
$SUBROUTINE   ADVAN13 TOL=9
$MODEL        NCOMPARTMENTS=1
              COMP=(CENTRAL)
$PK           BSA          = (WEIGHT**0.425)*(HEIGHT**0.725)*0.007184
              IF(SCr.LT.1)  SCT=1
              IF(SCr.GE.1)  SCT=SCr
              IF(SEX.EQ.1)  = ((1.00*(WEIGHT*(140-AGE)))/(72*SCT))*(1.73/BSA)
              CrCL          ; Males
              IF(SEX.EQ.0)  = ((0.85*(WEIGHT*(140-AGE)))/(72*SCT))*(1.73/BSA)
              CrCL          ; Females
              CLBSA         = CrCL
              TVCL          =THETA(1)*(CLBSA/125)
              CL             =TVCL*EXP(ETA(1))
              TVV1          = THETA(2)*WEIGHT
              V1             =TVV1*EXP(ETA(2))
              S1             =V1
              K10            =CL/V1
$DES          DADT(1)      = - K10*A(1)
$error        IPRED        =F
              IRES         =DV-IPRED
              DEL           =0
              IF (F.EQ.0) DEL =1
              W             =IPRED+DEL
              Y             =IPRED+W*ERR(1)
$THETA        6.54        ; TVCL
              0.51        ; TVV1
$OMEGA        0.071       ; CL
              0.057       ; V1
$SIGMA        0.035       ; Proportional error
$SIMULATION   METHOD=1 INTER MAXEVAL=0 PRINT=5 NOABORT POSTHOC
$TABLE        FILE=Output_data.csv

```

```

$PROBLEM    Goti et al. 2018 The Drug Monit – PMID: 29470227
$INPUT      ID TIME TAD AMT RATE DV MDV EVID CMT SEX AGE DIABETES WEIGHT
            HEIGHT BMI ICU SCr EGFR BUN ALBUMIN TOTAL_PROTEIN DIALYSIS
            CVVH FUROSEMIDE DOSE ETH
$DATA       data.csv IGNORE=C
$SUBROUTINE ADVAN13 TOL=9
$MODEL      NCOMPARTMENTS=2
            COMP=(CENTRAL)
            COMP=(PERIPH)
$PK         IF(SCr.LT.1.AND.AGE.GT.60) SCT = 1.0
            IF (SCr.GE.1.OR.AGE.LE.60) SCT = SCr
            IF(SEX.EQ.1) CrCL = (1.00*WEIGHT*(140-AGE))/(72*SCT) ; Male
            IF(SEX.EQ.0) CrCL = (0.85*WEIGHT*(140-AGE))/(72*SCT) ;
            Female
            IF (CrCL.GT.150) CCT = 150
            IF (CrCL.LE.150) CCT = CrCL
            TVCL =
            THETA(1)*((CCT/120)**THETA(2))*(THETA(3)**
            DIALYSIS)
            CL = TVCL*EXP(ETA(1))
            TVV1 =
            THETA(4)*(WEIGHT/70)*(THETA(5)**DIALYSIS)
            V1 = TVV1*EXP(ETA(2))
            TVQ = THETA(6)
            Q = TVQ
            TVV2 = THETA(7)*(WEIGHT/70)
            V2 = TVV2* EXP(ETA(3))
            S1 = V1
            S2 = V2
            K10 = CL/V1
            K12 = Q/V1
            K21 = Q/V2
$DES        DADT(1) = K21*A(2) - K12*A(1) - K10*A(1)
            DADT(2) = K12*A(1) - K21*A(2)
$ERROR      IPRED = F
            IRES = DV-IPRED
            Y = IPRED*(1+ERR(1))+ ERR(2)
$THETA      4.5 ; TVCL
            0.8 ; CrCL on CL
            0.7 ; Dialysis on CL
            58.4 ; TVV1
            0.5 ; Dialysis on V1
            6.5 ; TVQ

```

	38.4	; TVV2
\$OMEGA	0.158	; CL
	0.666	; V1
	0.326	; V2
\$SIGMA	11.56	; Proportional error
	0.515	; Additive error
\$SIMULATION	METHOD=1 INTER MAXEVAL=0 PRINT=5 NOABORT POSTHOC	
\$TABLE	FILE=Output_data.csv	

\$PROBLEM	Ji et al. 2018 <i>Acta Pharmacol Sin</i> PMID: 28836582	
\$INPUT	ID TIME TAD AMT RATE DV MDV EVID CMT SEX AGE DIABETES WEIGHT HEIGHT BMI ICU SCr EGFR BUN ALBUMIN TOTAL_PROTEIN DIALYSIS CVVH FUROSEMIDE DOSE ETH	
\$DATA	data.csv IGNORE=C	
\$SUBROUTINE	ADVAN13 TOL=9	
\$MODEL	NCOMPARTMENTS=2 COMP=(CENTRAL) COMP=(PERIPH)	
\$PK	IF(SEX.EQ.1) CrCL	= (1.00*WEIGHT*(140-AGE))/(72*SCr) ; Male
	IF(SEX.EQ.0) CrCL	= (0.85*WEIGHT*(140-AGE))/(72*SCr) ; Female
	IF (FUROSEMIDE.EQ.0)	= THETA(1)*CrCL*(60/1000)
	TVCL	
	IF (FUROSEMIDE.EQ.1)	= THETA(2)*CrCL*(60/1000)
	TVCL	
	CL	= TVCL*EXP(ETA(1))
	(AGE.LE.65) TVV1	= THETA(3)*WEIGHT
	(AGE.GT.65) TVV1	= THETA(4)*WEIGHT
	V1	= TVV1*EXP(ETA(2))
	TVQ	= THETA(4)
	Q	= TVQ
	TVV2	= THETA(5)*(WEIGHT)
	V2	= TVV2
	S1	= V1
	S2	= V2
	K10	= CL/V1
	K12	= Q/V1
	K21	= Q/V2
\$DES	DADT(1)	= K21*A(2) - K12*A(1) - K10*A(1)
	DADT(2)	= K12*A(1) - K21*A(2)
\$ERROR	IPRED	= F
	IRES	= DV-IPRED
	Y	= IPRED*(1+ERR(1))+ ERR(2)
\$THETA	0.49	; TVCL without Furosemide
	0.34	; TVCL with Furosemide
	0.7	; Dialysis on CL
	0.74	; TVV1 in young patients
	1.07	; TVV1 in older patients
	0.81	; TVQ
	5.86	; TVV2
\$OMEGA	0.137	; CL
	0.16	; V1
\$SIGMA	0.037	; Proportional error
	12.25	; Additive error

```
$SIMULATION METHOD=1 INTER MAXEVAL=0 PRINT=5 NOABORT POSTHOC  
$TABLE      FILE=Output_data.csv
```

```

$PROBLEM      Mangin et al. 2014 Clin Pharmacokinet – PMID: 25117184
$INPUT        ID TIME TAD AMT RATE DV MDV EVID CMT SEX AGE DIABETES WEIGHT HEIGHT BMI ICU
              TOTAL_PROTEIN DIALYSIS CVVH FUROSEMIDE DOSE ETH
$DATA         data.csv IGNORE=C
$SUBROUTINE   ADVAN13 TOL=9
$MODEL        NCOMPARTMENTS=2
              COMP=(CENTRAL)
              COMP=(PERIPH)
$PK           IF(SEX.EQ.1) CrCL  = (1.00*WEIGHT*(140-AGE))/(72*SCr); Males
              IF(SEX.EQ.0) CrCL  = (0.85*WEIGHT*(140-AGE))/(72*SCr); Females
              IF(SEX.EQ.1)      = THETA(1)
              F_FEM
              IF(SEX.EQ.0)      = 1
              F_FEM
              SCT                = SCr*88.4
              IF(DIABETES.EQ.1) = THETA(2)
              F_DM
              IF(DIABETES.EQ.0) = 1
              F_DM
              SAPSII             = 50
              TVCL               =THETA(3)*F_FEM**((WEIGHT/70)**0.75)*((SAPSII/50)**THETA(4)
              CL                 =TVCL*EXP(ETA(1))
              TVV1               = THETA(6)*(WEIGHT/70)
              V1                 =TVV1
              TVQ                =THETA(7)*F_DM*((WEIGHT/70)**0.75)
              Q                  =TVQ*EXP(ETA(2))
              TVV2               =THETA(8)*(WEIGHT/70)
              V2                 =TVV2*EXP(ETA(3))
              S1                 =V1
              S2                 =V2
              K10                = CL/V1
              K12                = Q/V1
              K21                = Q/V2
$DES          DADT(1)           = K21*A(2) - K12*A(1) - K10*A(1)
              DADT(2)           = K12*A(1) - K21*A(2)
$ERROR        IPRED             = F
              IRES              = DV-IPRED
              Y                  = IPRED + ERR(1)
$THETA        0.66              ; F_FEM on CL
              0.30              ; F_DM on Q
              1.91              ; TVCL
              -0.5              ; SAPSII on CL
              -0.9              ; SCT on CL
              21.9              ; TVV1
              5.71              ; TVQ
              68.0              ; TVV2

```

```
$OMEGA      0.084      ; CL
             1.020      ; Q
             0.271      ; V2
$SIGMA      53.58      ; Additive error
$SIMULATION  METHOD=1 INTER MAXEVAL=0 PRINT=5 NOABORT POSTHOC
$TABLE      FILE=Output_data.csv
```

\$PROBLEM	Medellin–Garibay <i>et al.</i> 2016 <i>J Antimicrob Chemother</i> – PMID: 26568565	
\$INPUT	ID TIME TAD AMT RATE DV MDV EVID CMT SEX AGE DIABETES WEIGHT HEIGHT BMI ICU SCr EGFR BUN ALBUMIN TOTAL_PROTEIN DIALYSIS CVVH FUROSEMIDE DOSE ETH	
\$DATA	data.csv IGNORE=C	
\$SUBROUTINE	ADVAN13 TOL=9	
\$MODEL	NCOMPARTMENTS=2 COMP=(CENTRAL) COMP=(PERIPH)	
\$PK	IF(SEX.EQ.1) CrCL	= (1.00*WEIGHT*(140-AGE))/(72*SCr) ; Male
	IF(SEX.EQ.0) CrCL	= (0.85*WEIGHT*(140-AGE))/(72*SCr) ; Female
	IF (FUROSEMIDE.EQ.0)	= THETA(1)*CrCL*(60/1000)
	TVCL	
	IF (FUROSEMIDE.EQ.1)	= THETA(2)*CrCL*(60/1000)
	TVCL	
	CL	= TVCL*EXP(ETA(1))
	(AGE.LE.65) TVV1	= THETA(3)*WEIGHT
	(AGE.GT.65) TVV1	= THETA(4)*WEIGHT
	V1	= TVV1*EXP(ETA(2))
	TVQ	= THETA(4)
	Q	= TVQ
	TVV2	= THETA(5)*(WEIGHT)
	V2	= TVV2
	S1	= V1
	S2	= V2
	K10	= CL/V1
	K12	= Q/V1
	K21	= Q/V2
\$DES	DADT(1)	= K21*A(2) - K12*A(1) - K10*A(1)
	DADT(2)	= K12*A(1) - K21*A(2)
\$ERROR	IPRED	= F
	IRES	= DV-IPRED
	Y	= IPRED*(1+ERR(1))+ ERR(2)
\$THETA	0.49	; TVCL without Furosemide
	0.34	; TVCL with Furosemide
	0.7	; Dialysis on CL
	0.74	; TVV1 in young patients
	1.07	; TVV1 in older patients
	0.81	; TVQ
	5.86	; TVV2
\$OMEGA	0.137	; CL
	0.16	; V1
\$SIGMA	0.037	; Proportional error
	12.25	; Additive error

```
$SIMULATION METHOD=1 INTER MAXEVAL=0 PRINT=5 NOABORT POSTHOC  
$TABLE      FILE=Output_data.csv
```

```

$PROBLEM  Revilla et al. 2010 Br J Clin Pharmacol – PMID: 20653673
$INPUT    ID TIME TAD AMT RATE DV MDV EVID CMT SEX AGE DIABETES WEIGHT
          HEIGHT BMI ICU SCr EGFR BUN ALBUMIN TOTAL_PROTEIN DIALYSIS
          CVVH FUROSEMIDE DOSE ETH
$DATA     data.csv IGNORE=C
$SUBROUT  ADVAN13 TOL=9
INE
$MODEL    NCOMPARTMENTS=1
          COMP=(CENTRAL)
$PK        BSA = (WEIGHT**0.425)*(HEIGHT**0.725)*0.007184
          IF(SEX.EQ.1.AND.ETH = 170*(SCr**(-0.999))*(AGE**(-0.176))*(BUN**(-
          .EQ.0) CrCL = 0.17))*((ALBUMIN**0.318)*1.000*1.00)*((BSA/1.
          73)/WEIGHT) ; Male Caucasian
          IF(SEX.EQ.1.AND.ETH = 170*(SCr**(-0.999))*(AGE**(-0.176))*(BUN**(-
          .EQ.1) CrCL = 0.17))*((ALBUMIN**0.318)*1.000*1.18)*((BSA/1.
          73)/WEIGHT) ; Male Black
          IF(SEX.EQ.0.AND.ETH = 170*(SCr**(-0.999))*(AGE**(-0.176))*(BUN**(-
          .EQ.0) CrCL = 0.17))*((ALBUMIN**0.318)*0.762*1.00)*((BSA/1.
          73)/WEIGHT) ; Female Caucasian
          IF(SEX.EQ.0.AND.ETH = 170*(SCr**(-0.999))*(AGE**(-0.176))*(BUN**(-
          .EQ.1) CrCL = 0.17))*((ALBUMIN**0.318)*0.762*1.18)*((BSA/1.
          73)/WEIGHT) ; Female Black
          IF (SCr.GE.1) AB = 1
          IF (SCr.LT.1) AB = 0
          TVCL = (THETA(1)*CrCL+(AGE**THETA(2)))*((60/1000)
          *WEIGHT)
          CL = TVCL*EXP(ETA(1))
          TVV1 = THETA(3)*(THETA(4)**AB)*WEIGHT
          V1 = TVV1*EXP(ETA(2))
          S1 = V1
          K10 = CL/V1
$DES      DADT(1) = - K10*A(1)
$ERROR    IPRED = F
          IRES = DV-IPRED
          Y = IPRED+ERR(1)
$THETA    0.67 ; TVCL
          0.82 ; TVV1
          -0.24 ; AGE on CL
          2.49 ; Serum creatinine categorical variable on V
$OMEGA    0.087 ; CL
          0.051 ; V1
$SIGMA    17.89 ; Additive error
$SIMULATI METHOD=1 INTER MAXEVAL=0 PRINT=5 NOABORT POSTHOC
ON
$TABLE    FILE=Output_data.csv

```

```

$PROBLEM      Roberts et al. 2011 Antimicrob Agents Chemother – PMID: 21402850
$INPUT        ID TIME TAD AMT RATE DV MDV EVID CMT SEX AGE DIABETES WEIGHT HEIGHT
              BMI ICU SCr EGFR BUN ALBUMIN TOTAL_PROTEIN DIALYSIS CVVH
              FUROSEMIDE DOSE ETH
$DATA         data.csv IGNORE=C
$SUBROUTINE   ADVAN13 TOL=9
$MODEL        NCOMPARTMENTS=1
              COMP=(CENTRAL)
$PK           UCrCL           =100
              TVCL            =THETA(1)*(UCrCL/100)
              CL              =TVCL*EXP(ETA(1))
              TVV1            = THETA(2)*WEIGHT
              V1              =TVV1*EXP(ETA(2))
              S1              =V1
              K10              =CL/V1
$DES          DADT(1)         = - K10*A(1)
$error        IPRED           =F
              IRES            =DV-IPRED
              Y               =IPRED*(1+ERR(1))+ ERR(2)
$THETA        4.58            ; TVCL
              1.53            ; TVV1
$OMEGA        0.151           ; CL
              0.140           ; V1
$SIGMA        0.039           ; Proportional error
              5.76            ; Additive error
$SIMULATION   METHOD=1 INTER MAXEVAL=0 PRINT=5 NOABORT POSTHOC
$TABLE        FILE=Output_data.csv

```

```

$PROBLEM      Thomson et al. 2009 J Antimicrob Chemother – PMID: 19299472
$INPUT        ID TIME TAD AMT RATE DV MDV EVID CMT SEX AGE DIABETES WEIGHT
              HEIGHT BMI ICU SCr EGFR BUN ALBUMIN TOTAL_PROTEIN DIALYSIS
              CVVH FUROSEMIDE DOSE ETH
$DATA         data.csv IGNORE=C
$SUBROUTINE   ADVAN13 TOL=9
$MODEL        NCOMPARTMENTS=2
              COMP=(CENTRAL)
              COMP=(PERIPH)
$PK           IF (SCr.LT.0.68) SCT      = 0.68
              IF (SCr.GE.0.68) SCT      = SCr
              IF (SEX.EQ.1) CrCL        = (1.00*WEIGHT*(140-AGE))/(72*SCT) ; Male
              IF (SEX.EQ.0) CrCL        = (0.85*WEIGHT*(140-AGE))/(72*SCT) ;
                                                Female
              TVCL                      = THETA(1)*THETA(2)*CrCL
              CL                        = TVCL*EXP(ETA(1))
              TVV1                      = THETA(3)*WEIGHT
              V1                        = TVV1*EXP(ETA(2))
              TVQ                       = THETA(4)
              Q                         = TVQ*EXP(ETA(4))
              TVV2                      = THETA(5)*WEIGHT
              V2                        = TVV2*EXP(ETA(3))
              S1                        = V1
              S2                        = V2
              K10                       = CL/V1
              K12                       = Q/V1
              K21                       = Q/V2
$DES          DADT(1)                   = K21*A(2) - K12*A(1) - K10*A(1)
              DADT(2)                   = K12*A(1) - K21*A(2)
$ERROR        IPRED                     = F
              IRES                      = DV-IPRED
              Y                         = IPRED*(1+ERR(1))+ ERR(2)
$THETA        2.99                      ; TVCL
              0.0154                    ; CrCL on CL
              0.675                      ; TVV1
              2.28                       ; TVQ
              0.732                      ; TVV2
$OMEGA        0.073                     ; CL
              0.023                      ; V1
              1.690                      ; V2
              0.240                      ; Q
$SIGMA        0.023                     ; Proportional error
              2.56                       ; Additive error
$SIMULATION   METHOD=1 INTER MAXEVAL=0 PRINT=5 NOABORT POSTHOC
$TABLE        FILE=Output_data.csv

```

```

$PROBLEM      Zhou et al. 2019 Eur J Drug Metab Pharmacokinet – PMID: 30506225
$INPUT        ID TIME TAD AMT RATE DV MDV EVID CMT SEX AGE DIABETES WEIGHT HEIGHT
              BMI ICU SCr EGFR BUN ALBUMIN TOTAL_PROTEIN DIALYSIS CVVH
              FUROSEMIDE DOSE ETH
$DATA         data.csv IGNORE=C
$SUBROUTINE   ADVAN13 TOL=9
$MODEL        NCOMPARTMENTS=1
              COMP=(CENTRAL)
$PK           IF(SEX.EQ.1)      = (1.00*(WEIGHT*(140-AGE))/(72*SCr); Males
              CrCL
              IF(SEX.EQ.0)      = (0.85*(WEIGHT*(140-AGE))/(72*SCr); Females
              CrCL
              TVCL               =THETA(1)* * ((CrCL/56.28)**THETA(2))
              CL                 =TVCL*EXP(ETA(1))
              TVV1               = THETA(3)
              V1                 =TVV1*EXP(ETA(2))
              S1                 =V1
              K10                =CL/V1
$DES          DADT(1)           = - K10*A(1)
$error        IPRED              =F
              IRES               =DV-IPRED
              Y                  =IPRED*(1+ERR(1))
$THETA        2.45              ; TVCL
              0.54              ; CrCL on CL
              154                ; TVV1
$OMEGA        0.030             ; CL
              0.115             ; V1
$SIGMA        0.004             ; Proportional error
$SIMULATION   METHOD=1 INTER MAXEVAL=0 PRINT=5 NOABORT POSTHOC
$TABLE        FILE=Output_data.csv

```


General discussion

1. Main findings and implications for clinical practice

In geriatric patients and older adults, the primary goal of medical interventions is to achieve therapeutic outcomes, prevent risks or adverse event and enhance the quality of life in their remaining years. Drug therapy plays a central role in achieving this objective. However, clinical pharmacology, drug efficacy, and safety can be markedly different in geriatric patients compared to younger adults. These differences often stem from the variability in drug exposure and response, commonly referred to as PK and PD, which describes the relationship between dosage, drug exposure and clinical outcomes.

Driven by a prior PK study conducted by our research group, which demonstrated a high interindividual variability in exposure to paracetamol, a commonly used drug, among geriatric patients (1), this PhD thesis focuses on PK of high risk and narrow therapeutic window drugs in this population. The observed variability, especially pronounced during the absorption phase, could not be accounted for by any of the tested covariates in the study (2). Hence, this thesis investigates the impact of age-related changes in gastrointestinal physiology, specifically gastrointestinal transit and intraluminal pH, on pharmacokinetics. A key aim is to determine whether this variability, not only during absorption but also during distribution, metabolism and excretion, extends to other frequently used high-risk e.g., direct oral anticoagulants and narrow therapeutic window medications e.g. vancomycin.

In summary, this thesis addresses several key areas of PK in the aging population. **Chapter 1** characterized GI transit and pH during fasted state in three different groups: healthy young volunteers, community-dwelling older adults, and hospitalized geriatric patients. **Chapter 2** provided a comprehensive review of the PK and PD of direct oral DOACs in older patients. Using a systematic search, PK/PD changes and important covariates that influence these processes in older adults were identified. This review informed the design of the prospective edoxaban PK study by clarifying which covariate data needed to be collected. In **Chapter 3**, a prospective PK study of edoxaban was conducted in 17 geriatric patients to develop a popPK model and perform simulations. **Chapter 4** evaluated vancomycin exposure in older patients compared to younger individuals, noting that single-point exposure may not adequately reflect total exposure, with the Area Under the Curve (AUC) being more closely associated with treatment outcomes. However, the most accurate model for predicting exposure, i.e. AUC, in older patients remains uncertain. To address this, **Chapter 5** focused on evaluating the predictive performance of published

vancomycin popPK models in older adults. The findings of this PhD thesis are summarized in **Figure 29**.

Key finding 1

Frail geriatric patients had delayed gastric emptying which could explain slower edoxaban absorption that was observed.

Implication for clinical practice

Potential delays in drug absorption could be anticipated. Although it is less important for edoxaban, it matters for drugs with C_{max} as target.

In **Chapter 1**, the comparison between healthy young volunteers and non-frail community-dwelling older adults did not reveal any difference in gastric emptying. Meanwhile, frail geriatric patients who were hospitalized during the course of the study demonstrated significantly longer gastric emptying times than robust older adults and young volunteers. This delay in gastric emptying may explain the finding observed in **Chapter 3**, where the absorption rate constant of edoxaban in geriatric patients was much slower than that reported in popPK studies using data from pivotal clinical trials that did not include geriatric patients or data from other populations.

Gastrointestinal physiology, particularly gastric emptying, plays a critical role in drug absorption, especially for drugs like edoxaban, which falls under Biopharmaceutical Classification System Class IV, where solubility and permeability are the rate-limiting factors. Although our prospective study did not directly assess this relationship, previous research on paracetamol and atorvastatin has shown that delayed gastric emptying can significantly slow the absorption (3). However, directly investigating the impact of gastric emptying on drug absorption *in vivo* is challenging, especially in geriatric patients, a highly complex population. Many factors, such as comorbidities like diabetes or thyroid disorders, along with co-medications that can affect GI transit, make it difficult to isolate the specific impact of gastric emptying on drug absorption. Furthermore, conducting interventional studies in this patient group presents additional challenges due to the high disease burden they already face. Despite these difficulties, the relationship between drug absorption and gastric emptying could potentially be assessed using surrogate markers, such as frailty. For instance, individuals with a clinical frailty score of ≥ 5 may have slower drug absorption as they tend to have prolonged gastric emptying, as shown in **Chapter 1**.

By assessing a patient's frailty, clinicians could anticipate potential delays in drug absorption and adjust dosing strategies accordingly. While the delay in drug absorption observed in our study may be less relevant for repeated dosing of edoxaban, the absorption rate constant remains significant in certain contexts. A lower absorption rate can lead to lower peak concentrations (C_{max}), which may influence therapeutic outcomes, particularly for drugs where C_{max} is critical markers of efficacy or safety. In such cases, adjustments to dosing strategies may be necessary. The absorption rate is typically assessed based on equivalent C_{max} and T_{max} . For drugs such as fluoroquinolones, dosing adjustments may be needed in the event of delayed gastric emptying. Studies have shown that inadequate C_{max}/MIC ratios can contribute to fluoroquinolone resistance, making timely achievement of maximum concentration critical for efficacy (4-7). However, in the case of edoxaban, a slower absorption rate may not require dose adjustment because, based on data from the ENGAGE-AF trial, clinical outcomes were associated with trough concentrations, and their association with C_{max} has not been studied yet (8). Nevertheless, in addition to DOACs, opioids also require rapid absorption, highlighting the need for further research to evaluate the clinical impact of altered absorption rates, therapeutic outcomes, and the potential necessity of a loading dose for initial administration.

Taking all into account, clinicians must remain vigilant when treating geriatric patients with drugs with narrow therapeutic window. TDM or MIPD can play a pivotal role in geriatric patients who may experience symptoms of gastrointestinal motility issues such as nausea, vomiting, or constipation. While the benefits of TDM or MIPD in these specific scenarios have yet to be confirmed in randomized controlled trials, these approaches can provide valuable guidance by tailoring doses based on measured drug levels and predicted pharmacokinetics.

Key finding 2

Large interindividual variability and alterations in PK of edoxaban was observed in geriatric patients.

Implication for clinical practice:

In the absence of the benefit of lower edoxaban dose, TDM may be necessary for geriatric patients in certain cases such as the history of bleeding, or temporary anticoagulant therapy interruption before a high bleeding risk procedure.

We observed large interindividual variability in edoxaban PK among geriatric patients, despite the population studied in **Chapter 3** being relatively homogeneous in terms of age and consisting of a small sample size. This finding aligns with the literature review in **Chapter 2**, which highlighted the high interindividual variability in DOAC exposure among older adults. However, when focusing specifically on frail geriatric patients, our systematic search found no studies comparing DOAC PK between geriatric patients and non-frail older adults. Only the studies by Bendayan et al. and Eidam et al. reported a significant relationship between frailty—measured using physical frailty criteria like Fried’s and sarcopenia, a major component of frailty—and DOAC concentrations. This makes our popPK study of edoxaban the first to provide data specifically observed in geriatric patients. Unfortunately, although our review summarized important and potential covariates in geriatric patients that can explain these PK changes and variability, such as such as renal and liver function, lean mass, and other comorbidities, our sample size was too small to observe the effect of these covariates on PK.

When comparing the typical PK parameters from our popPK model in geriatric patients (**Chapter 3**) with models from the literature (**Chapter 2**) that included older adults but not specifically geriatric patients, we found notable differences. In geriatric patients, total clearance was reduced by 2–3-fold, the volume of distribution decreased by 3–4-fold, and the absorption rate constant was significantly lower, being 5–10 times lower. These PK changes result in higher trough concentrations. This raises the question of whether a lower dose of 15 mg might be appropriate for at least some geriatric patients. Although robust evidence from randomized trials has shown that unmonitored DOACs therapy is at least as effective and safer than warfarin, with lower rates of intracranial hemorrhage and reduced mortality (9), plasma monitoring might still be informative in certain cases. Given the higher

drug exposure in geriatric patients and the uncertain benefits of the 15 mg dose, TDM or even model-informed precision dosing could offer additional benefits in those with a history of bleeding or undergoing high-risk procedures by providing more personalized dosing.

Key finding 3

Age may reflect underlying physiological changes, which can be captured by popPK models.

Implication for clinical practice

Model-informed precision dosing can be used to guide dosing in older patients by incorporating popPK models tailored to geriatric patients.

Even though in **Chapter 3**, we failed to see the relationship between renal function and edoxaban PK, we could show how decreased renal function was associated with higher concentrations for another substance, vancomycin, in **Chapter 4**. Although, age was not a significant predictor in the multivariable analysis in **Chapter 4** once renal function (eGFR) was considered, **Chapter 5** showed that older patients' data were best fitted and predicted using popPK models developed from the same age group. This likely reflects the greater heterogeneity in (patho-)physiological and PK parameters among older adults, allowing these models to provide more flexible predictions tailored to individual patients. Therefore, when using model-informed precision dosing to guide treatment in older and geriatric patients, it is crucial to select models that account for higher interindividual variability and are specifically developed for older populations, where available.

Potential relationship between frailty and drug PK

Frailty is commonly defined as a reduction in physiological reserves and diminished capacity to respond to internal or external stressors (10). Among the various domains of frailty, physical impairment appears directly related to PK alterations, affecting absorption, distribution, metabolism, and elimination. While age alone generally does not affect drug absorption, frailty can impair gastric motility, potentially leading to delayed absorption (**Chapter 1 and 3**). Based on a previous study, distribution is also altered in frail older adults due to changes in body composition, such as increased body fat and reduced lean body mass (11). This can increase the volume of distribution for lipophilic drugs, resulting in lower plasma concentrations and prolonged half-lives (12). For hydrophilic drugs, the volume of distribution may decrease, though findings on this have been inconsistent in frail

populations (13). Metabolic changes are particularly pronounced in frailty, with reduced hepatic blood flow and liver size further impairing drug metabolism for flow-limited drugs (14). As summarized by Hilmer et al., the metabolism of such drugs, including morphine, propranolol, verapamil, and amitriptyline, is reduced by approximately 40%, aligning with the reduction in hepatic blood flow observed in older adults (15). In **Chapter 3**, we attempted to test the relationship between frailty with distribution and metabolism using biomarkers of lean mass and liver function, but these relationships were not confirmed, likely due to the small sample size. Elimination is significantly impacted by frailty, as declining renal function and reduced clearance of renally-excreted drugs are common (**Chapter 4**). Beside physiological changes, these PK alterations are often worsened by polypharmacy (co-medication with PPI [**Chapter 3**]) and comorbidities (e.g., heart failure and diabetes), which are prevalent in frail individuals. As a result, drug dosing in this vulnerable population must be carefully re-evaluated to ensure it is optimal, avoiding toxicity while maintaining efficacy.

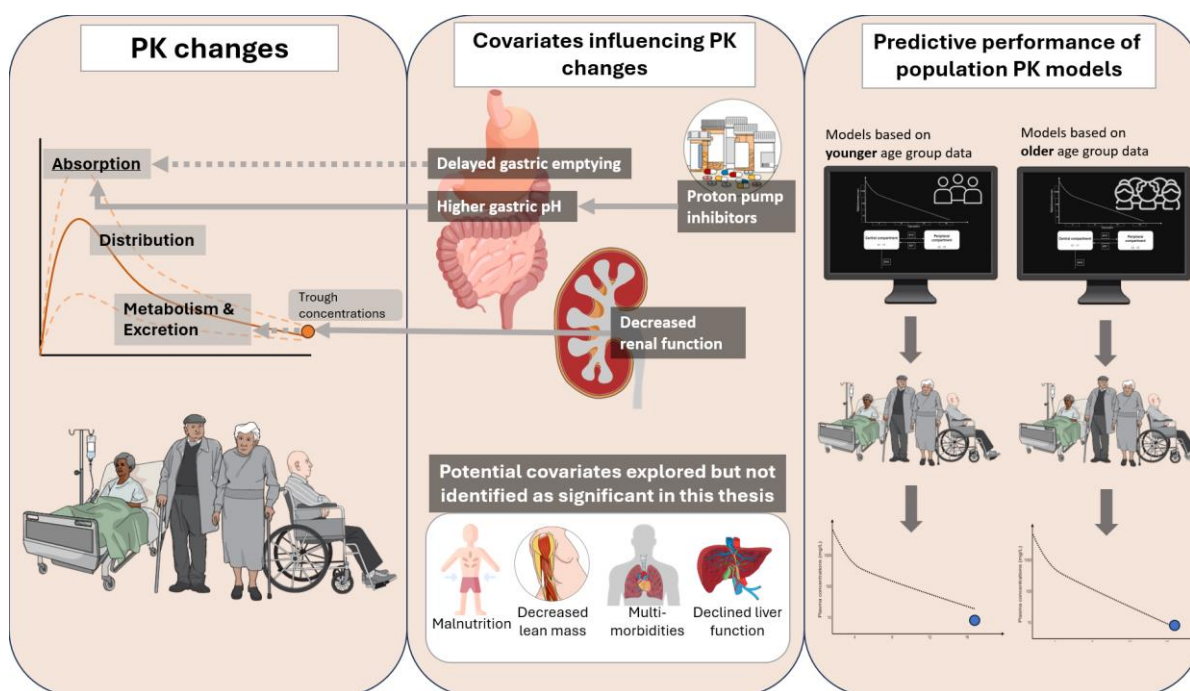


Figure 29. The summary of the main findings of this PhD thesis. PK changes in geriatric patients were influenced by decreased renal function and, most likely, by significant delays in gastric emptying and elevated gastric pH. Other patient characteristics, such as malnutrition, reduced lean body mass, multimorbidity, and impaired liver function, were explored but not identified as significant contributors in this thesis. Additionally, popPK models developed using data from older age groups demonstrated better fit for the geriatric population studied, enabling more accurate and precise dosing predictions.

2. Limitations

The main limitation of our prospective studies (**Chapters 1** and **3**) was the low number of geriatric patients who could be successfully included. Several barriers contributed to this, primarily the burden of pre-existing illnesses, as well as concerns from families, and perceived cognitive impairments that hindered participation. Because participants needed to fully understand and consent to the study, cognitive adequacy was a requirement to ensure they could understand the process of the study. This was essential for participants to make informed decisions based on comprehensive explanations. Given that cognitive limitations affect approximately 60% of individuals in our geriatric wards, we were only able to enroll a small percentage of patients. This limited sample size also restricted our ability to assess the in vivo impact of gastrointestinal transit on edoxaban PK, as swallowing a

wireless motility capsule and collecting repeated blood sampling were too burdensome for geriatric patients who already dealt with significant health challenges.

The second limitation of our studies in **Chapters 3 and 4** is the use of therapeutic targets for edoxaban and vancomycin derived from the general adult population, which may not necessarily result in optimal outcomes for geriatric patients. For edoxaban, geriatric patients are known to be more susceptible to both bleeding and stroke, raising concerns about whether the observed range of 12–43 ng/mL, based on data from the ENGAGE-AF trial, is appropriate for this population. This is because geriatric patients were underrepresented in the trial, and it is possible that a narrower therapeutic window may be required to balance efficacy and safety, minimizing both bleeding risks and treatment failure in geriatric patients.

Similarly, for vancomycin, the therapeutic targets used in our study may not fully account for the increased vulnerability of older patients to both drug toxicity and infections, given their age-related immune decline (i.e., immunosenescence). The older population is particularly prone to vancomycin-induced nephrotoxicity, and their altered immune response may affect treatment outcomes. Current evidence is only based on a Japanese study suggesting that a more specific AUC/MIC target of 250–450 $\mu\text{g}\cdot\text{h}/\text{mL}$ may improve efficacy while reducing toxicity risks in older patients (16). However, further research is needed to externally validate these ranges.

3. Future perspectives

Given these limitations, the next logical step after focusing on PK in this PhD thesis is to evaluate the relationship with clinical outcomes, i.e. PD. The findings from this thesis strongly suggest that PD variability, despite limited evidence in geriatric patients, could be as pronounced as PK variability in this population, further complicating drug therapy management. A joint PK/PD modeling effort is urgently needed for this population to better understand the relationship between drug exposure and therapeutic response, as well as to identify the drivers of variability. Such models would enable more precise dosing strategies, optimizing efficacy while minimizing adverse effects in geriatric patients. By incorporating real-world data and accounting for the physiological changes and comorbidities common

in geriatric patients, PK/PD models can guide individualized treatment. This is particularly critical for drugs with narrow therapeutic windows, such as anesthetics and opioids, which pose a higher risk of adverse effects, including delirium, cognitive impairment, and other neurotoxic effects. Antibiotics also warrant close attention, not only because geriatric patients are more prone to infections due to immunosenescence, but also because they are more susceptible to toxicities like diarrhea, allergic reactions, delirium and nephrotoxicity. These side effects can have a significant impact on the quality of life as well as clinical outcomes, particularly in frail, older patients with multiple comorbidities. Furthermore, the growing issue of antibiotic resistance adds another layer of complexity, making it essential to balance drug efficacy with safety in this vulnerable group. Ultimately, future models should adopt a holistic approach, as PK alone may not be sufficient to predict efficacy and safety. Integrating PK/PD modeling with clinically relevant and easily measurable endpoints such as these antibiotic toxic effects will enhance their applicability and utility in this population.

As specified above, geriatric patients present numerous interrelated covariates that affect both PK and PD with a particular emphasis on frailty for future PK/PD modeling. While other covariates are also important, frailty warrants special attention because it has been largely understudied, yet our findings in **Chapter 1** suggest it may play a significant role, especially in how it affects drug absorption rates and the overall extent of absorption. Artificial intelligence (AI), particularly machine learning (ML) algorithms combined with pharmacometrics modeling, offers a promising approach to efficiently identify patterns and relationships between these variables. For example, Ma et al. demonstrated that integrating ML algorithms with popPK models enables rapid and cost-effective predictions of vancomycin concentrations (17). However, the challenge with employing ML lies in the need for large datasets, and collecting prospective data from geriatric patients is particularly difficult. A viable solution is to use retrospective data from electronic health records to train these algorithms, allowing the models to continuously improve with new data. One of the key advantages of ML is its ability to handle complex models efficiently. Unlike population PK models, ML does not require solving differential equations, meaning its performance remains robust regardless of model complexity. A potential approach is to use ML for an initial covariate screening, identifying the most important variables, which can then be

incorporated into a final model-building step using established pharmacometrics methods like stepwise covariate modeling. Keutzer et al. demonstrated that ML could provide fast predictions of both full concentration-time series and PK exposure indices, e.g., AUC_{24h} or trough concentrations, with acceptable precision (18). However, they recommended that ML be used as an assistive tool in pharmacometrics modeling, rather than a primary decision-making tool, at this stage. While ML can streamline and enhance traditional approaches, its results can be more challenging to interpret, making it essential to combine ML with pharmacometrics models to ensure accuracy and clinical relevance.

As highlighted in the introduction, PBPK modeling provides a complementary approach to evaluating the impact of physiological changes in geriatric patients. The findings from Chapter 1 of this thesis, which reveal significant alterations in GI physiology, can be incorporated into PBPK models to predict drug disposition and optimize dosing strategies more accurately. Conducting clinical studies to directly test the effects of GI physiology alterations on the PK of edoxaban is challenging. However, PBPK modeling offers a valuable alternative by allowing us to explore these relationships *in silico*. By integrating the data from Chapter 1 and validating the models with the clinical data presented in Chapter 3, PBPK models can bridge the gap created by the limited availability of clinical data in geriatric patients. Importantly, PBPK models are not limited to changes in a single organ system, such as the GI tract. They can also be used to simulate drug-drug interaction which is common in geriatric patients. This approach enhances our ability to predict the complex interplay of physiological changes in geriatric patients, ultimately leading to more accurate dose recommendations.

In addition to mathematical models, preclinical systems, including both *in vitro* and *in vivo* (animal) models, can provide valuable insights to inform clinical study designs, such as identifying and selecting relevant covariates. For example, the TNO Intestinal Model is a dynamic *in vitro* system that can simulate intraluminal drug behavior under conditions mimicking the GI environment. Parameters such as GI transit time, pH, and the effects of fed or fasted states can be adjusted to reflect physiological traits commonly observed in geriatric patients, such as delayed gastric emptying or increased gastric pH. Additionally, animal models provide a complementary approach to *in vitro* systems, allowing for the

exploration of PK/PD under controlled in vivo conditions. For instance, aged rat models can be used in preclinical PK studies to identify potential covariates that warrant further investigation in clinical PK studies.

References

1. Hias J, Van der Linden L, Walgraeve K, Gijssen M, Mian P, Koch BCP, et al. Pharmacokinetics of 2 oral paracetamol formulations in hospitalized octogenarians. *British Journal of Clinical Pharmacology*. 2021.
2. Van Der Heijden LT, Mian P, Hias J, De Winter BCM, Tournoy J, Van Der Linden L, et al. Highly Variable Paracetamol Pharmacokinetics After Multiple Oral Dosing in Frail Older People: A Population Pharmacokinetic Analysis. *Drugs & Aging*. 2022;39(1):83-95.
3. Langeskov EK, Kristensen K. Population pharmacokinetic of paracetamol and atorvastatin with co-administration of semaglutide. *Pharmacol Res Perspect*. 2022;10(4):e00962.
4. Drusano GL, Johnson DE, Rosen M, Standiford HC. Pharmacodynamics of a fluoroquinolone antimicrobial agent in a neutropenic rat model of *Pseudomonas* sepsis. *Antimicrob Agents Chemother*. 1993;37(3):483-90.
5. Blaser J, Stone BB, Groner MC, Zinner SH. Comparative study with enoxacin and netilmicin in a pharmacodynamic model to determine importance of ratio of antibiotic peak concentration to MIC for bactericidal activity and emergence of resistance. *Antimicrob Agents Chemother*. 1987;31(7):1054-60.
6. Ambrose PG, Bhavnani SM, Owens RC, Jr. Clinical pharmacodynamics of quinolones. *Infect Dis Clin North Am*. 2003;17(3):529-43.
7. Zelenitsky S, Ariano R, Harding G, Forrest A. Evaluating ciprofloxacin dosing for *Pseudomonas aeruginosa* infection by using clinical outcome-based Monte Carlo simulations. *Antimicrob Agents Chemother*. 2005;49(10):4009-14.
8. Ruff CT, Giugliano RP, Braunwald E, Morrow DA, Murphy SA, Kuder JF, et al. Association between edoxaban dose, concentration, anti-Factor Xa activity, and outcomes: an analysis of data from the randomised, double-blind ENGAGE AF-TIMI 48 trial. *Lancet*. 2015;385(9984):2288-95.
9. Eikelboom JW, Quinlan DJ, Hirsh J, Connolly SJ, Weitz JI. Laboratory Monitoring of Non-Vitamin K Antagonist Oral Anticoagulant Use in Patients With Atrial Fibrillation. *JAMA Cardiology*. 2017;2(5):566.
10. Hoogendijk EO, Suanet B, Dent E, Deeg DJ, Aartsen MJ. Adverse effects of frailty on social functioning in older adults: Results from the Longitudinal Aging Study Amsterdam. *Maturitas*. 2016;83:45-50.
11. Cesari M, Leeuwenburgh C, Lauretani F, Onder G, Bandinelli S, Maraldi C, et al. Frailty syndrome and skeletal muscle: results from the Invecchiare in Chianti study. *Am J Clin Nutr*. 2006;83(5):1142-8.
12. Delafuente JC. Pharmacokinetic and pharmacodynamic alterations in the geriatric patient. *Consult Pharm*. 2008;23(4):324-34.
13. Hilmer SN, Tran K, Rubie P, Wright J, Gnjdic D, Mitchell SJ, et al. Gentamicin pharmacokinetics in old age and frailty. *Br J Clin Pharmacol*. 2011;71(2):224-31.
14. Kim IH, Kisseleva T, Brenner DA. Aging and liver disease. *Curr Opin Gastroenterol*. 2015;31(3):184-91.
15. Hilmer SN, Shenfield GM, Le Couteur DG. Clinical implications of changes in hepatic drug metabolism in older people. *Ther Clin Risk Manag*. 2005;1(2):151-6.
16. Mizokami F, Shibasaki M, Yoshizue Y, Noro T, Mizuno T, Furuta K. Pharmacodynamics of vancomycin in elderly patients aged 75 years or older with methicillin-resistant *Staphylococcus aureus* hospital-acquired pneumonia. *Clin Interv Aging*. 2013;8:1015-21.

17. Ma P, Ma H, Liu R, Wen H, Li H, Huang Y, et al. Prediction of vancomycin plasma concentration in elderly patients based on multi-algorithm mining combined with population pharmacokinetics. *Sci Rep.* 2024;14(1):27165.
18. Keutzer L, You H, Farnoud A, Nyberg J, Wicha SG, Maher-Edwards G, et al. Machine Learning and Pharmacometrics for Prediction of Pharmacokinetic Data: Differences, Similarities and Challenges Illustrated with Rifampicin. *Pharmaceutics.* 2022;14(8).

Summary

The global population is aging, with 11% of Belgium's population in 2022 aged 75 and older. Geriatric patients, often multimorbid and polymedicated, undergo physiological changes that affect drug pharmacokinetics (PK) and pharmacodynamics (PD). Our group identified high PK variability, particularly in drug absorption, which remains unexplained. Changes in gastrointestinal (GI) physiology, such as intraluminal pH and transit times, may contribute to this variability, although data on GI alterations in older adults is lacking. This variability is especially concerning for high-risk medications like oral anticoagulants. Direct oral anticoagulants (DOACs) are preferred over vitamin K antagonists for stroke prevention in geriatric atrial fibrillation (AF) patients, but data on their use in this population is limited. Vancomycin, another high-risk drug due to its narrow therapeutic window, requires therapeutic drug monitoring (TDM). Recent TDM guidelines now focus on AUC_{24h}/MIC , complicating monitoring. Model-informed precision dosing using population pharmacokinetics (popPK) models facilitates vancomycin dosing, but its predictive performance in older patients with high variability remains uncertain. The aim of this PhD thesis is to investigate the pharmacokinetics and GI physiology in geriatric patients and older adults aged 75 years and older, to ultimately improve the safety and efficacy of commonly used, high-risk medications with narrow therapeutic windows in this vulnerable population.

In **Chapter 1**, a prospective study was conducted in geriatric inpatients and community-dwelling adults to assess GI transit and intraluminal pH using a wireless motility capsule system. Data from these participants were combined with those from a previous study involving healthy young volunteers. Multivariable linear regression models were developed to identify factors influencing GI transit time and pH. We recruited 30 subjects (median age 80 [IQR=77–83] years). When compared to young volunteers (median age 21 [20–33] years), geriatric inpatients (N=8) exhibited prolonged gastric emptying times ($p<0.01$). Community-dwelling older adults (N=22) did not demonstrate delayed gastric emptying ($p=0.85$). Proton pump inhibitors (PPI) use was associated with increased gastric pH ($p<0.01$) in the pooled analysis.

In **Chapter 2**, a systematic search was conducted to identify PK/PD studies of apixaban, dabigatran, edoxaban, and rivaroxaban in older adults. The review found that age alone did not significantly impact the exposure of edoxaban, rivaroxaban, or dabigatran, while apixaban peak concentrations were 40% higher in older adults compared to young volunteers. However, considerable interindividual variability in DOAC exposure was observed in older adults. This variability can be attributed to factors such as impaired kidney function, changes in body composition (particularly reduced muscle mass), and the use of P-gp inhibitors, aligning with current dose reduction guidelines for apixaban, edoxaban, and rivaroxaban. Dabigatran exhibited the greatest variability, as its dose adjustment criterion is based solely on age, making it a less preferable option for older adults. Moreover, lower and higher DOAC exposure were strongly associated with stroke and bleeding events, respectively, although no specific therapeutic range for these outcomes have been established. The insights from this review guided the design of the prospective PK study in the following chapter.

In the prospective PK study (**Chapter 3**), edoxaban PK was evaluated in geriatric patients. A population PK (popPK) model was developed using a nonlinear mixed-effects approach to simulate various dosing regimens. The model, based on data from frail geriatric patients (median age 87, IQR 83-90), was best described by a two-compartment structure, with proton pump inhibitors decreasing relative bioavailability. Nearly half of the patients exhibited higher-than-expected trough concentrations, and edoxaban concentrations were strongly correlated with anti-Xa activity levels. Simulations demonstrated that a 15 mg dose had a 72.3% probability of achieving the same range trough concentrations observed in ENGAGE-AF TIMI 48 study, compared to 54.7% for the 30 mg dose and 15.7% for the 60 mg dose.

In **Chapter 4**, we conducted a retrospective analysis of early vancomycin concentrations in older and younger patients. Multivariable analyses were used to evaluate the relationship between baseline characteristics and supratherapeutic exposure (logistic regression) as well as dose-normalized concentrations (linear regression). The study included 449 patients aged ≥ 75 years (median age 80) and 1,609 patients aged < 75 years (median age 61). Univariable analysis revealed that older adults had significantly higher

initial vancomycin concentrations ($p < 0.001$) and more frequently reached supratherapeutic levels. In the multivariable analysis, decreased eGFR (CKD-EPI) was associated with supratherapeutic concentrations, while administering lower loading and maintenance doses was inversely related to supratherapeutic concentrations. Predictors of higher dose-normalized concentrations included decreased eGFR (CKD-EPI), lower body weight, elevated blood urea nitrogen, and delayed timing of therapeutic drug monitoring.

Finally in **Chapter 5**, published popPK models were evaluated for their predictive performance using retrospective data from older patients on vancomycin, each with at least three vancomycin concentration measurements. The performance of individual models and a multi-model algorithm—incorporating model selection and averaging—was assessed through (i) a priori predictions based solely on patient covariates and (ii) a posteriori predictions using both covariates and prior vancomycin concentrations. Data from 255 older adults (median age: 79, IQR: 77–84) contributing 765 vancomycin concentrations were analyzed. The most accurate dosing predictions in this population were achieved when using the most recent sample, with popPK models developed specifically for older populations performing best. The Thomson model demonstrated the highest accuracy, followed closely by the Ji model.

In conclusion, edoxaban pharmacokinetics were significantly altered in frail geriatric patients compared to those observed in pivotal trials, with lower clearance, reduced volume of distribution, and a slower absorption rate. The delayed gastric emptying time in frail elderly patients likely contributed to the slower absorption. Advanced age did not directly influence PK. The effect of age was confounded by covariates, such as declining renal and liver function, as shown in the vancomycin study. These changes can be effectively captured and modeled using popPK approaches.

Curriculum Vitae

Angela **Elma** Edwina

PHD RESEARCHER

Born on 21 January 1995 in Semarang, Indonesia

Nationality: Indonesian

Noormannenstraat 68, Leuven

a.e.edwina@gmail.com

Gerontology & Geriatrics, Department of Public Health and Primary Care

KU Leuven

Herestraat 49 – PO box 7003, UZ Gasthuisberg, 3000 Leuven, Belgium

elma.edwina@kuleuven.be

Education

2021-present	KU Leuven, Belgium Doctoral training in Biomedical Sciences within Marie Skłodowska-Curie Doctoral Network AGePOP (Drug Absorption in Geriatric Patients and Older People) agepop.eu
2019-2021	University of Groningen, the Netherlands MSc Medical Pharmaceutical Sciences Toxicology and Drug Disposition track Graduated Cum Laude
2017-2018	Gadjah Mada University Doctor of Pharmacy Graduated Cum Laude Minored in Clinical and Community Pharmacy
2013-2017	Gadjah Mada University Bachelor of Pharmacy Graduated Cum Laude Minored in Clinical and Community Pharmacy

Research experience

2021-present	PhD thesis KU Leuven, Belgium "Pharmacokinetics in Aging: Optimizing Drug Therapies for Geriatric Patients" Gerontology & Geriatrics, Department of Public Health and Primary Care, KU Leuven Supervisor: Prof. dr. Jos Tournoy – co-supervisors: Prof. Apr. Isabel Spriet, Prof. Apr. Lorenz Van der Linden, Prof. Apr. Erwin Dreesen
Jan – March 2024	AbbVie Deutschland GmbH & Co KG , Germany Industrial internship as a part of AGePOP training network Project: Biorelevant in vitro tools to simulate edoxaban dissolution in geriatric patients Supervisors: Dr. Mirko Koziolok, Dr. Alvaro Lopez Marmol

Sep 2020 – May 2021

Erasmus Medical Center, Rotterdam, the Netherlands
Intern, Department of Pharmacy, Clinical Pharmacometrics group

Projects:

Model-Informed Precision Dosing of Antibiotics in Pediatric Patients: A Narrative Review

Population plasma and urine pharmacokinetics and the probability of target attainment of fosfomycin in healthy male volunteers

Supervisors: Prof. Birgit Koch, Dr. Sebastiaan Sassen, Dr. Alan Abdulla

Nov 2019 – Jun 2020

University Medical Center Groningen, the Netherlands
Intern, Department of Clinical Pharmacy and Pharmacology

Project:

Ganciclovir Therapeutic Drug Monitoring in Organ Transplant Recipients

Supervisors: Prof. Daan Touw, Dr. Anne-Grete Märtsen

Working experience

2021-present

PhD researcher KU Leuven, Belgium

2018-2019

Gadjah Mada University, Yogyakarta, Indonesia
Research assistant, Laboratory of Pharmacology and Toxicology, Department of Clinical Pharmacy and Pharmacology

Grants

2024

FWO travel grant for short course abroad
Uppsala Pharmacometrics summer School 2024

2023

Daiichi-Sankyo research grant
Edoxaban PK study in geriatric patients
Supervisor: Prof. Jos Tournoy

March 2023

Student/Trainee grant, American Society of Clinical Pharmacology and Therapeutics

Teaching experience

2022-2024

KU Leuven, Belgium
PhD researcher, Gerontology and geriatrics

Master's students advised

Louise Debruyne, "Edoxaban exposure in geriatric patients with atrial fibrillation"

Hanne Boon, "Geriatric patients attitudes towards participation in clinical studies"

Peer-reviewed publications

Included in this PhD thesis

Edwina, A.E.*, Van der Linden, L.*, Debruyne, L., Foerster, K., Burhenne, J., Spriet, I., Dreesen, E. †, Tournoy, J. † († shared last authors, *shared first author) (2024)

Assessing the Suitability of Fixed-Dose Edoxaban in Frail Geriatric Patients: A Population Pharmacokinetics Study. Submitted

Edwina, A.E.*, Dia, N.*, Gijssen, M., Van der Linden, L., Spriet, I., Tournoy, J.†, Dreesen, E.† (* shared first authors, † shared last authors) (2024). **Optimizing vancomycin dosing in older patients: predictive performance of pharmacometrics models.** In preparation

Edwina AE, Koziol M, Steenackers N, Vanuytsel T, López Mármol Á, Matthys C, Dreesen E, Van der Linden L, Spriet I, Tournoy J. **Tracking the journey: gastrointestinal transit and intraluminal pH dynamics in aging populations using wireless motility capsule.** International Journal of Pharmaceutics, 2024 Dec 5;669:125016

Edwina, A.E., Dreesen, E., Gijssen, M., van den Hout, H.C., Desmet, S., Flamaing, J., van der Linden, L., Spriet, I., Tournoy, J. (2024). **Decreased Kidney Function Explains Higher Vancomycin Exposure in Older Adults.** DRUGS & AGING, 41 (9), 753-762

Demeester, C.†, Robins, D.†, **Edwina, A.E.†**, Tournoy, J., Augustijns, P., Ince, I., Lehmann, A., Vertzoni, M., Schlender, J.F. († shared first authors) (2023). **Physiologically based pharmacokinetic (PBPK) modelling of oral drug absorption in older adults-an AGePOP review.** European Journal of Pharmaceutical Sciences, 188.

Edwina, A.E., et al. (2023). **Insights into the Pharmacokinetics and Pharmacodynamics of Direct Oral Anticoagulants in Older Adults with Atrial Fibrillation: A Structured Narrative Review.** Clinical Pharmacokinetics.

Other publications

Edwina, A.E., Koch, B., Muller, A., al Jalali, V., Matzneller, P., Zeitlinger, M., Sassen, S. (2023). **Population plasma and urine pharmacokinetics and the probability of target attainment of fosfomycin in healthy male volunteers.** European Journal of Clinical Pharmacology.

Edwina, A.E., Dreesen, E., Hias, J., Koch, B.C P., Van den Eede, N., Pauwels, S., Allegaert, K., Van der Linden, L., Spriet, I., Tournoy, J (2024). **Agreement Between a Colorimetric Assay and Ultra-Performance Liquid Chromatography-Tandem Mass Spectrometry for Quantifying Paracetamol Plasma Concentrations.** AAPS Journal, 26 (1)

Murwanti, R., Nurrochmad, A., Gani, A.P., Sasmito, E., **Edwina, A.E.**, Chandra, M.K., Suryawan, F.H., Wardana, A.R., Natalia, , Budiningsih, J.L S R. (2023). **Acute and Subchronic Oral Toxicity Evaluation of Herbal Formulation: Piper crocatum Ruiz and Pav., Typhonium flagelliforme (Lodd.) Blume, and Phyllanthus niruri L. in Sprague-Dawley Rats.** J Toxicol, 2023.

Oktora, M.P., **Edwina, A.E.**, Denig, P. (2022). **Differences in Older Patients' Attitudes**

Toward Deprescribing at Contextual and Individual Level. *Frontiers in Public Health.*

Martson, A-G., **Edwina, A.E.**, Kim, H.Y., Knoester, M., Touw, D.J., Sturkenboom, M.G G., Alffenaar, J-W.C. (2022). **Therapeutic Drug Monitoring of Ganciclovir: Where Are We?** *Therapeutic Drug Monitoring*, 44 (1).

Märtson, A-G., Sturkenboom, M.G G., Knoester, M., van der Werf, T.S., **Edwina, A.E.**, Burgerhof, J.G M., Berger, S.P., de Joode, A., Damman, K., Verschuuren, E.A M., Blokzijl, H., Bakker, M., Touw, D., Alffenaar, J-W.C., Hope, W. (2021). **Standard ganciclovir dosing results in slow decline of cytomegalovirus viral loads.** *Journal Of Antimicrobial Chemotherapy*, 77 (2).

Martson, A-G., **Edwina, A.E.**, Burgerhof, J.G M., Berger, S.P., de Joode, A., Damman, K., Verschuuren, E.A M., Blokzijl, H., Bakker, M., Span, L.F., van der Werf, T.S., Touw, D.J., Sturkenboom, M.G G., Knoester, M., Alffenaar, J.W C. (2021). **Ganciclovir therapeutic drug monitoring in transplant recipients.** *Journal of Antimicrobial Chemotherapy*, 76 (9).

Abdulla, A., **Edwina, A.E.**, Flint, R.B., Allegaert, K., Wildschut, E.D., Koch, B.C P., de Hoog, M. (2021). **Model-Informed Precision Dosing of Antibiotics in Pediatric Patients: A Narrative Review.** *Frontiers in Pediatrics*, 9

Presentations at congresses

March 2023	Poster presentation "Comparison of a Colorimetric Assay and UPLC-MS/MS for Quantifying Paracetamol Plasma Concentrations: Design of a Geriatric Pharmacokinetics Study". Presented at the Annual Meeting of the American-Society-for-Clinical-Pharmacology-and-Therapeutics (ASCPT)
Sept 2021	Oral presentation "Population plasma and urine pharmacokinetics and the probability of target attainment of fosfomycin in healthy male volunteers", Presented at the International Association of Therapeutic Drug Monitoring & Clinical Toxicology
July 2021	Oral presentation "Population plasma and urine pharmacokinetics and the probability of target attainment of fosfomycin in healthy male volunteers", Presented at the European Congress of Clinical Microbiology & Infectious Diseases

Training

2024	Uppsala Pharmacometrics Summer School
2021-2024	AGePOP training https://www.agepop.eu/training/
2022 and 2023	Scientific integrity
2022	Applied pharmacometrics
2022	An open-source Pharmacometrician's workflow in R: from exploration (xGx) to model building (nlmixr) and diagnostics (ggPMX)
2021	Good Clinical Practice

Academic involvement

Peer-reviewer

European Journal of Clinical Pharmacology
British Journal of Clinical Pharmacology
Clinical Pharmacokinetics

2021-present

**Member, American Society of Clinical Pharmacology and
Therapeutics**

Acknowledgements, Personal Contribution and Conflict of Interest Statements

Acknowledgements

Chapter 1

Elma Edwina, Jos Tournoy, Isabel Spriet, Lorenz Van der Linden, Erwin Dreesen designed the research; Elma Edwina collected data, performed the analysis and wrote the manuscript; Elma Edwina, Mirko Koziolk and Jos Tournoy interpreted the motility capsule data; Jos Tournoy supervised the data collection; Erwin Dreesen supervised the statistical analysis; Mirko Koziolk, Nele Steenackers, Tim Vanuytsel, Álvaro López Mármol, Christophe Matthys, Erwin Dreesen, Lorenz Van der Linden, Isabel Spriet, Jos Tournoy critically reviewed the manuscript. The authors would like to acknowledge the contributions of Louise Debruyne and Hanne Boon who assisted with data collection for this study.

Chapter 2

Elma Edwina, Jos Tournoy, Lorenz Van der Linden provided the concept for this review. Elma Edwina and Nada Dia performed the literature search and data analysis. Elma Edwina drafted the original manuscript. Nada Dia, Erwin Dreesen, Thomas Vanassche, Peter Verhamme, Isabel Spriet, Lorenz Van der Linden and Jos Tournoy critically reviewed the manuscript. The authors wish to thank Chayenne Van Meel and Eline Vancoppenolle, the biomedical reference librarians of the KU Leuven Libraries—2Bergen—learning Centre Désiré Collen (Leuven, Belgium), for their help in providing the assistance with the search strategy.

Chapter 3

Elma Edwina, Jos Tournoy, Isabel Spriet, Lorenz Van der Linden, Erwin Dreesen designed the research. Jos Tournoy, Lorenz Van der Linden provided the concept. Elma Edwina, Louise Debruyne collected the data. Jos Tournoy supervised the data collection. Elma Edwina analyzed the data and performed the modeling and simulation. Erwin Dreesen supervised the modeling. Kathrin Foerster, Jürgen Burhenne performed the bioanalysis. Louise Debruyne, Kathrin Foerster, Jürgen Burhenne, Peter Verhamme, Thomas Vanassche, Isabel Spriet, Lorenz Van der Linden, Erwin Dreesen, Jos Tournoy critically reviewed the manuscript.

Chapter 4

Elma Edwina, Jos Tournoy, Isabel Spriet made substantial contributions to the concept, design, and interpretation of data. Elma Edwina performed the data analysis and wrote the original draft. Erwin Dreesen supervised the data analysis and statistics. Matthias Gijssen provided the dataset. Erwin Dreesen, Matthias Gijssen, Helena Cornelia van den Hout, Stefanie Desmet, Johan Flamaing, Lorenz Van der Linden, Isabel Spriet and Jos Tournoy critically reviewed the manuscript.

Chapter 5

Erwin Dreesen, Isabel Spriet, Jos Tournoy made substantial contributions to the concept and design. Data preparation and analysis were performed by Elma Edwina and Nada Dia. The first draft of the manuscript was written by Elma Edwina and Nada Dia. Erwin Dreesen supervised the project. Matthias Gijssen, Lorenz Van der Linden, Isabel Spriet, Jos Tournoy, Erwin Dreesen critically reviewed the manuscript.

Personal contribution

This thesis was conceived by the author and her promotor, Prof. Jos Tournoy, and co-promotors, Prof. Isabel Spriet, Prof. Lorenz Van der Linden and Prof. Erwin Dreesen. The contribution of the author to this PhD thesis are including:

- the development and design of the clinical studies protocols
- recruitment, screening and inclusion of the volunteers and patients
- performing the clinical study visits
- collecting data
- analyzing data
- performing population PK modeling and simulation
- writing the manuscripts
- writing the thesis

Financial supports



This project has received funding from the European Union's Horizon 2020 research and innovation programme under the Marie Skłodowska-Curie grant agreement No 956146.



Chapter 3 was co-funded by the investigator-initiated grant from Daiichi Sankyo, Inc.

Conflict of interest

The authors report no proprietary or commercial interest in any product mentioned or discussed in this manuscript.